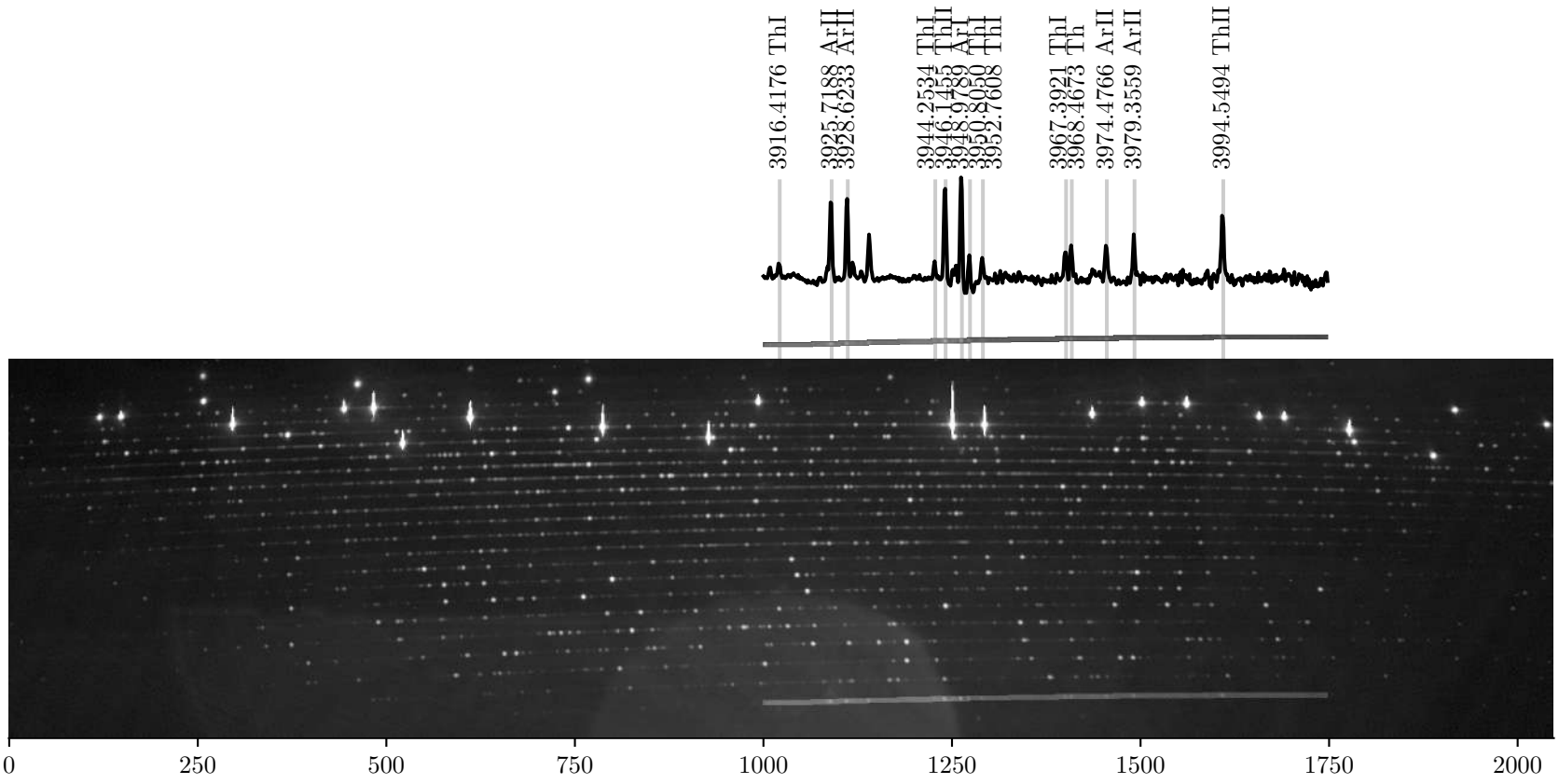
Guide to Medium Resolution ThAr Spectra

Echelle Spectrograph for the Nanshan Observatory

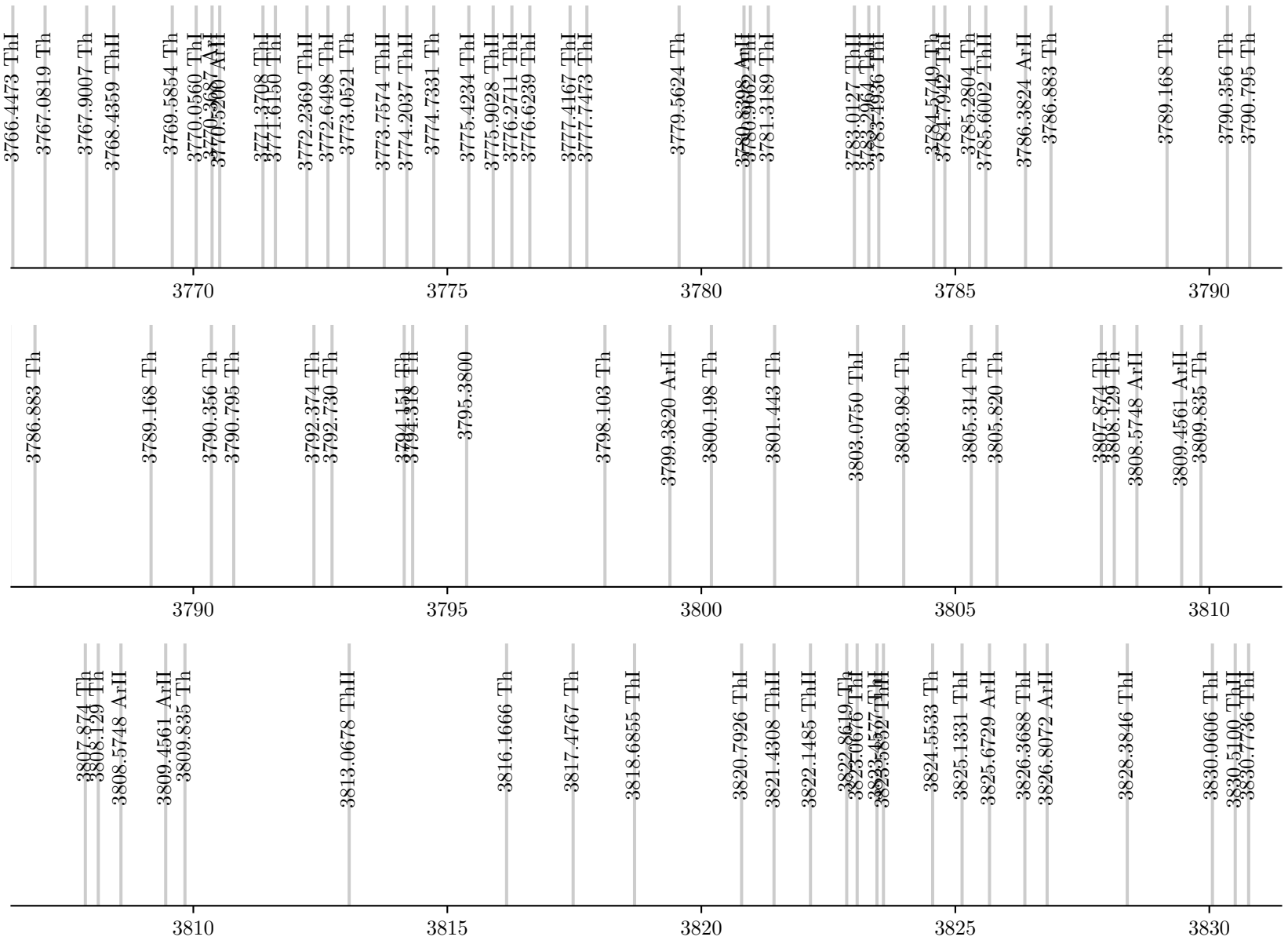
Astronomical Consultants & Equipment, Inc.

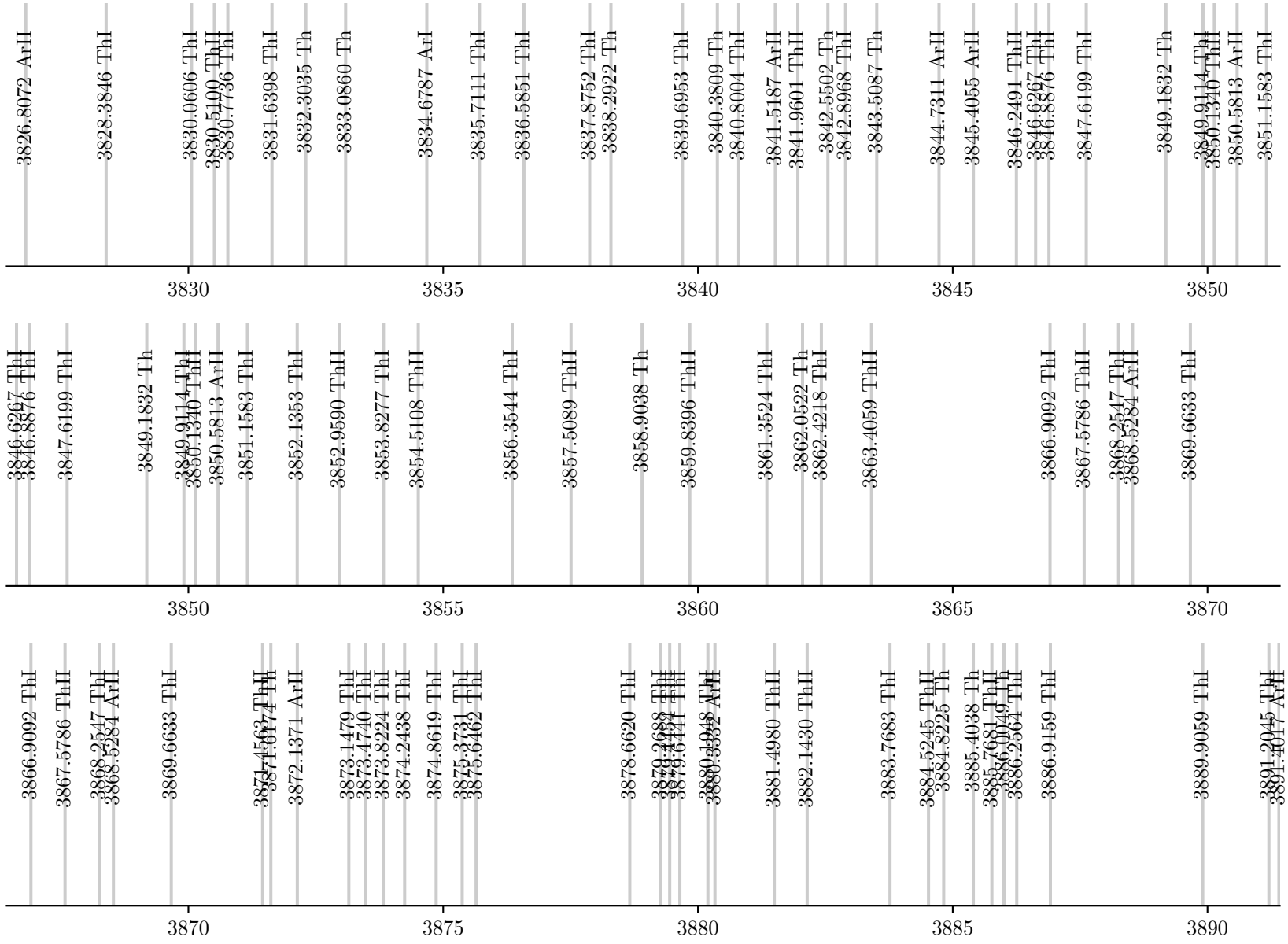
November 9, 2020

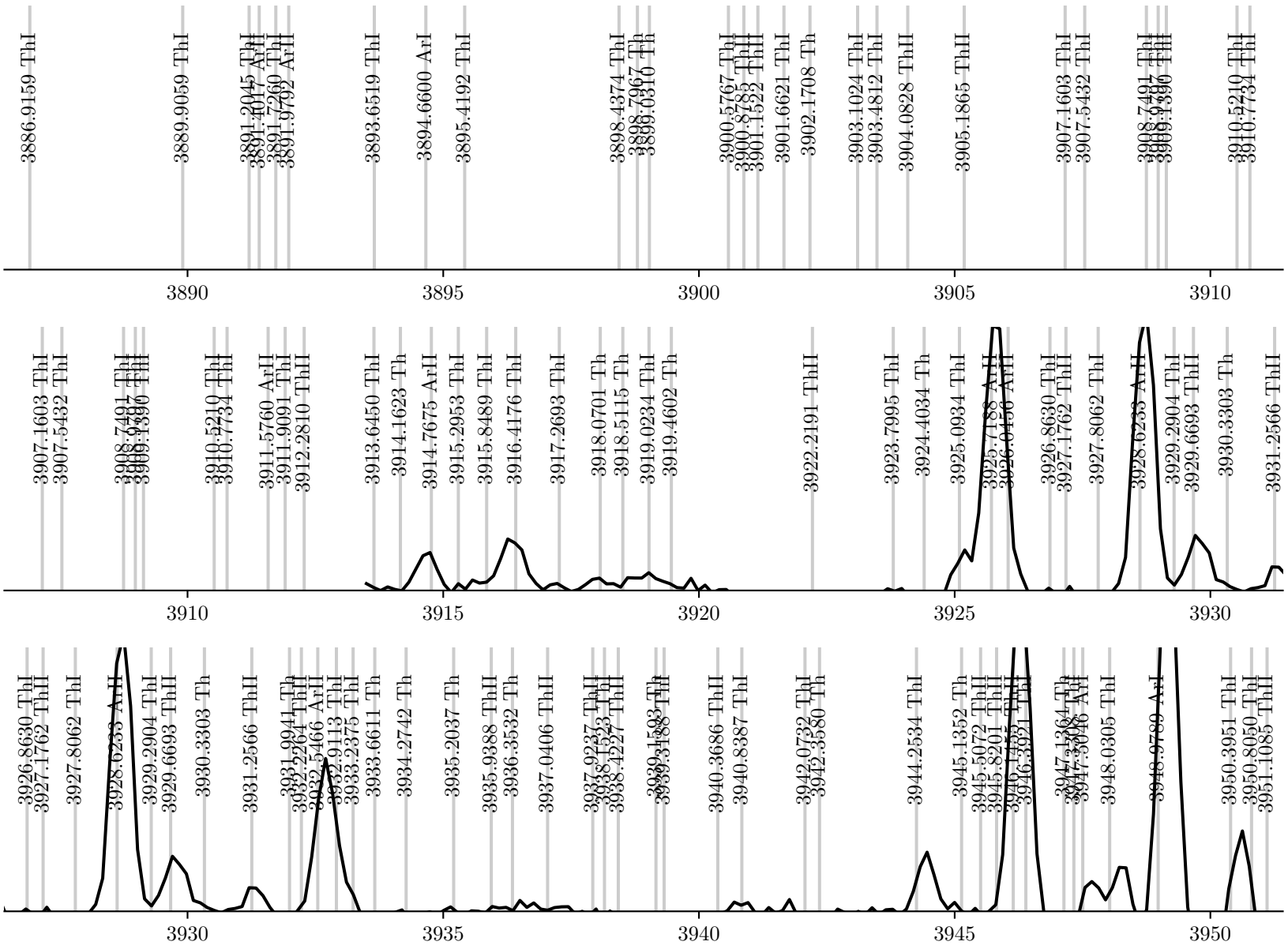
Order 38

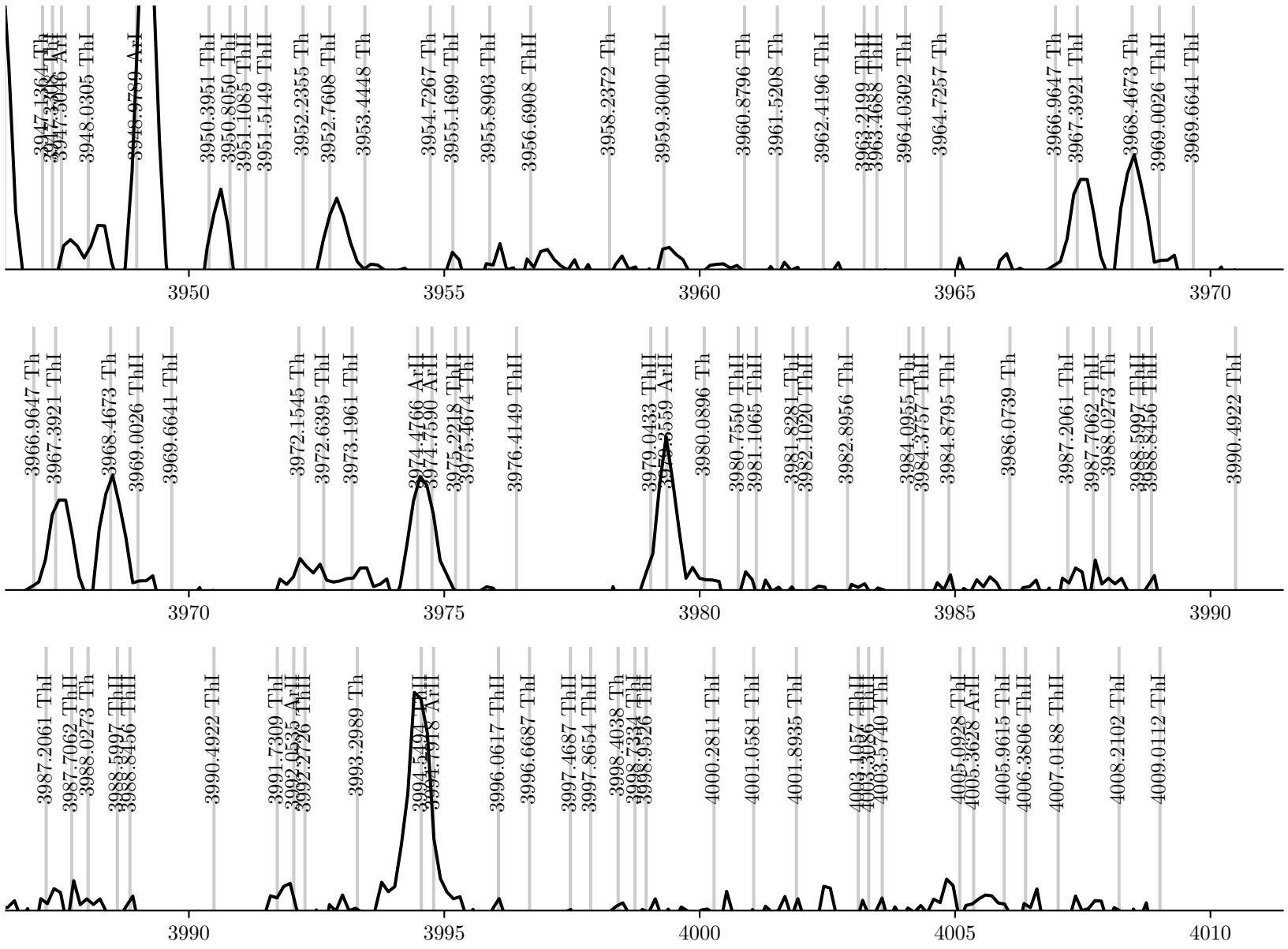


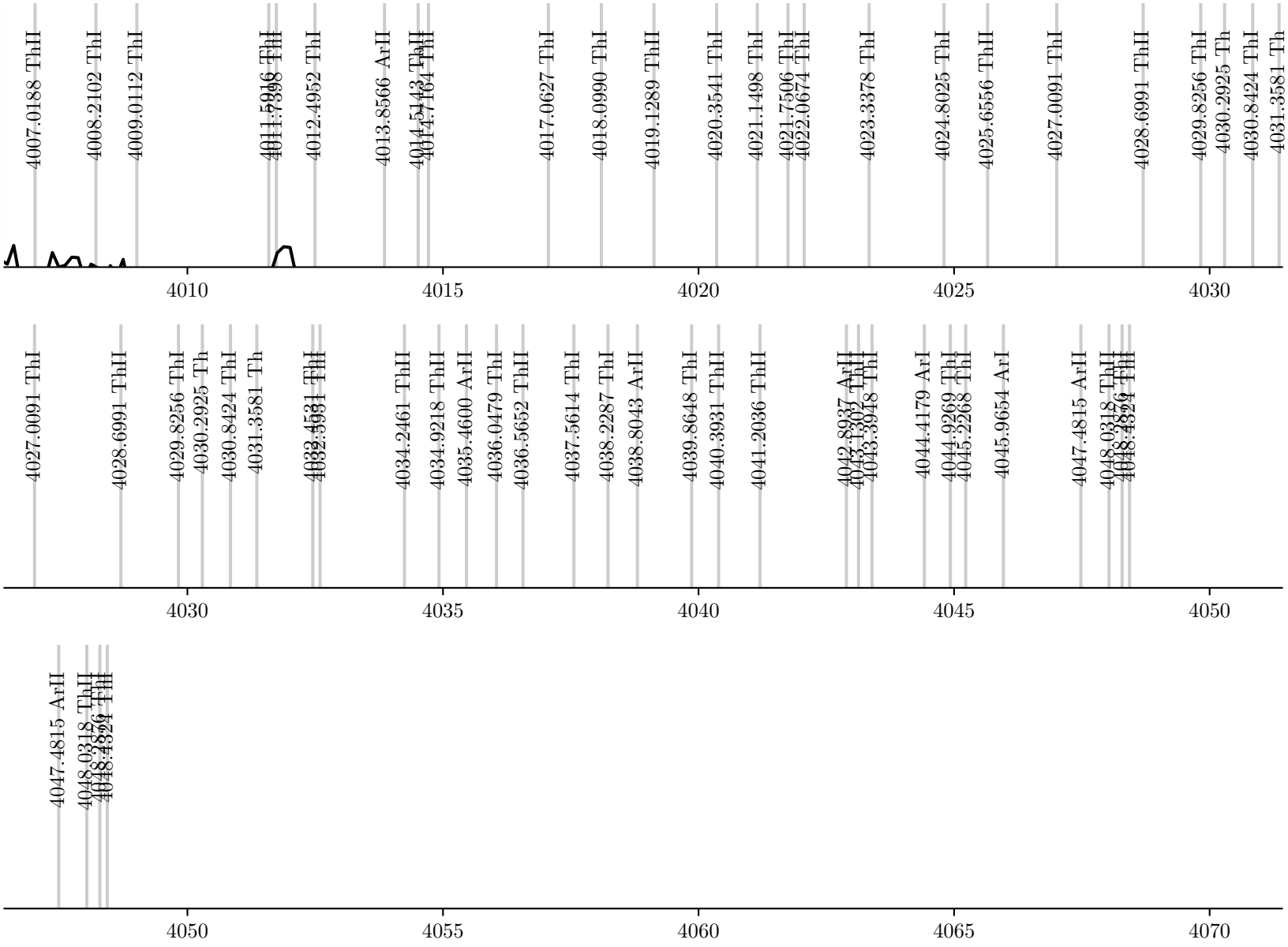
Dispersion:	0.137 Å pix^{-1}
Central wavelength:	3917 Å
Wavelength range:	3769 Å – 4050 Å


3766.4473 ThI
3767.0819 Th
3767.9007 Th
3768.4359 ThII
3769.5854 Th
3770.0560 ThI
3770.3687 ArII
3770.5200 ArII
3771.3708 ThI
3771.8150 ThI
3772.2369 ThII
3772.6498 ThI
3773.0521 Th
3773.7574 ThII
3774.2037 ThII
3774.7331 Th
3775.4234 ThI
3775.9028 ThII
3776.2711 ThI
3776.6239 ThI
3777.4167 ThI
3777.7473 ThI
3779.5624 Th
3780.8308 ArII
3781.3189 ThI
3783.0127 ThII
3783.3964 ThII
3783.4936 ThII
3784.5749 Th
3784.7012 ThI
3785.2804 Th
3785.6002 ThII
3786.3824 ArII
3786.883 Th
3789.168 Th
3790.356 Th
3790.795 Th
3770
3775
3780
3785
3790
3786.883 Th
3789.168 Th
3790.356 Th
3790.795 Th
3792.374 Th
3792.730 Th
3795.3800
3798.103 Th
3799.3820 ArII
3800.198 Th
3801.443 Th
3803.0750 ThI
3803.984 Th
3805.314 Th
3805.820 Th
3807.874 Th
3808.129 Th
3808.5748 ArII
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3809.835 Th
3790
3795
3800
3805
3810
3807.874 Th
3808.129 Th
3808.5748 ArII
3809.4561 ArII
3809.835 Th
3813.0678 ThII
3816.1666 Th
3817.4767 Th
3818.6855 ThI
3820.7926 ThI
3821.4308 ThII
3822.1485 ThII
3824.5533 Th
3825.1331 ThI
3825.6729 ArII
3826.3688 ThI
3826.8072 ArII
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3825
3830

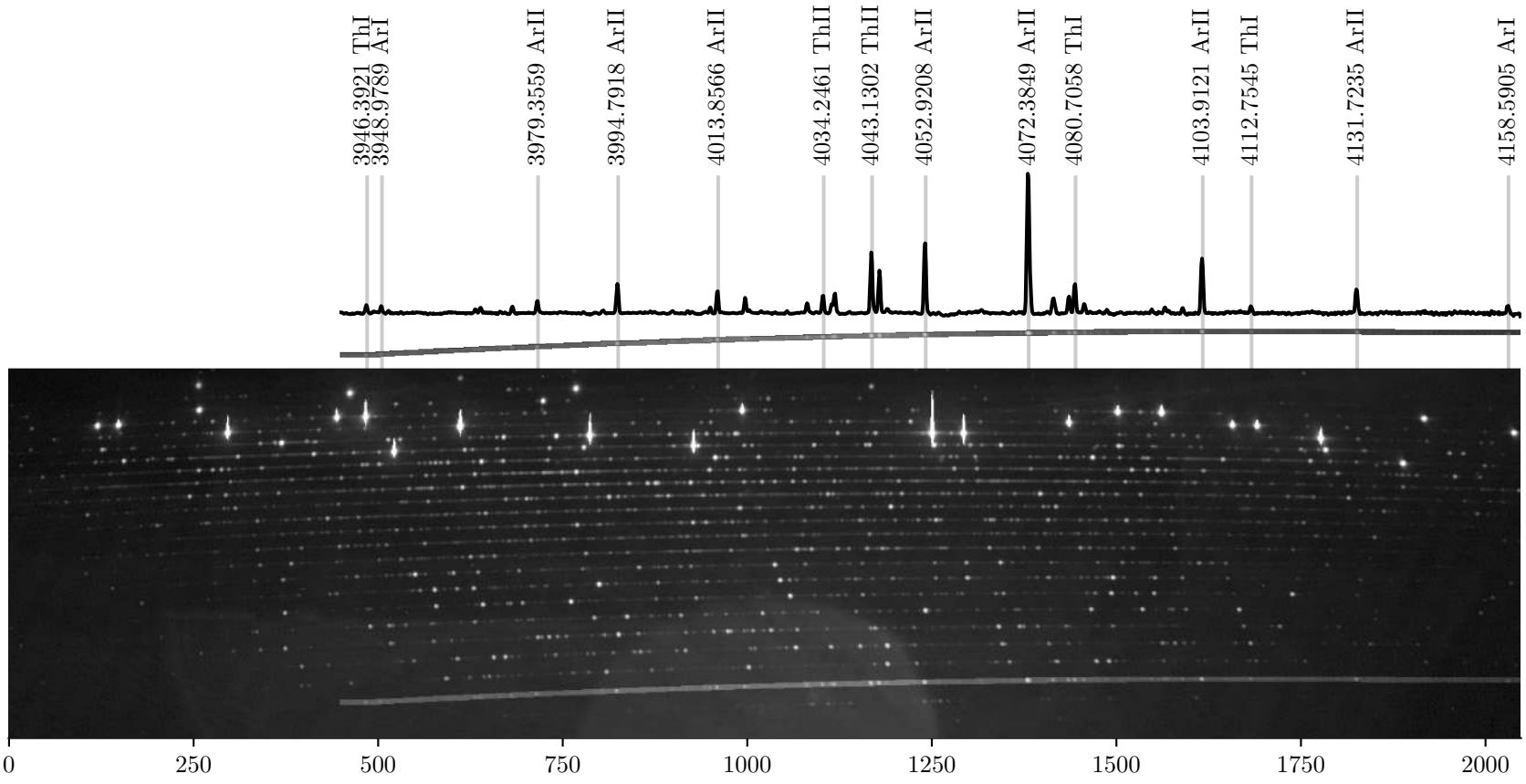

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3833.0860 Th
3834.6787 ArI
3835.7111 ThI
3836.5851 ThI
3837.8752 ThI
3838.2922 Th
3839.6953 ThI
3840.3809 Th
3840.8004 ThI
3841.5187 ArII
3841.9601 ThII
3842.5502 Th
3842.8968 ThI
3843.5087 Th
3844.7311 ArII
3845.4055 ArII
3846.2491 ThII
3846.6267 ThI
3846.8876 ThI
3847.6199 ThI
3849.1832 Th
3849.9114 ThI
3850.1340 ThII
3850.5813 ArII
3851.1583 ThI
3846.6267 ThI
3846.8876 ThI
3847.6199 ThI
3849.1832 Th
3849.9114 ThI
3850.1340 ThII
3850.5813 ArII
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3852.1353 ThI
3852.9590 ThII
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3854.5108 ThII
3856.3544 ThI
3857.5089 ThII
3858.9038 Th
3859.8396 ThII
3861.3524 ThI
3862.0522 Th
3862.4218 ThI
3863.4059 ThII
3866.9092 ThI
3867.5786 ThII
3868.2547 ThI
3868.5284 ArII
3869.6633 ThI
3866.9092 ThI
3867.5786 ThII
3868.2547 ThI
3868.5284 ArII
3869.6633 ThI
3871.4563 ThII
3871.6174 ThI
3872.1371 ArII
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3873.8224 ThI
3874.2438 ThI
3874.8619 ThI
3875.3731 ThI
3875.6462 ThI
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3881.4980 ThII
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3889.9059 ThI
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3870
3875
3880
3885
3890
3850
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3860
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3845
3850


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3900.8785 ThII
3901.1533 ThII
3901.6621 ThI
3902.1708 Th
3903.1024 ThI
3903.4812 ThI
3904.0828 ThII
3905.1865 ThII
3907.1603 ThI
3907.5432 ThI
3908.7491 ThI
3909.1390 ThII
3910.5210 ThI
3910.7734 ThI
3910
3915
3920
3925
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3907.1603 ThI
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3910.5210 ThI
3910.7734 ThI
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3917.2693 ThI
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3925.0934 ThI
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3927.8062 ThI
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3929.2904 ThI
3929.6693 ThII
3930.3303 Th
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3935
3940
3945
3950
3926.8630 ThI
3927.1762 ThII
3927.8062 ThI
3928.6233 ArII
3929.2904 ThI
3929.6693 ThII
3930.3303 Th
3931.2566 ThII
3931.9941 Th
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3932.5466 ArII
3932.9113 ThI
3933.2375 ThI
3933.6611 Th
3934.2742 Th
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3935.9388 ThII
3936.3532 Th
3937.0406 ThII
3937.9237 ThII
3938.1523 ThI
3938.4227 ThII
3939.1593 Th
3939.3188 ThI
3940.3686 ThII
3940.8387 ThI
3942.0732 ThI
3942.3580 Th
3944.2534 ThI
3945.1352 Th
3945.5072 ThII
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3947.1364 Th
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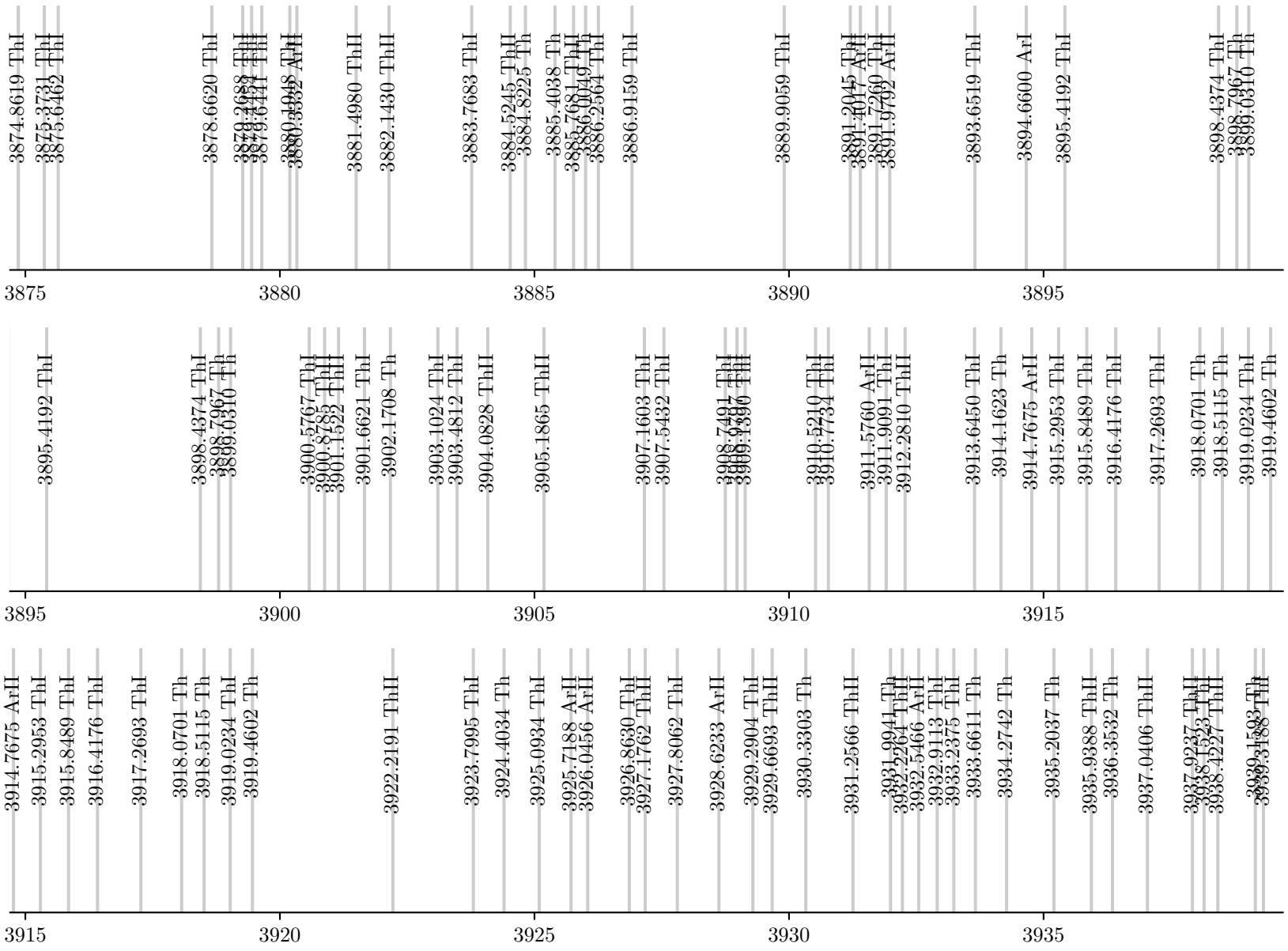

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3948.9789 ArI
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3951.1085 ThII
3951.5149 ThII
3952.2355 Th
3952.7608 ThI
3953.4448 Th
3954.7267 Th
3955.1699 ThI
3955.8903 ThI
3956.6908 ThII
3958.2372 Th
3959.3000 ThI
3960.8796 Th
3961.5208 Th
3962.4196 ThI
3963.2199 ThII
3963.4688 ThII
3964.0302 ThI
3964.7257 Th
3966.9647 Th
3967.3921 ThI
3968.4673 Th
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3969.6641 ThI
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3955
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3965
3970
3966.9647 Th
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3968.4673 Th
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3970
3975
3980
3985
3990
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3992.2726 ThII
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3994.7918 ArII
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3996.6687 ThI
3997.4687 ThII
3997.8654 ThII
3998.4038 Th
3998.7334 ThI
3998.9526 ThI
4000.2811 ThI
4001.0581 ThI
4001.8935 ThI
4003.1057 ThII
4003.3526 ThI
4003.5740 ThI
4005.0028 ThI
4005.3628 ArII
4005.9615 ThI
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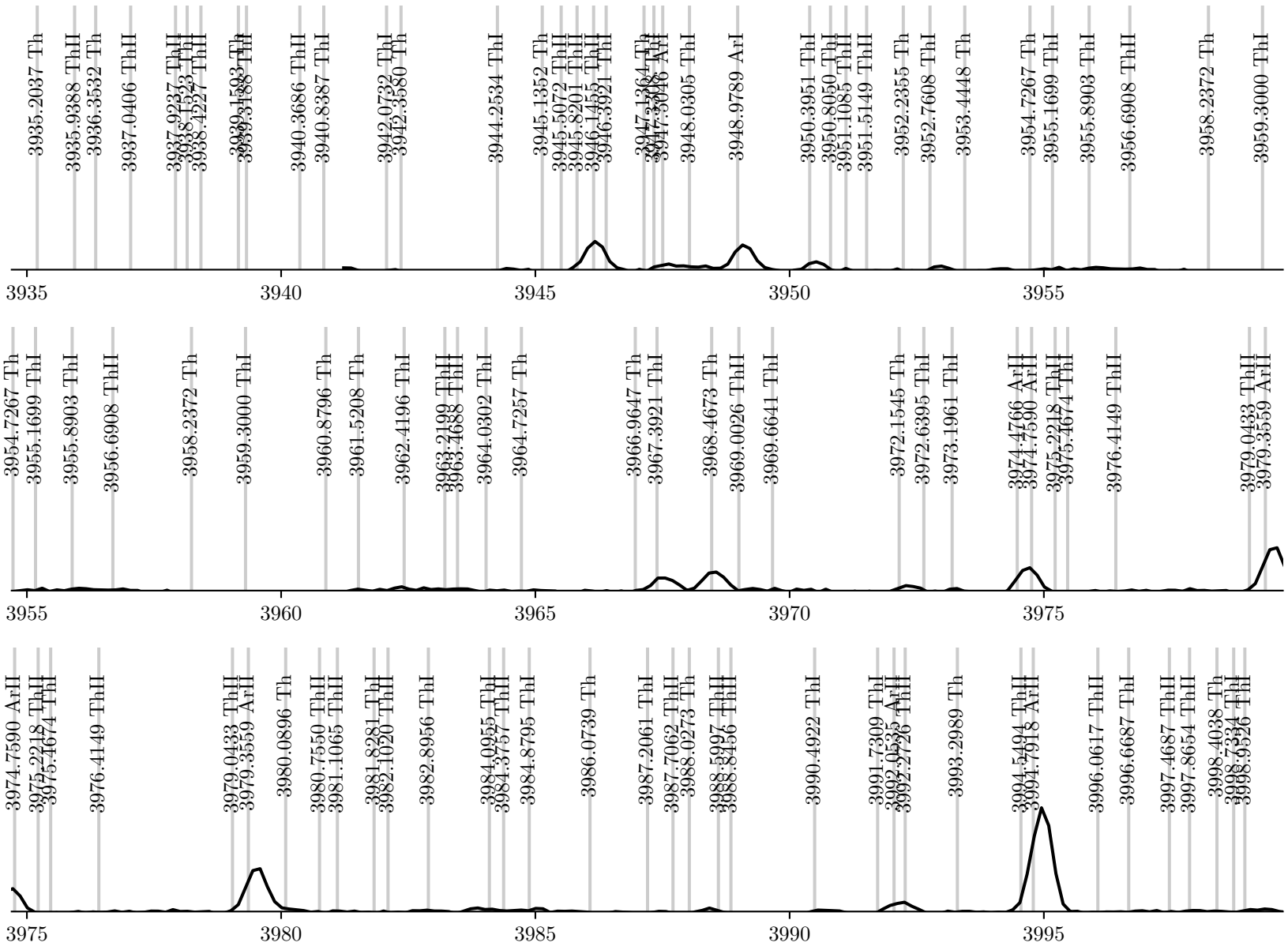

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4018.0990 ThI
4019.1289 ThII
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4021.1498 ThI
4021.7506 ThI
4022.0674 ThI
4023.3378 ThI
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4027.0091 ThI
4028.6991 ThII
4029.8256 ThI
4030.2925 Th
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4031.3581 Th
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4015
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4029.8256 ThI
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4045.2268 ThI
4045.9654 ArI
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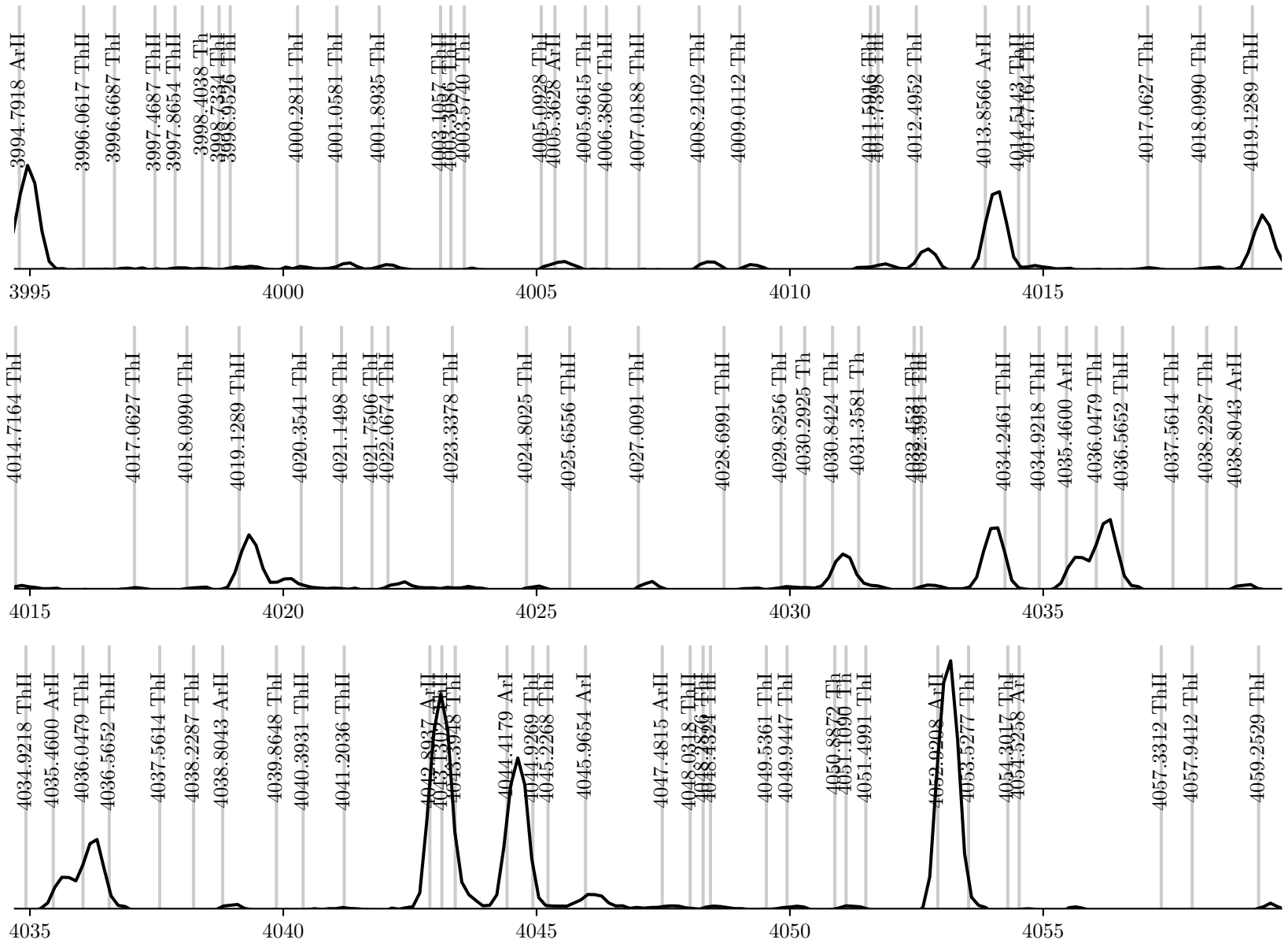
Order 37

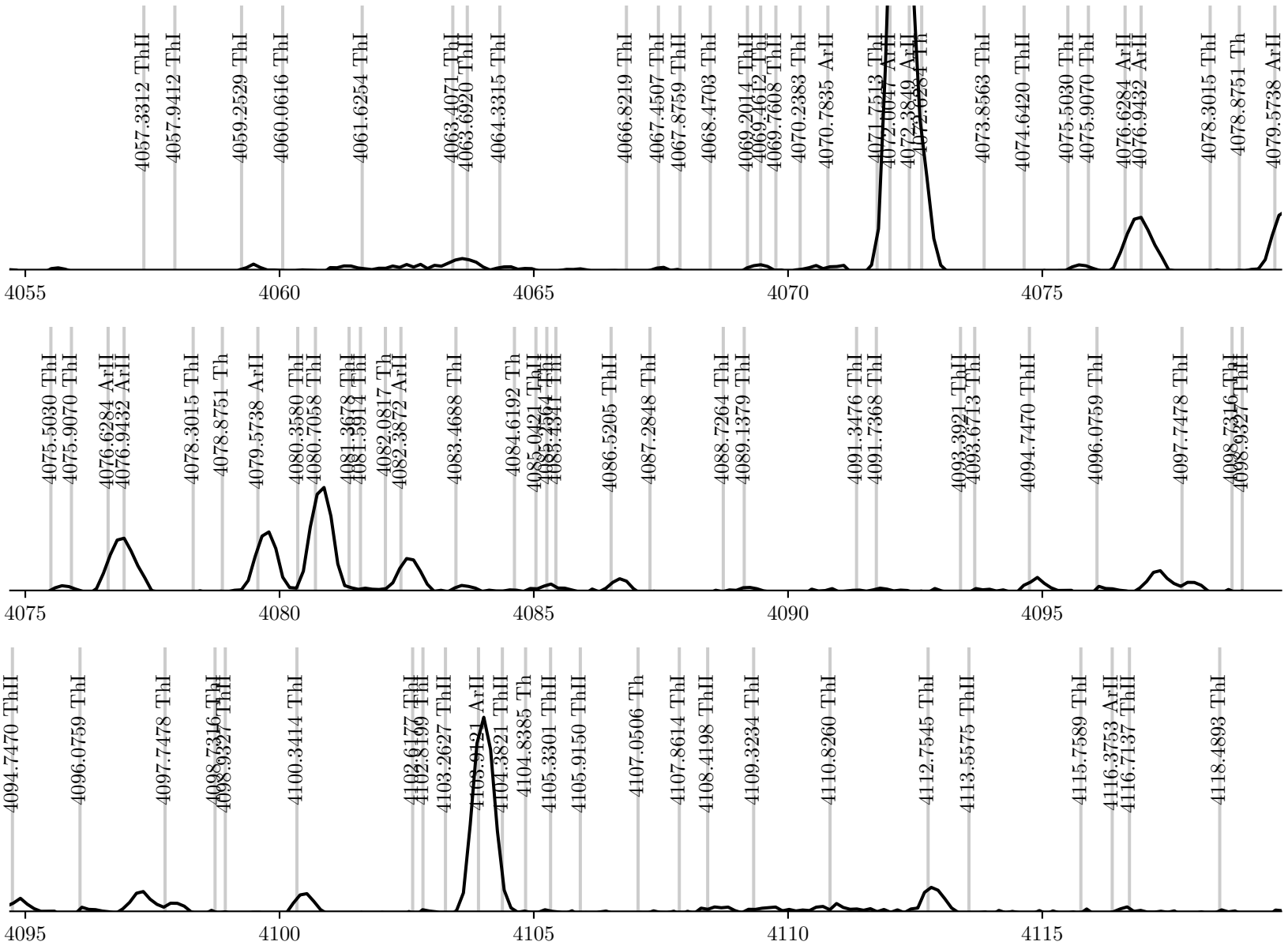


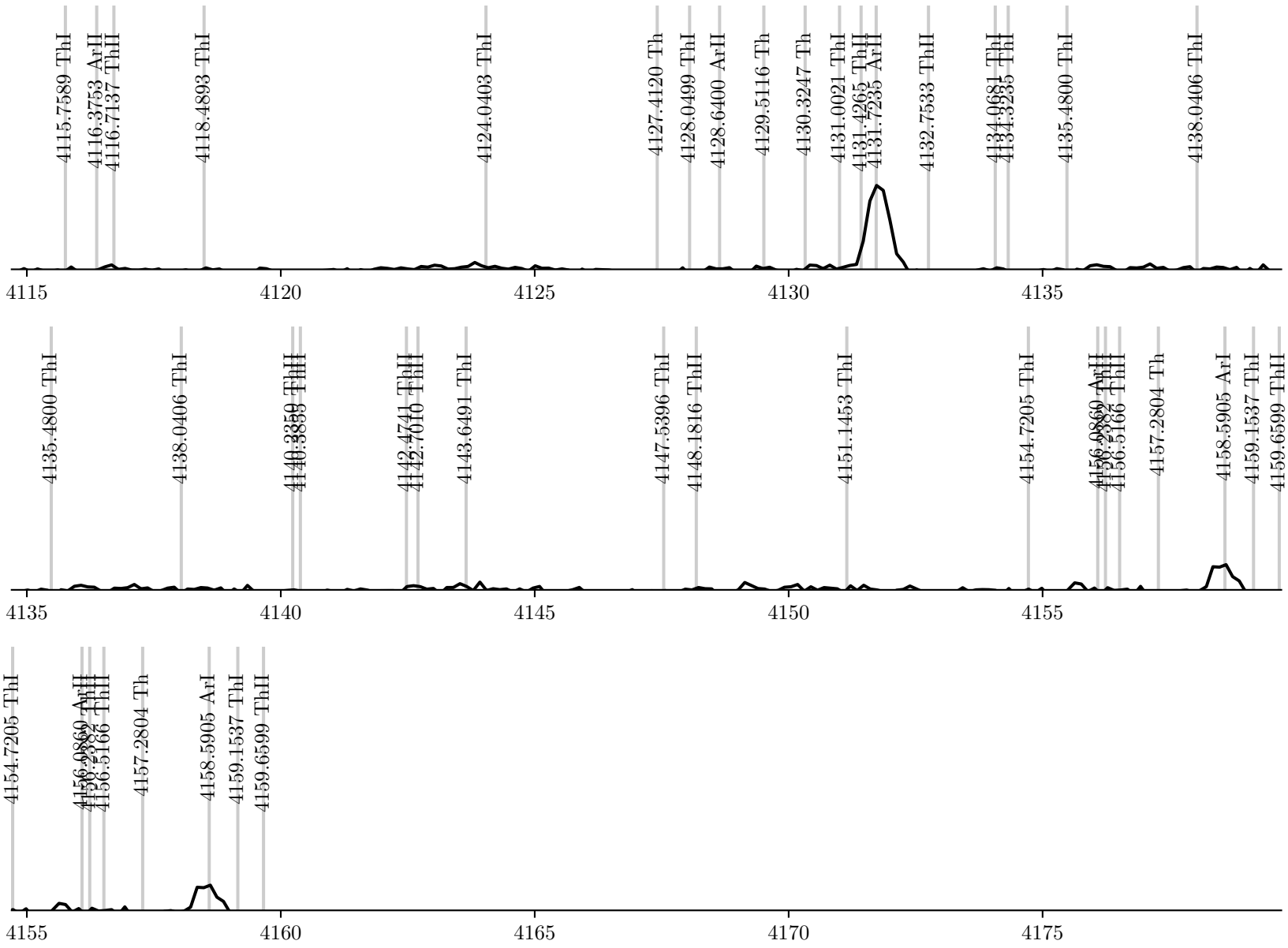
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Central wavelength:	4023 Å
Wavelength range:	3875 Å – 4161 Å


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3878.6620 ThI
3881.4980 ThII
3882.1430 ThII
3883.7683 ThI
3884.5245 ThII
3884.8225 Th
3885.4038 Th
3886.2564 Th
3886.9159 ThI
3889.9059 ThI
3893.6519 ThI
3894.6600 ArI
3895.4192 ThI
3898.4374 ThI
3899.0310 Th
3875
3880
3885
3890
3895
3895.4192 ThI
3898.4374 ThI
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3915.2953 ThI
3915.8489 ThI
3916.4176 ThI
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3918.0701 Th
3918.5115 Th
3919.0234 ThI
3919.4602 Th
3895
3900
3905
3910
3915
3914.7675 ArII
3915.2953 ThI
3915.8489 ThI
3916.4176 ThI
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3925.0934 ThI
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3931.2566 ThII
3931.9941 Th
3932.2264 ThII
3932.5466 ArII
3932.9113 ThI
3933.2375 ThI
3933.6611 Th
3934.2742 Th
3935.2037 Th
3935.9388 ThII
3936.3532 Th
3937.0406 ThII
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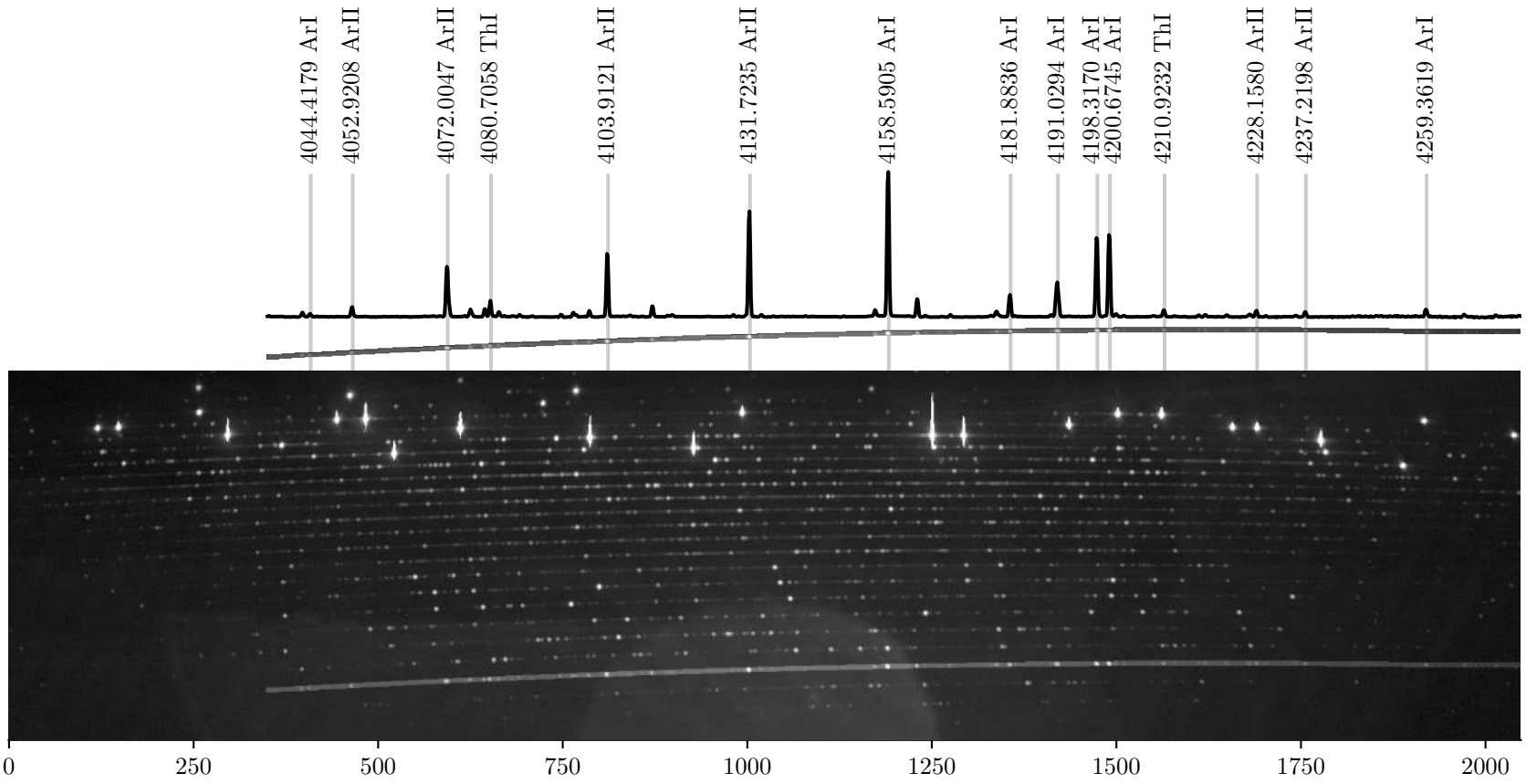

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3945.8201 ThI
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3950.3951 ThI
3950.8050 ThI
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3956.6908 ThII
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3956.6908 ThII
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3979.0433 ThII
3979.3559 ArII
3980.0896 Th
3980.7550 ThII
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3981.8281 ThI
3982.1020 ThII
3982.8956 ThI
3984.0955 ThI
3984.3757 ThII
3984.8795 ThI
3986.0739 Th
3987.2061 ThI
3987.7062 ThII
3988.0273 Th
3990.4922 ThI
3991.7309 ThI
3993.2989 Th
3994.5494 ThII
3996.0617 ThII
3996.6687 ThI
3997.4687 ThII
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3998.7334 ThI
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3990
3995


3994.7918 ArII
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3996.6687 ThI
3997.4687 ThII
3997.8654 ThII
3998.4038 Th
4000.2811 ThI
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4009.0112 ThI
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4015
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4035.4600 ArII
4036.0479 ThI
4036.5652 ThII
4037.5614 ThI
4038.2287 ThI
4038.8043 ArII
4015
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4025
4030
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4034.9218 ThII
4035.4600 ArII
4036.0479 ThI
4036.5652 ThII
4037.5614 ThI
4038.2287 ThI
4038.8043 ArII
4039.8648 ThI
4040.3931 ThII
4041.2036 ThII
4044.4179 ArI
4045.2268 Th
4045.9654 ArI
4047.4815 ArII
4049.5361 ThI
4049.9447 ThI
4051.4991 ThI
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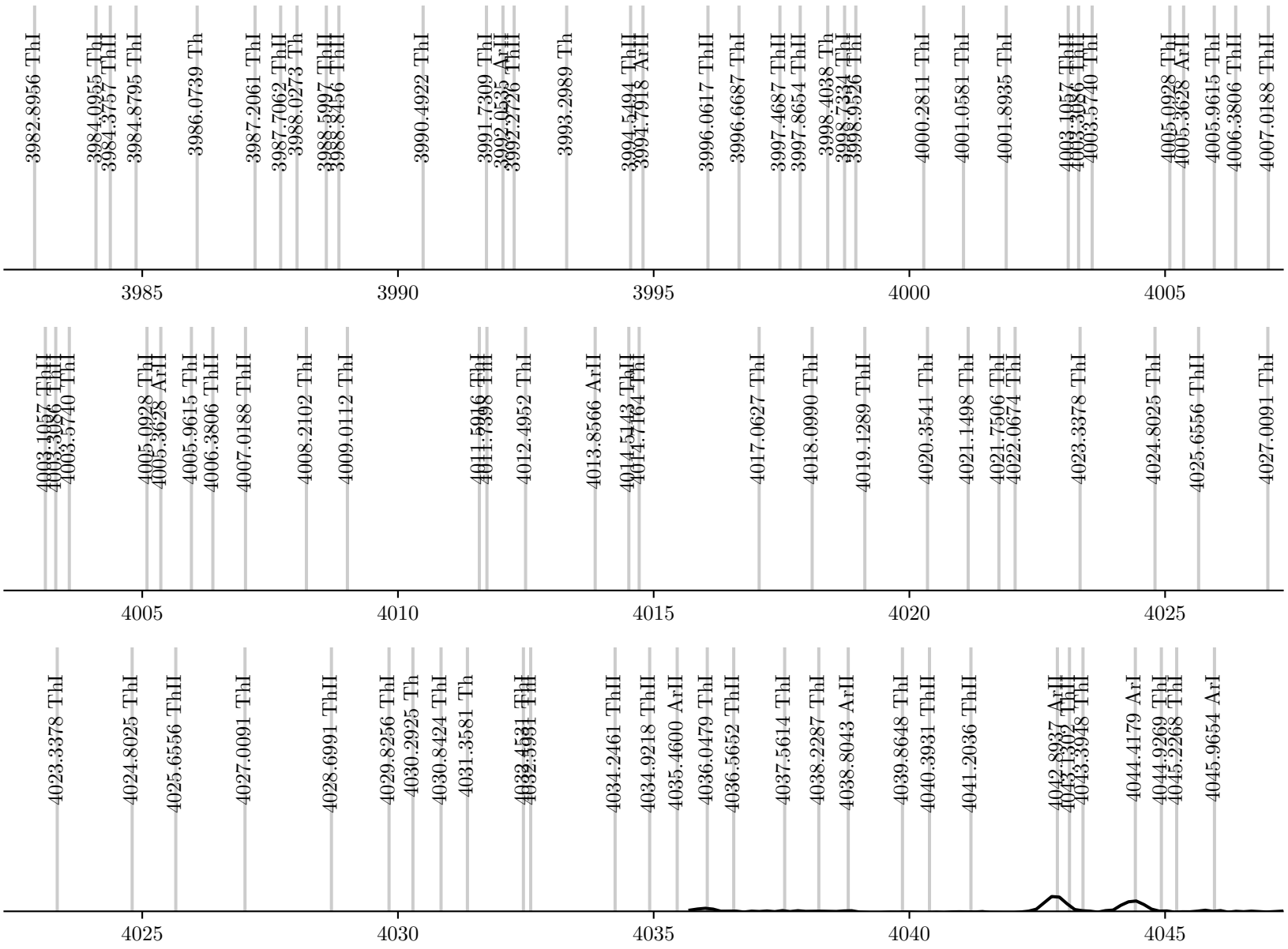

4057.3312 ThII
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4063.4071 ThI
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4064.3315 ThI
4066.8219 ThI
4067.4507 ThI
4067.8759 ThII
4068.4703 ThI
4069.2014 ThII
4069.4612 ThI
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4072.0047 ArII
4072.3849 ArII
4072.6284 ThI
4073.8563 ThI
4074.6420 ThII
4075.5030 ThI
4075.9070 ThI
4076.6284 ArII
4076.9432 ArII
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4078.8751 Th
4079.5738 ArII
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4075.5030 ThI
4075.9070 ThI
4076.6284 ArII
4076.9432 ArII
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4078.8751 Th
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4081.3678 ThI
4081.3914 ThI
4082.0817 Th
4082.3872 ArII
4083.4688 ThI
4084.6192 Th
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4087.2848 ThI
4088.7264 ThI
4089.1379 ThI
4091.3476 ThI
4091.7368 ThI
4093.3921 ThII
4093.6713 ThI
4094.7470 ThII
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4097.7478 ThI
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4094.7470 ThII
4096.0759 ThI
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4098.7316 ThI
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4102.8169 ThI
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4103.9121 ArII
4104.3821 ThII
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4105.3301 ThII
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4110.8260 ThI
4112.7545 ThI
4113.5575 ThII
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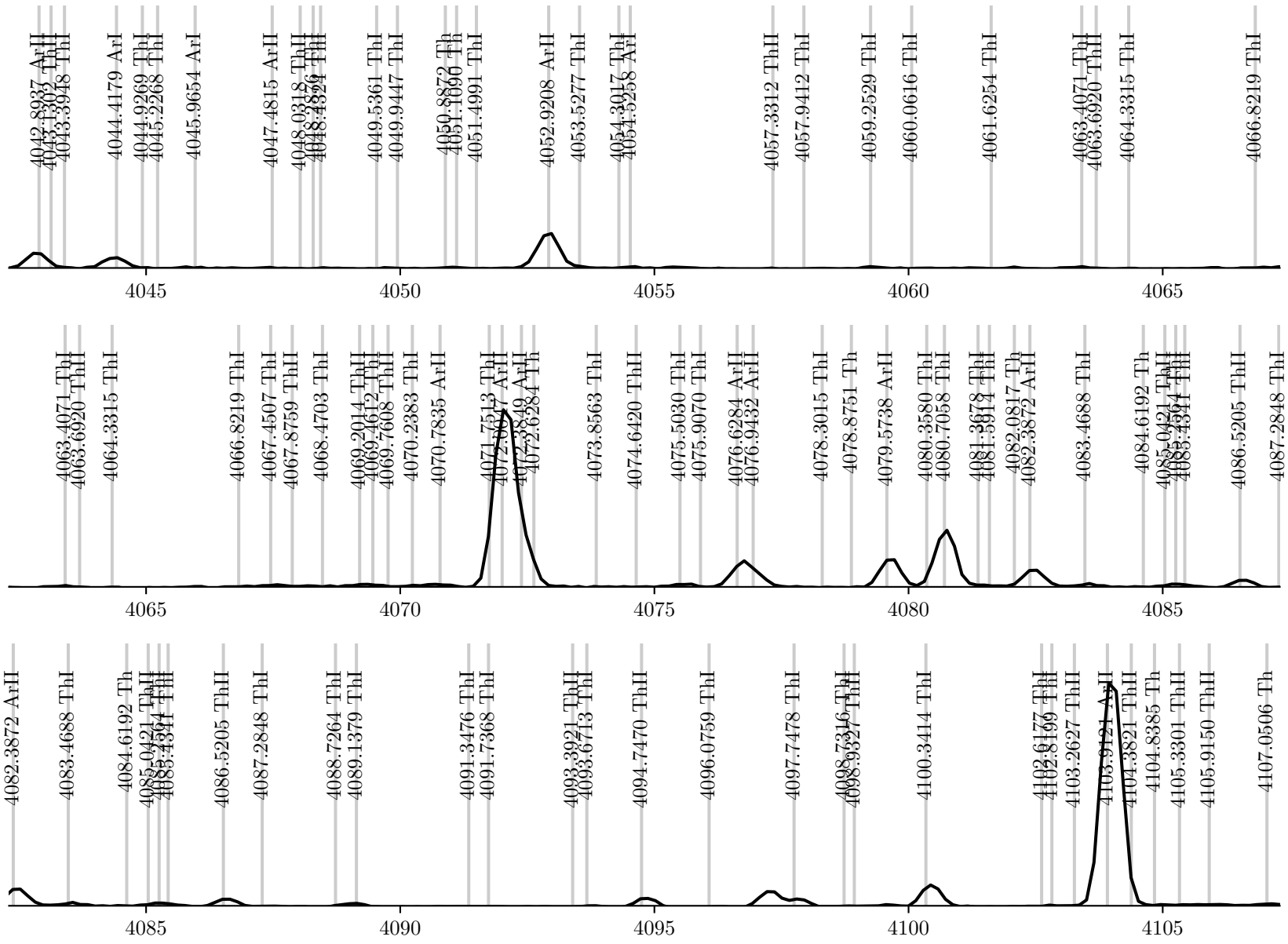

4115.7589 ThI
4116.3753 ArII
4116.7137 ThII
4118.4893 ThI
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4128.6400 ArII
4129.5116 Th
4130.3247 Th
4131.0021 ThI
4131.4265 ThII
4131.7235 ArII
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4138.0406 ThI
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4156.2382 ThII
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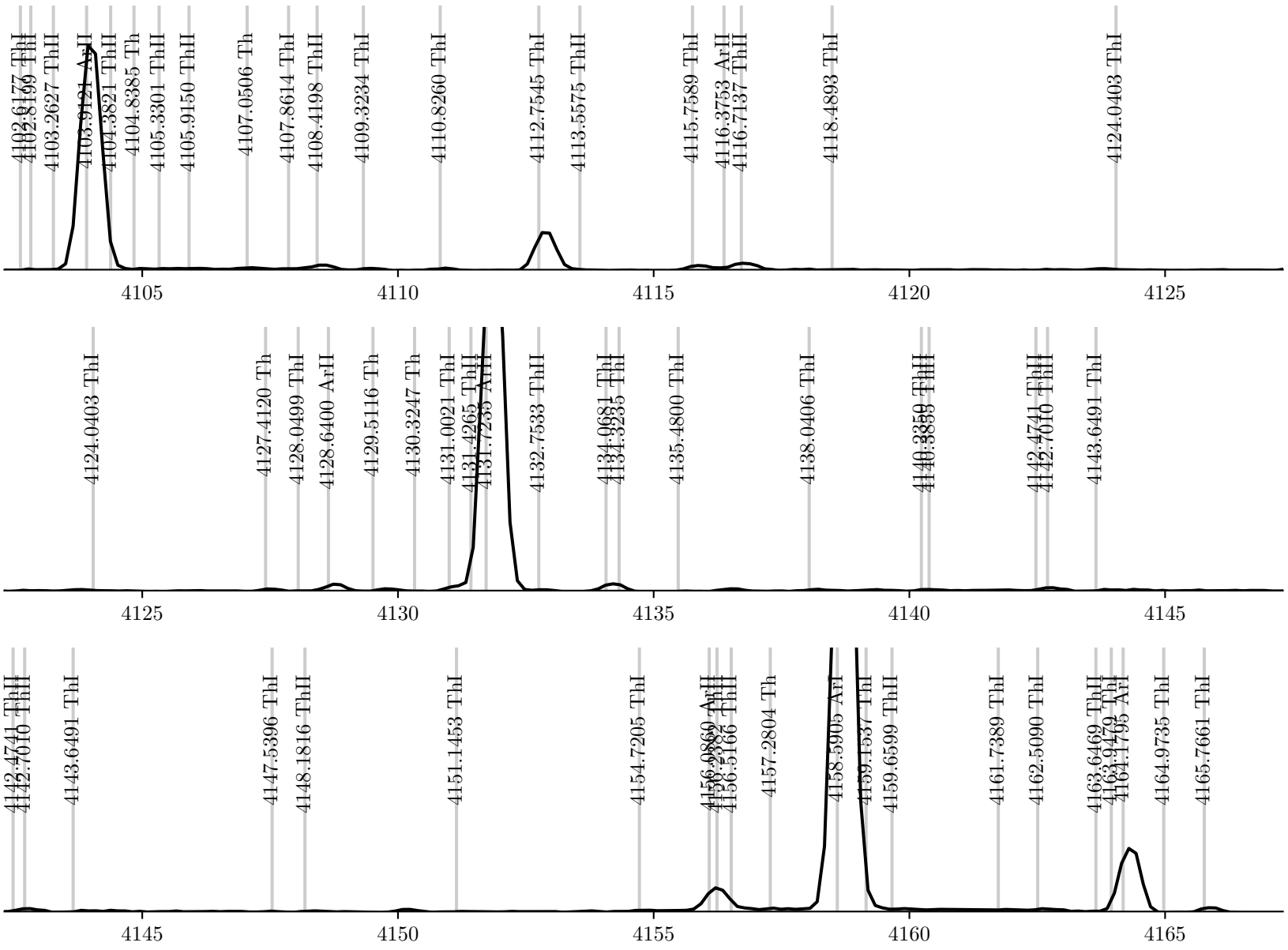
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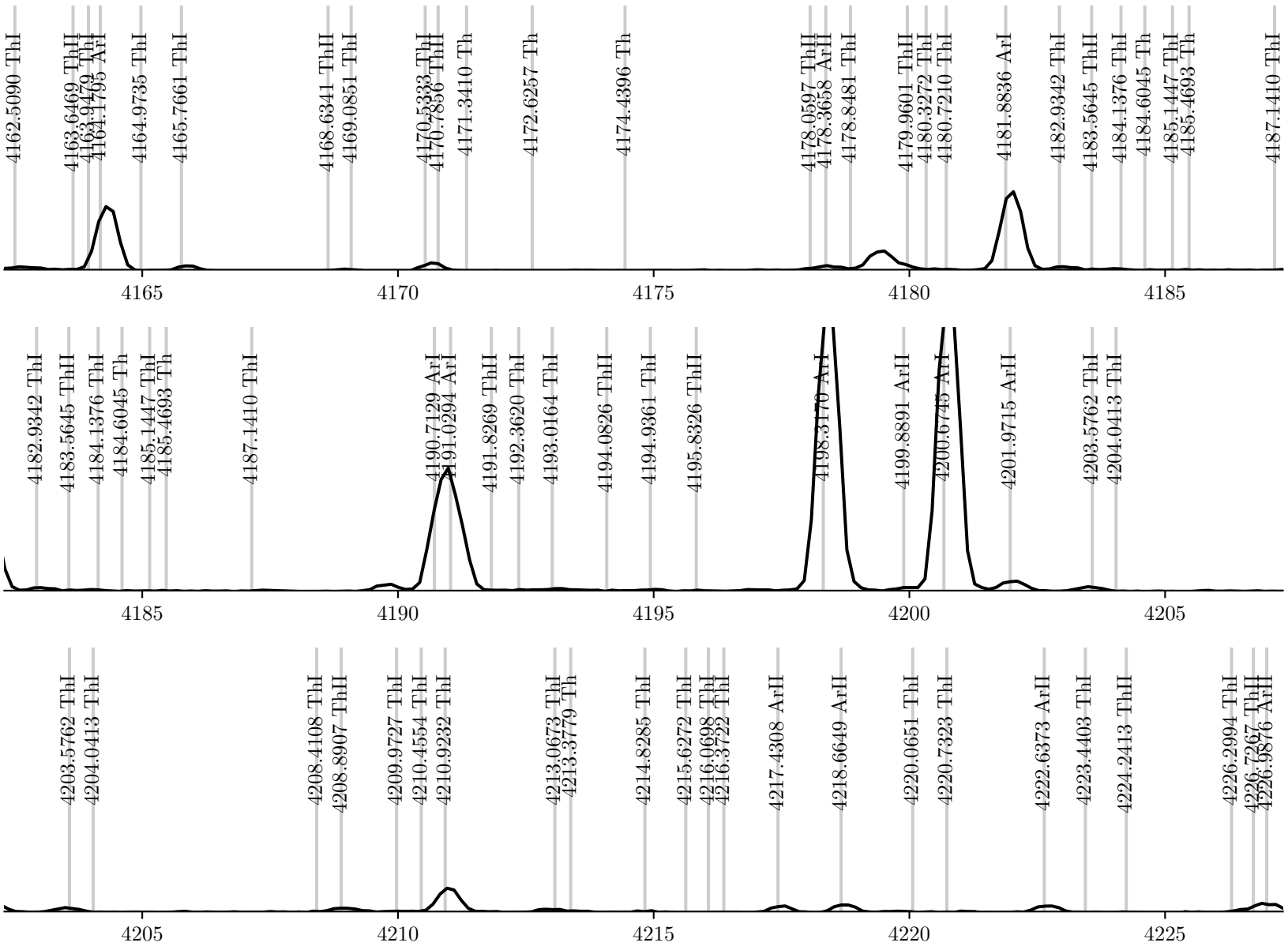


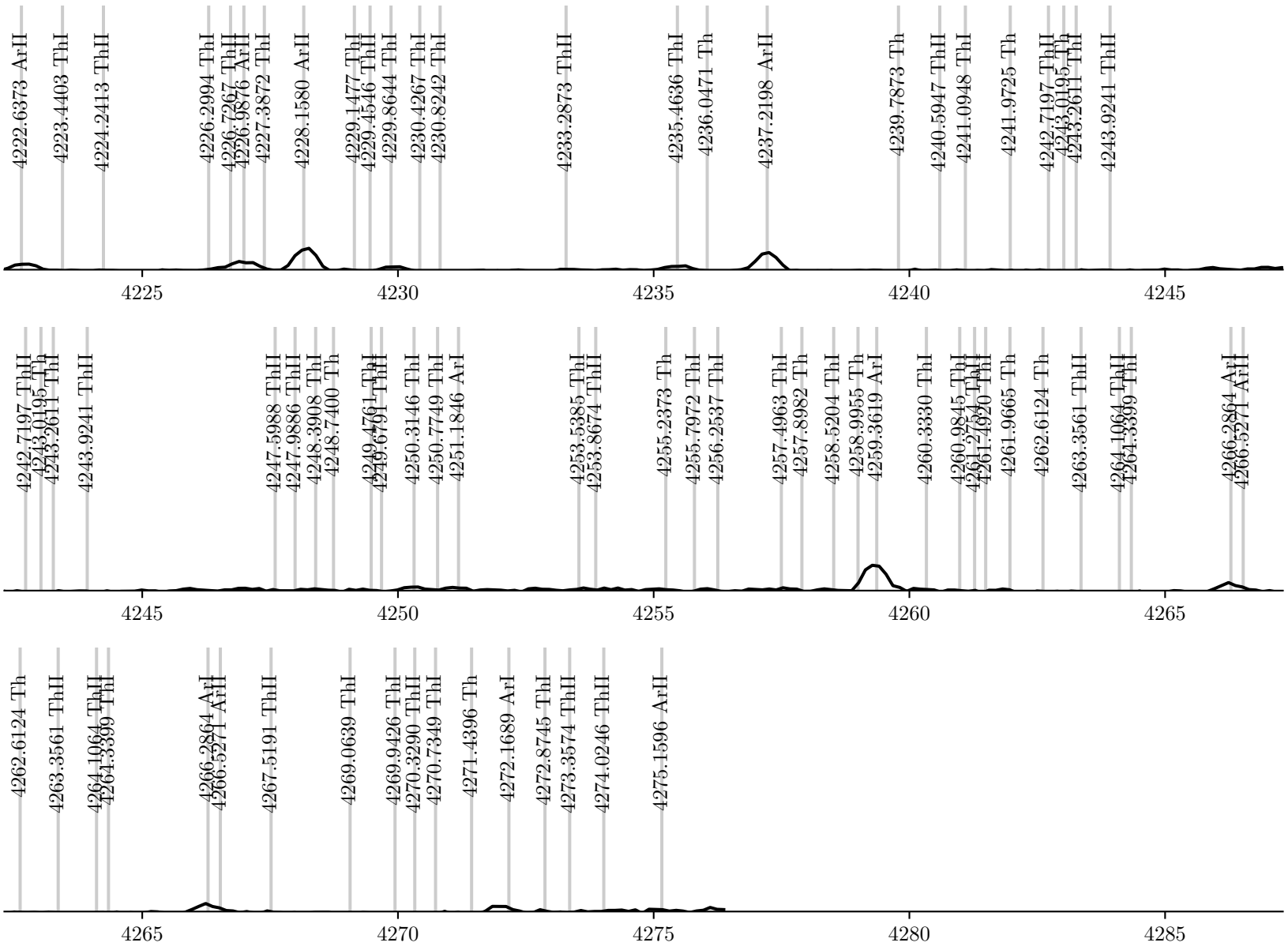
Dispersion:	0.144 Å pix^{-1}
Central wavelength:	4135 Å
Wavelength range:	3983 Å – 4276 Å


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3984.8795 ThI
3986.0739 Th
3987.2061 ThI
3987.7062 ThII
3988.0273 Th
3988.5997 ThII
3988.8456 ThII
3990.4922 ThI
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3992.0535 ArII
3992.2726 ThII
3993.2989 Th
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3997.8654 ThII
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3998.9526 ThI
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4001.8935 ThI
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4003.3086 ThII
4003.5740 ThI
4005.0928 ThI
4005.3628 ArII
4005.9615 ThI
4006.3806 ThII
4007.0188 ThII
3985
3990
3995
4000
4005
4003.1057 ThII
4003.3086 ThII
4003.5740 ThI
4005.0928 ThI
4005.3628 ArII
4005.9615 ThI
4006.3806 ThII
4007.0188 ThII
4008.2102 ThI
4009.0112 ThI
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4014.5443 ThII
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4017.0627 ThI
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4023.3378 ThI
4024.8025 ThI
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4010
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4025
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4040.3931 ThII
4041.2036 ThII
4042.8937 ArII
4043.1307 ThII
4043.3948 ThI
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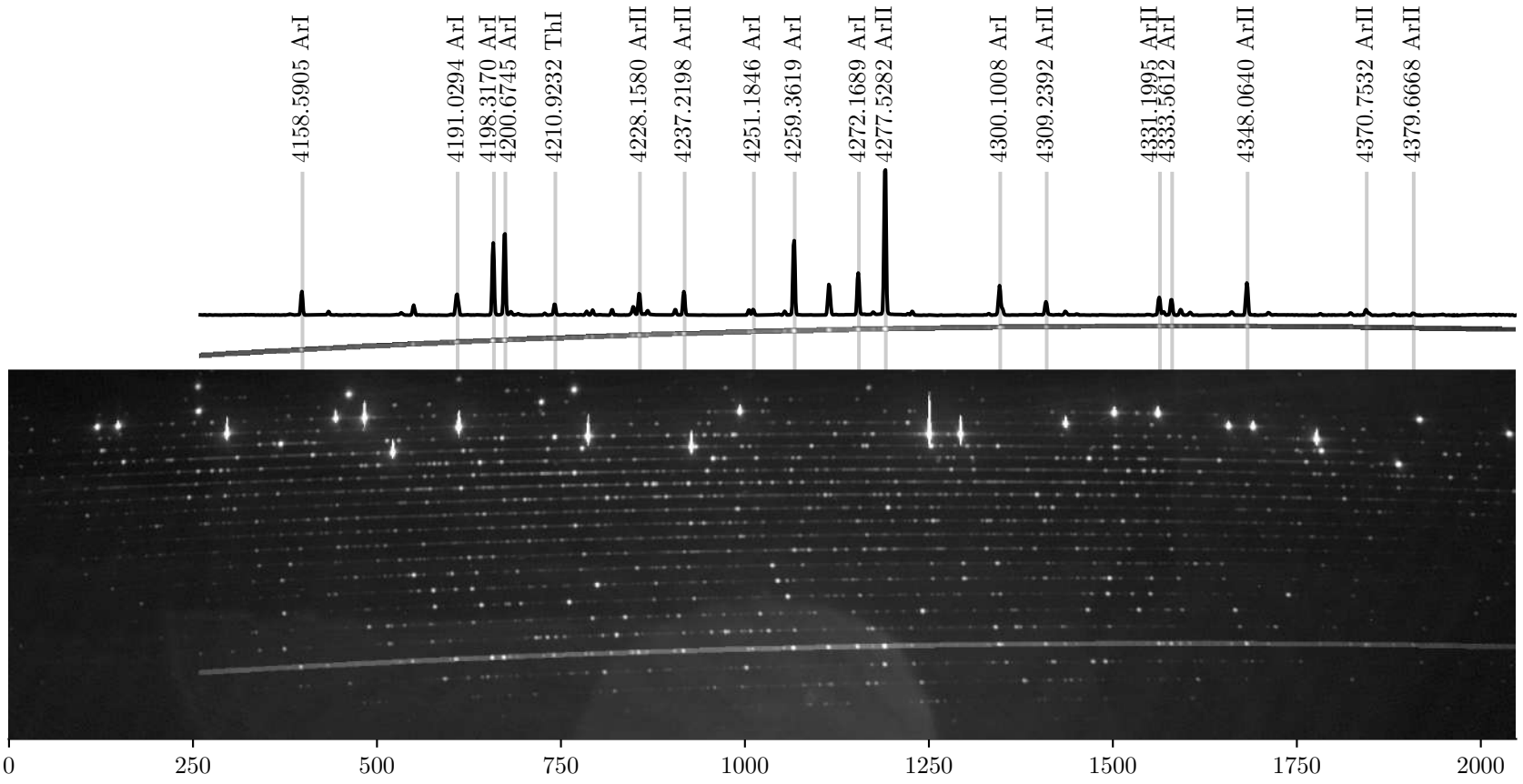

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4048.0318 ThII
4048.2824 ThI
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4049.9447 ThI
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4051.1090 Th
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4054.5258 ArI
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4060.0616 ThI
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4067.8759 ThII
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4069.2014 ThII
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4070.7835 ArII
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4072.0677 ArI
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4104.8385 Th
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4105.9150 ThII
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4102.8199 ThI
4103.2627 ThII
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4124.0403 ThI
4127.4120 Th
4128.0499 ThI
4128.6400 ArII
4129.5116 Th
4130.3247 Th
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4131.4265 ThII
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4154.7205 ThI
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4156.2382 ThII
4156.5166 ThII
4157.2804 Th
4158.5905 ArI
4159.1537 ThI
4159.6599 ThII
4161.7389 ThI
4162.5090 ThI
4163.6469 ThII
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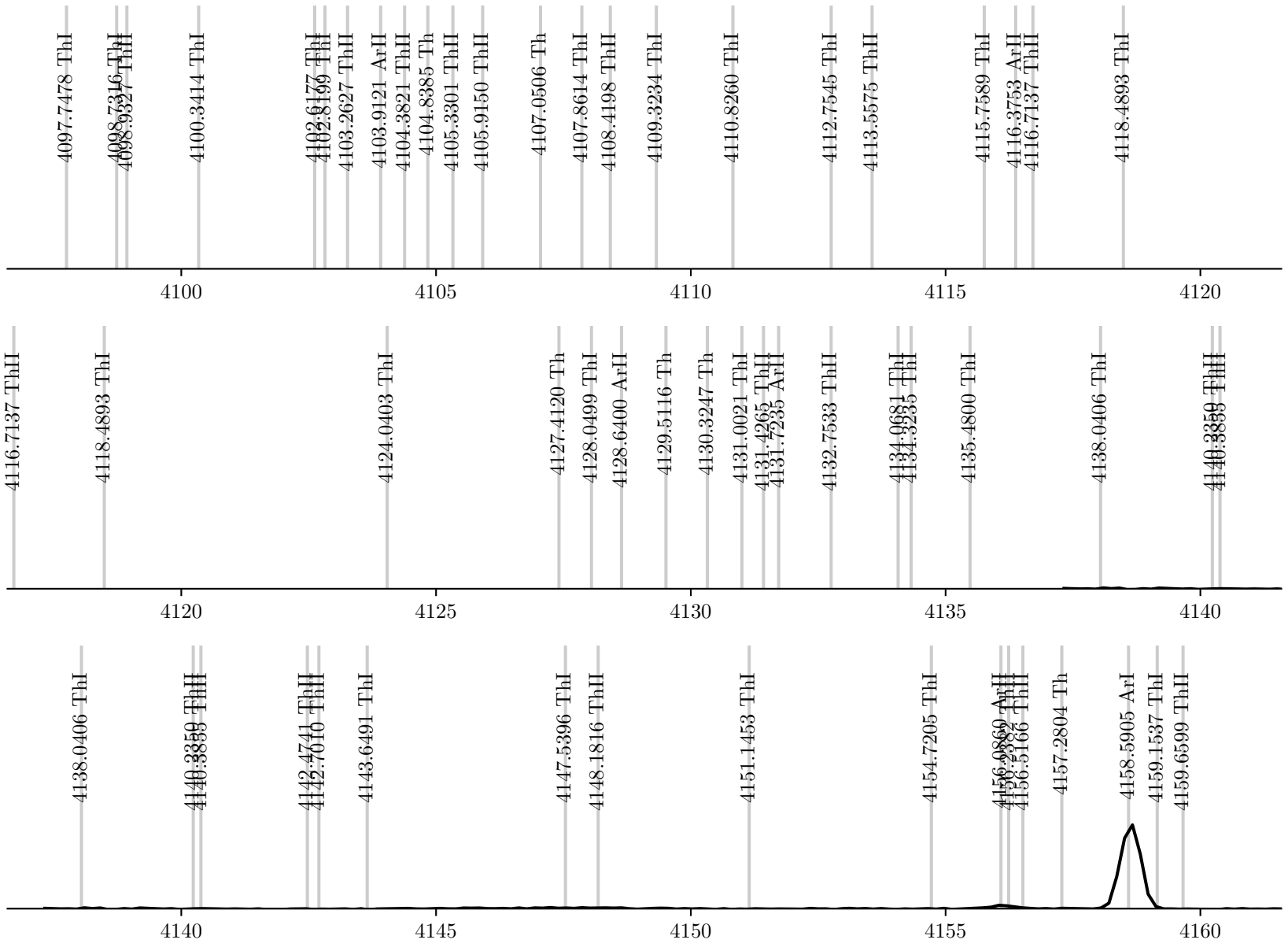

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4170
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4180
4185
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4184.1376 ThI
4184.6045 Th
4185.1447 ThI
4185.4693 Th
4187.1410 ThI
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4216.0698 ThI
4216.3722 ThI
4217.4308 ArII
4218.6649 ArII
4220.0651 ThI
4220.7323 ThI
4222.6373 ArII
4223.4403 ThI
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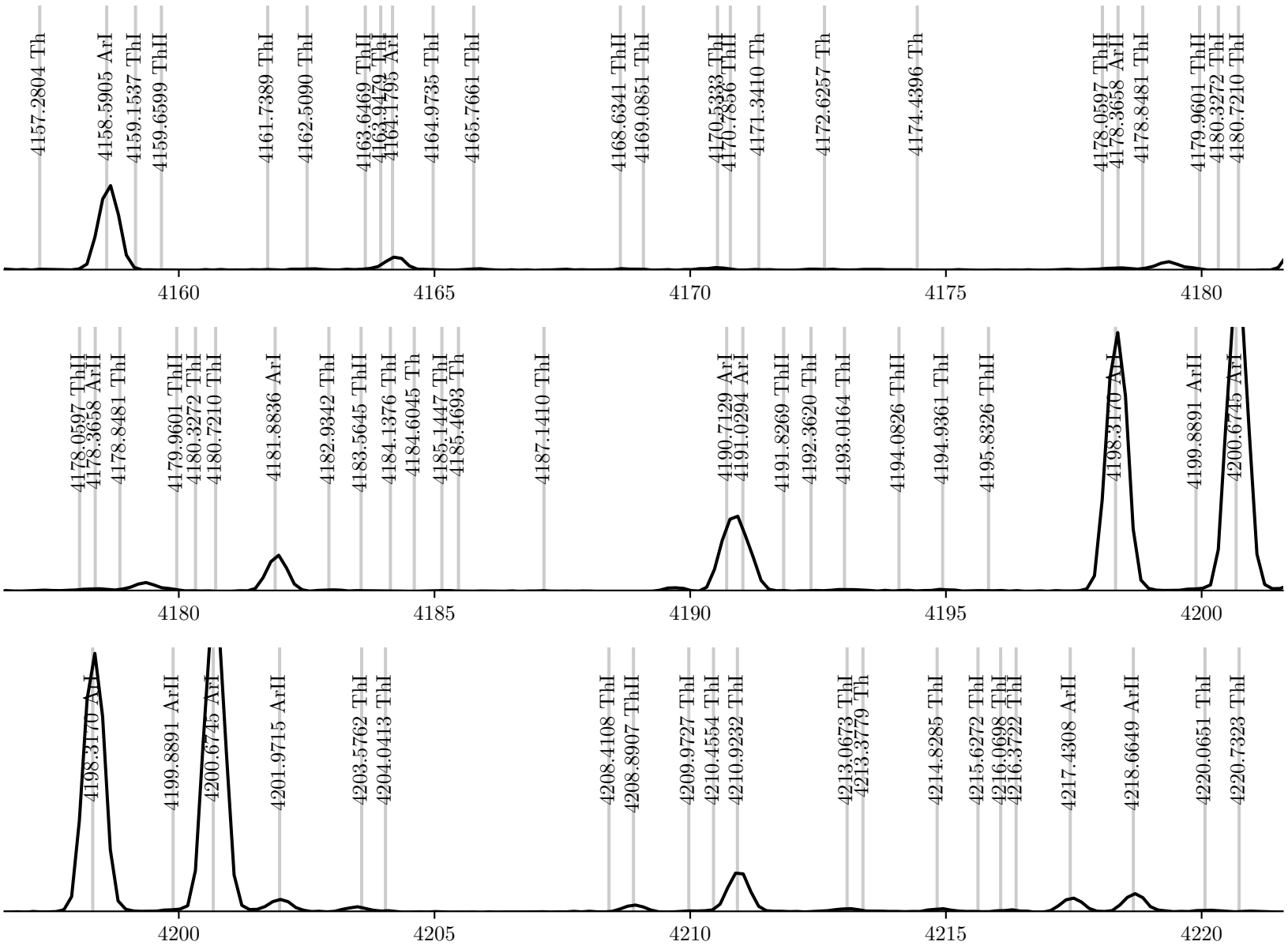

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4227.3872 ThI
4228.1580 ArII
4229.1477 ThI
4229.4546 ThII
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4230.8242 ThI
4233.2873 ThII
4235.4636 ThI
4236.0471 Th
4237.2198 ArII
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4241.0948 ThI
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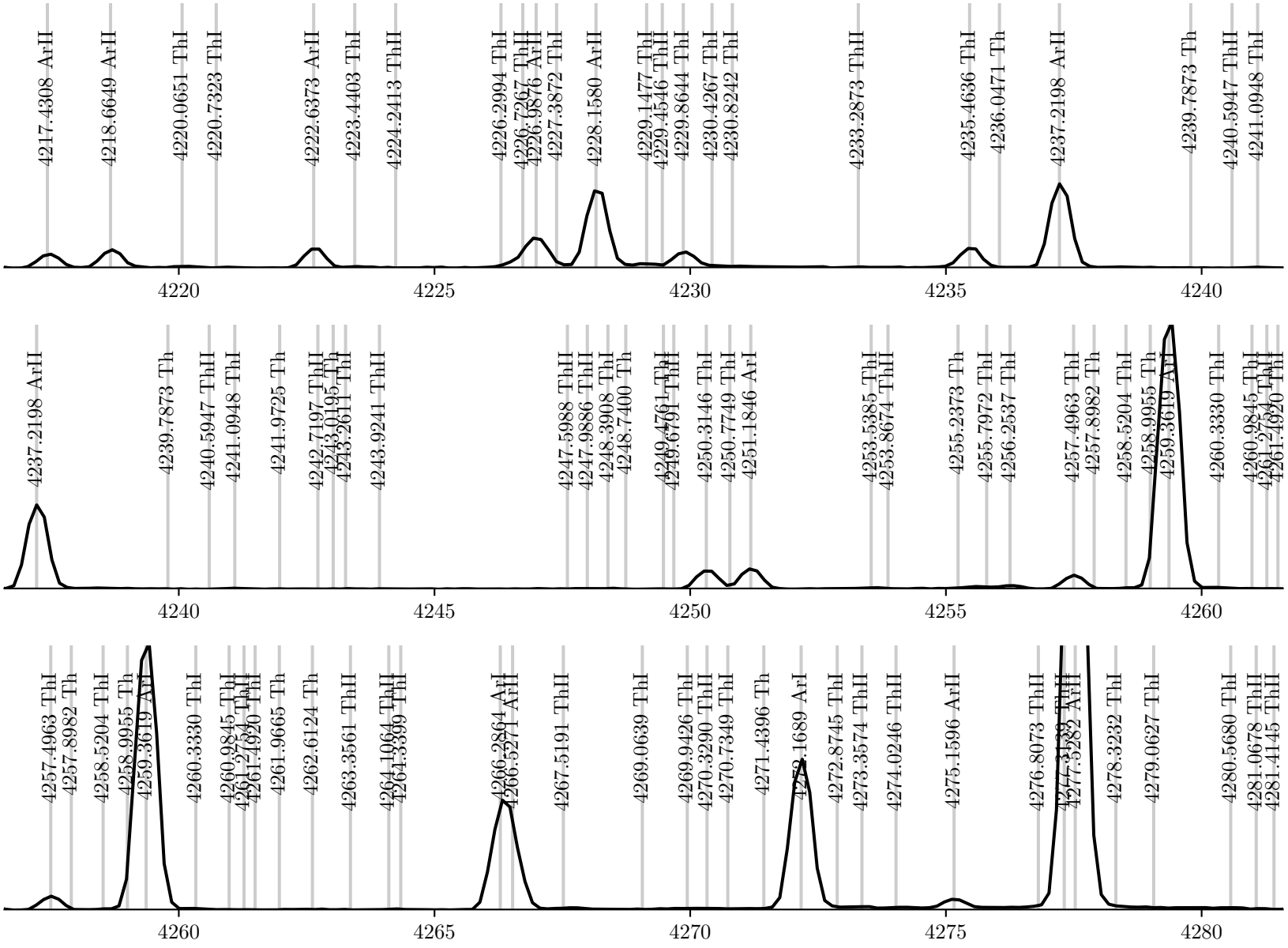
Order 35

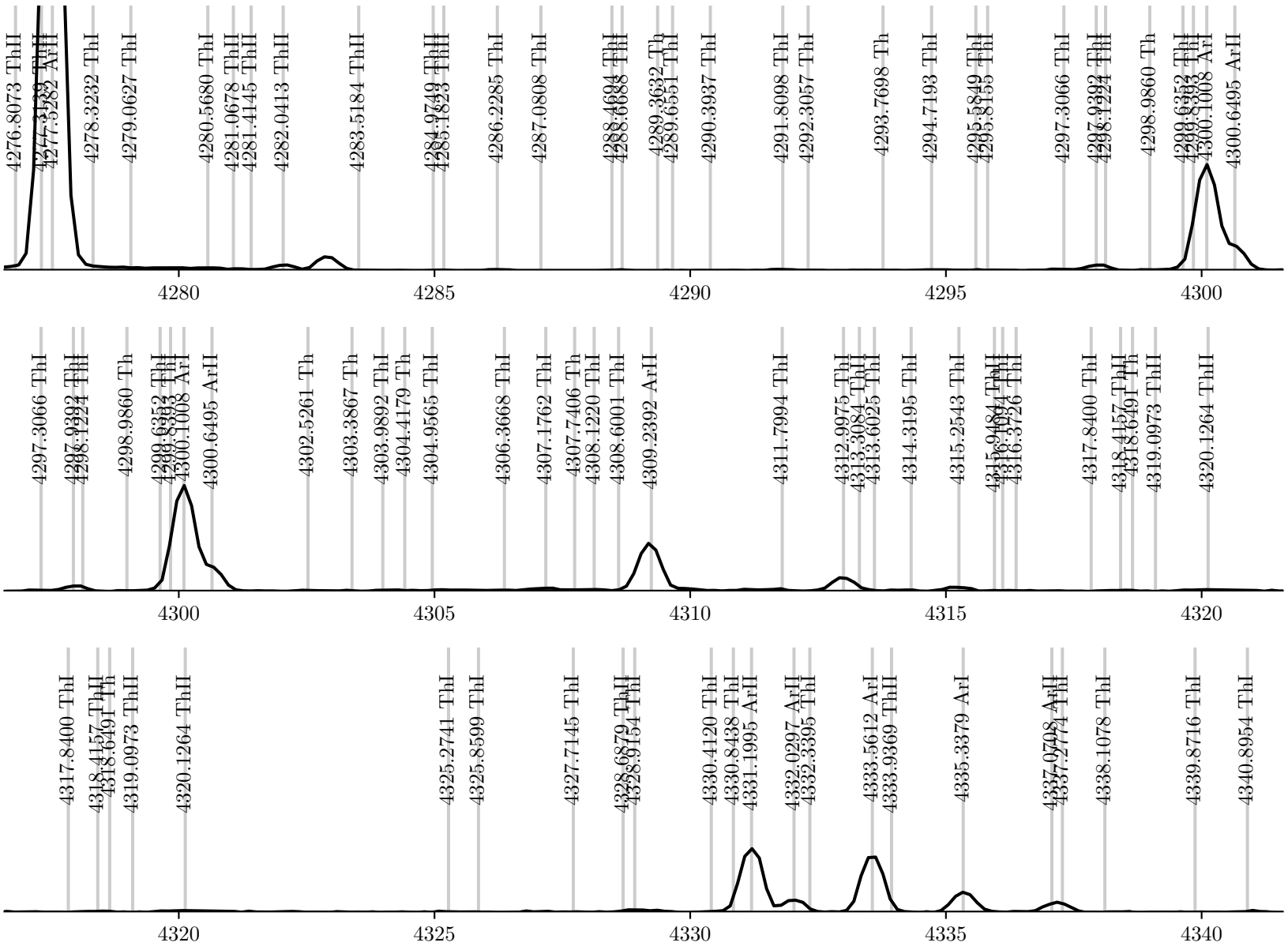


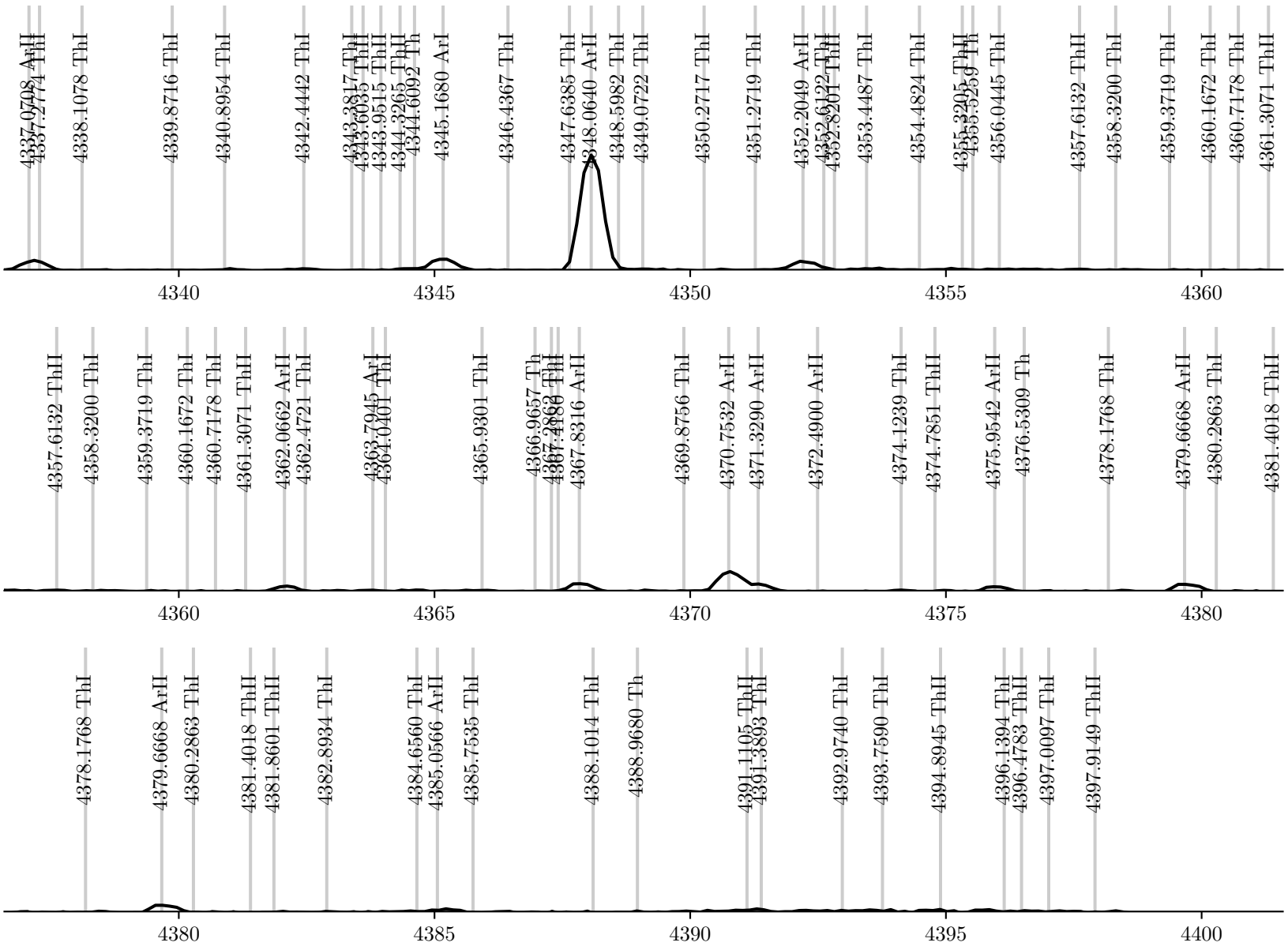
Dispersion:	0.148 Å pix^{-1}
Central wavelength:	4253 Å
Wavelength range:	4097 Å – 4399 Å

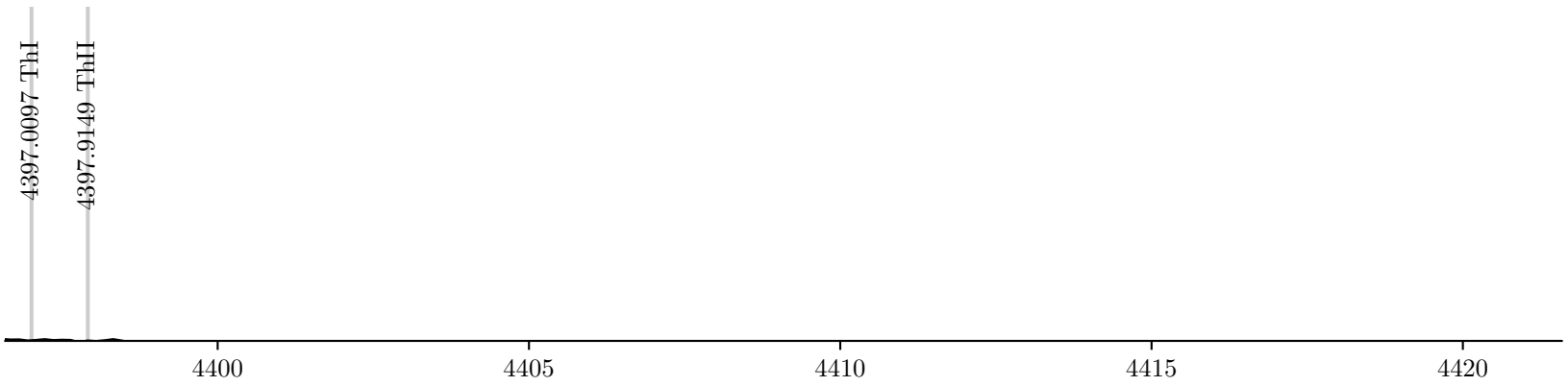

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4102.8169 ThI
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4104.3821 ThII
4104.8385 Th
4105.3301 ThII
4105.9150 ThII
4107.0506 Th
4107.8614 ThI
4108.4198 ThII
4109.3234 ThI
4110.8260 ThI
4112.7545 ThI
4113.5575 ThII
4115.7589 ThI
4116.3753 ArII
4116.7137 ThII
4118.4893 ThI
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4159.1537 ThI
4159.6599 ThII
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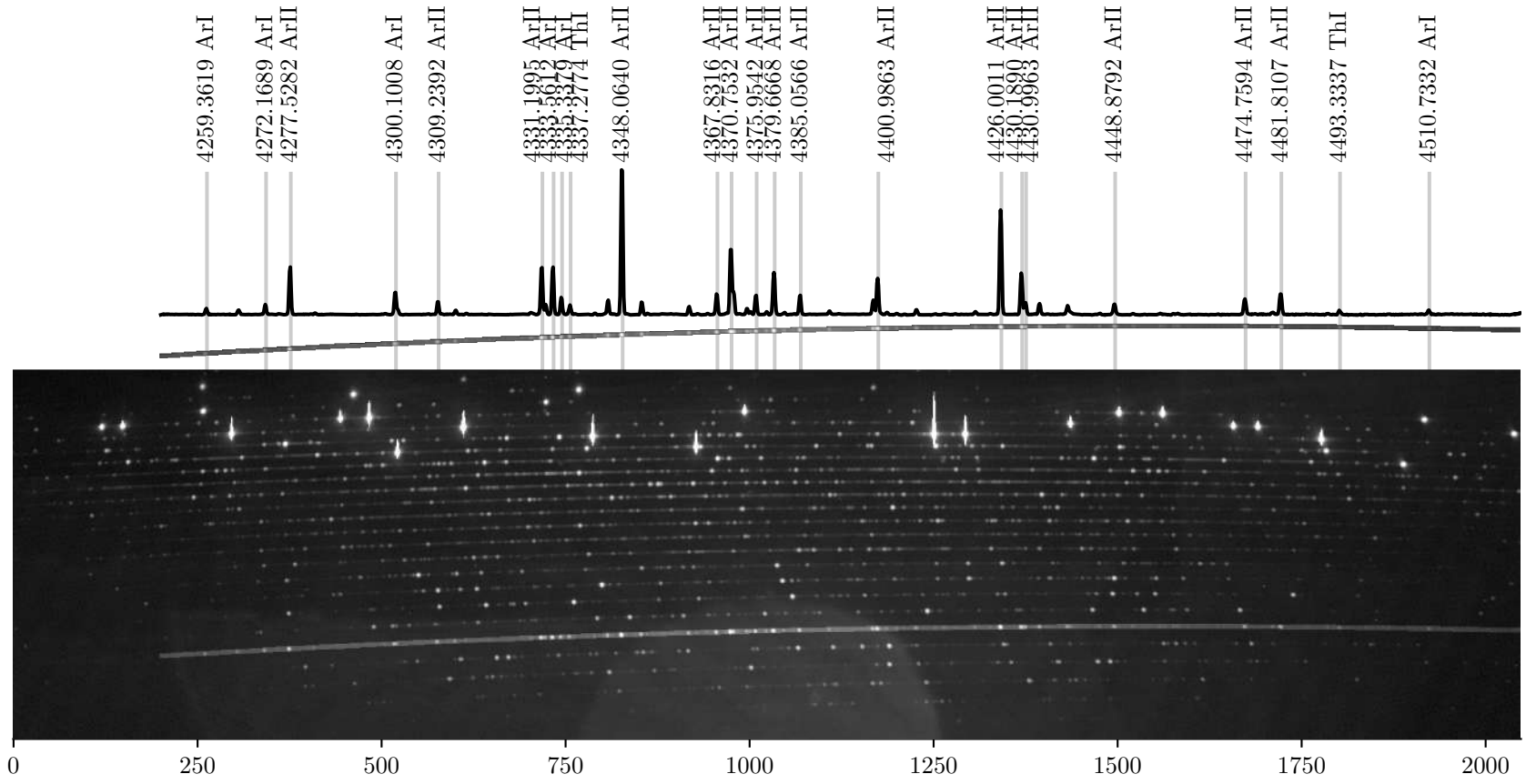

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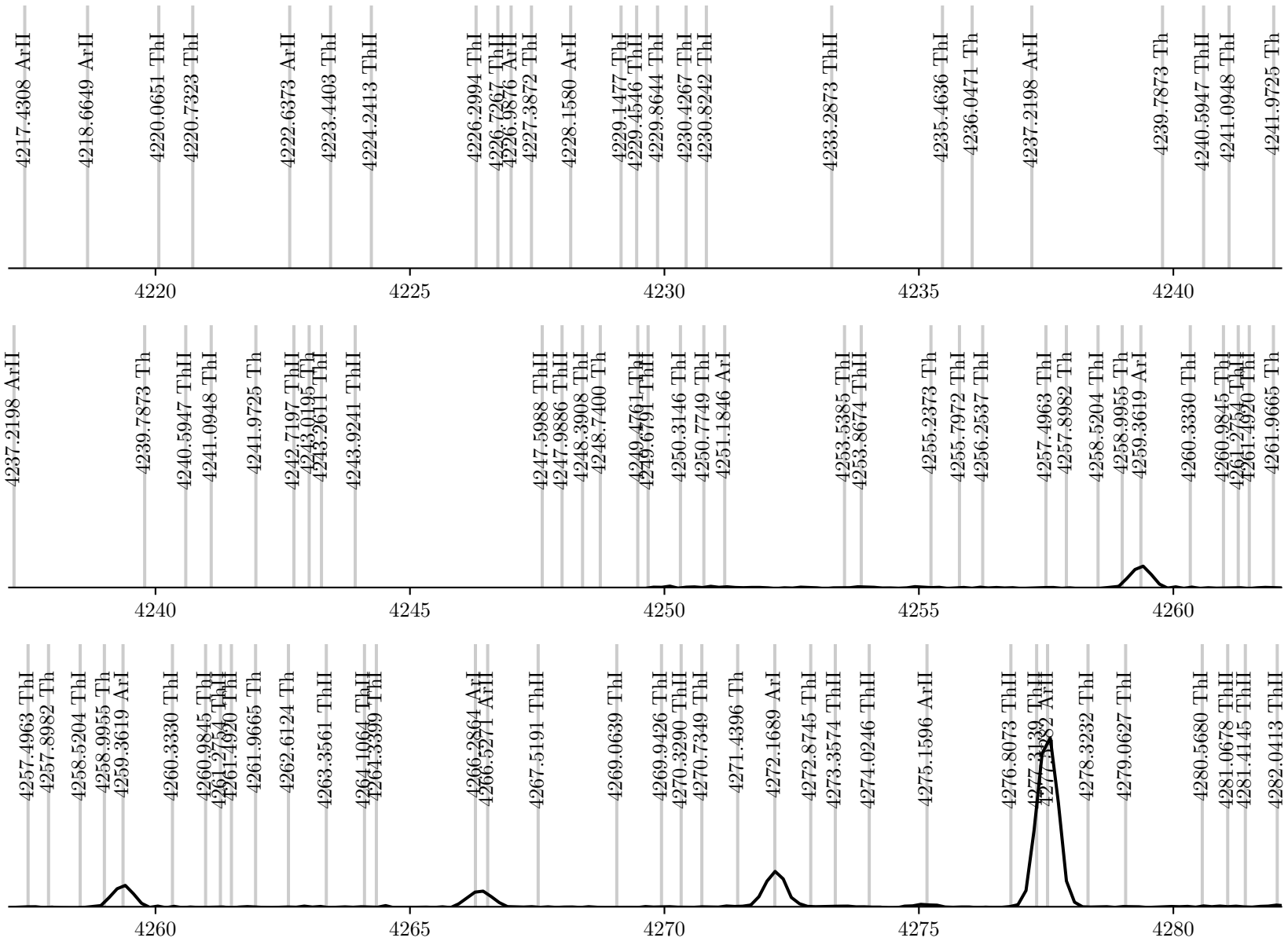

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4352.2049 ArII
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4354.4824 ThI
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4355.5259 Th
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4357.6132 ThII
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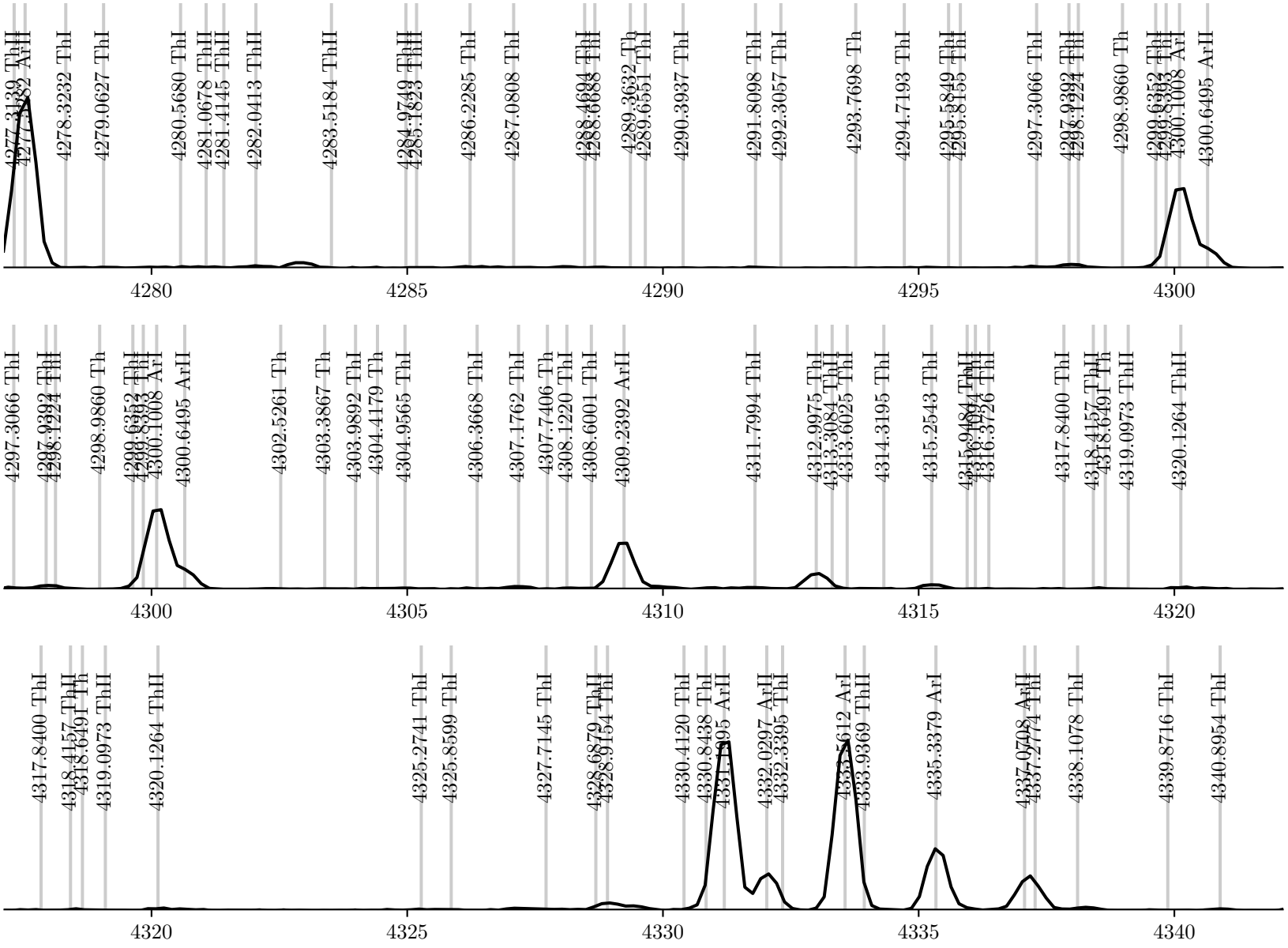

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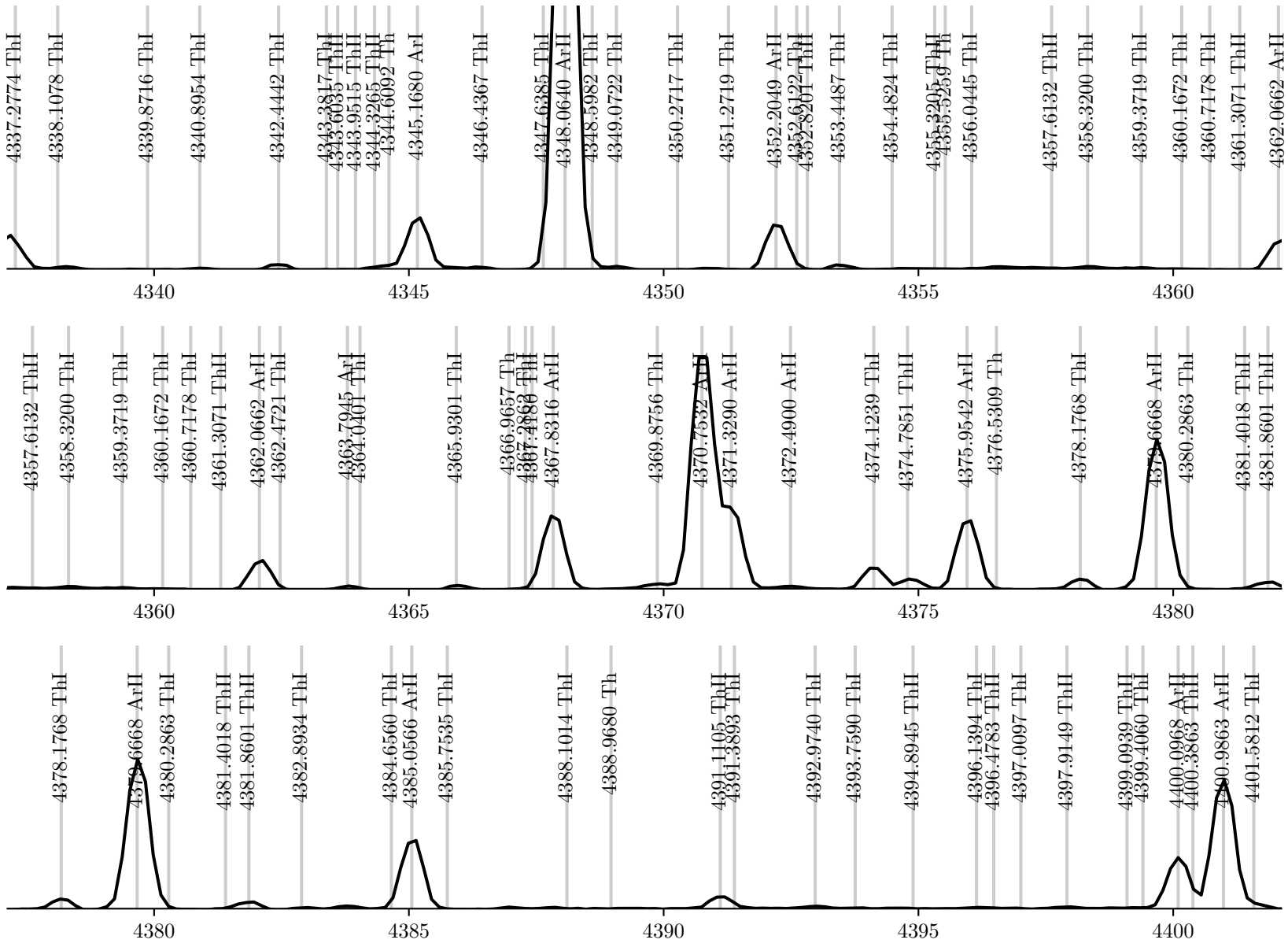
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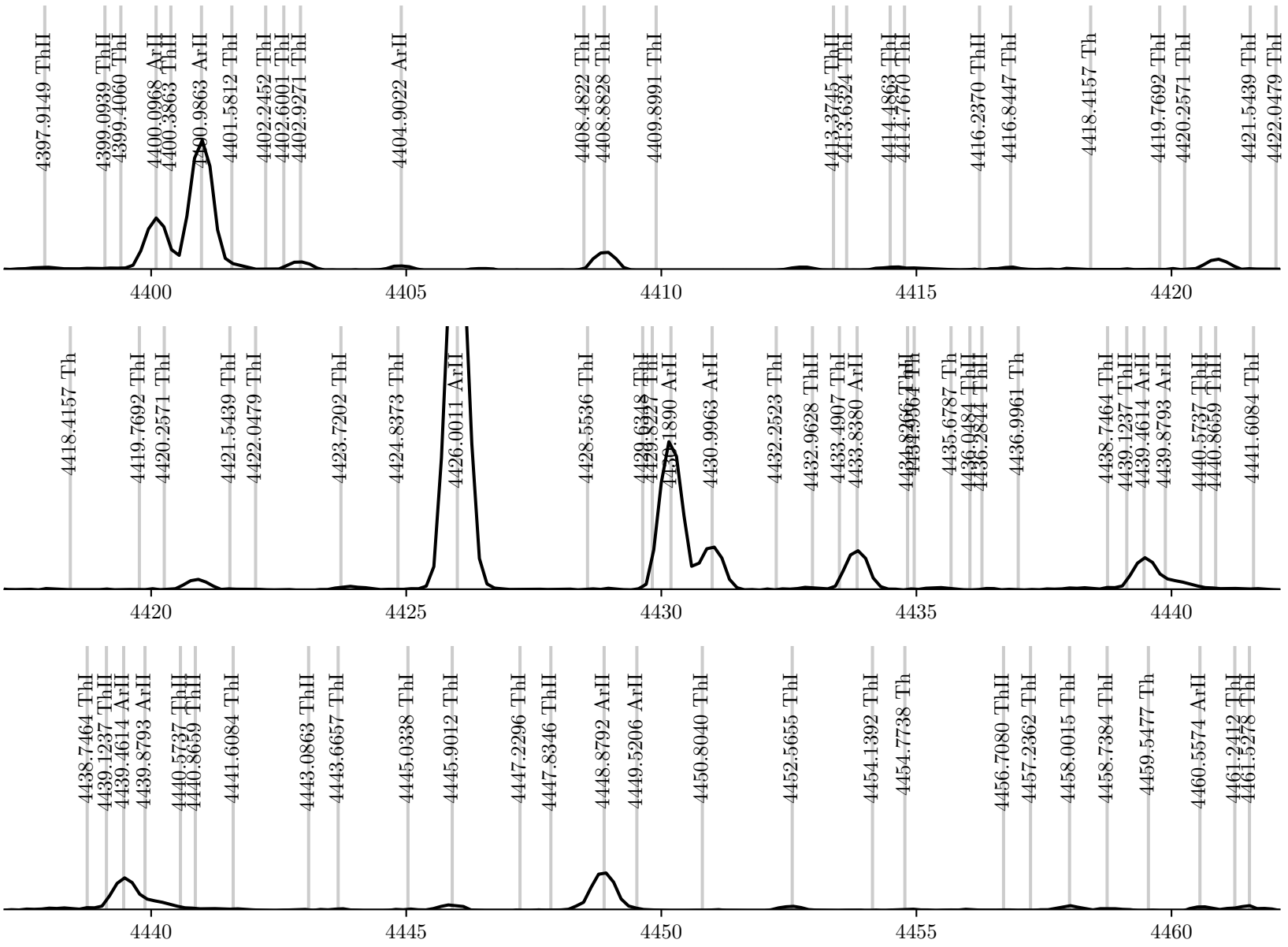


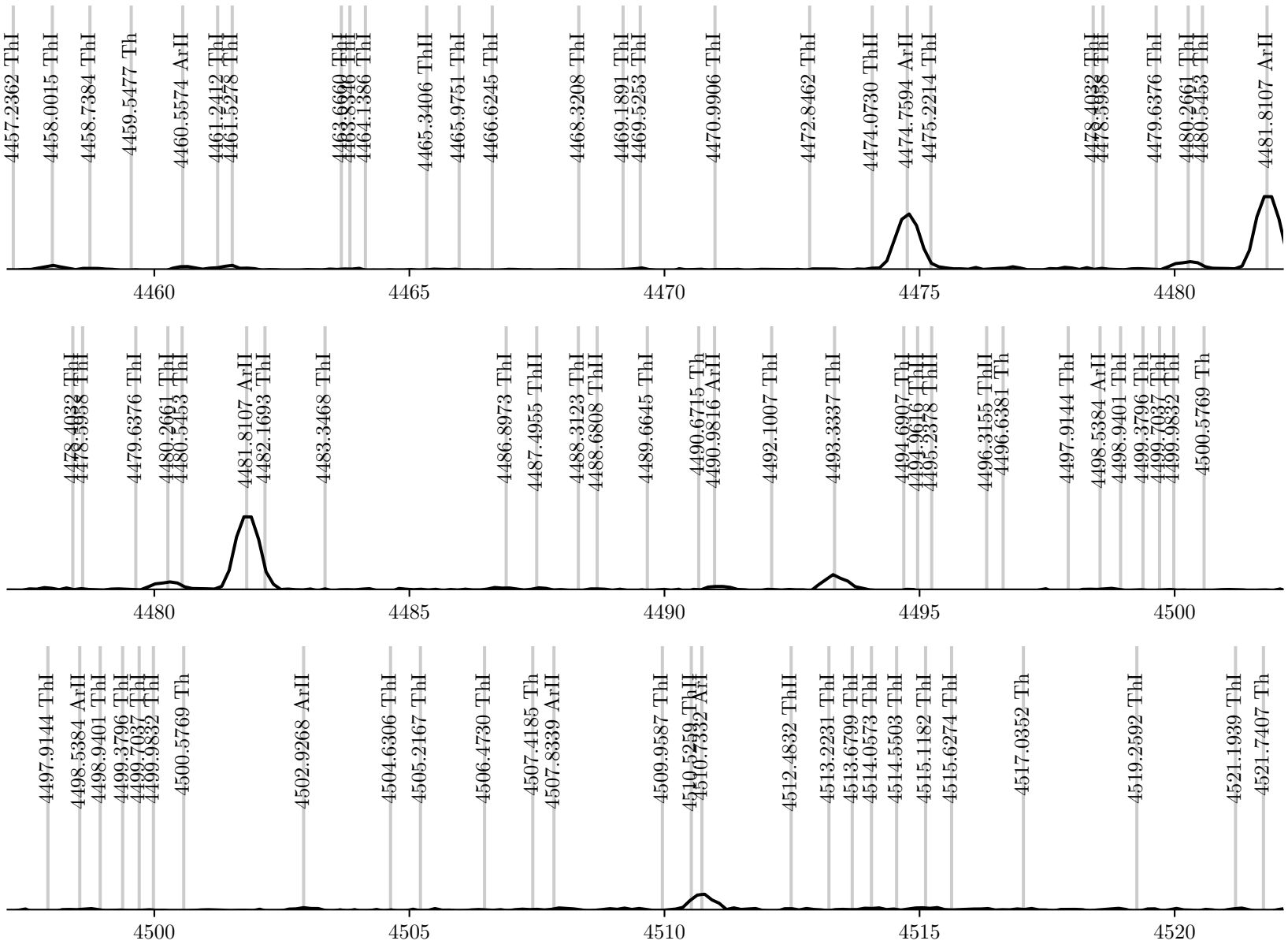
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Wavelength range:	4217 Å – 4528 Å

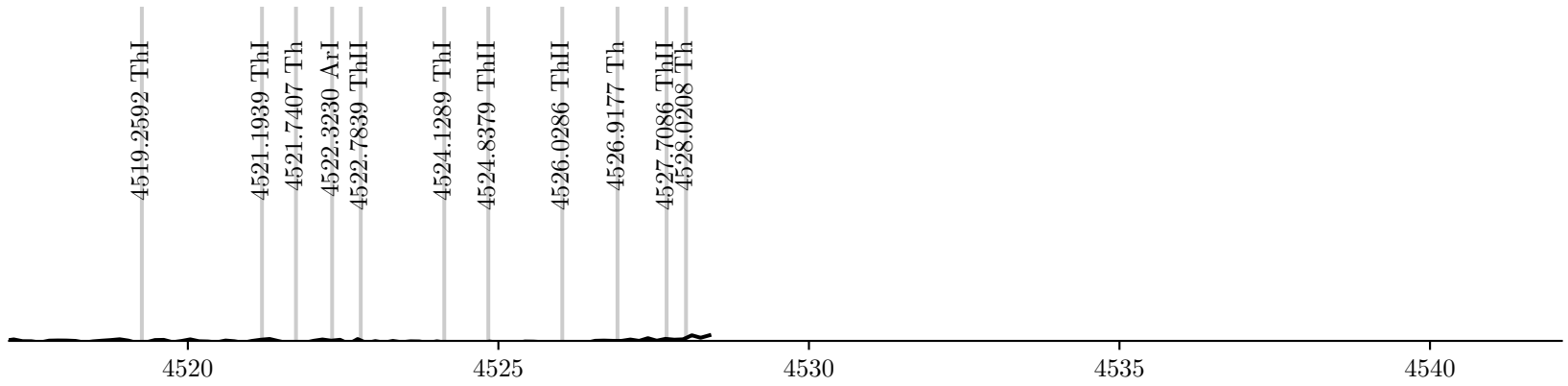

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4230.8242 ThI
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4235.4636 ThI
4236.0471 Th
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4239.7873 Th
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4241.9725 Th
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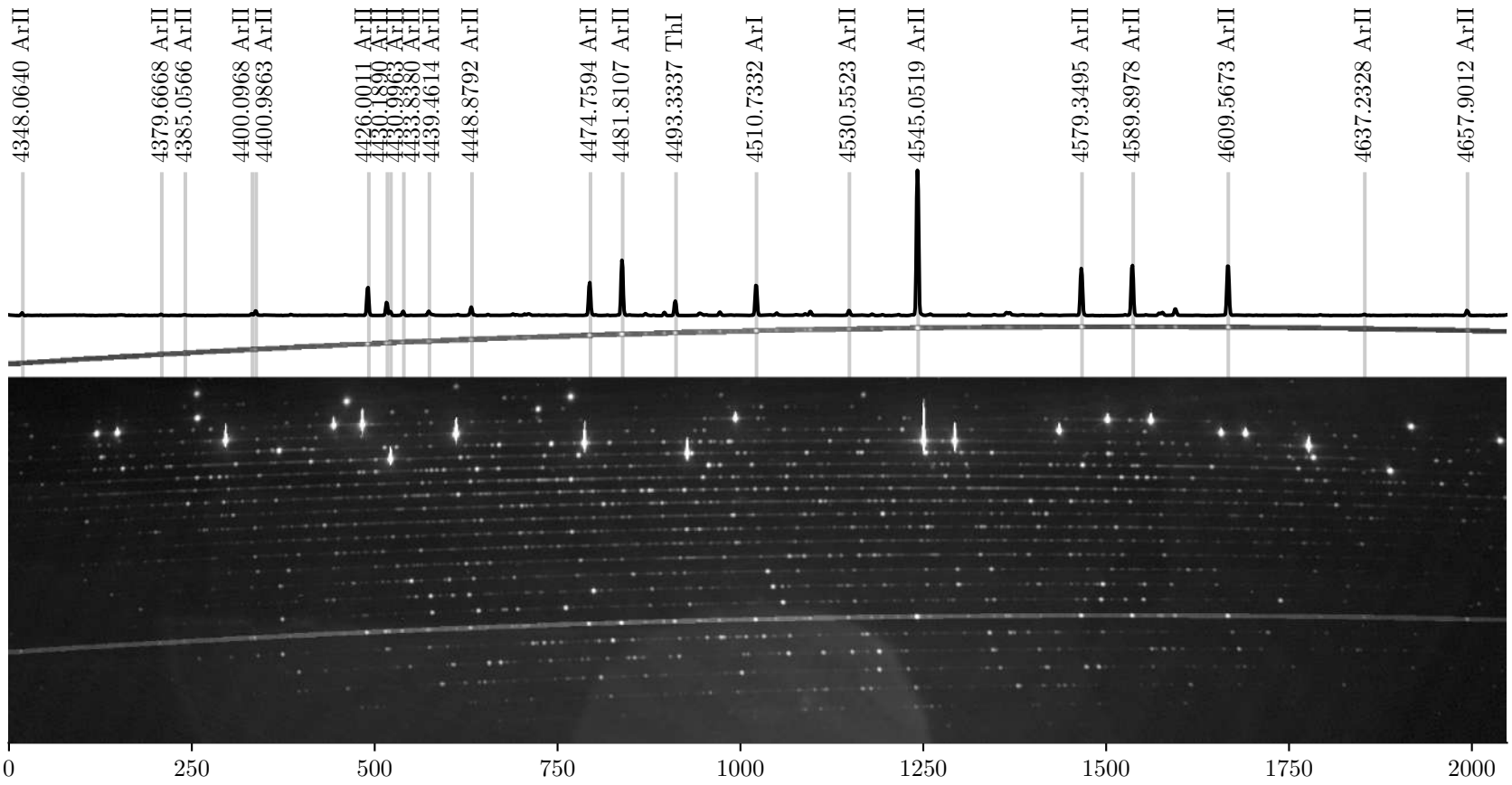

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4359.3719 ThI
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4360.7178 ThI
4361.3071 ThII
4362.0662 ArII
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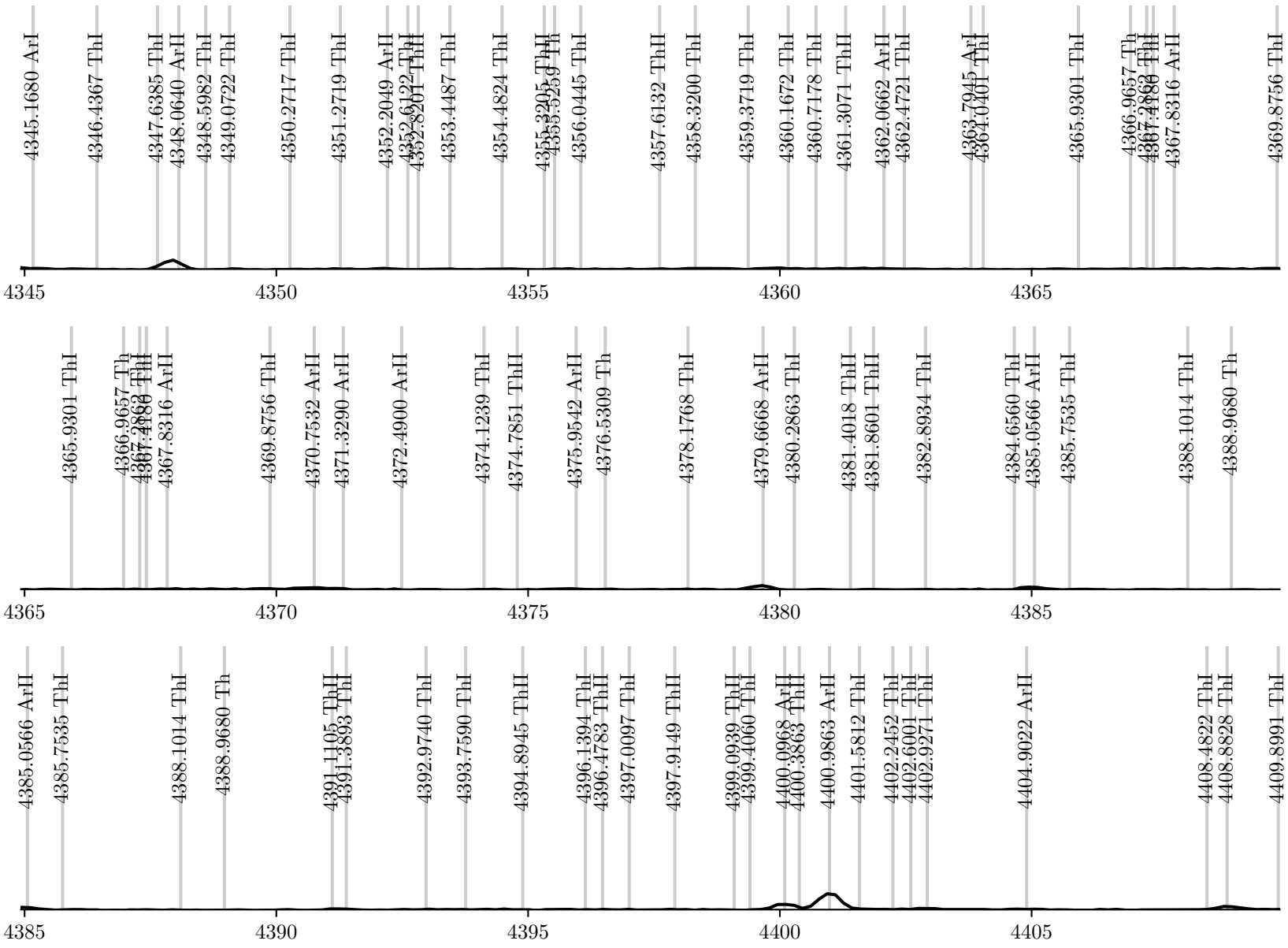

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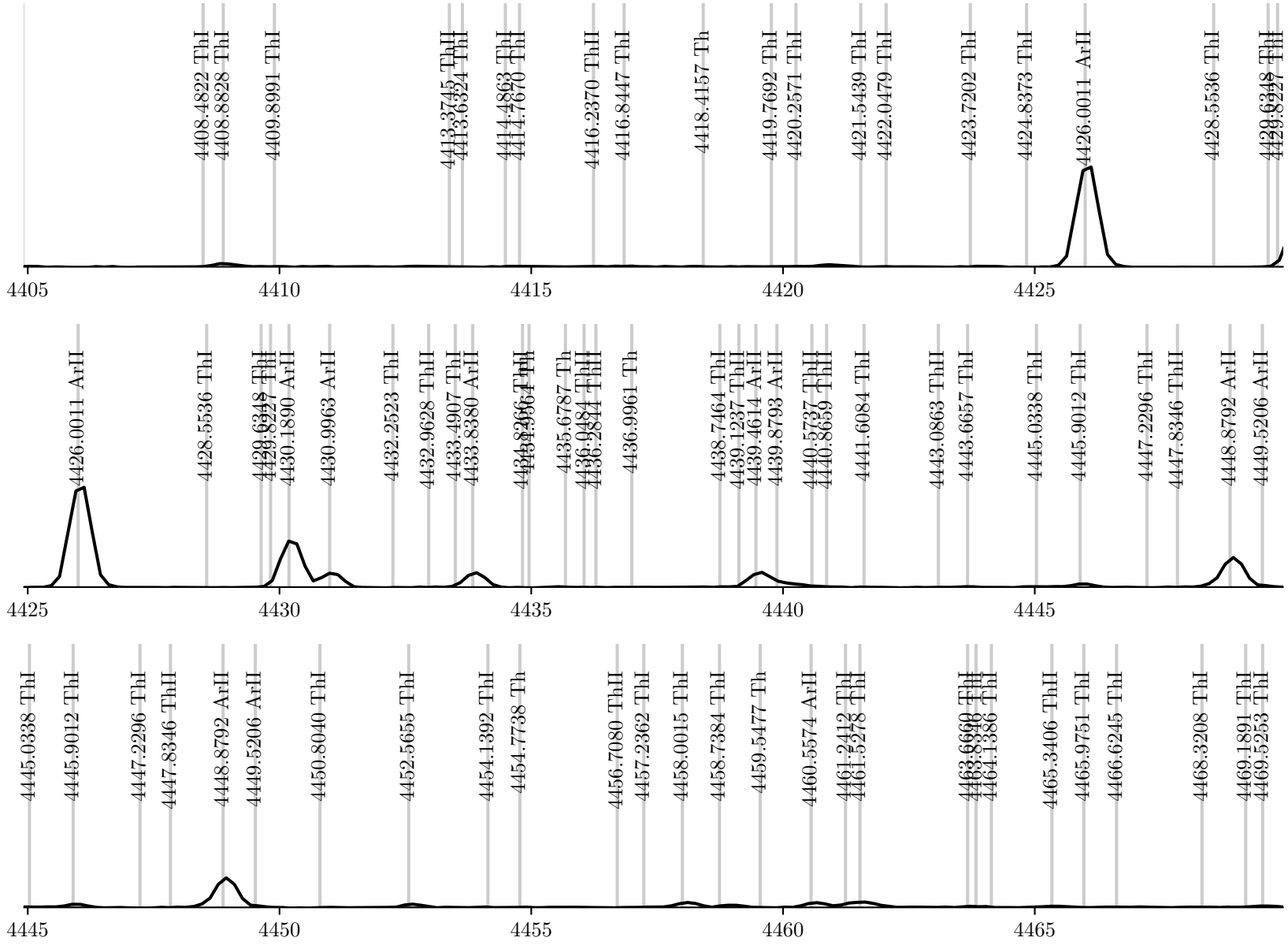

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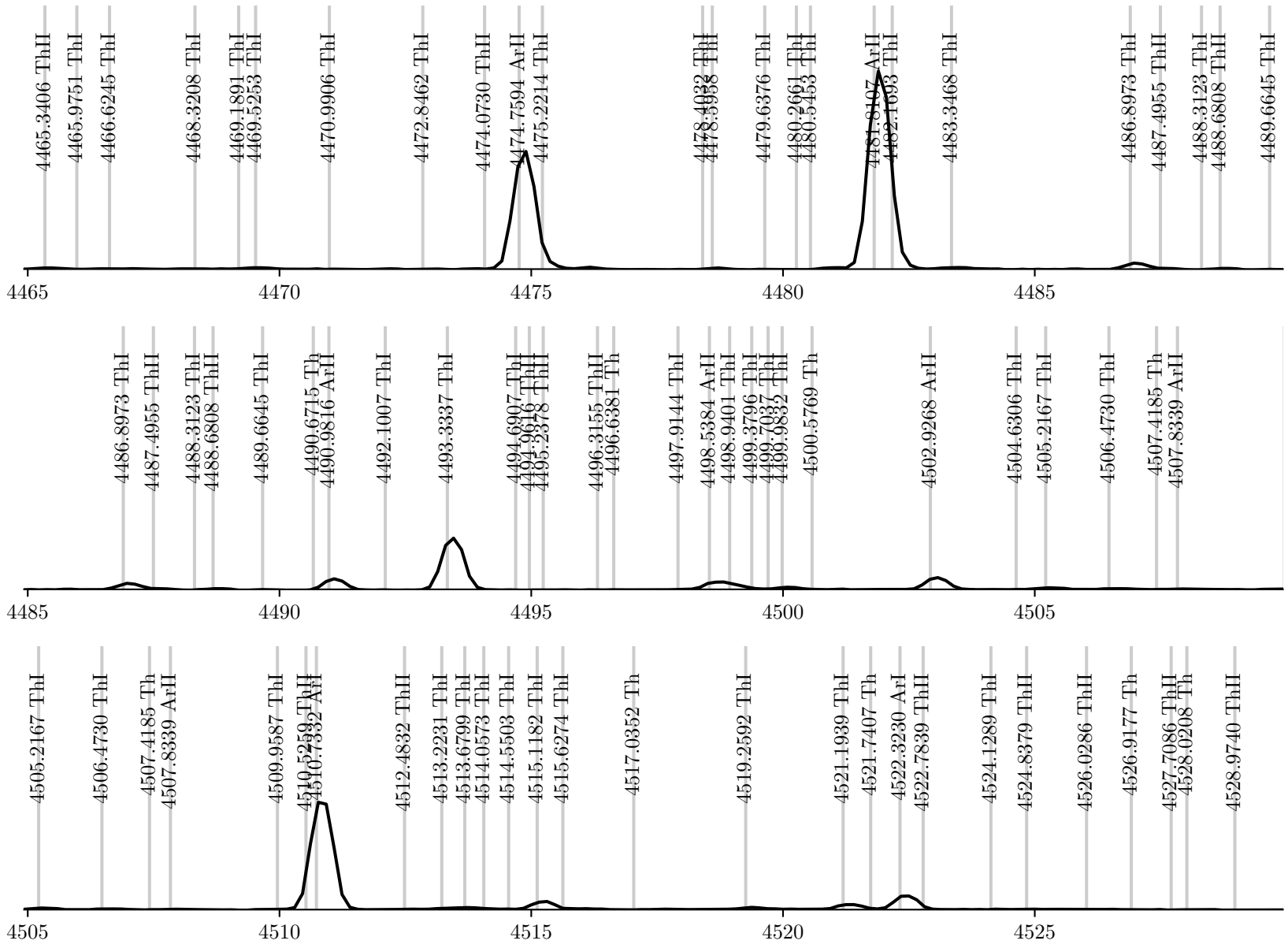
Order 33

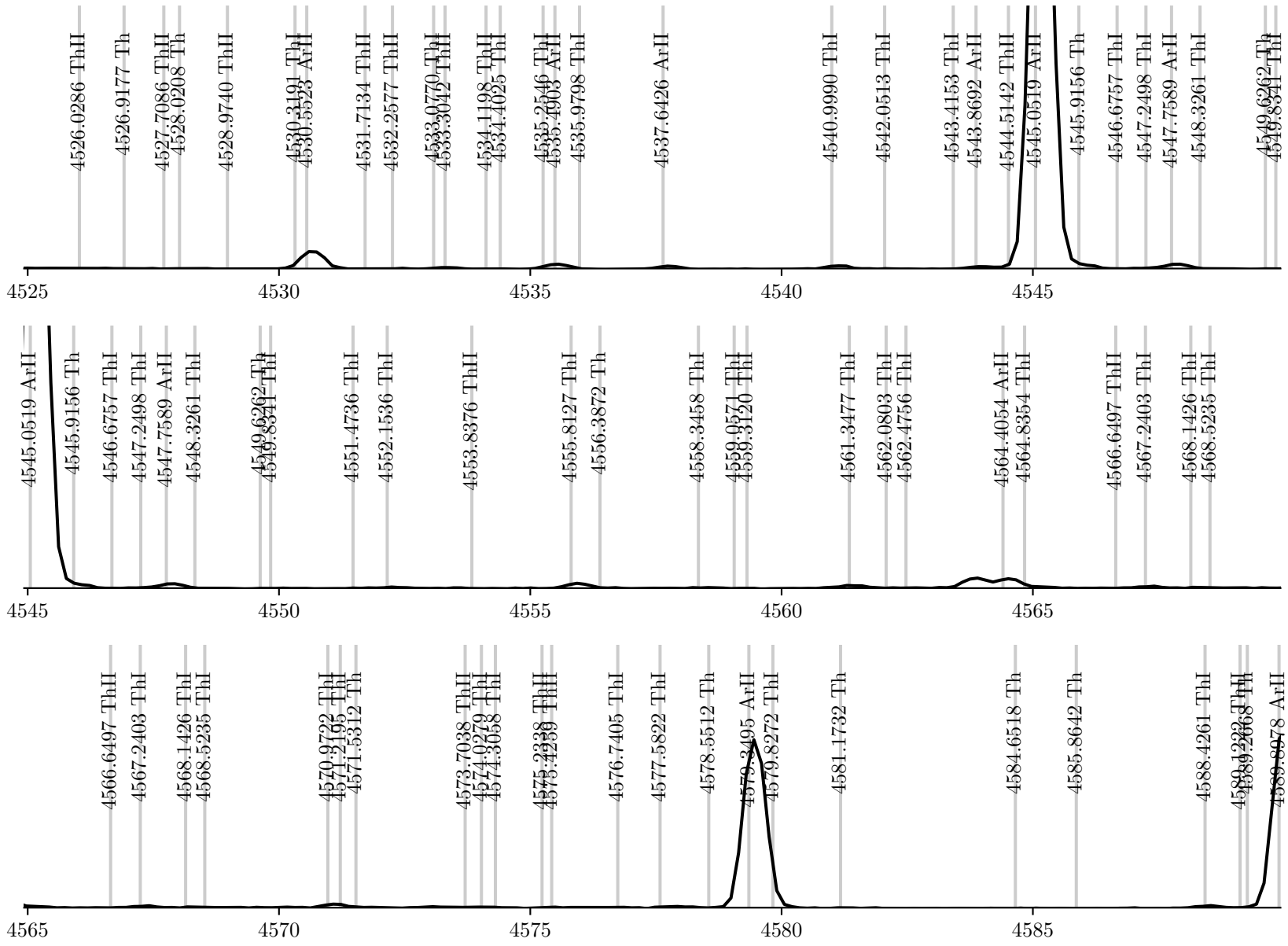


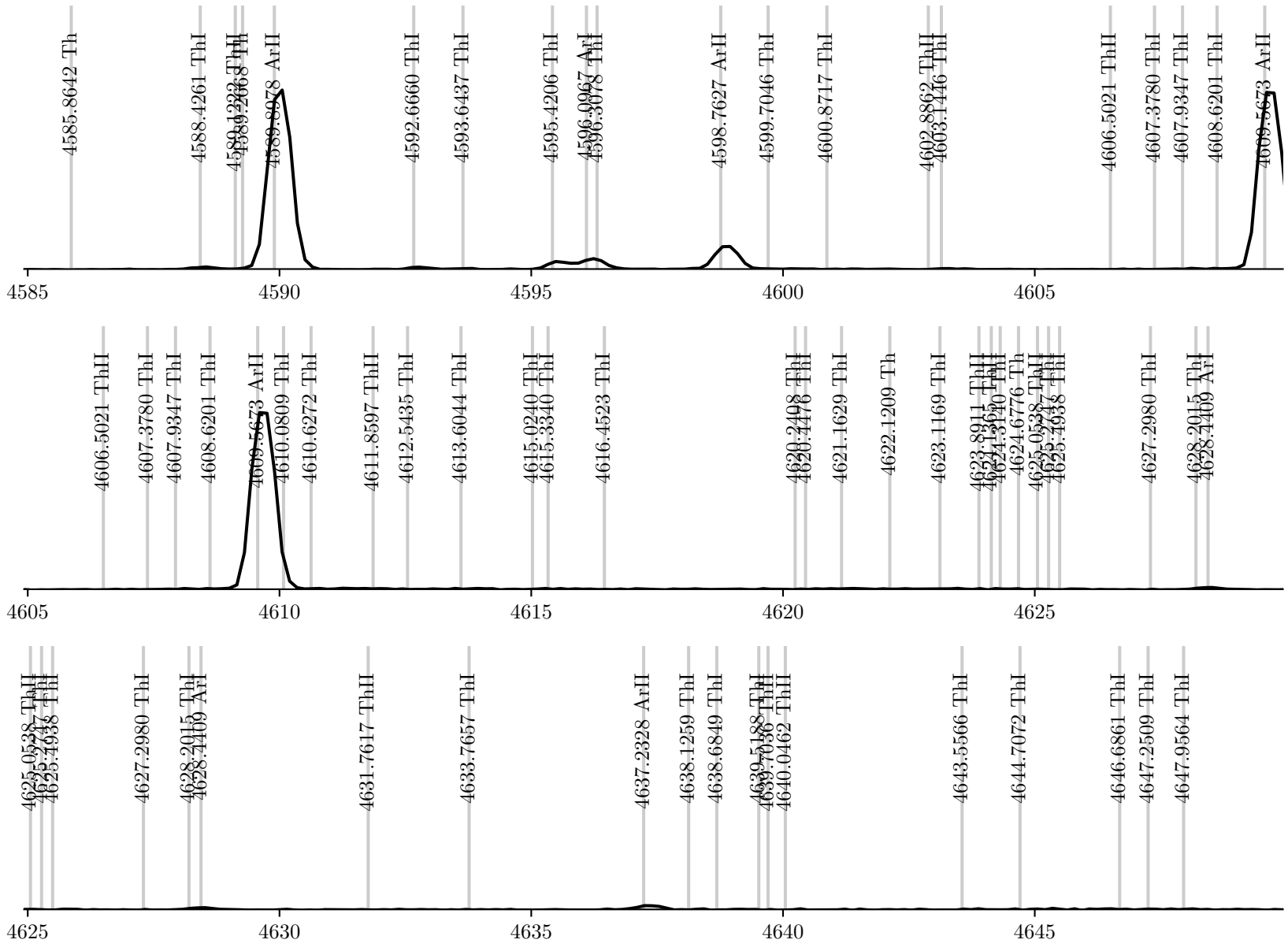
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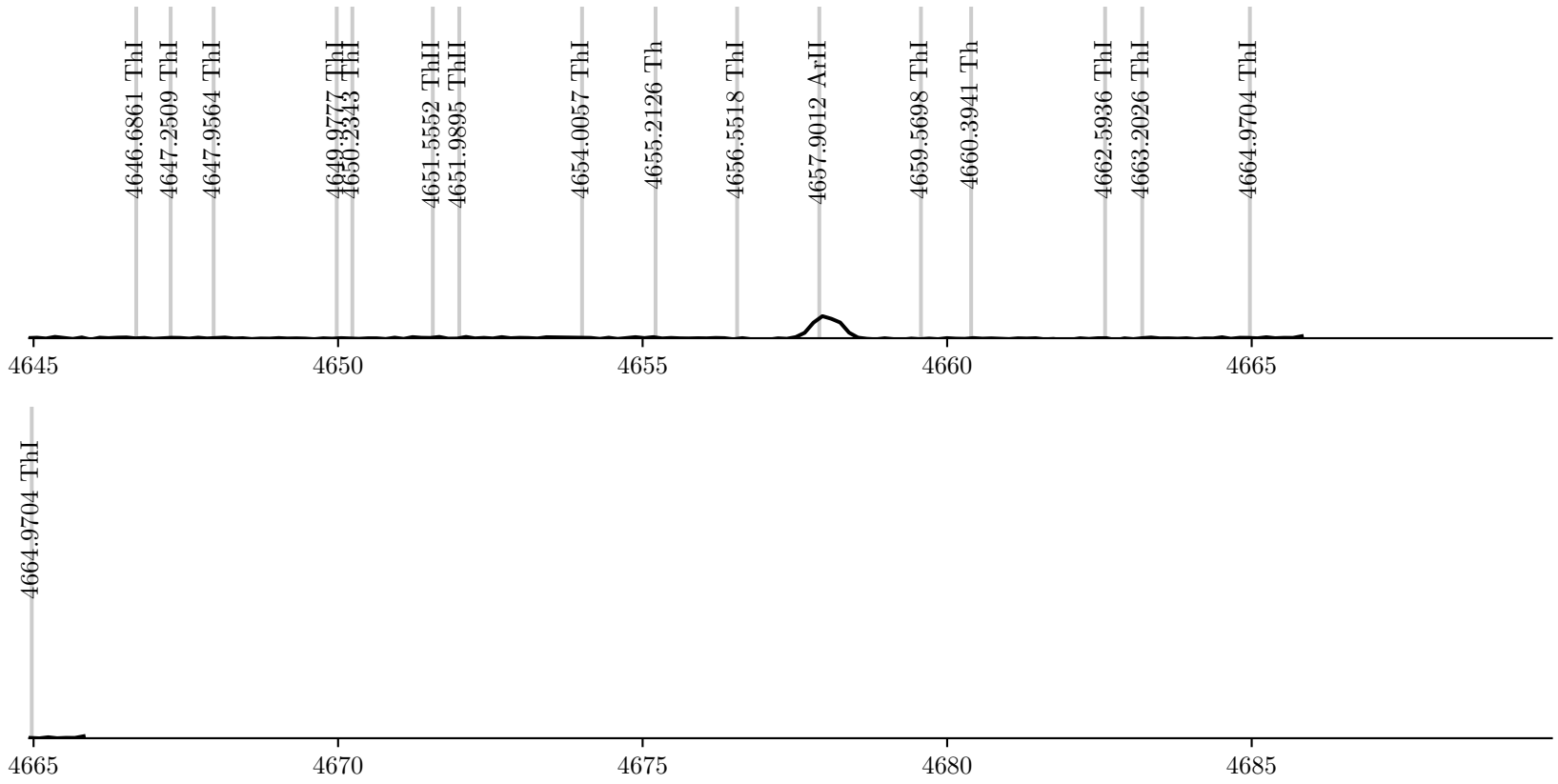

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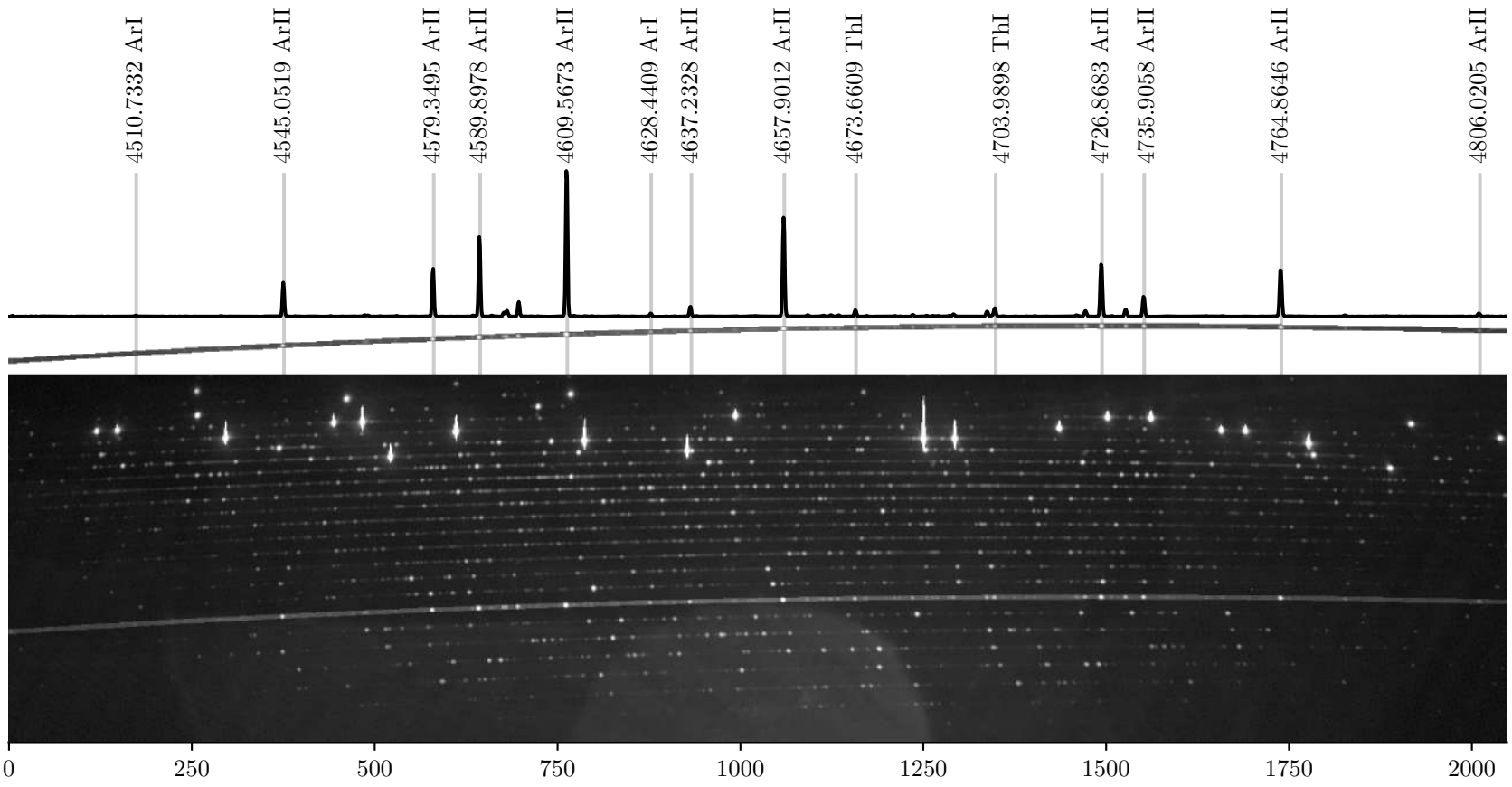

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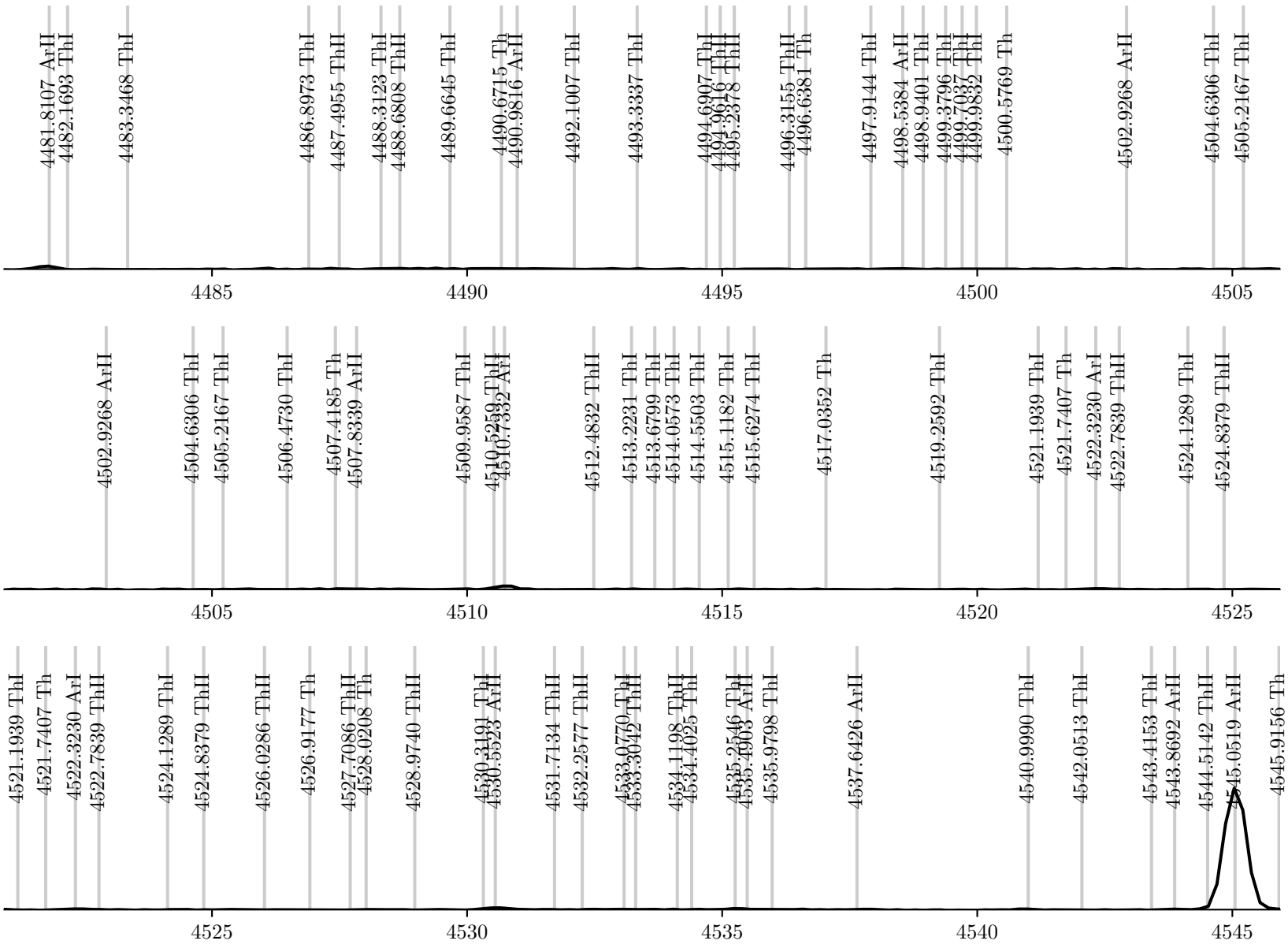

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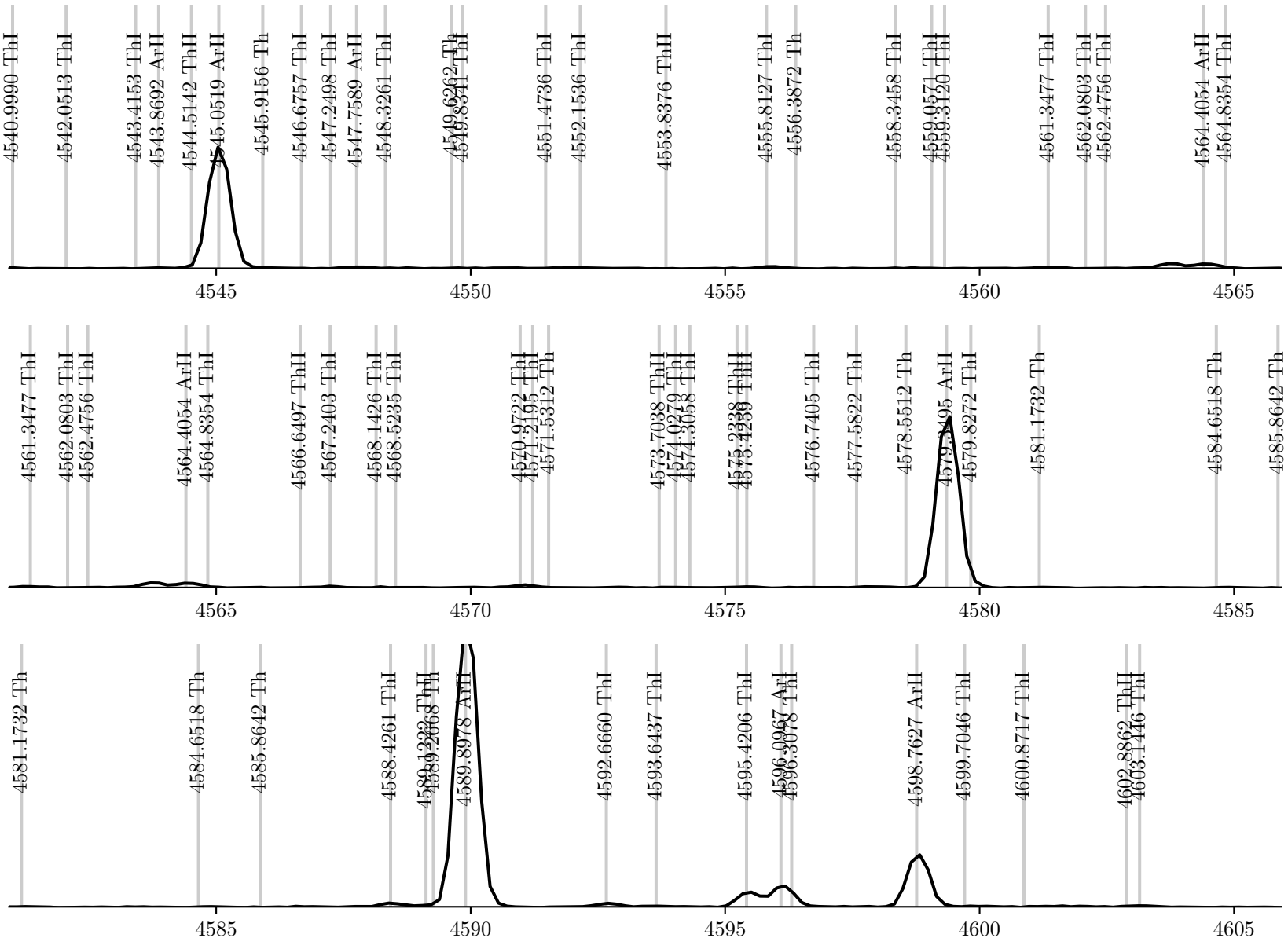

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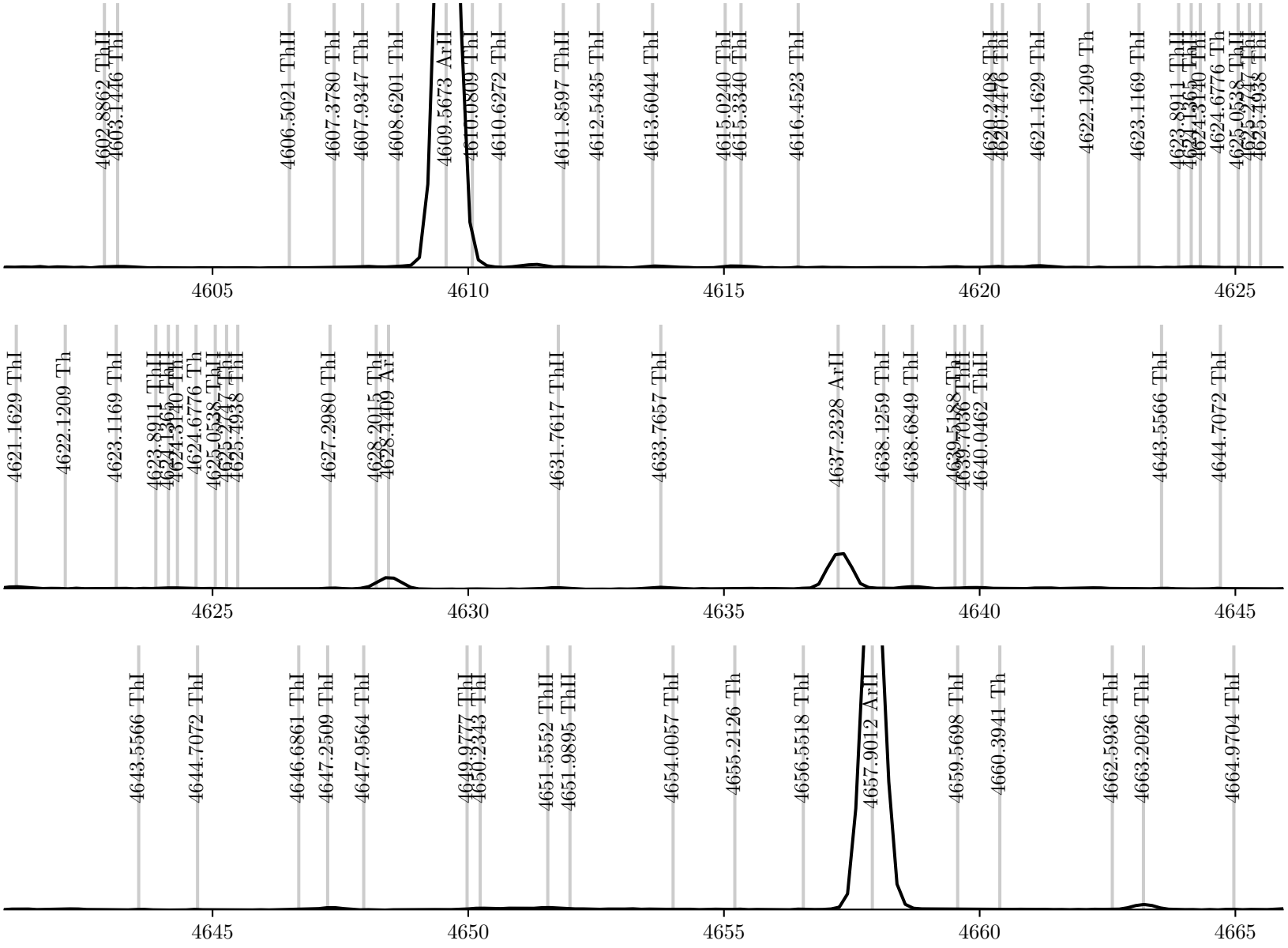
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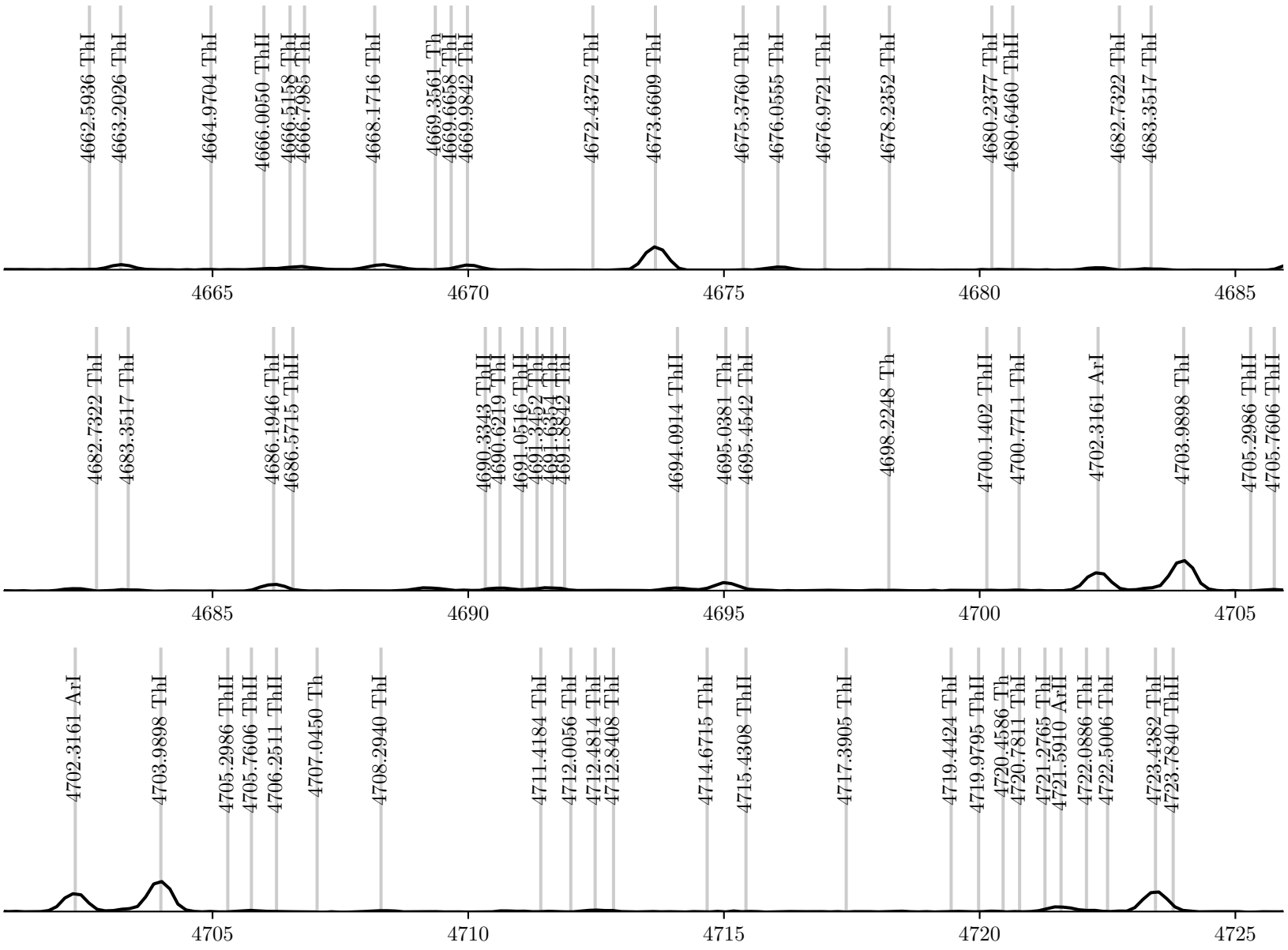


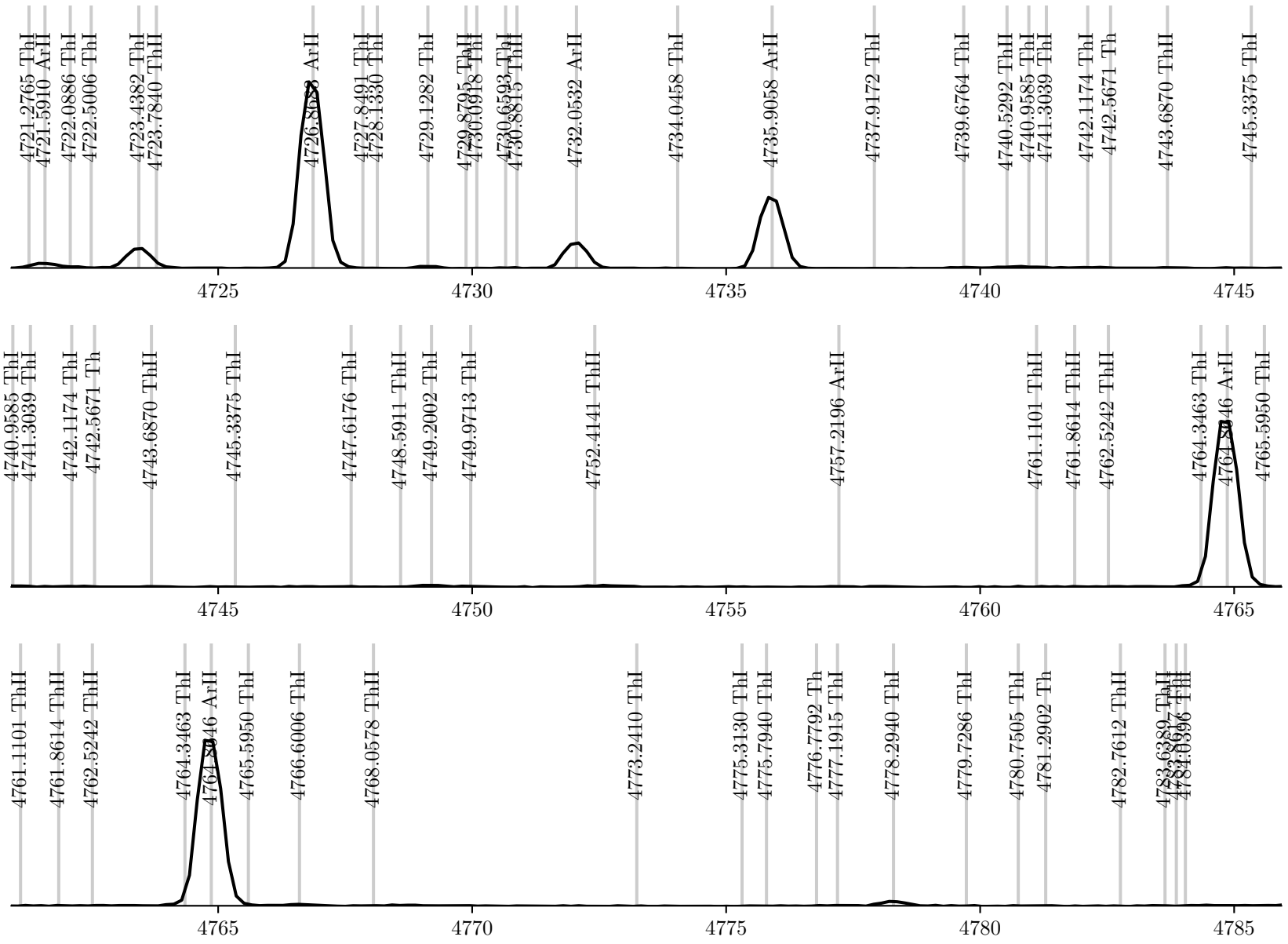
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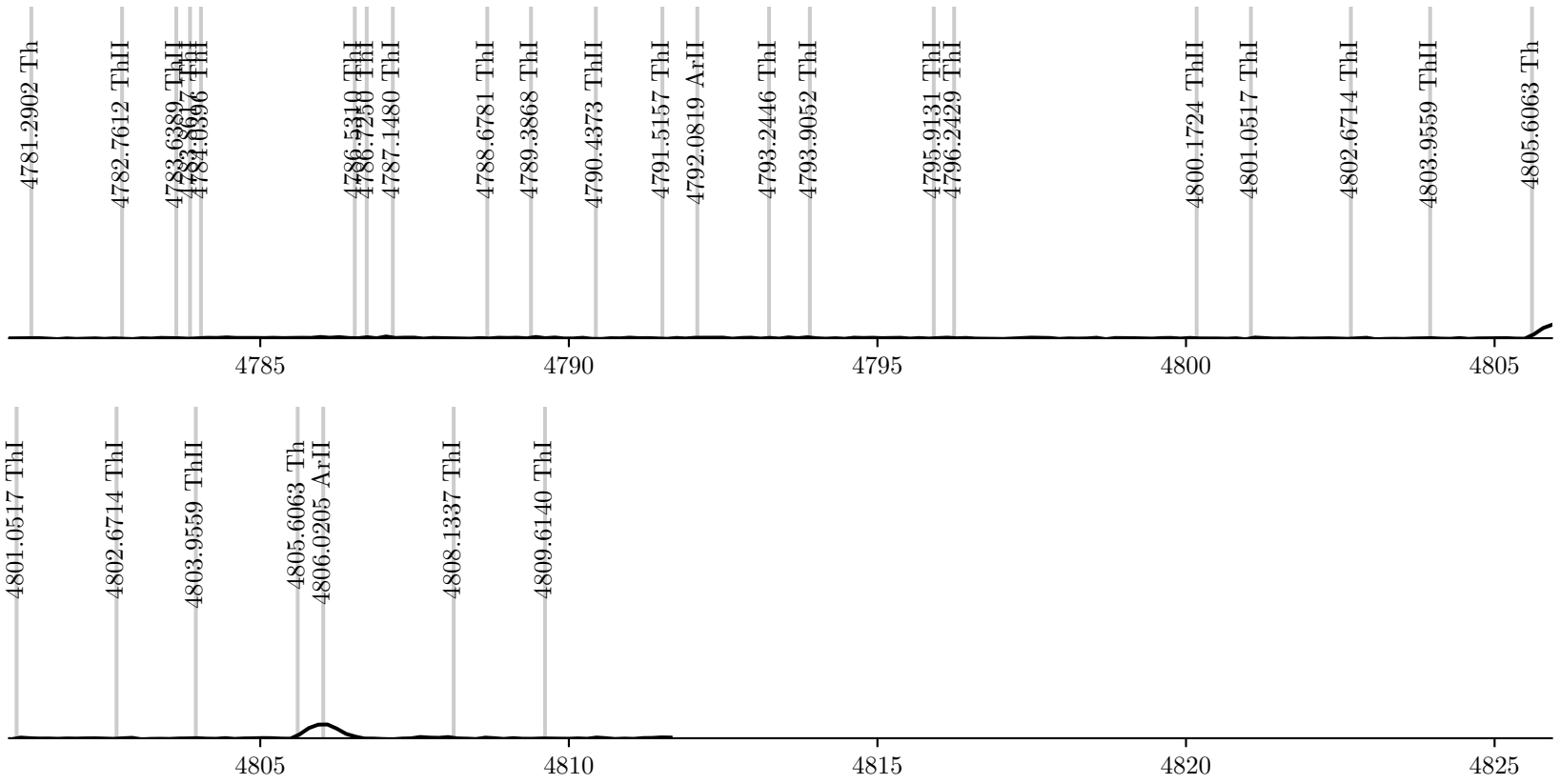

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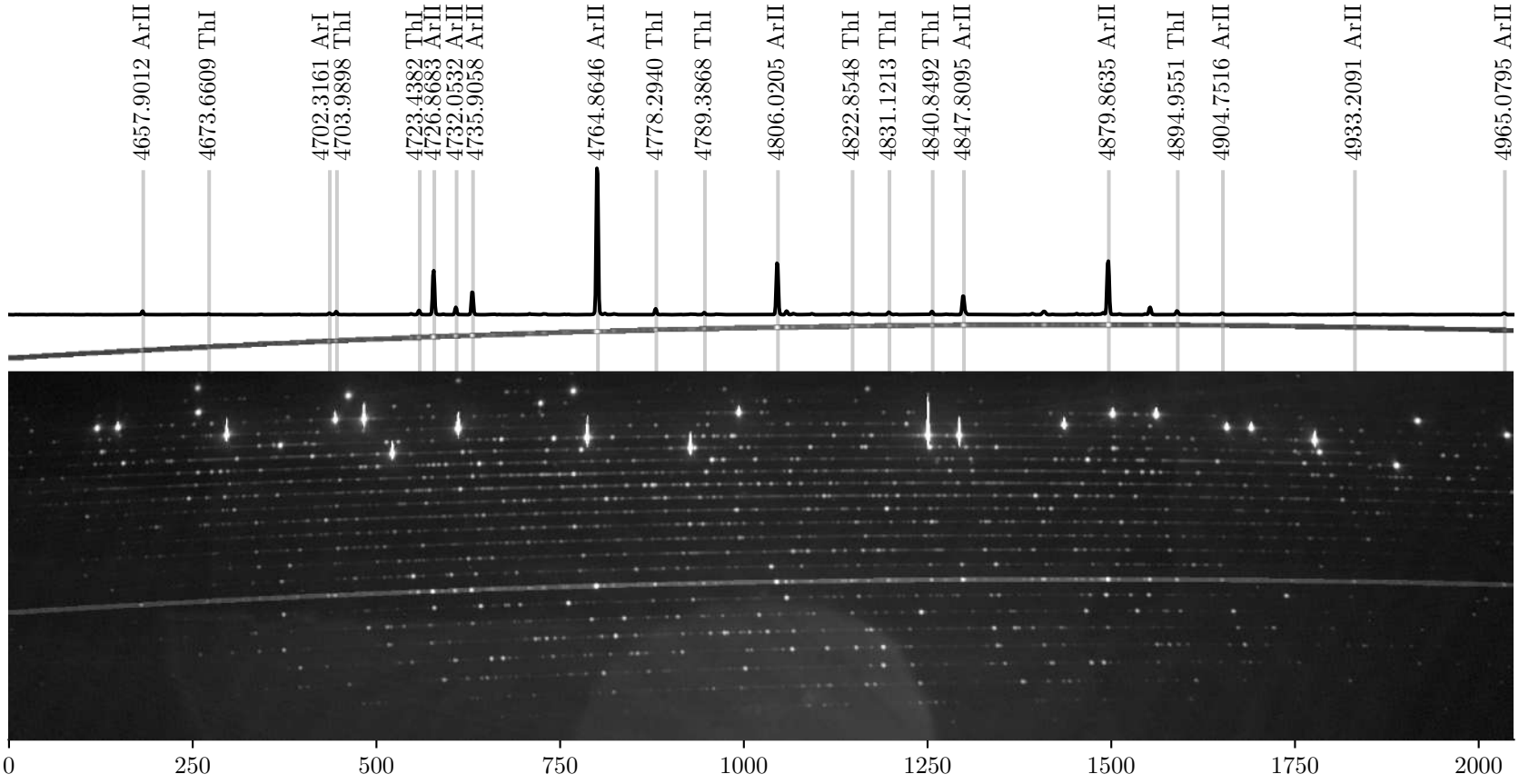

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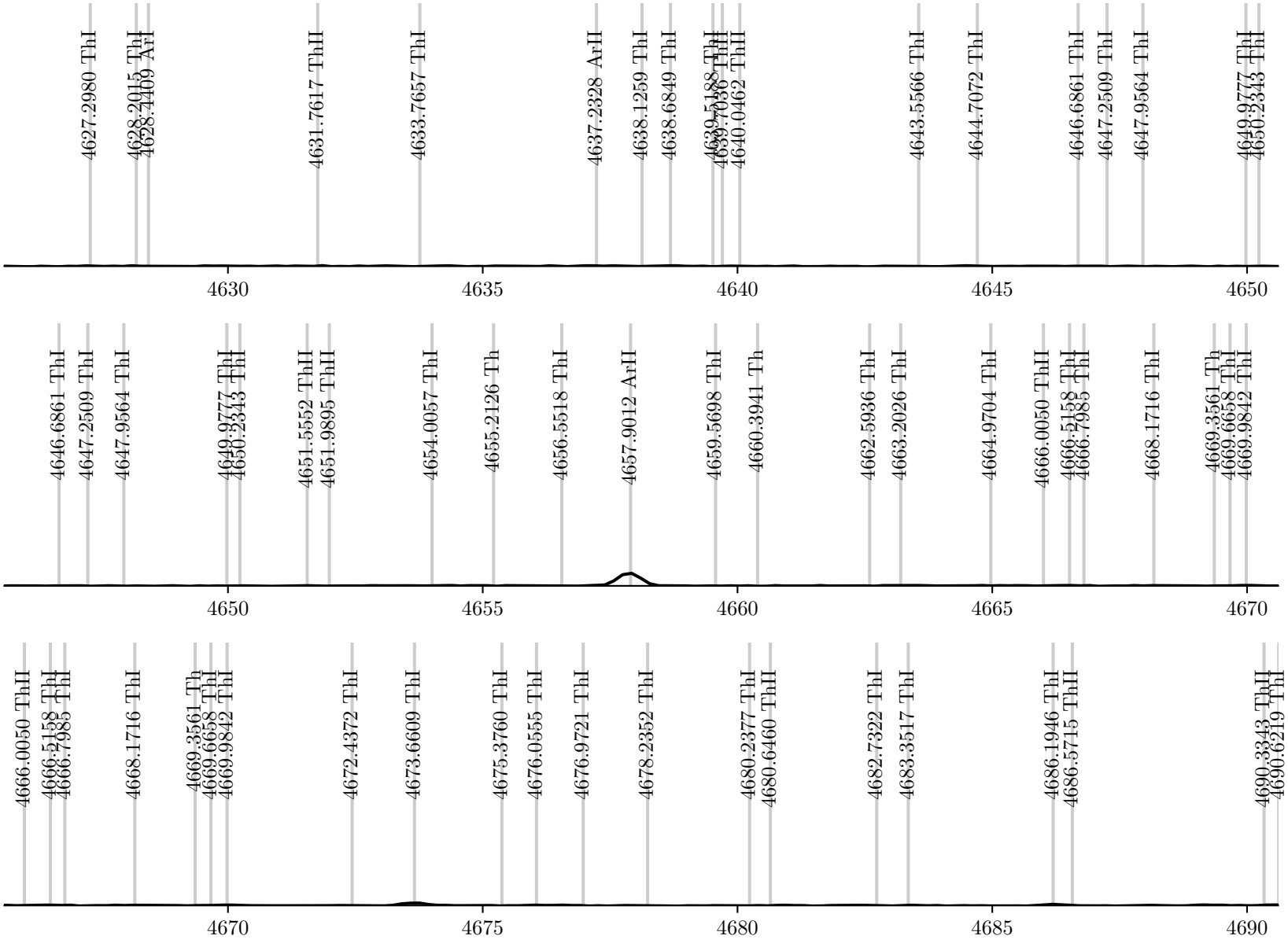

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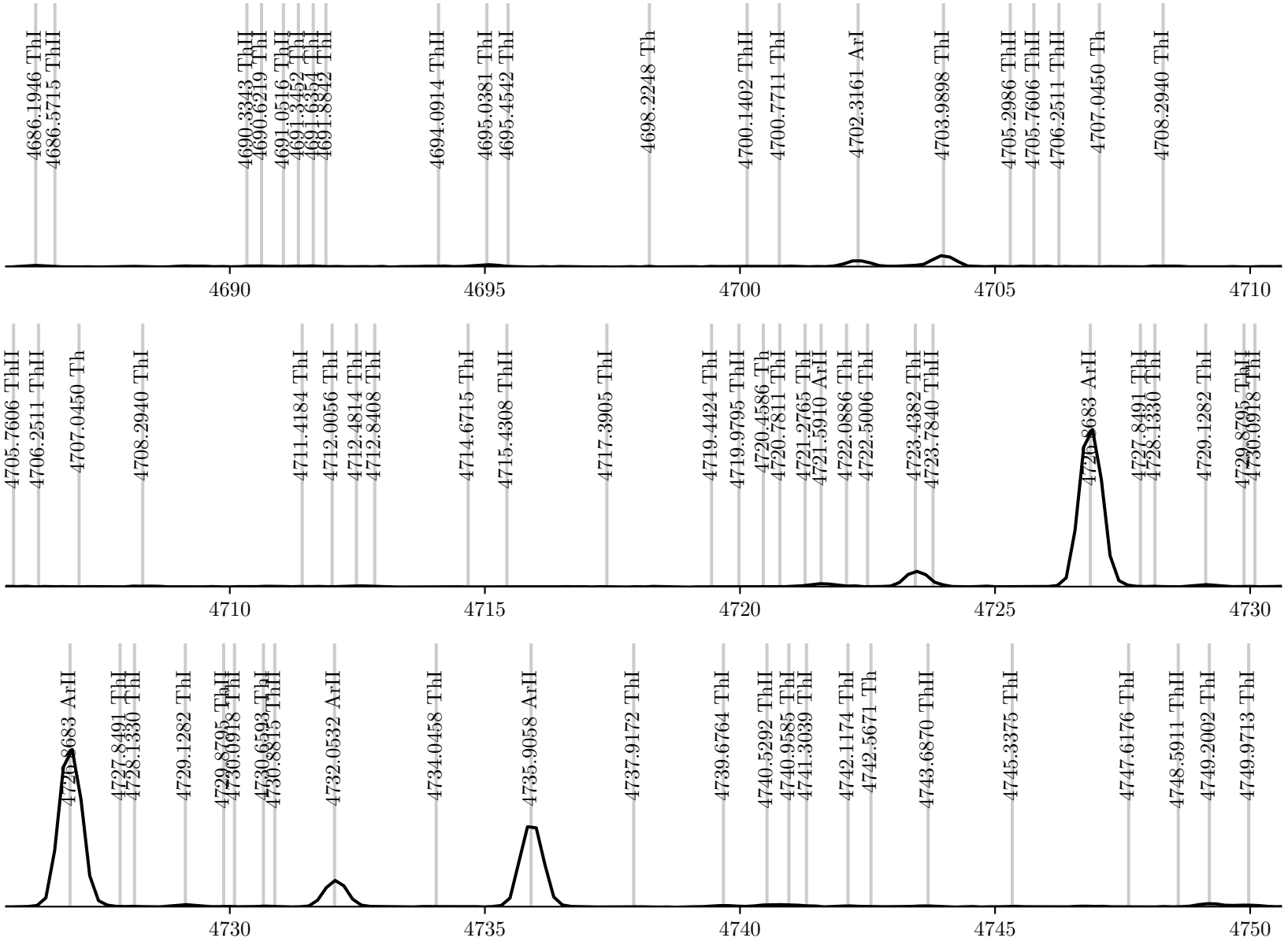

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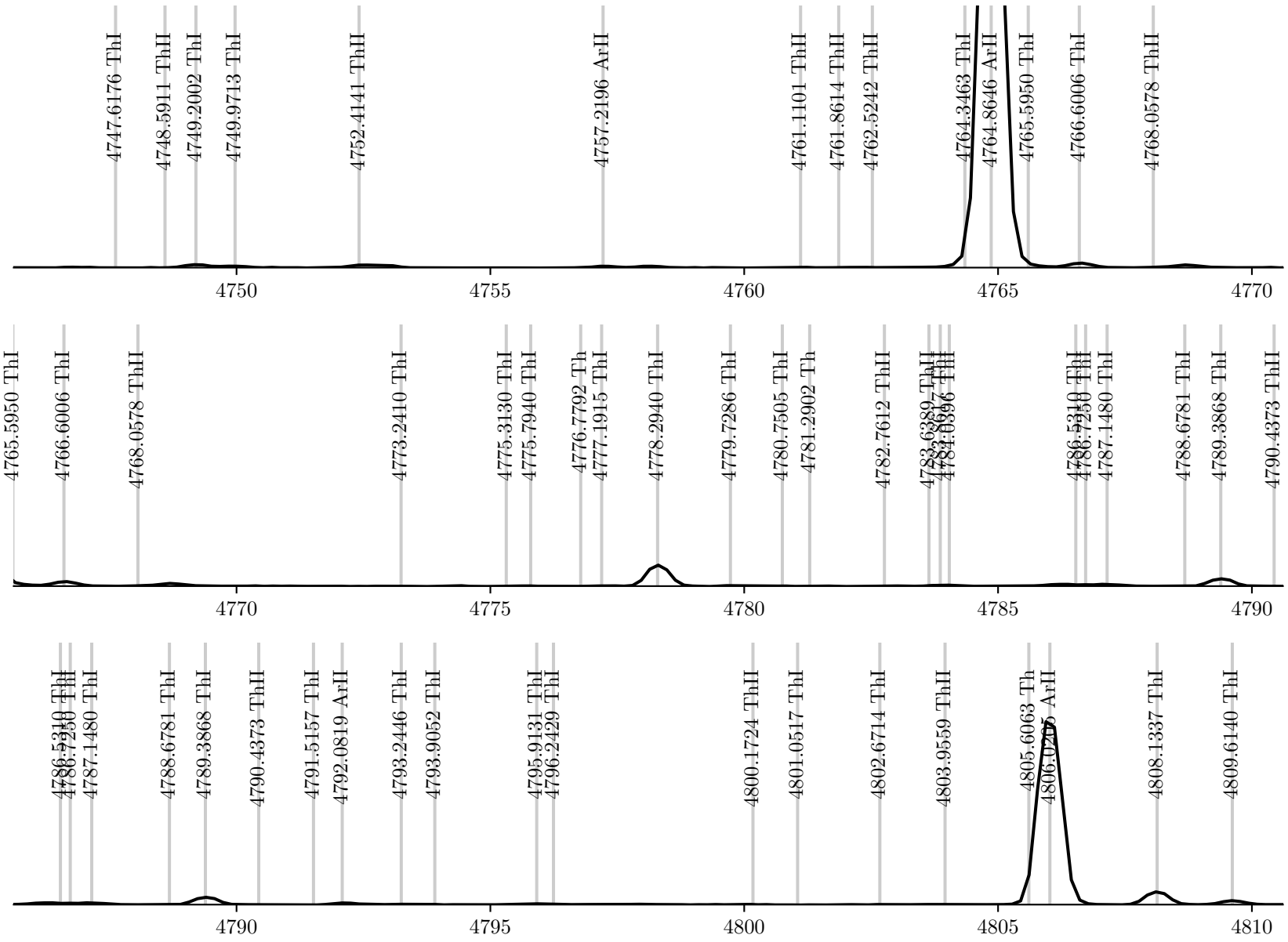
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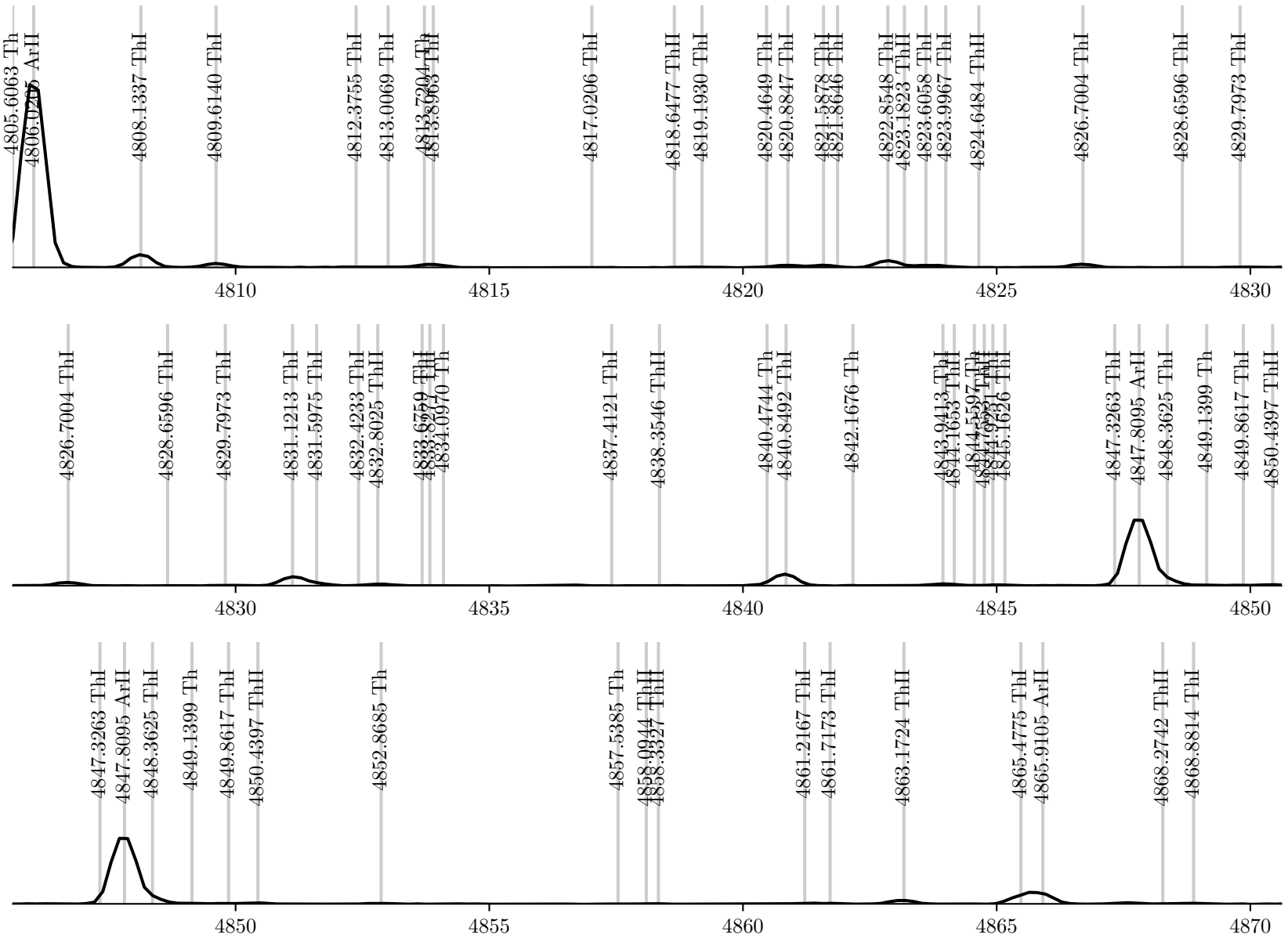


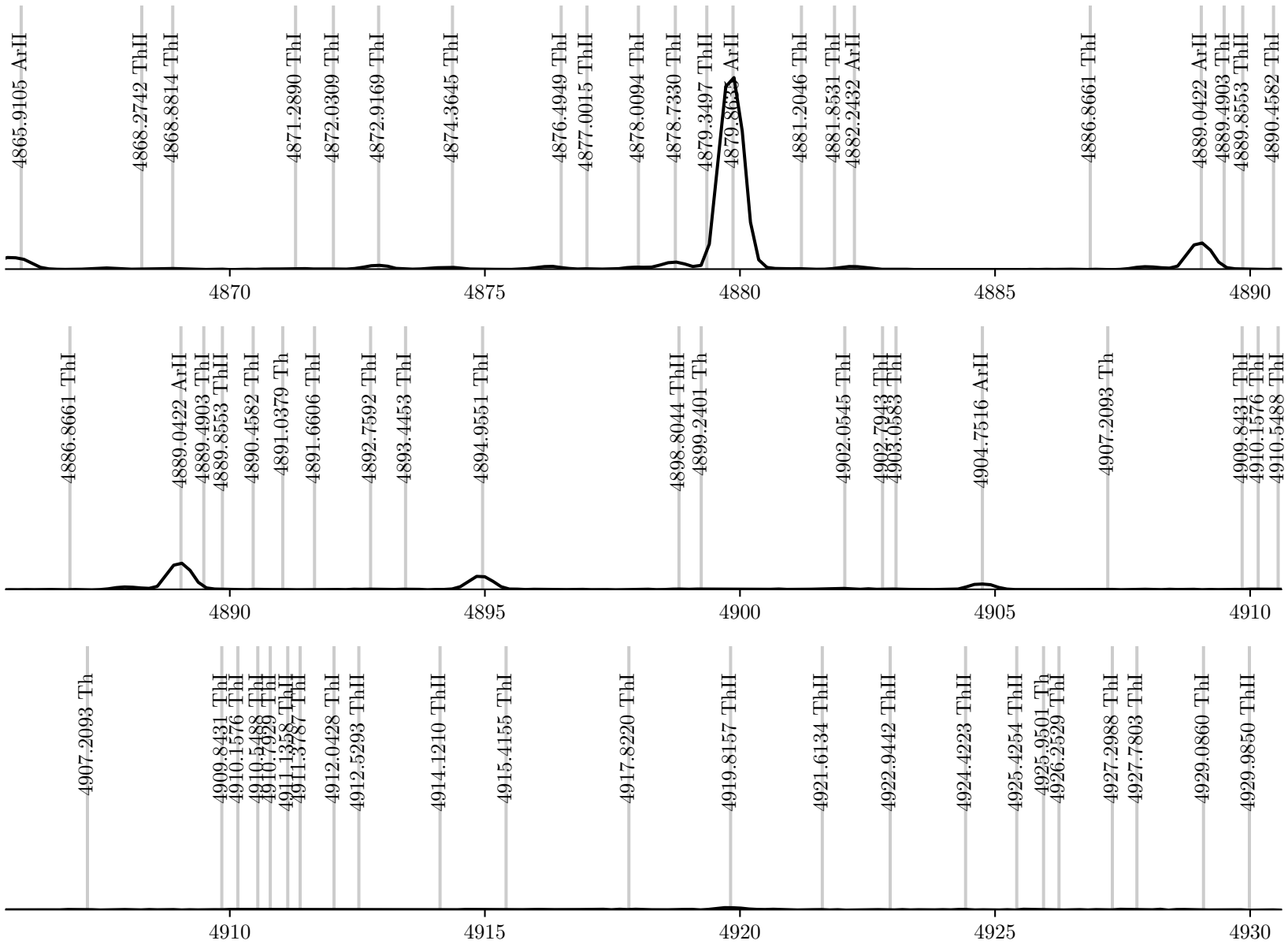
Dispersion:	0.167 Å pix^{-1}
Central wavelength:	4802 Å
Wavelength range:	4626 Å – 4967 Å

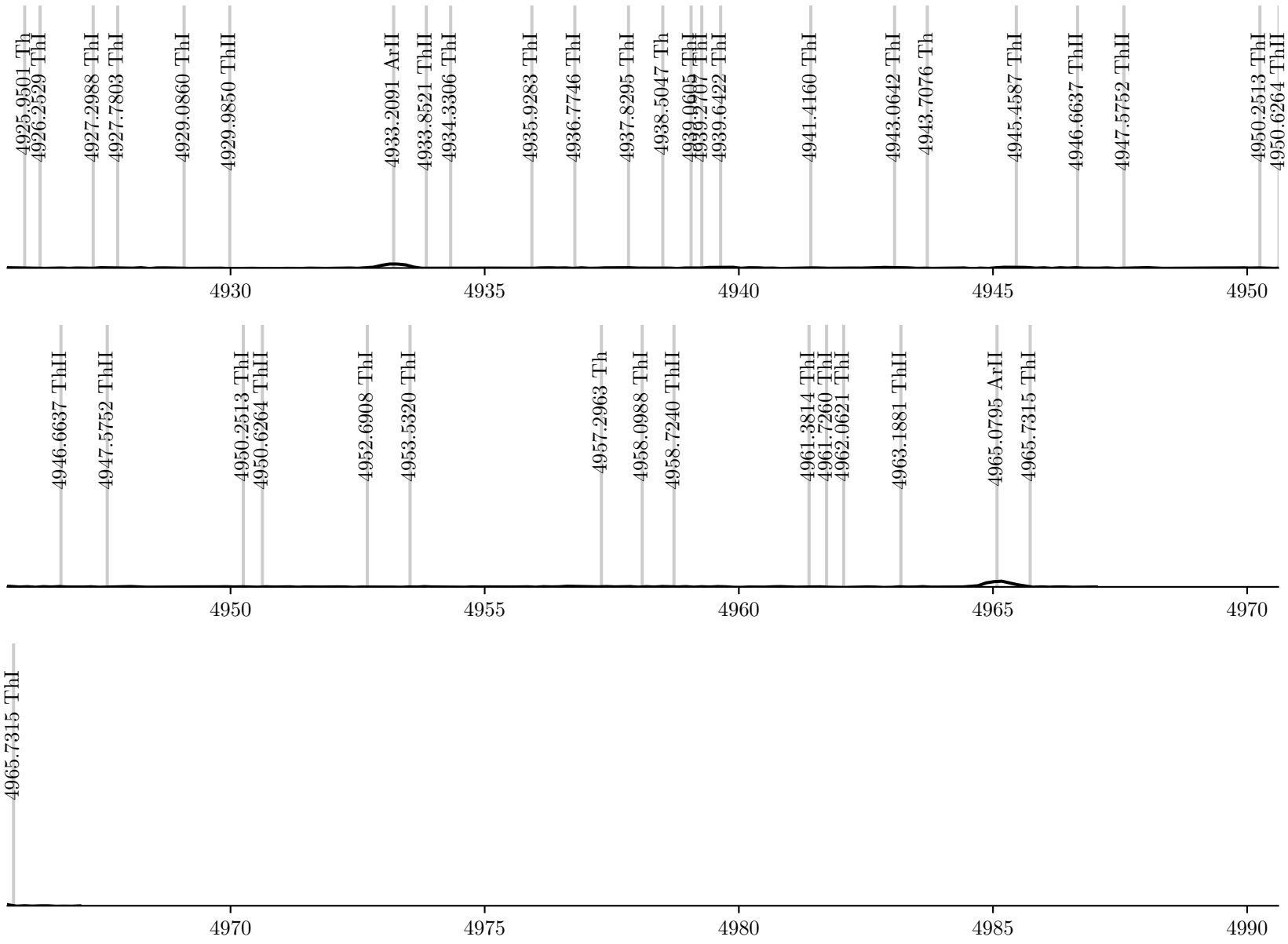

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4638.1259 ThI
4638.6849 ThI
4639.5188 ThI
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4640.0462 ThII
4643.5566 ThI
4644.7072 ThI
4646.6861 ThI
4647.2509 ThI
4647.9564 ThI
4649.9777 ThI
4650.2343 ThI
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4635
4640
4645
4650
4646.6861 ThI
4647.2509 ThI
4647.9564 ThI
4649.9777 ThI
4650.2343 ThI
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4651.9895 ThII
4654.0057 ThI
4655.2126 Th
4656.5518 ThI
4657.9012 ArII
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4662.5936 ThI
4663.2026 ThI
4664.9704 ThI
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4666.7985 ThI
4668.1716 ThI
4669.3561 Th
4669.6658 ThI
4669.9842 ThI
4650
4655
4660
4665
4670
4666.0050 ThII
4666.5158 ThI
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4668.1716 ThI
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4669.6658 ThI
4669.9842 ThI
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4673.6609 ThI
4675.3760 ThI
4676.0555 ThI
4676.9721 ThI
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4683.3517 ThI
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4690.6219 ThI
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4680
4685
4690


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4691.6354 ThI
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4695.4542 ThI
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4707.0450 Th
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4719.4424 ThI
4719.9795 ThII
4720.4586 Th
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4721.2765 ThI
4721.5910 ArII
4722.0886 ThI
4722.5006 ThI
4723.4382 ThI
4723.7840 ThII
4720.8683 ArII
4727.8491 ThI
4728.1330 ThI
4729.1282 ThI
4729.8795 ThII
4730.0918 ThI
4710
4715
4720
4725
4730
4720.8683 ArII
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4742.1174 ThI
4742.5671 Th
4743.6870 ThII
4745.3375 ThI
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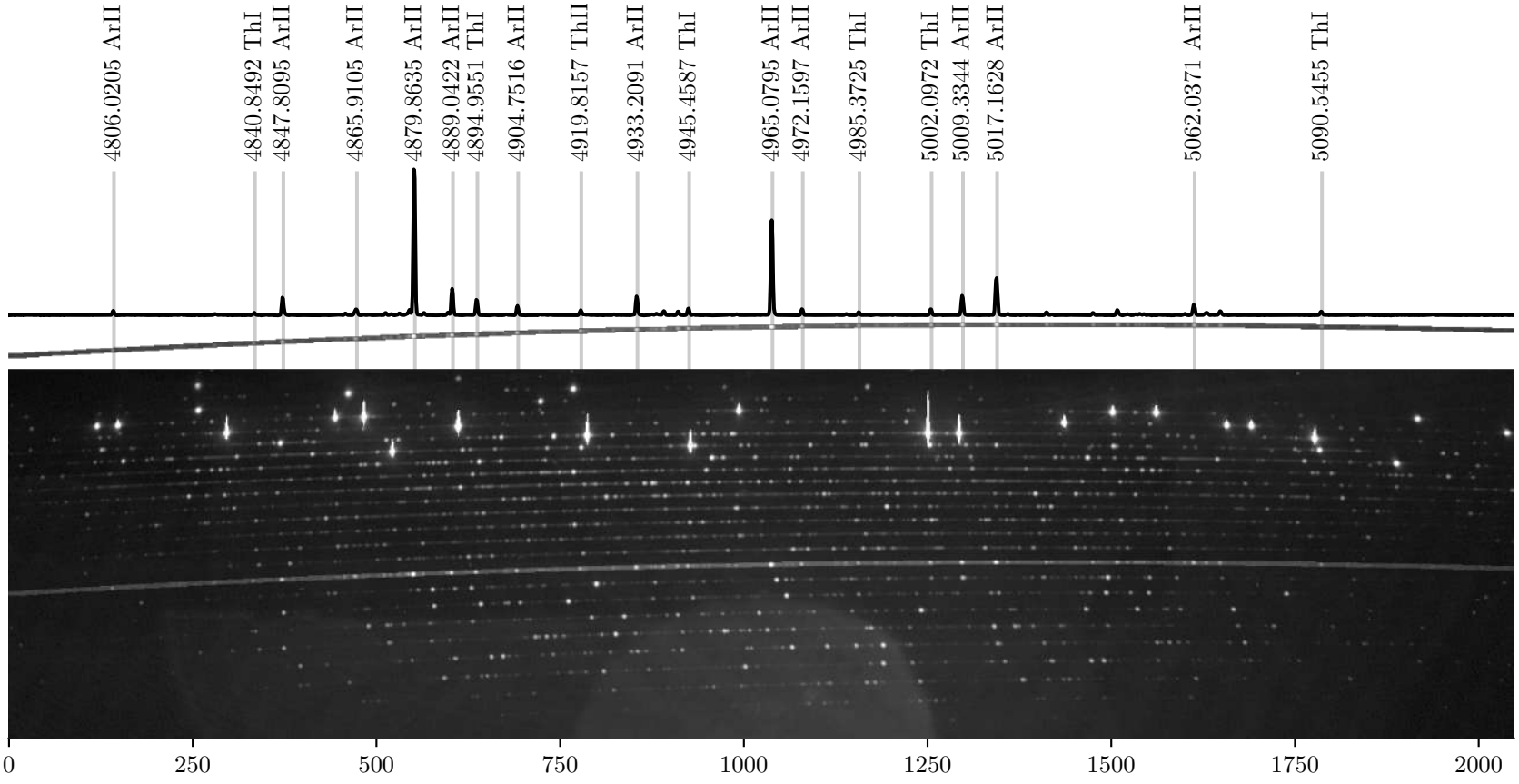

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4752.4141 ThII
4757.2196 ArII
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4761.8614 ThII
4762.5242 ThII
4764.3463 ThI
4764.8646 ArII
4765.5950 ThI
4766.6006 ThI
4768.0578 ThII
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4766.6006 ThI
4768.0578 ThII
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4775.3130 ThI
4775.7940 ThI
4776.7792 Th
4777.1915 ThI
4778.2940 ThI
4779.7286 ThI
4780.7505 ThI
4781.2902 Th
4782.7612 ThII
4786.5310 ThI
4787.1480 ThI
4788.6781 ThI
4789.3868 ThI
4790.4373 ThII
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4775
4780
4785
4790
4786.5310 ThI
4787.1480 ThI
4788.6781 ThI
4789.3868 ThI
4790.4373 ThII
4791.5157 ThI
4792.0819 ArII
4793.2446 ThI
4793.9052 ThI
4795.9131 ThI
4796.2429 ThI
4800.1724 ThII
4801.0517 ThI
4802.6714 ThI
4803.9559 ThII
4805.6063 Th
4806.0205 ArII
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4809.6140 ThI
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4795
4800
4805
4810


4805.6063 Th
4806.0210 ArII
4808.1337 ThI
4809.6140 ThI
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4813.7204 Th
4815.8963 ThI
4817.0206 ThI
4818.6477 ThII
4819.1930 ThI
4820.4649 ThI
4820.8847 ThI
4821.5878 ThI
4821.8646 ThI
4822.8548 ThI
4823.1823 ThII
4823.6058 ThI
4823.9967 ThI
4824.6484 ThII
4826.7004 ThI
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4829.7973 ThI
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4815
4820
4825
4830
4826.7004 ThI
4828.6596 ThI
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4838.3546 ThII
4840.4744 Th
4840.8492 ThI
4842.1676 Th
4843.9413 ThI
4844.1653 ThI
4847.3263 ThI
4847.8095 ArII
4848.3625 ThI
4849.1399 Th
4849.8617 ThI
4850.4397 ThII
4830
4835
4840
4845
4850
4847.3263 ThI
4847.8095 ArII
4848.3625 ThI
4849.1399 Th
4849.8617 ThI
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4863.1724 ThII
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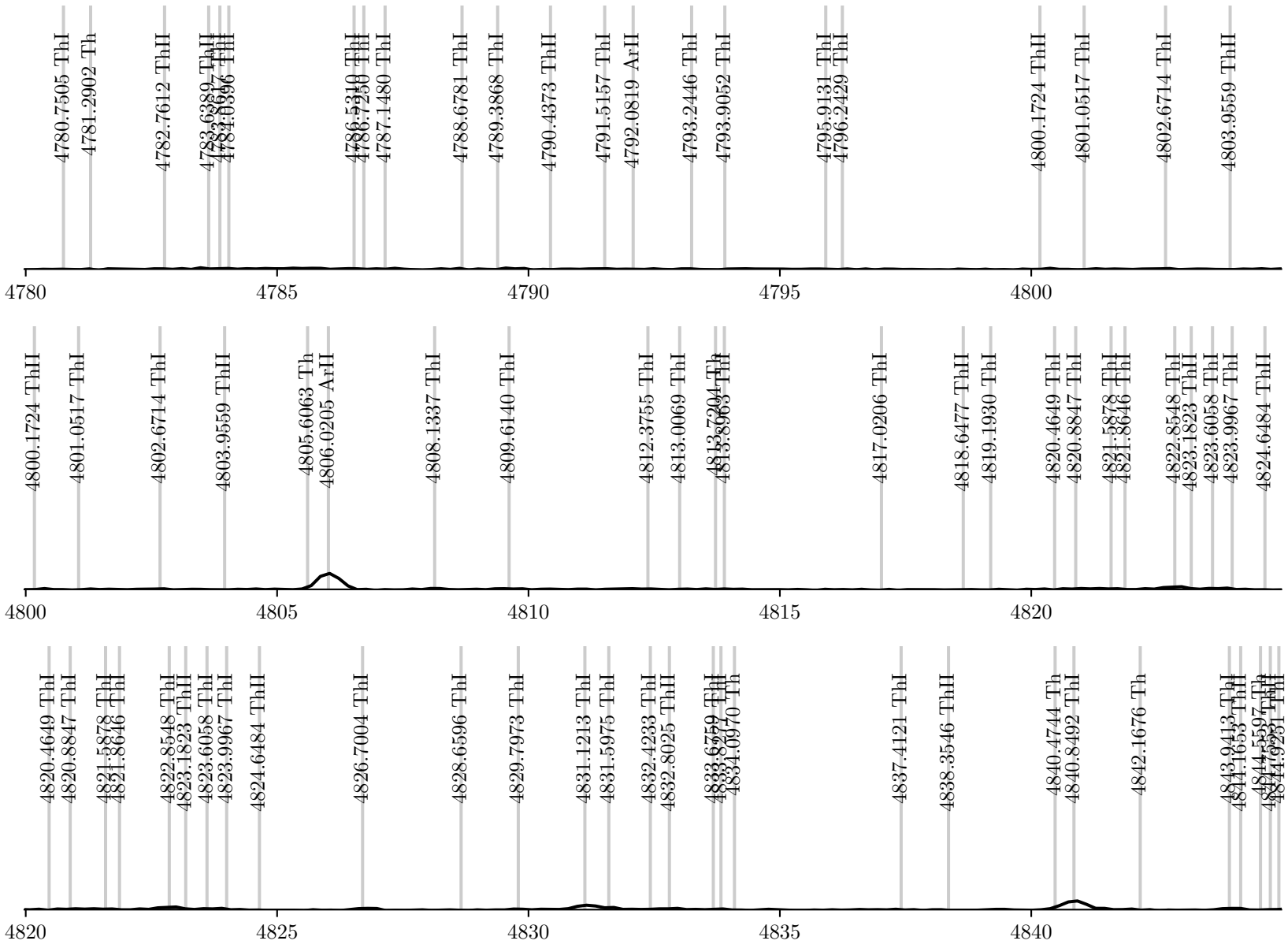

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4872.9169 ThI
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4876.4949 ThI
4877.0015 ThII
4878.0094 ThI
4878.7330 ThI
4879.3497 ThII
4879.8635 ArII
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4882.2432 ArII
4886.8661 ThI
4889.0422 ArII
4889.4903 ThI
4889.8553 ThII
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4885
4890
4886.8661 ThI
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4889.4903 ThI
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4893.4453 ThI
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4902.7943 ThI
4903.0583 ThI
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4909.8431 ThI
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4907.2093 Th
4909.8431 ThI
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4910.5488 ThI
4910.7959 ThI
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4911.3787 ThI
4912.0428 ThI
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4915.4155 ThI
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4919.8157 ThII
4921.6134 ThII
4922.9442 ThII
4924.4223 ThII
4925.4254 ThII
4925.9501 Th
4926.2529 ThI
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4927.7803 ThI
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4929.9850 ThII
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4920
4925
4930

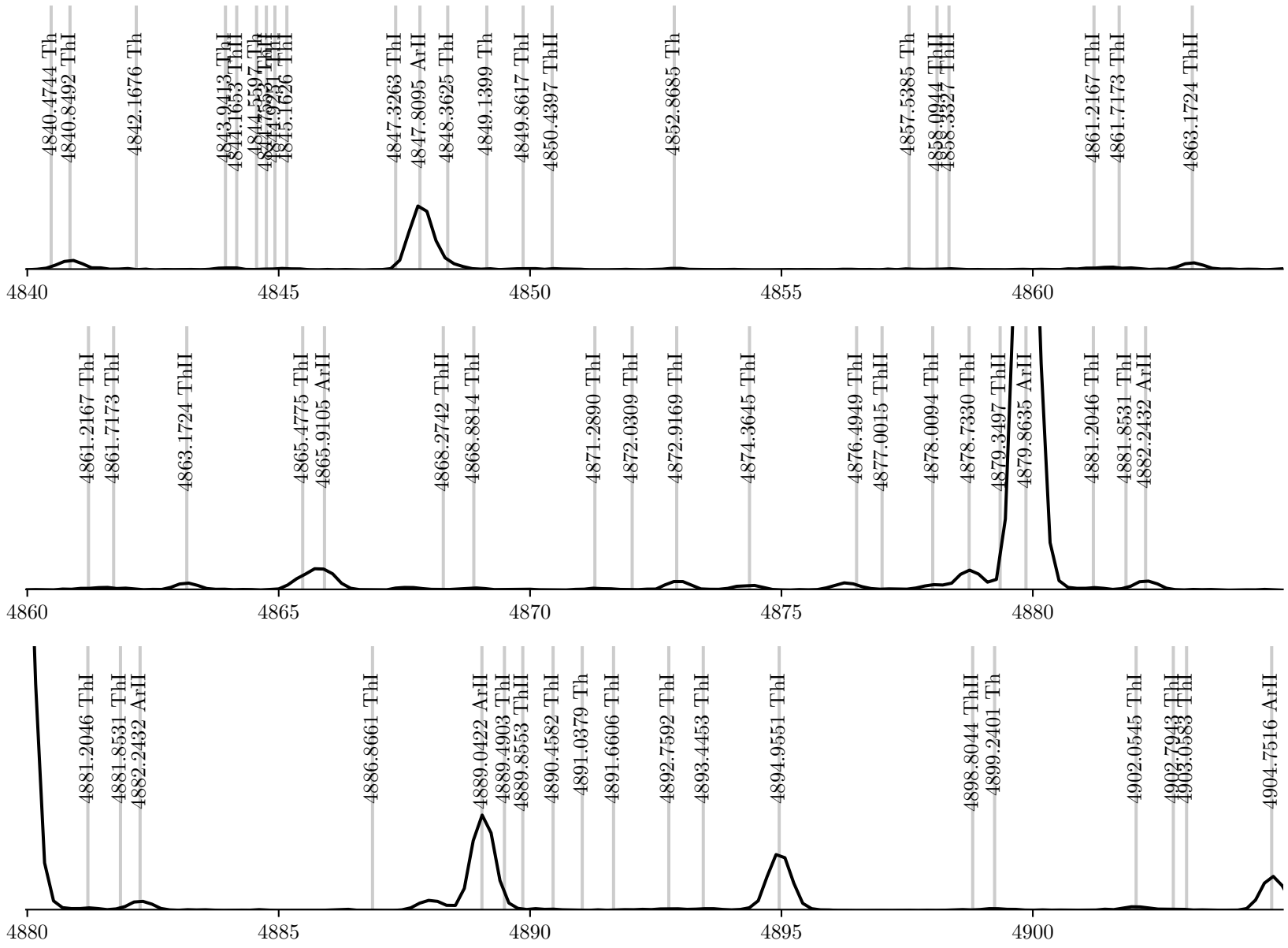

4925.9501 Th
4926.2529 ThI
4927.2988 ThI
4927.7803 ThI
4929.0860 ThI
4929.9850 ThII
4933.2091 ArII
4933.8521 ThII
4934.3306 ThI
4935.9283 ThI
4936.7746 ThI
4937.8295 ThI
4938.5047 Th
4939.0605 ThI
4939.2707 ThI
4939.6422 ThI
4941.4160 ThI
4943.0642 ThI
4943.7076 Th
4945.4587 ThI
4946.6637 ThII
4947.5752 ThII
4950.2513 ThI
4950.6264 ThII
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4935
4940
4945
4950
4946.6637 ThII
4947.5752 ThII
4950.2513 ThI
4950.6264 ThII
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4953.5320 ThI
4957.2963 Th
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4990

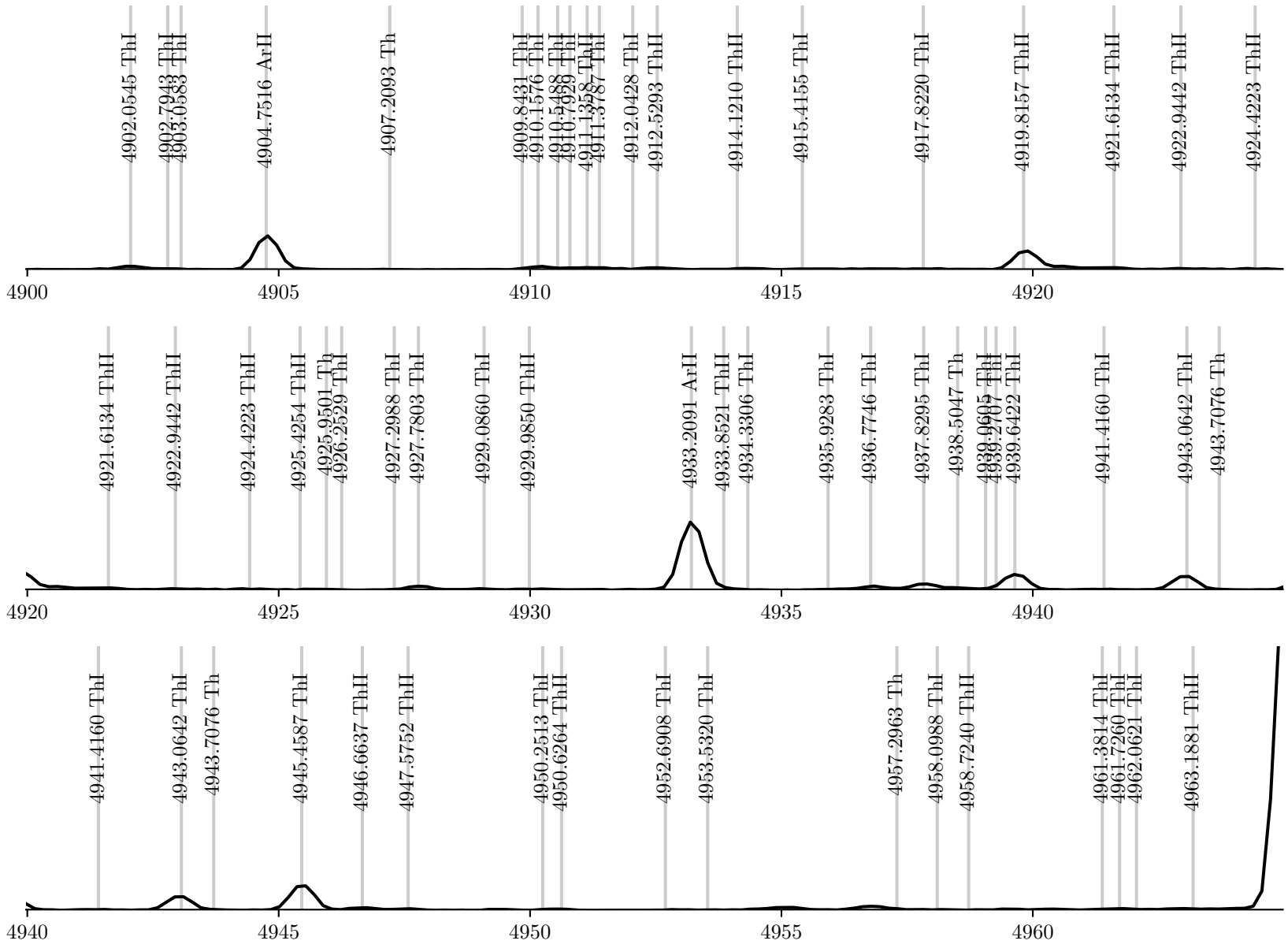
Order 30

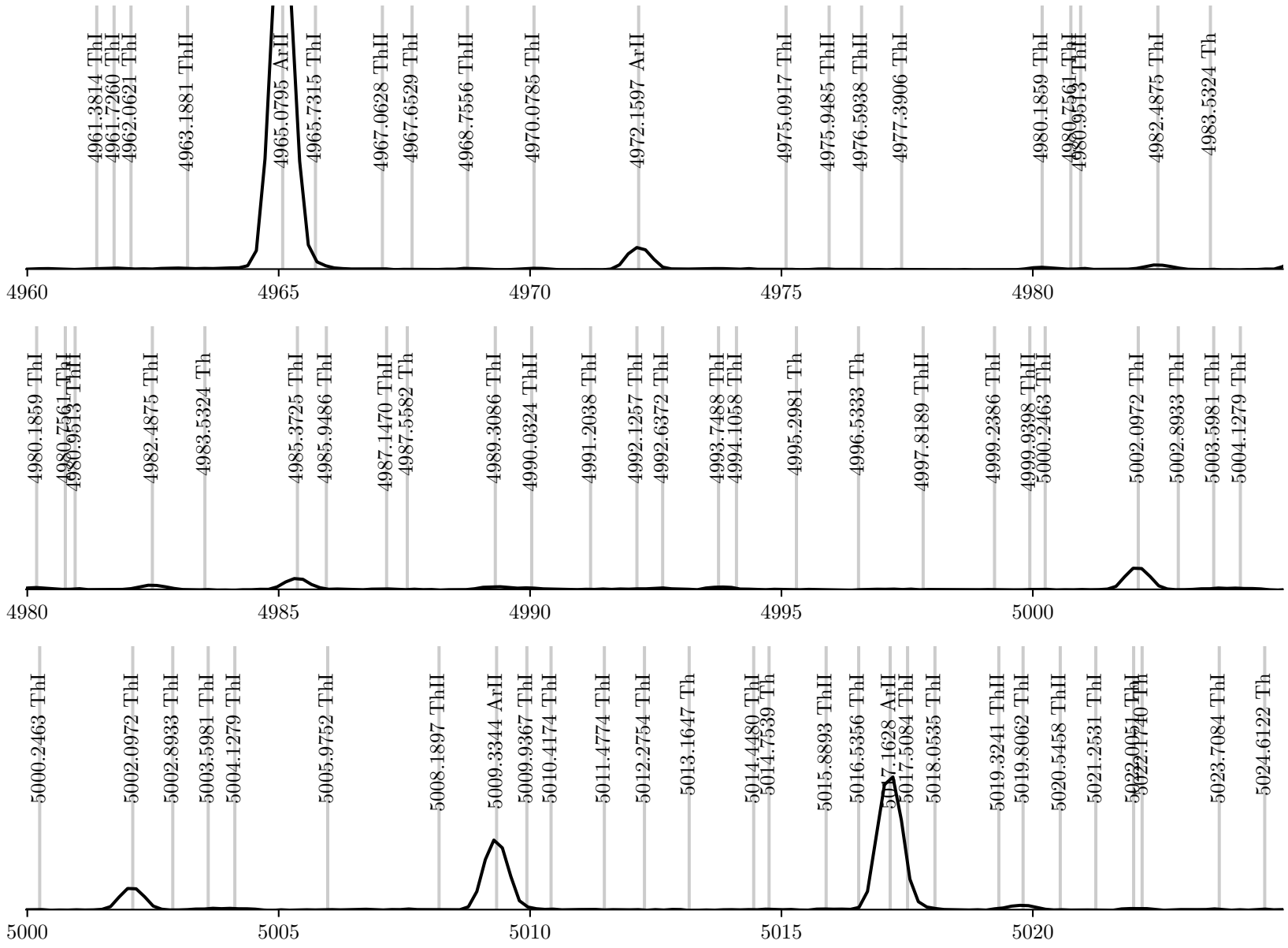


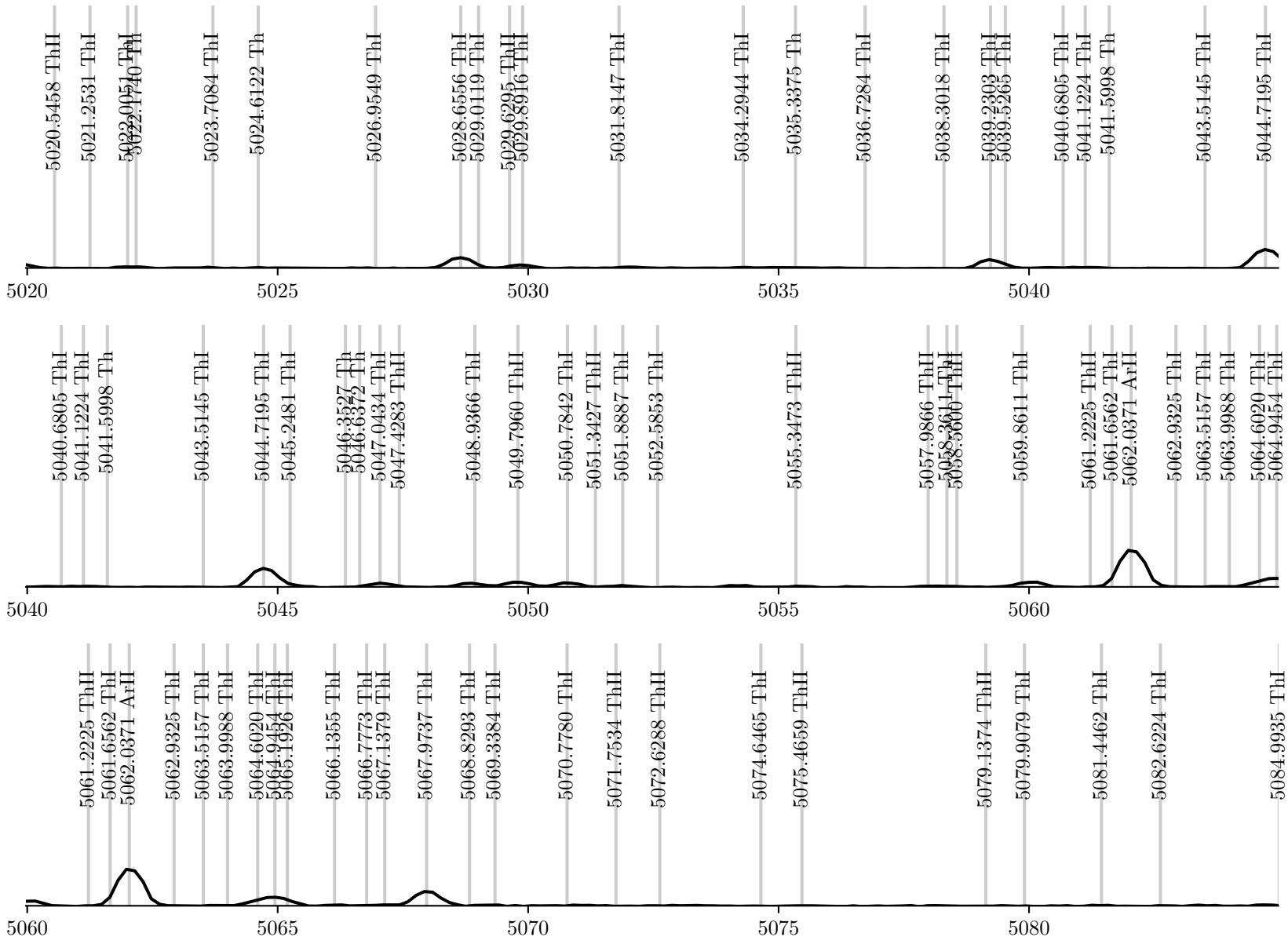
Dispersion:	0.172 Å pix^{-1}
Central wavelength:	4962 Å
Wavelength range:	4780 Å – 5133 Å

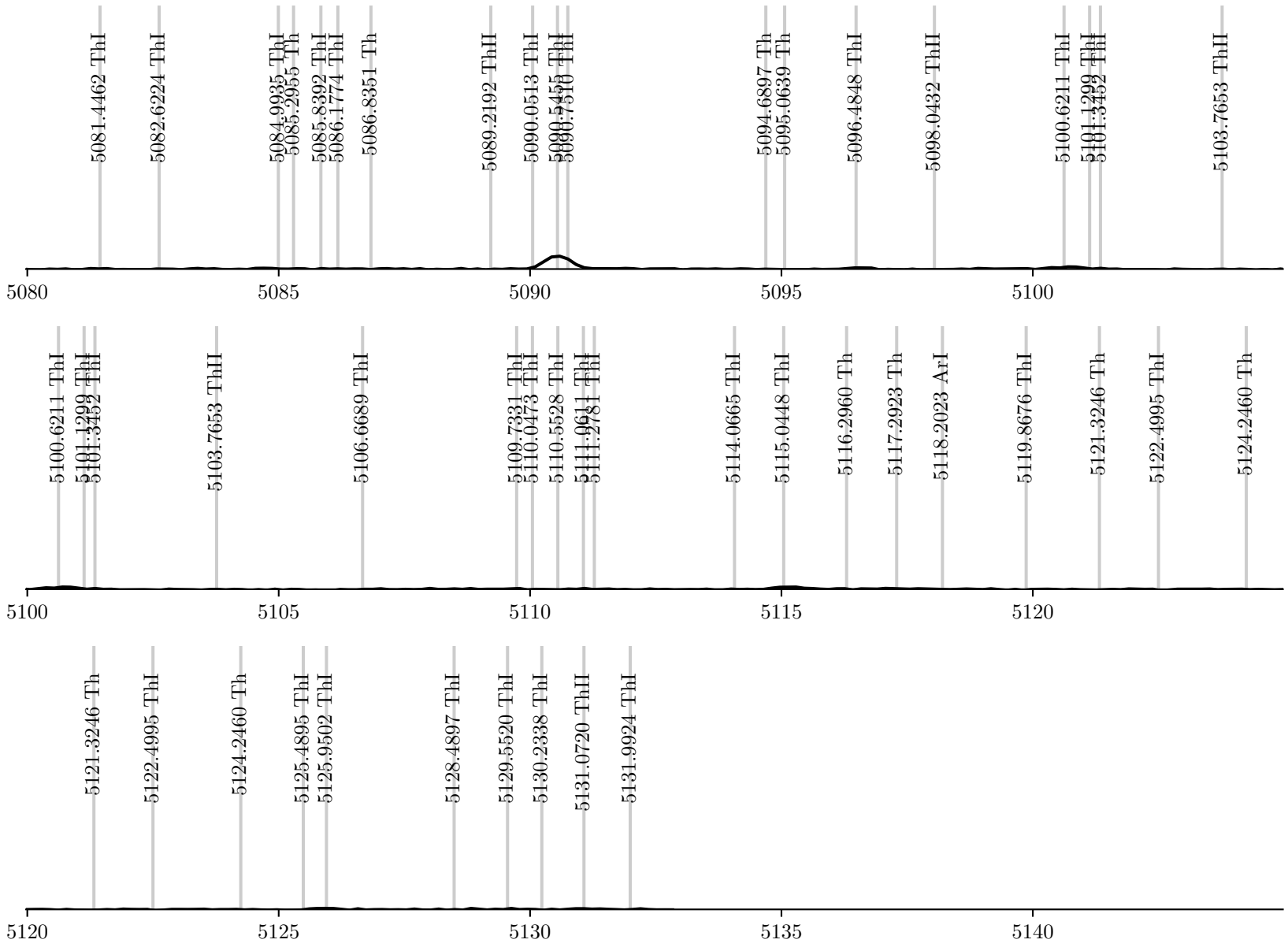

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4795.9131 ThI
4796.2429 ThI
4800.1724 ThII
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4790
4795
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4803.9559 ThII
4805.6063 Th
4806.0205 ArII
4808.1337 ThI
4809.6140 ThI
4812.3755 ThI
4813.0069 ThI
4813.7204 Th
4813.8963 ThI
4817.0206 ThI
4818.6477 ThII
4819.1930 ThI
4820.4649 ThI
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4821.5878 ThI
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4822.8548 ThI
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4823.6058 ThI
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4824.6484 ThII
4800
4805
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4815
4820
4820.4649 ThI
4820.8847 ThI
4821.5878 ThI
4821.8646 ThI
4822.8548 ThI
4823.1823 ThII
4823.6058 ThI
4823.9967 ThI
4824.6484 ThII
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4852.8685 Th
4857.5385 Th
4861.2167 ThI
4861.7173 ThI
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4900

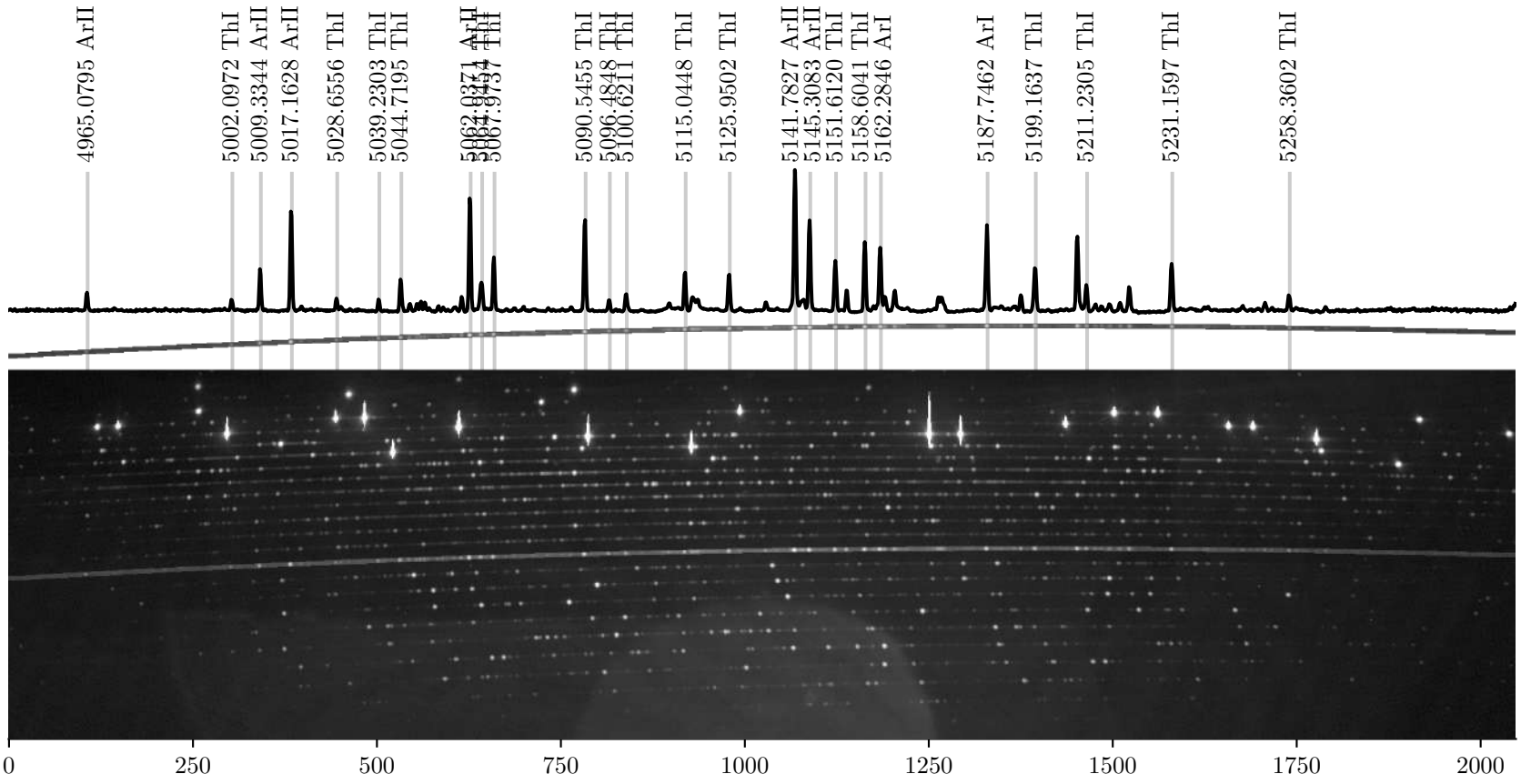

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4924.4223 ThII
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4921.6134 ThII
4922.9442 ThII
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4925.4254 ThII
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4961.3814 ThI
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4999.9398 ThII
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5018.0535 ThI
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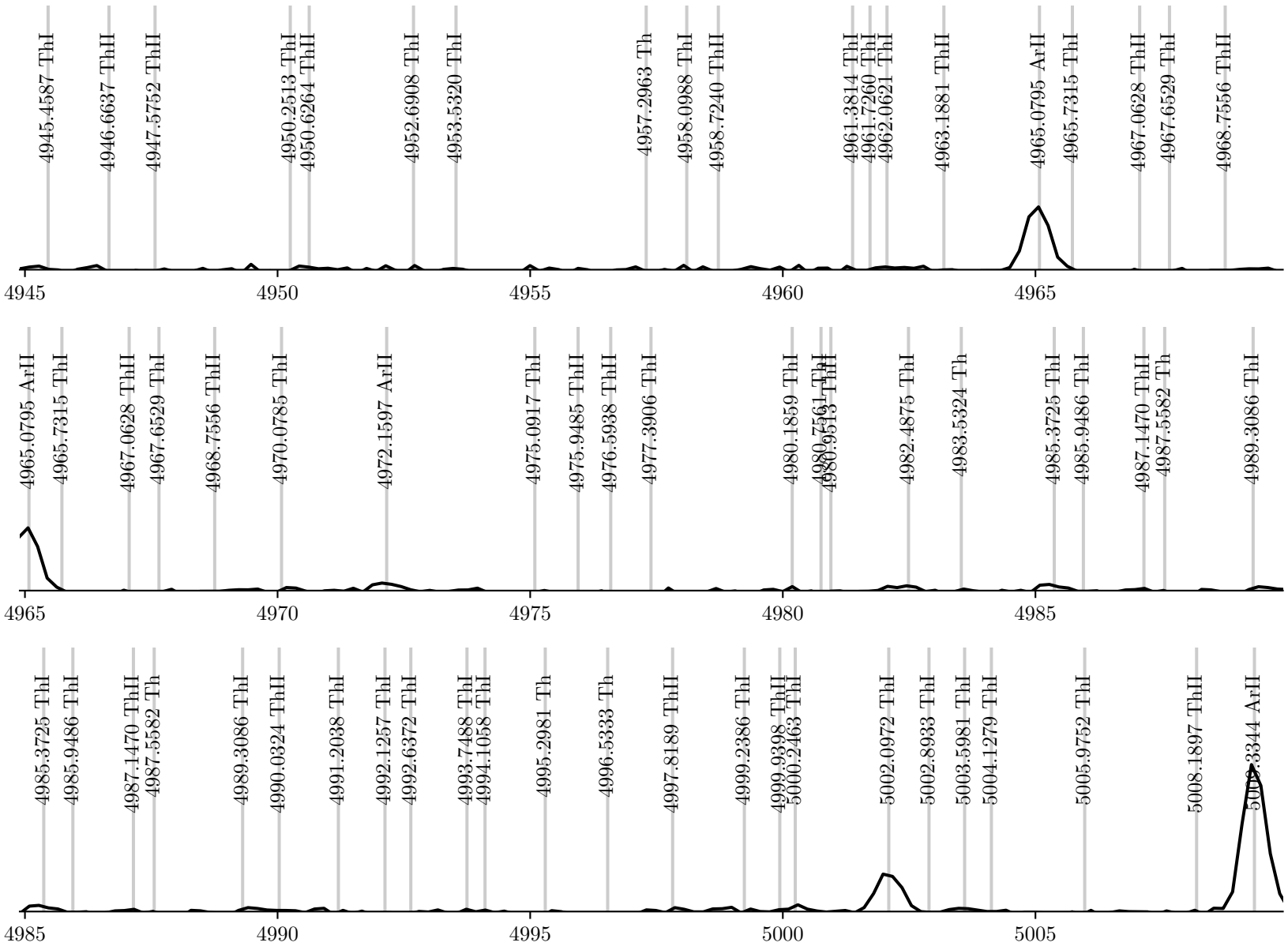

5020.5458 ThII
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5022.1740 ThI
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5024.6122 Th
5026.9549 ThI
5028.6556 ThI
5029.0119 ThI
5029.6295 ThII
5029.8016 ThI
5031.8147 ThI
5034.2944 ThI
5035.3375 Th
5036.7284 ThI
5038.3018 ThI
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5039.5265 ThI
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5061.6562 ThI
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5061.6562 ThI
5062.0371 ArII
5062.9325 ThI
5063.5157 ThI
5063.9988 ThI
5064.6020 ThI
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5065.1926 ThI
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5067.1379 ThI
5067.9737 ThI
5068.8293 ThI
5069.3384 ThI
5070.7780 ThI
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5072.6288 ThII
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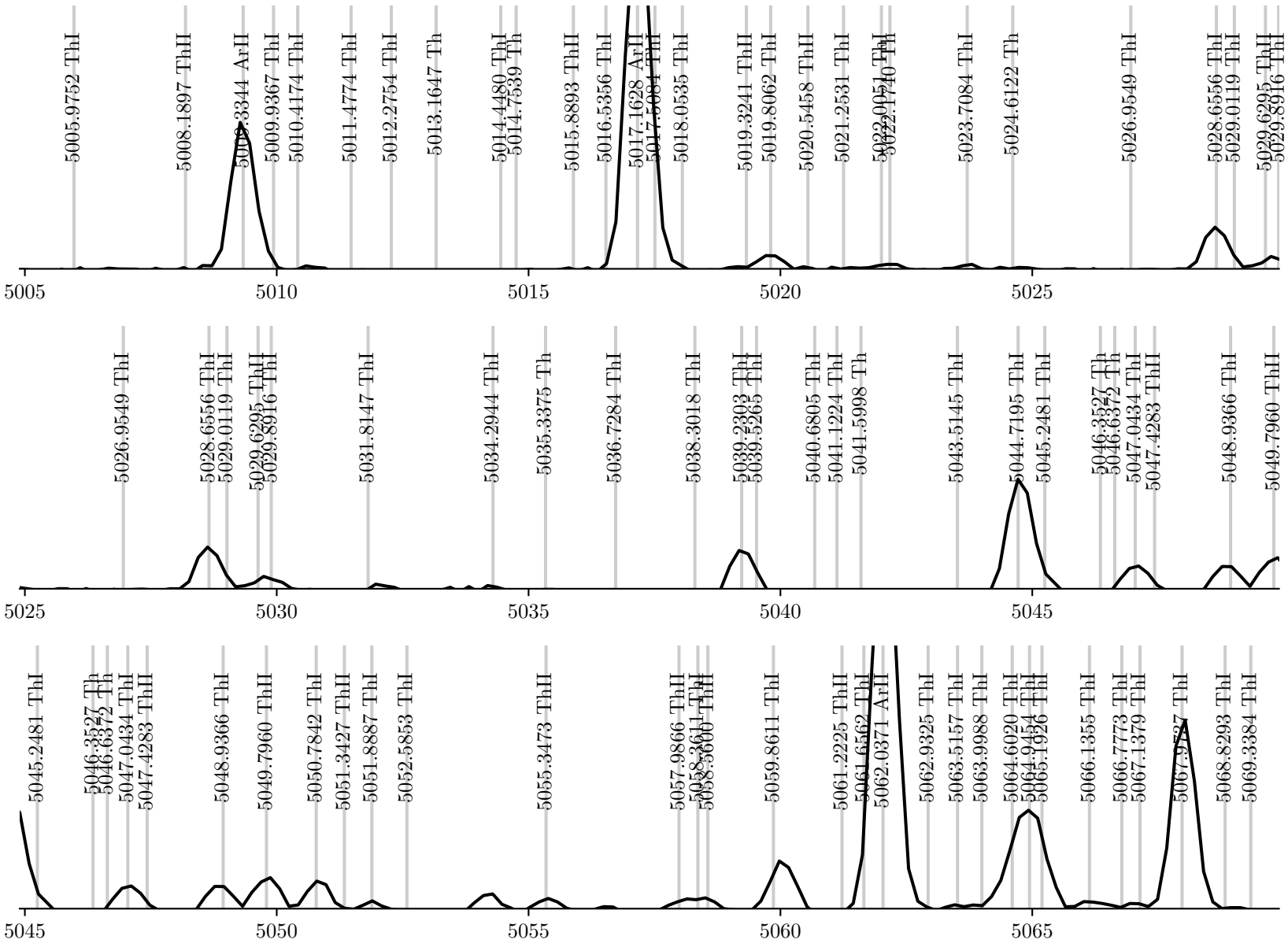

5081.4462 ThI
5082.6224 ThI
5084.9935 ThI
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5086.1774 ThI
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5090.5455 ThI
5090.7510 ThI
5094.6897 Th
5095.0639 Th
5096.4848 ThI
5098.0432 ThII
5100.6211 ThI
5101.1299 ThI
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5124.2460 Th
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5130.2338 ThI
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5131.9924 ThI
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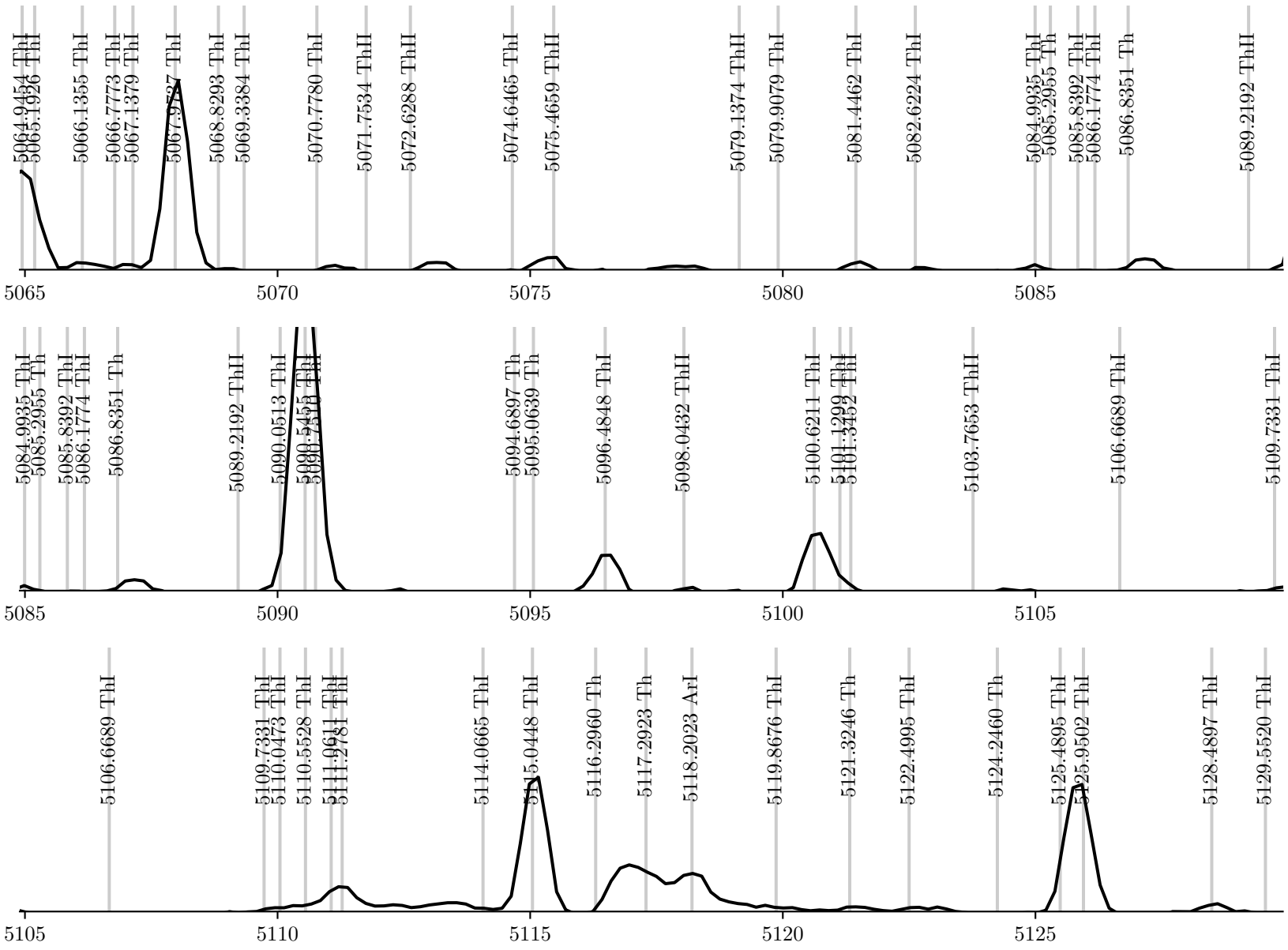
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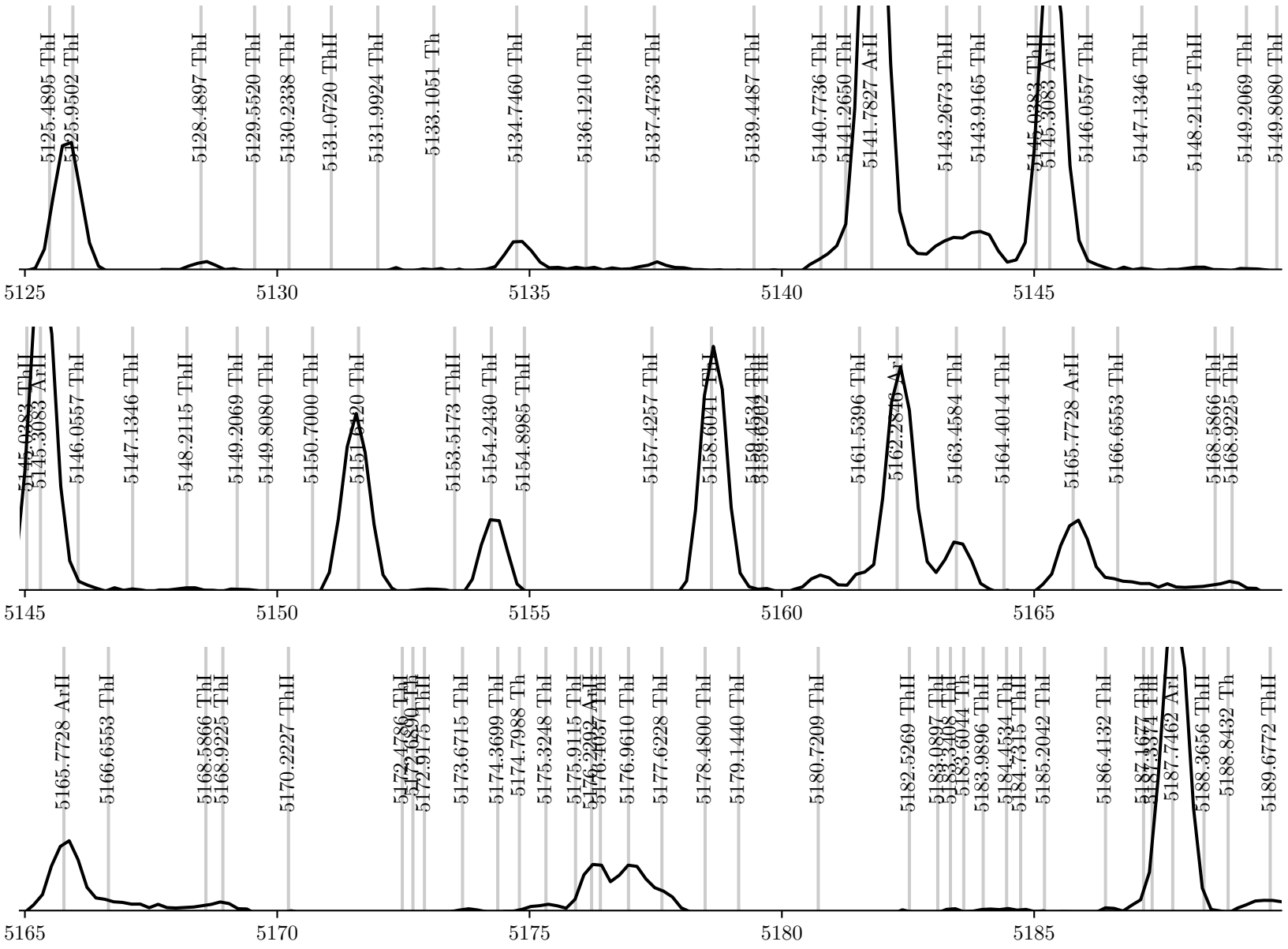


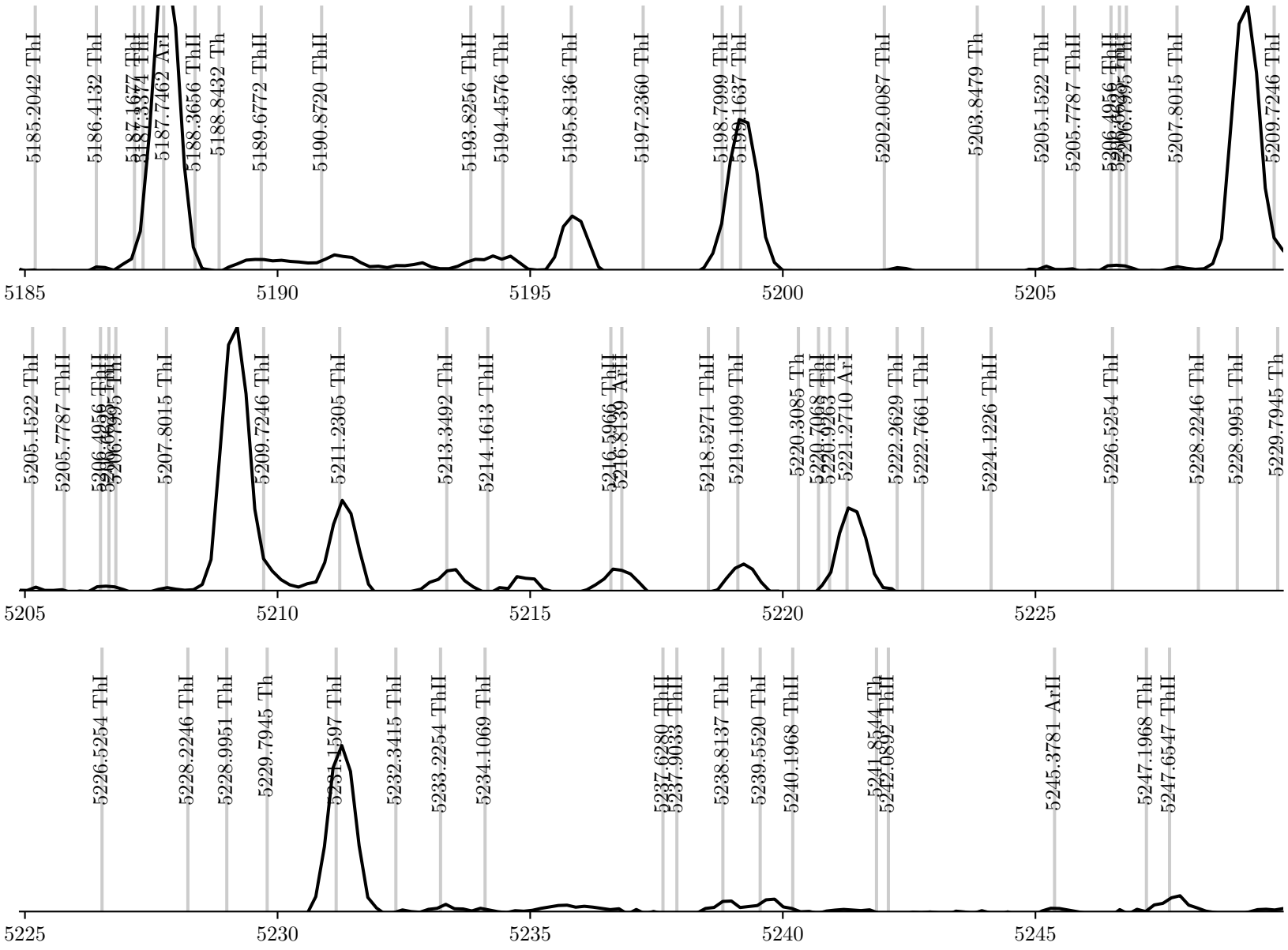
Dispersion:	0.178 Å pix^{-1}
Central wavelength:	5134 Å
Wavelength range:	4945 Å – 5310 Å

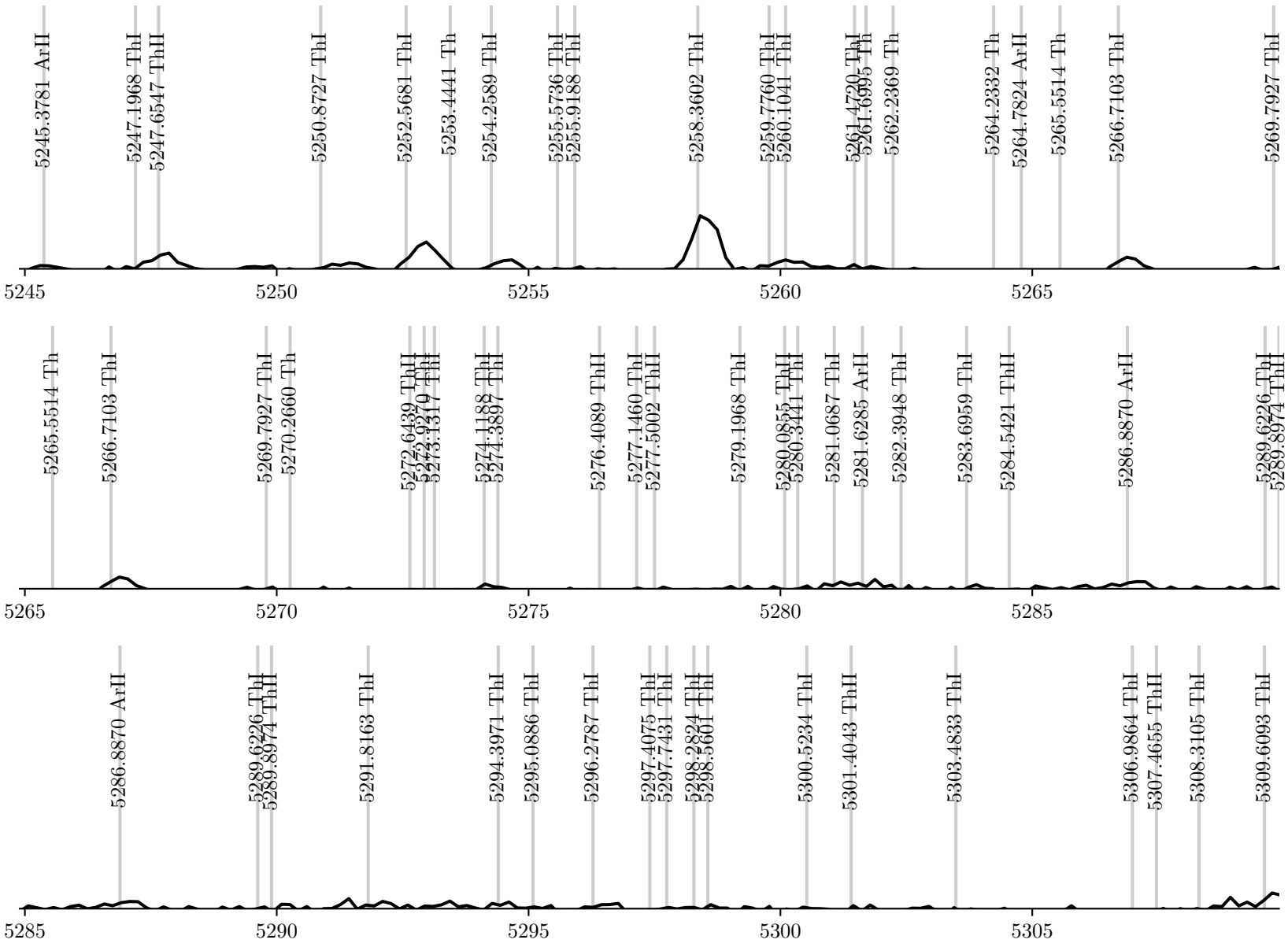

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4957.2963 Th
4958.0988 ThI
4958.7240 ThII
4961.3814 ThI
4961.7260 ThI
4962.0621 ThI
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4965.0795 ArII
4965.7315 ThI
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4968.7556 ThII
4945
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4965.0795 ArII
4965.7315 ThI
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4967.6529 ThI
4968.7556 ThII
4970.0785 ThI
4972.1597 ArII
4975.0917 ThI
4975.9485 ThII
4976.5938 ThII
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5005

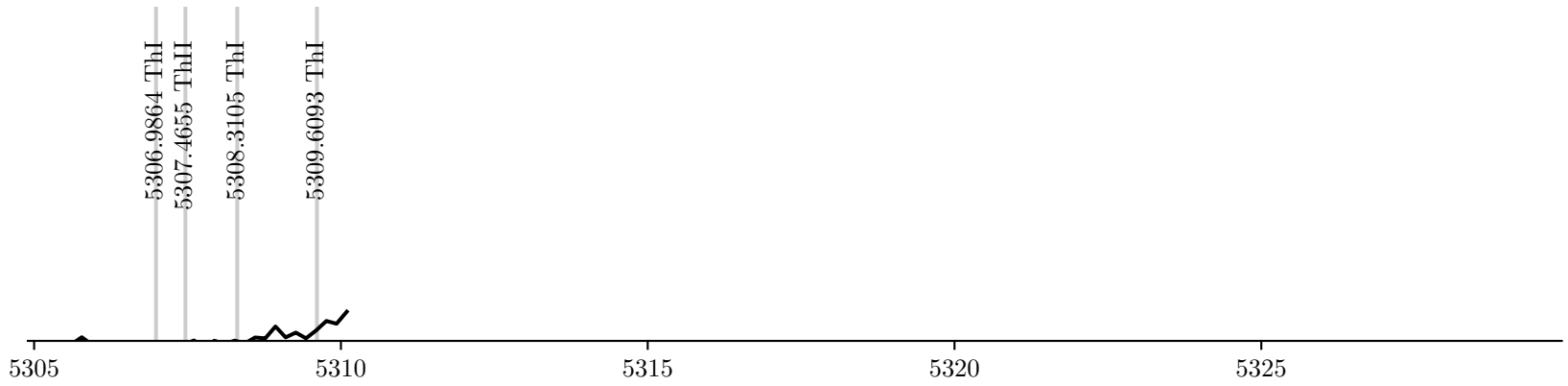

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5014.7539 Th
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5017.1628 ArII
5017.5084 ThI
5018.0535 ThI
5019.3241 ThII
5019.8062 ThI
5020.5458 ThII
5021.2531 ThI
5023.7084 ThI
5024.6122 Th
5026.9549 ThI
5028.6556 ThI
5029.0119 ThI
5029.6295 ThII
5029.8916 ThI
5005
5010
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5026.9549 ThI
5028.6556 ThI
5029.0119 ThI
5029.6295 ThII
5029.8916 ThI
5031.8147 ThI
5034.2944 ThI
5035.3375 Th
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5039.3265 ThI
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5041.5998 Th
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5044.7195 ThI
5045.2481 ThI
5046.3527 Th
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5047.4283 ThII
5048.9366 ThI
5049.7960 ThII
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5084.9935 ThI
5085.2955 Th
5085.8392 ThI
5086.1774 ThI
5086.8351 Th
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5101.3452 ThI
5103.7653 ThII
5106.6689 ThI
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5106.6689 ThI
5109.7331 ThI
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5110.5528 ThI
5111.0611 ThI
5111.2781 ThI
5114.0665 ThI
5115.0448 ThI
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5117.2923 Th
5118.2023 ArI
5119.8676 ThI
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5122.4995 ThI
5124.2460 Th
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5125.9502 ThI
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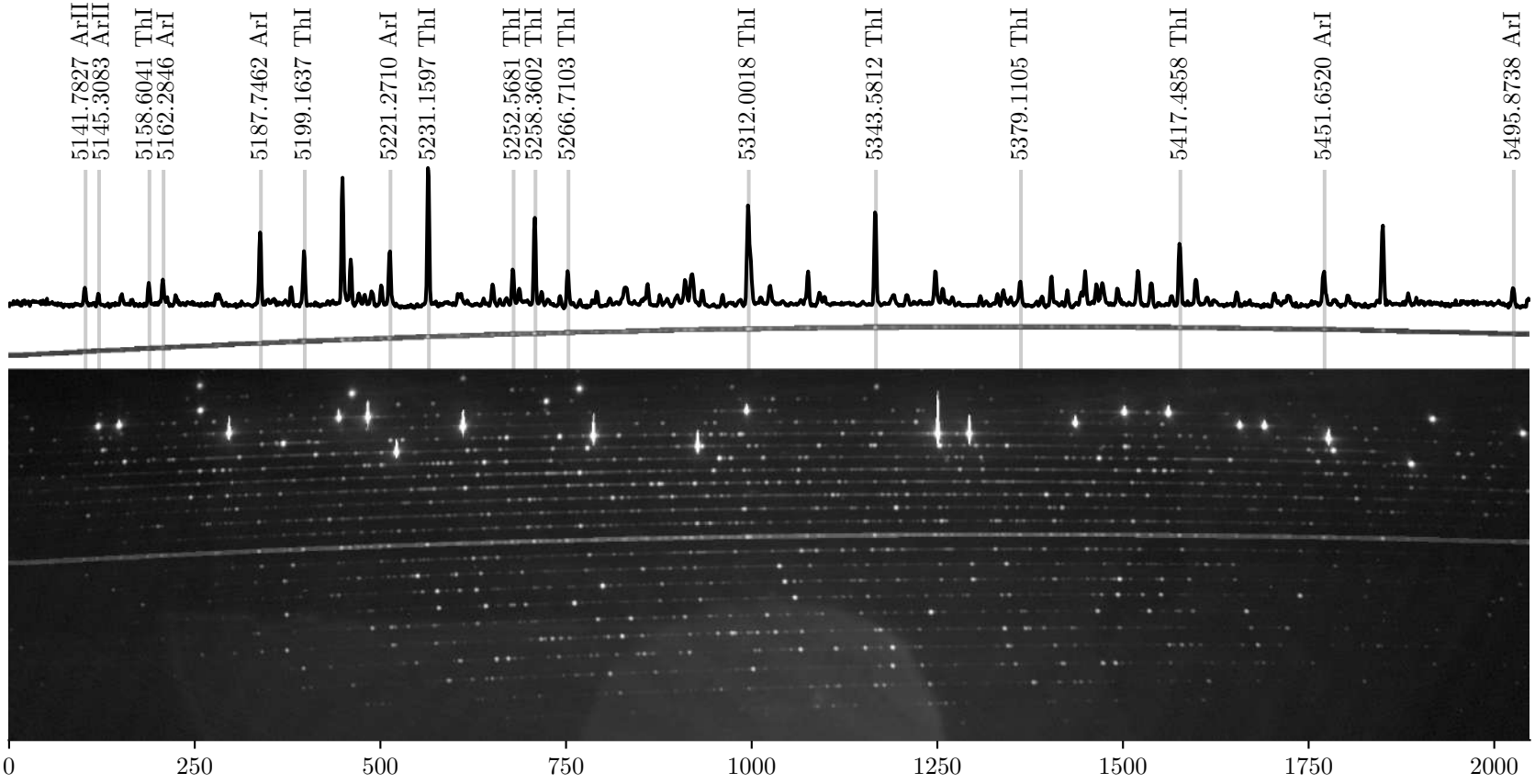

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5131.0720 ThII
5131.9924 ThI
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5137.4733 ThI
5139.4487 ThI
5140.7736 ThI
5141.2650 ThI
5141.7827 ArII
5143.2673 ThII
5143.9165 ThI
5145.0383 ThII
5145.3083 ArII
5146.0557 ThI
5147.1346 ThI
5148.2115 ThII
5149.2069 ThI
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5145.3083 ArII
5146.0557 ThI
5147.1346 ThI
5148.2115 ThII
5149.2069 ThI
5149.8080 ThI
5150.7000 ThI
5151.3220 ThI
5153.5173 ThII
5154.2430 ThI
5154.8985 ThII
5157.4257 ThI
5158.6041 ThI
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5165.7728 ArII
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5165.7728 ArII
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5175.9115 ThI
5176.9610 ThI
5177.6228 ThI
5178.4800 ThI
5179.1440 ThI
5180.7209 ThI
5182.5269 ThII
5184.4534 ThI
5185.2042 ThI
5186.4132 ThI
5187.7462 ArI
5188.3656 ThII
5188.8432 Th
5189.6772 ThII
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5185.2042 ThI
5186.4132 ThI
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5187.7462 ArI
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5188.8432 Th
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5203.8479 Th
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5222.7661 ThI
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5228.2246 ThI
5228.9951 ThI
5229.7945 Th
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5226.5254 ThI
5228.2246 ThI
5228.9951 ThI
5229.7945 Th
5231.1597 ThI
5232.3415 ThI
5233.2254 ThII
5234.1069 ThI
5238.8137 ThI
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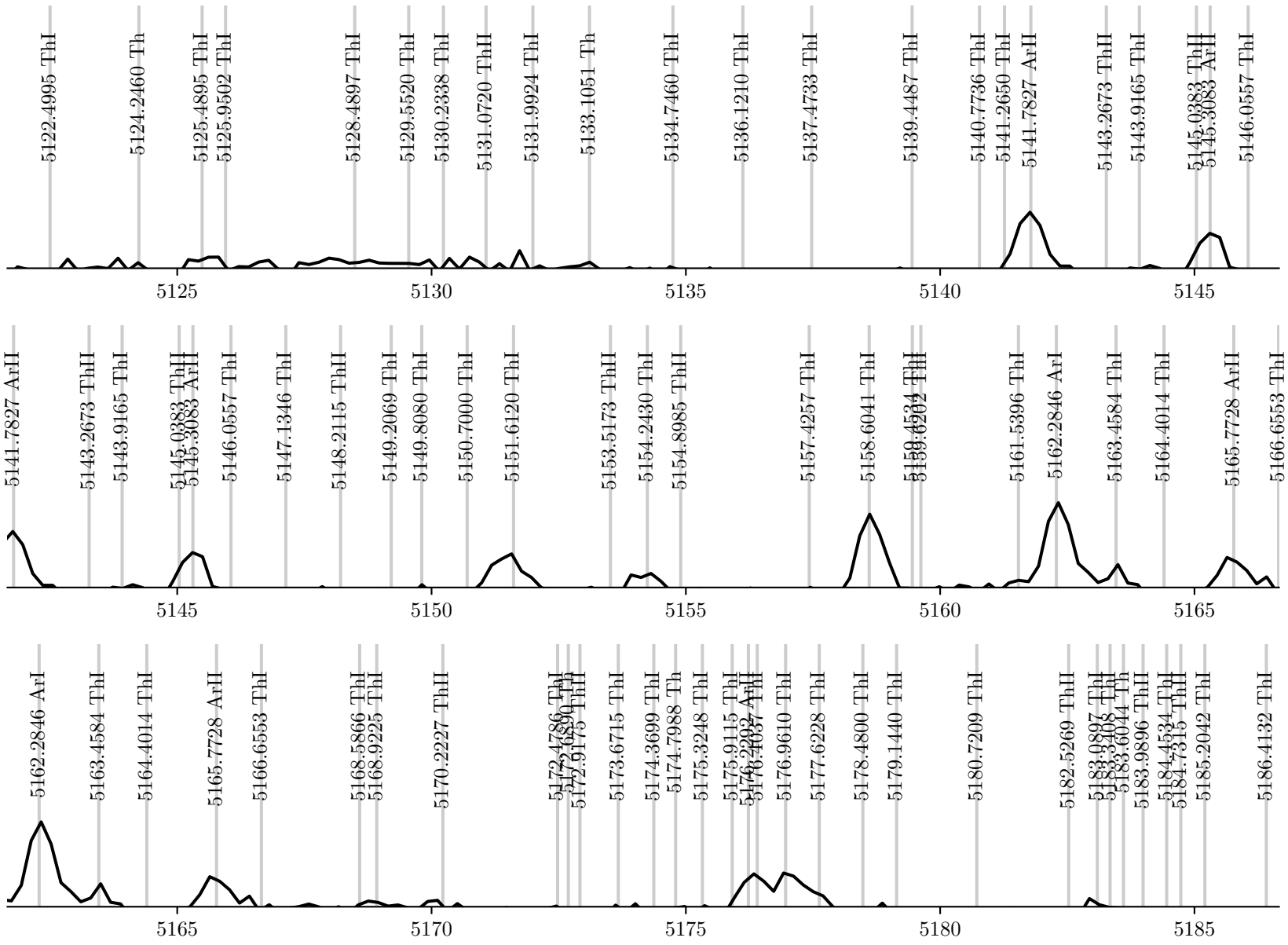

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5274.3807 ThI
5276.4089 ThII
5277.1460 ThI
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5283.6959 ThI
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5289.8974 ThII
5291.8163 ThI
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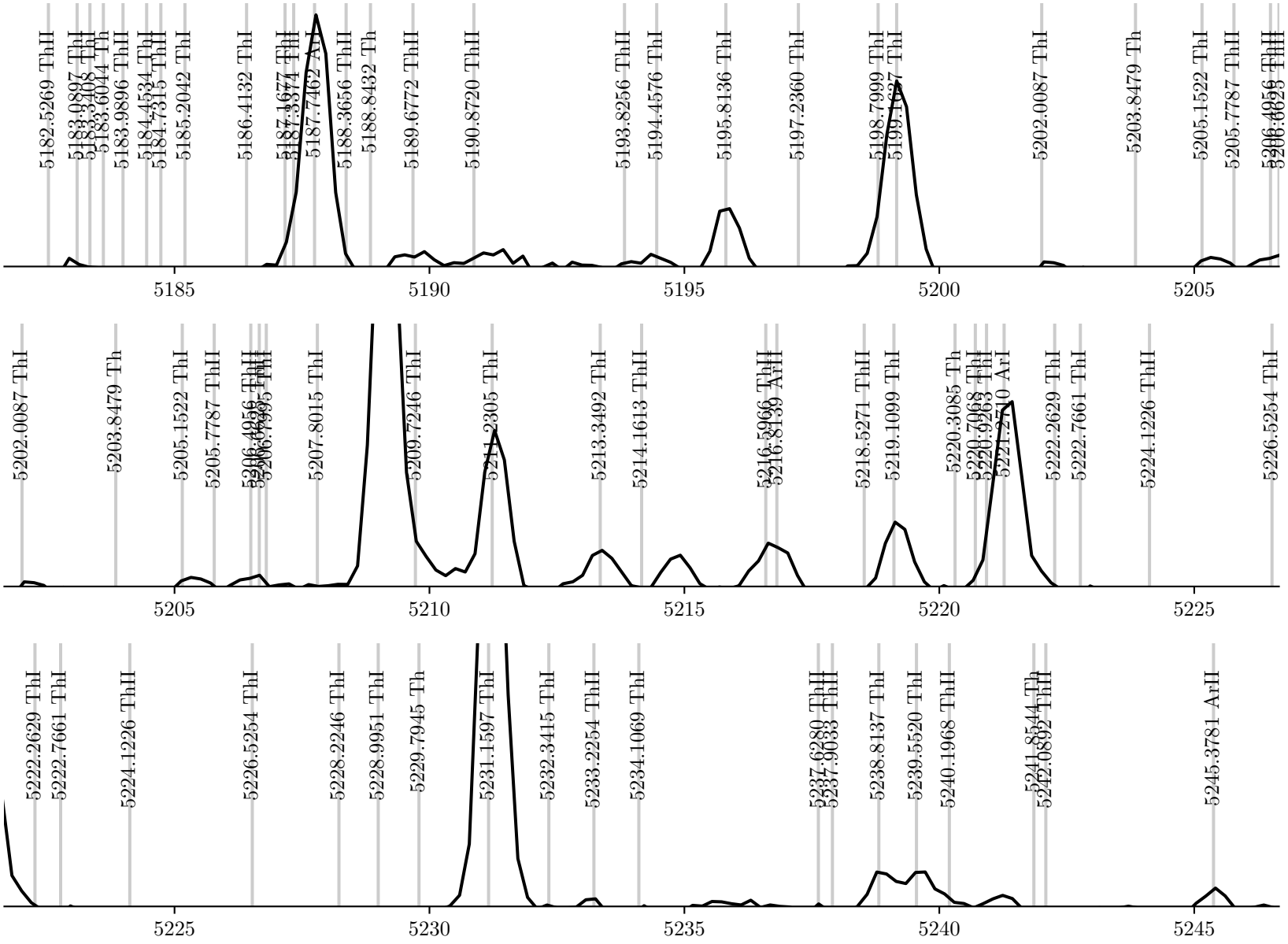

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5308.3105 ThI
5309.6093 ThI
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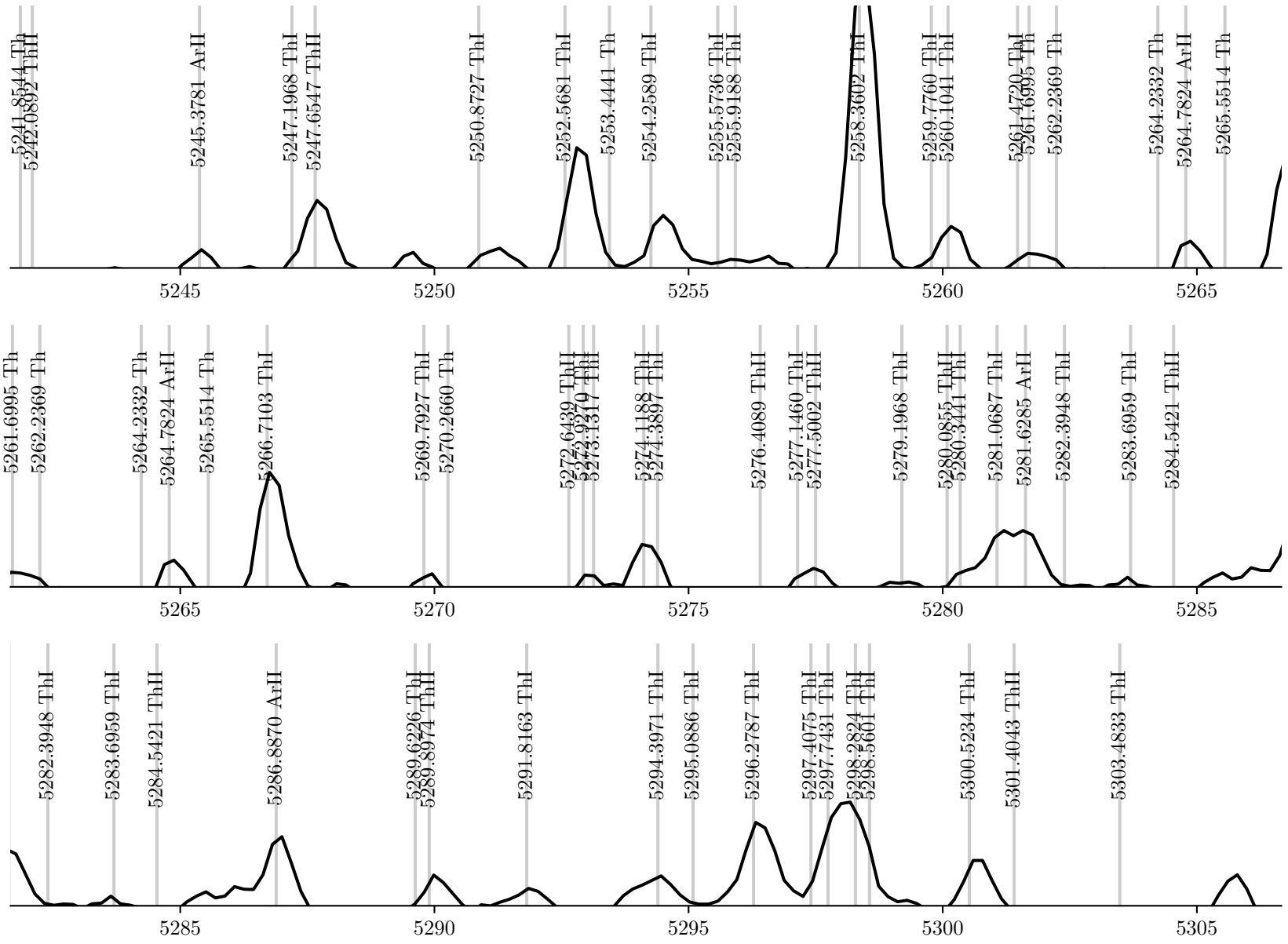
Order 28

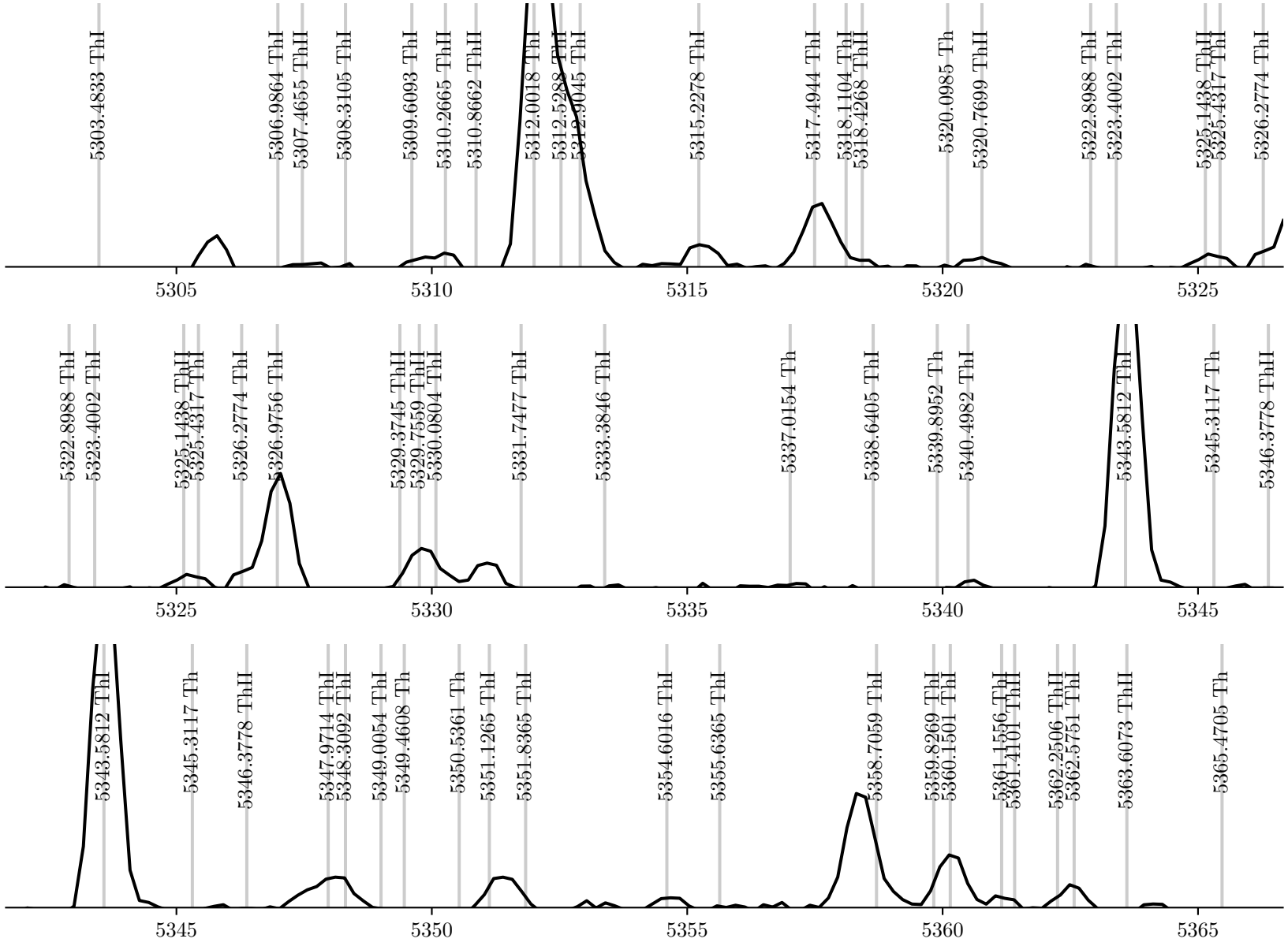


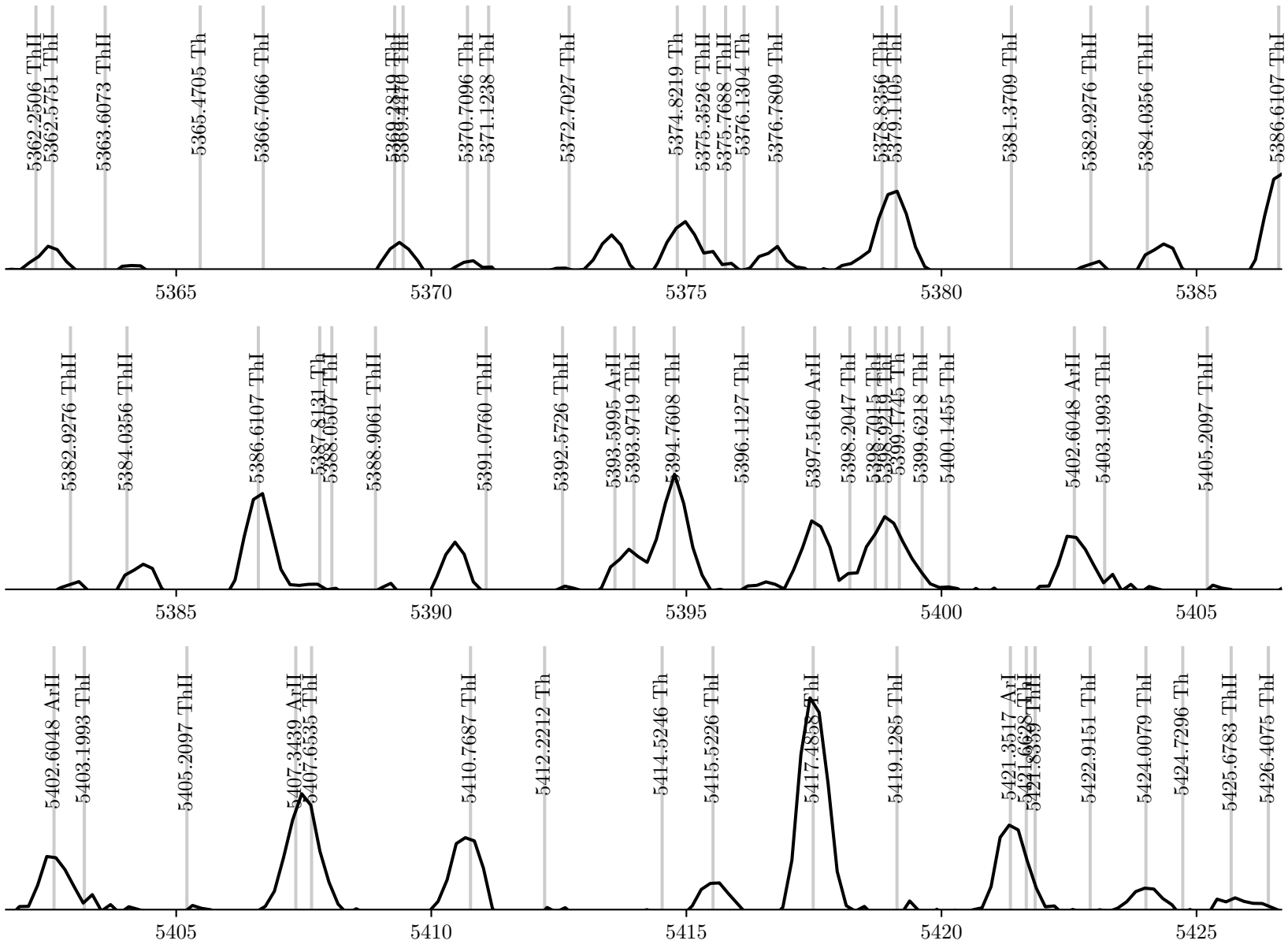
Dispersion:	0.185 Å pix^{-1}
Central wavelength:	5317 Å
Wavelength range:	5122 Å – 5500 Å

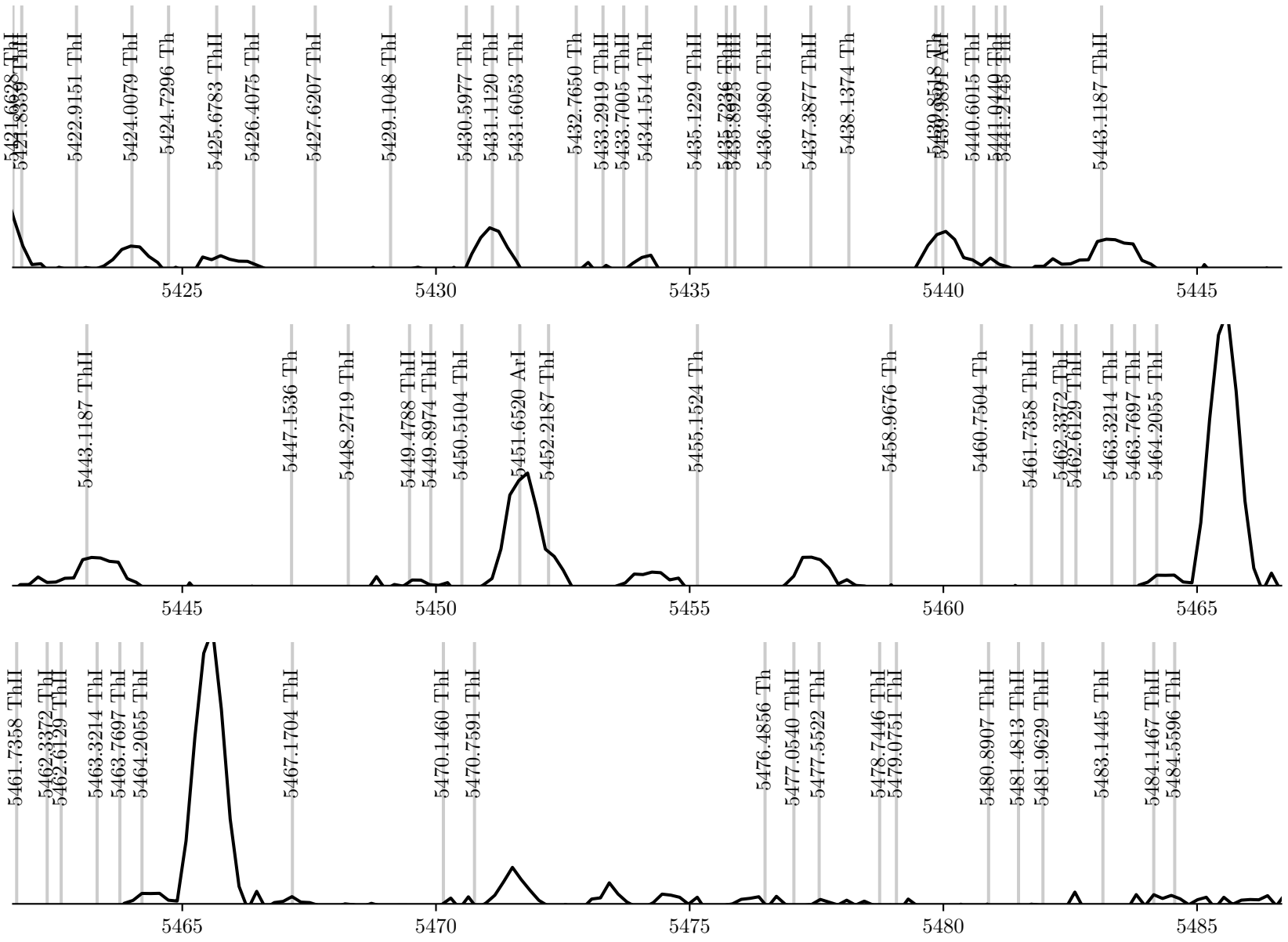

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5137.4733 ThI
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5140.7736 ThI
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5149.2069 ThI
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5165.7728 ArII
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5175.3248 ThI
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5176.9610 ThI
5177.6228 ThI
5178.4800 ThI
5179.1440 ThI
5180.7209 ThI
5182.5269 ThII
5183.0897 ThI
5183.6044 Th
5183.9896 ThII
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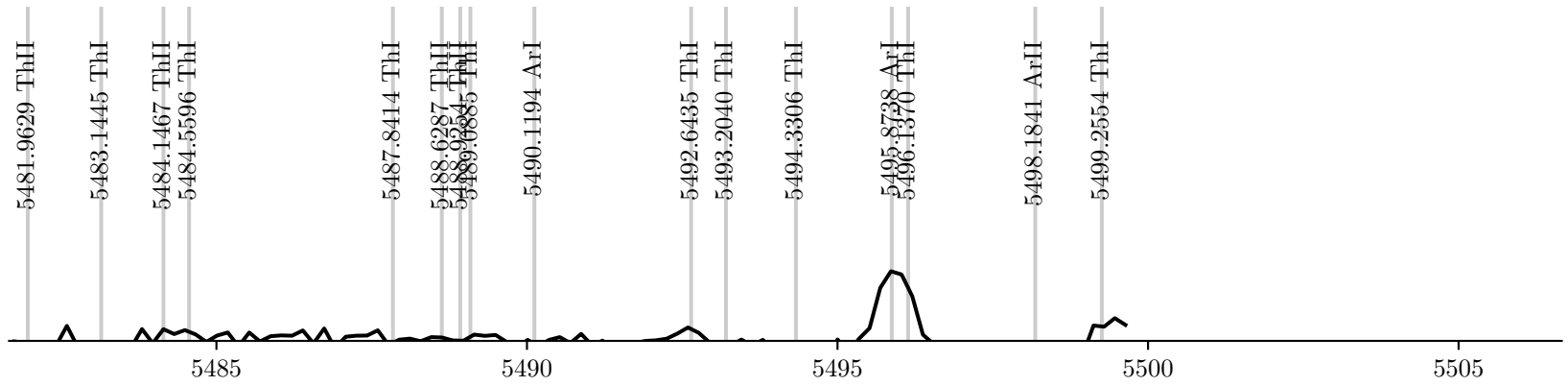

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5187.1677 ThI
5187.3374 ThI
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5188.3656 ThII
5188.8432 Th
5189.6772 ThII
5190.8720 ThII
5193.8256 ThII
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5197.2360 ThI
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5199.1627 ThI
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5203.8479 Th
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5202.0087 ThI
5203.8479 Th
5205.1522 ThI
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5207.8015 ThI
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5219.1099 ThI
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5281.6285 ArII
5282.3948 ThI
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5284.5421 ThII
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5294.3971 ThI
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5296.2787 ThI
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5298.2824 ThI
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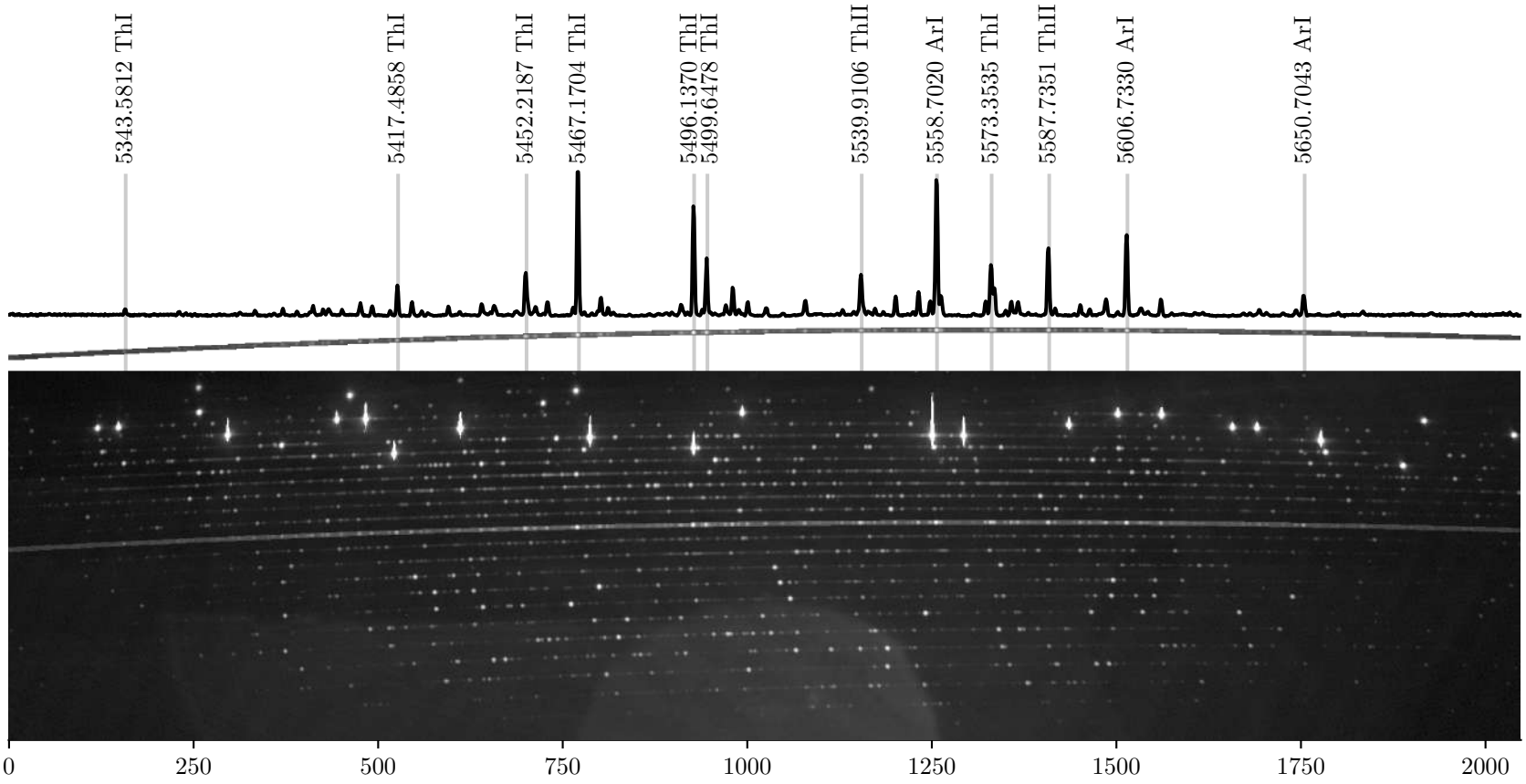

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5365


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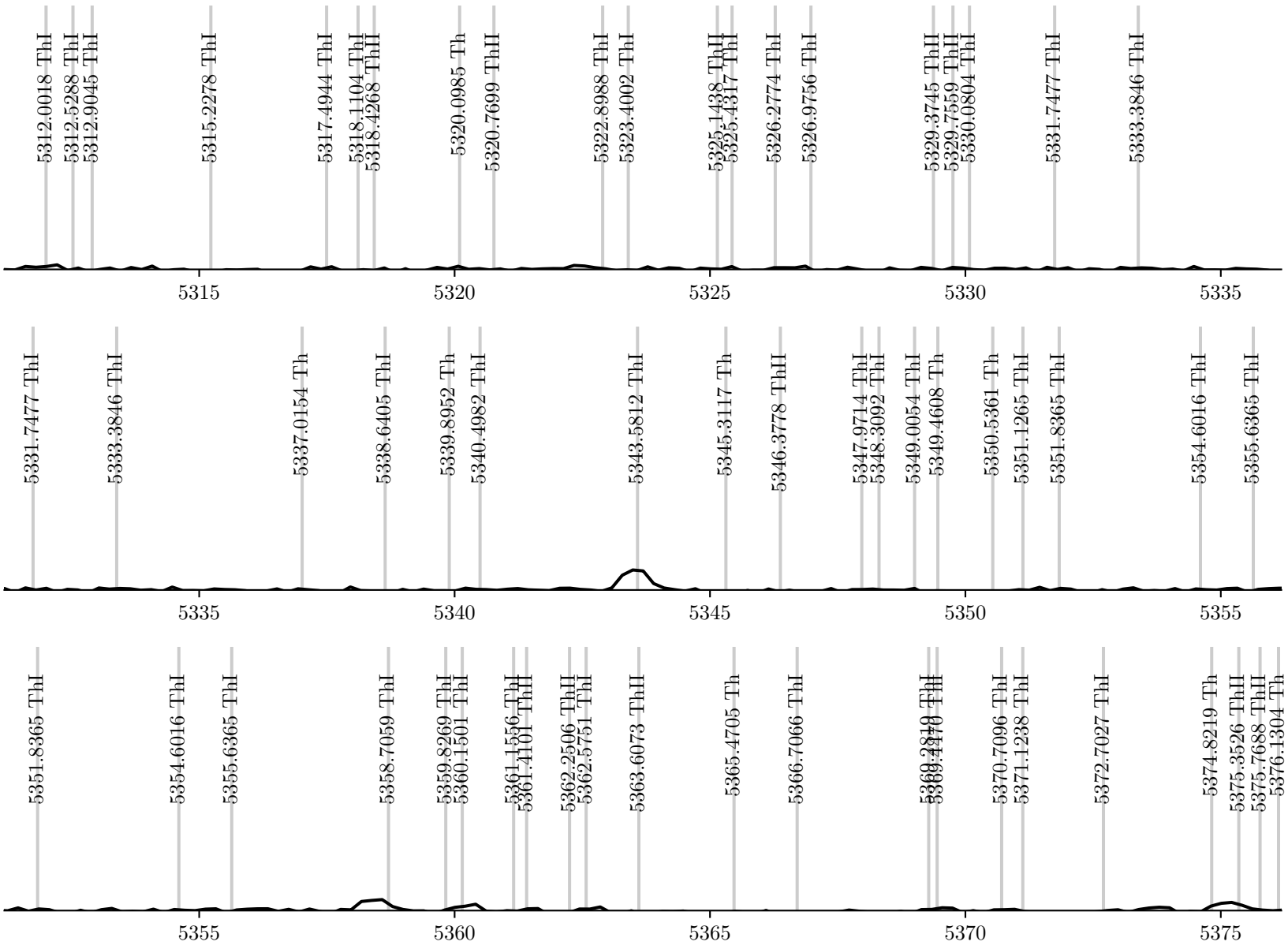

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5430.5977 ThI
5431.1120 ThI
5431.6053 ThI
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5433.7005 ThII
5434.1514 ThI
5435.1229 ThII
5435.7236 ThII
5435.8925 ThII
5436.4980 ThII
5437.3877 ThII
5438.1374 Th
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5439.9891 ThI
5440.6015 ThI
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5451.6520 ArI
5452.2187 ThI
5455.1524 Th
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5460.7504 Th
5461.7358 ThII
5462.3372 ThI
5462.6129 ThII
5463.3214 ThI
5463.7697 ThI
5464.2055 ThI
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5455
5460
5465
5461.7358 ThII
5462.3372 ThI
5462.6129 ThII
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5464.2055 ThI
5467.1704 ThI
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5476.4856 Th
5477.0540 ThII
5477.5522 ThI
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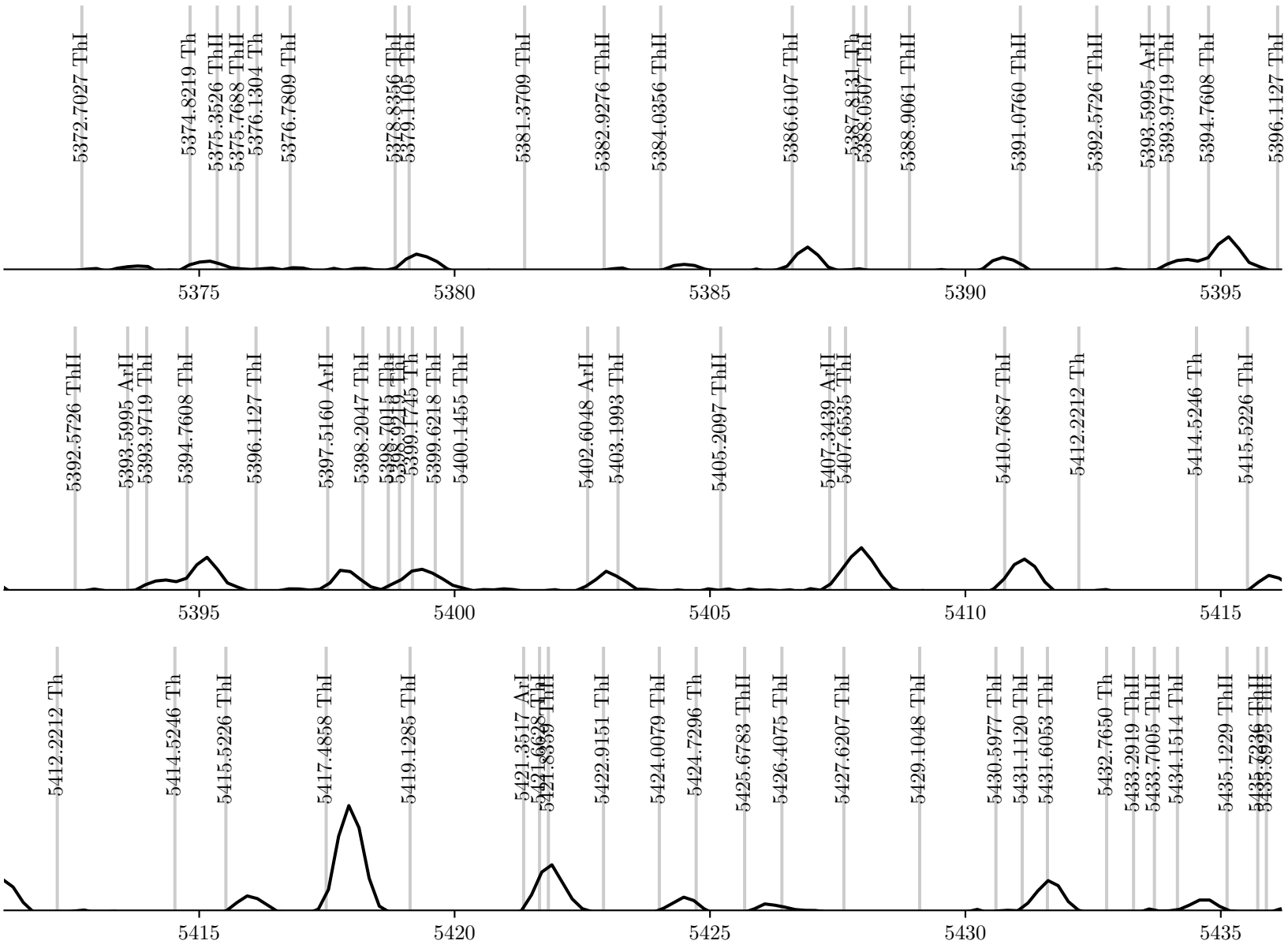

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5494.3306 ThI
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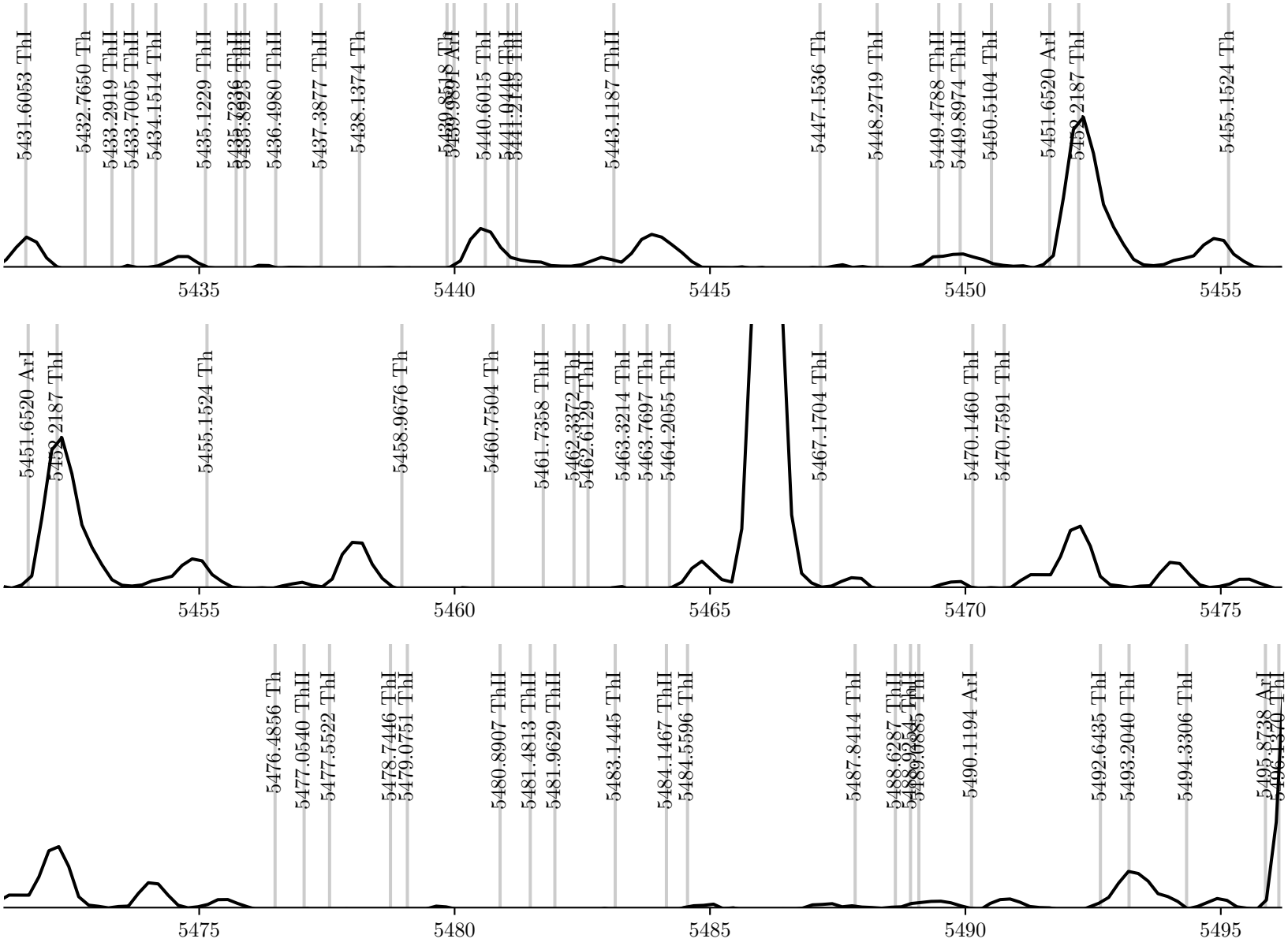
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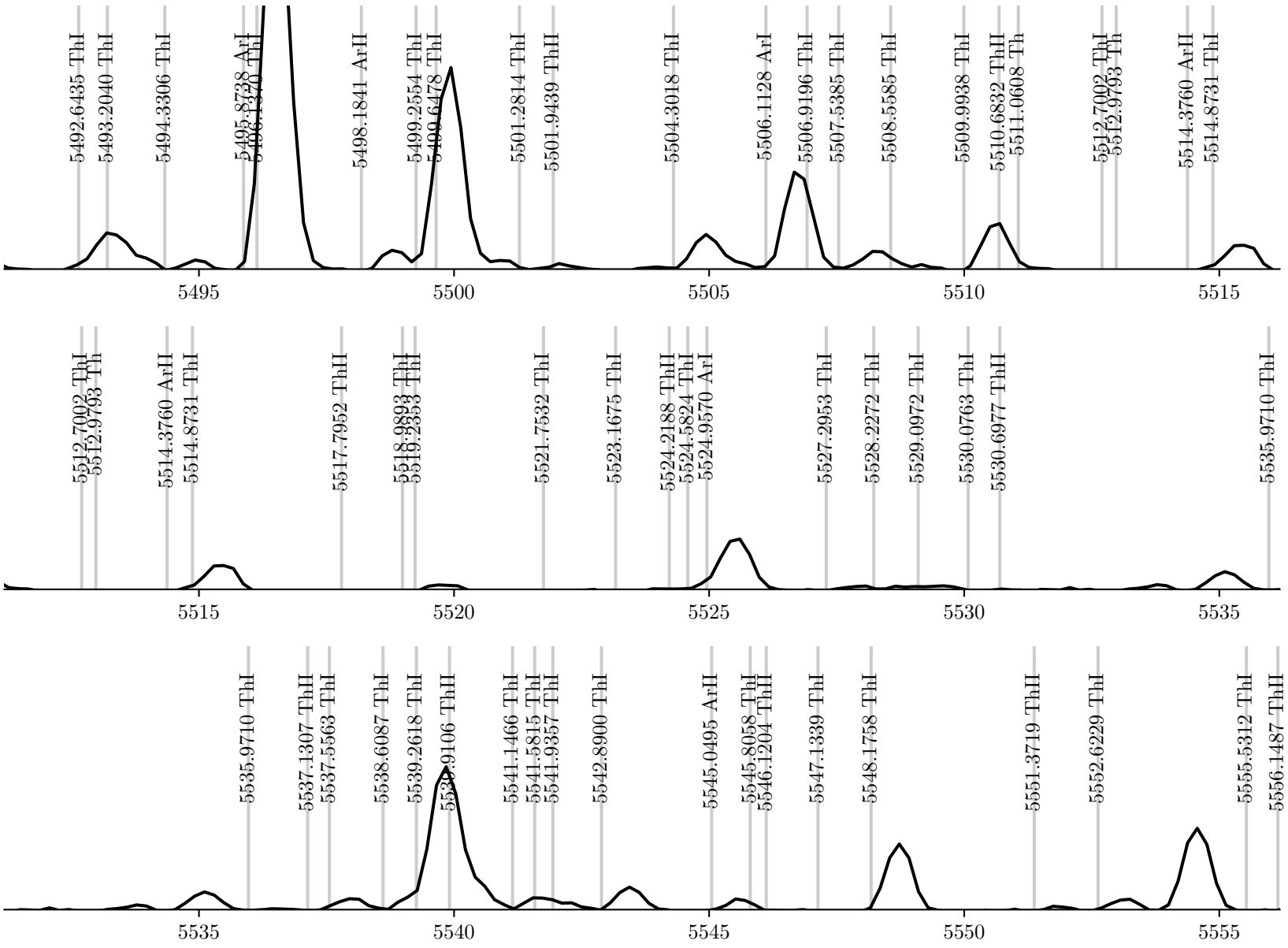


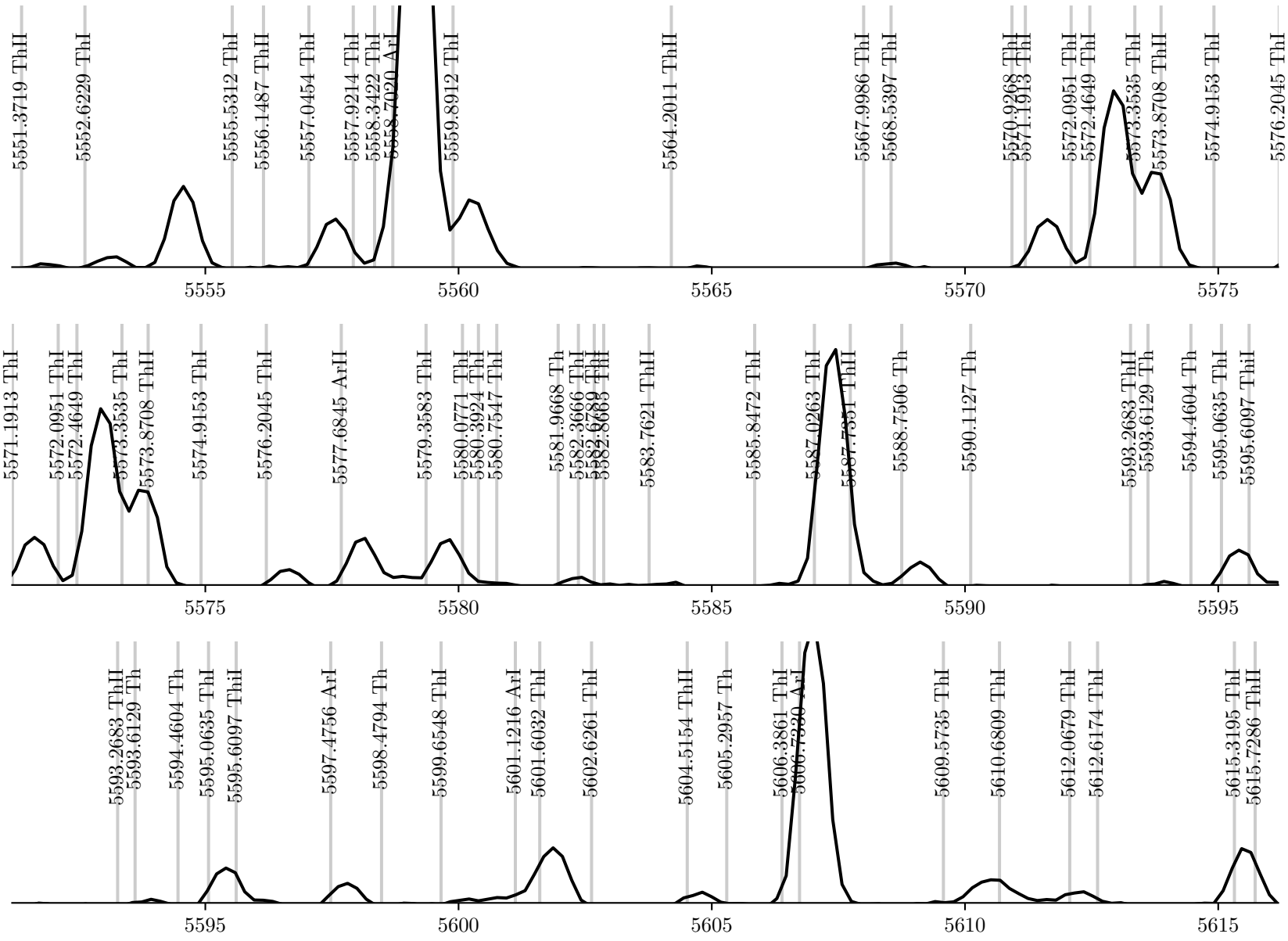
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Central wavelength:	5515 Å
Wavelength range:	5311 Å – 5703 Å

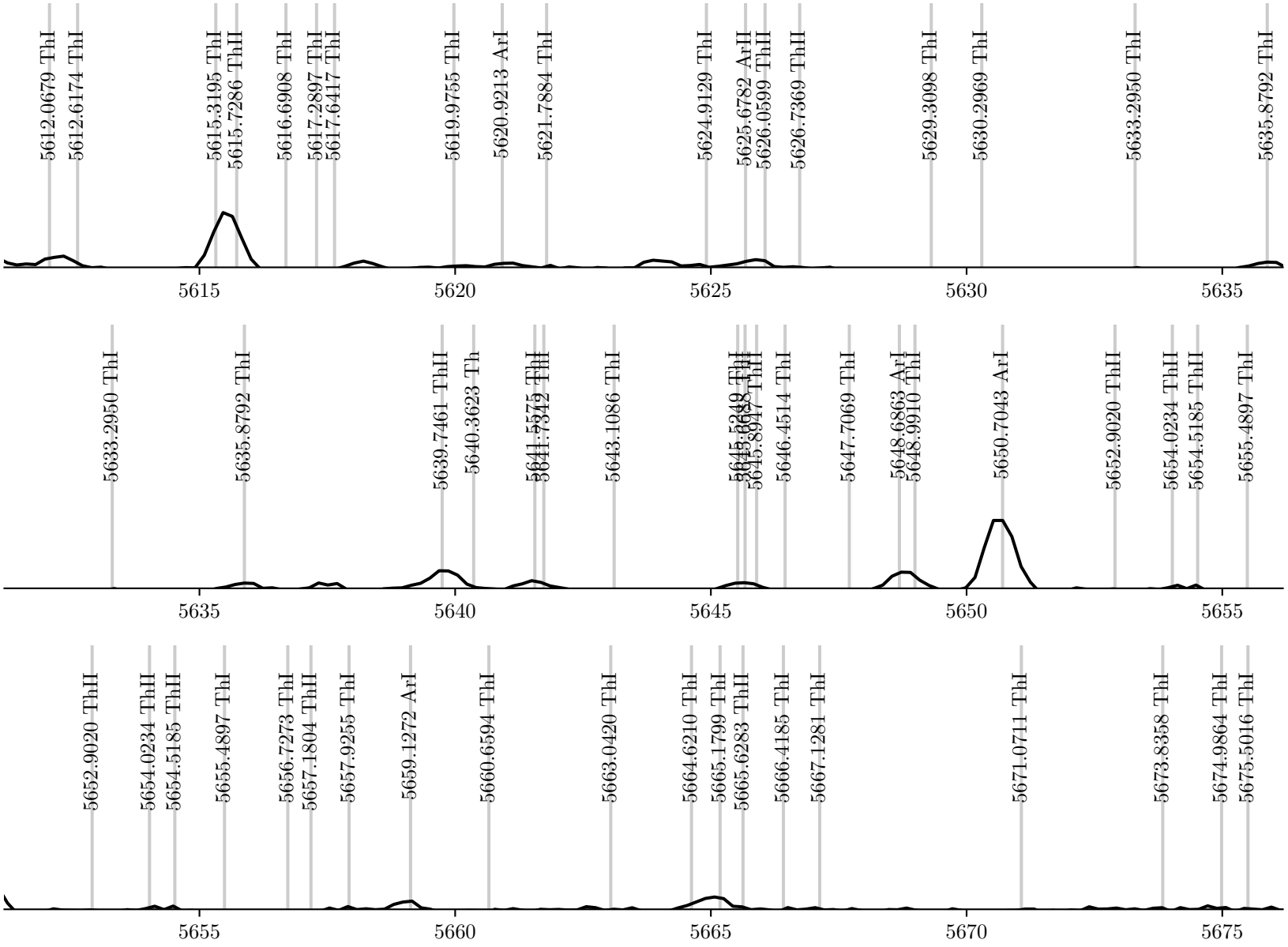

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5320.0985 Th
5320.7699 ThII
5322.8988 ThI
5323.4002 ThI
5325.1438 ThII
5325.4317 ThI
5326.2774 ThI
5326.9756 ThI
5329.3745 ThII
5329.7559 ThII
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5315
5320
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5330
5335
5331.7477 ThI
5333.3846 ThI
5337.0154 Th
5338.6405 ThI
5339.8952 Th
5340.4982 ThI
5343.5812 ThI
5345.3117 Th
5346.3778 ThII
5347.9714 ThI
5348.3092 ThI
5349.0054 ThI
5349.4608 Th
5350.5361 Th
5351.1265 ThI
5351.8365 ThI
5354.6016 ThI
5355.6365 ThI
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5351.8365 ThI
5354.6016 ThI
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5359.8269 ThI
5360.1501 ThI
5361.1556 ThI
5361.4101 ThII
5362.2506 ThII
5362.5751 ThI
5363.6073 ThII
5365.4705 Th
5366.7066 ThI
5370.7096 ThI
5371.1238 ThI
5372.7027 ThI
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5375.7688 ThII
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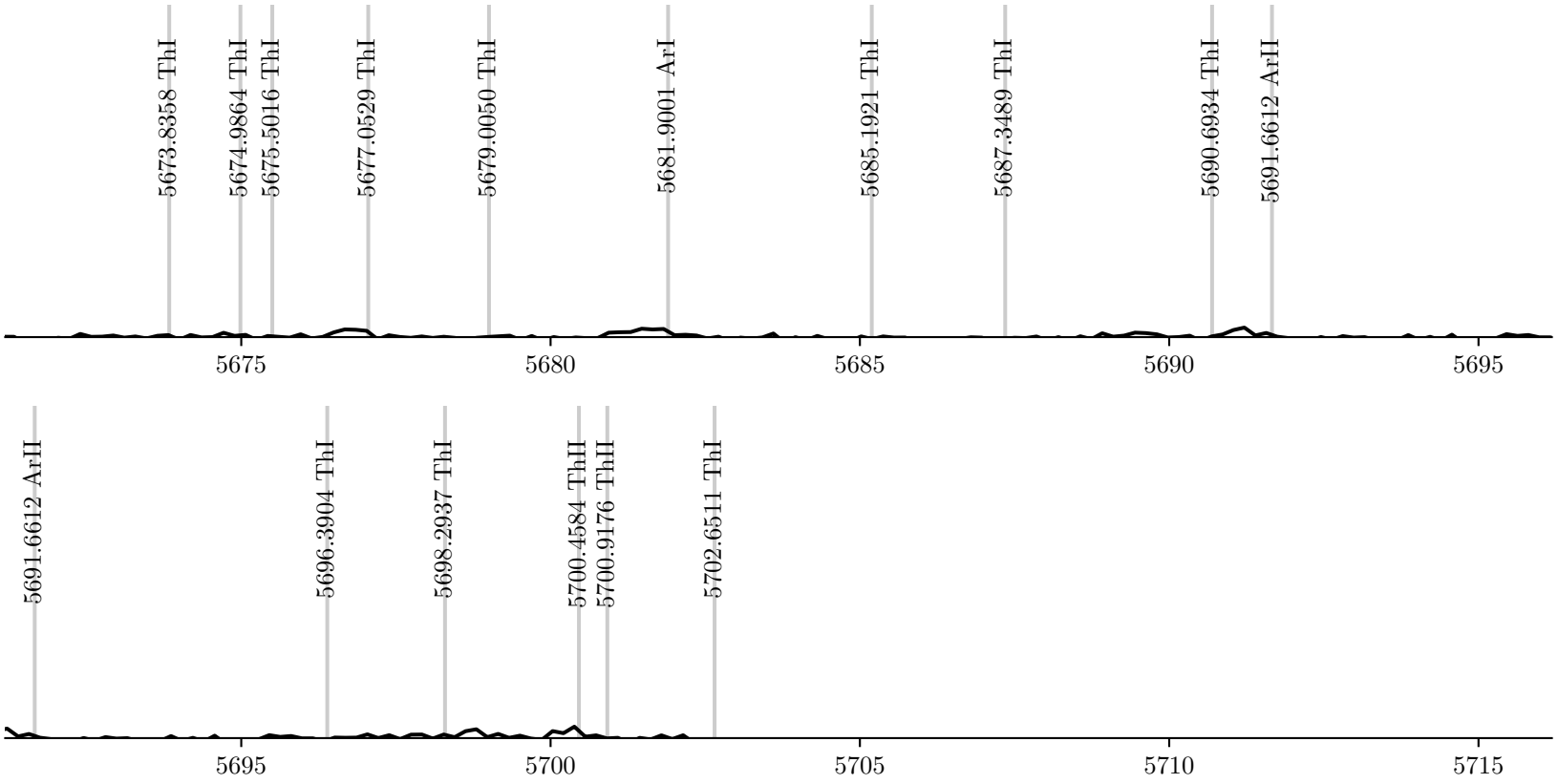

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5395
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5393.5995 ArII
5393.9719 ThI
5394.7608 ThI
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5399.6218 ThI
5400.1455 ThI
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5403.1993 ThI
5405.2097 ThII
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5426.4075 ThI
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5425
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5435


5431.6053 ThI
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5458.9676 Th
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5462.3372 ThI
5462.6129 ThII
5463.3214 ThI
5463.7697 ThI
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5470.1460 ThI
5470.7591 ThI
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5465
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5475
5476.4856 Th
5477.0540 ThII
5477.5522 ThI
5478.7446 ThI
5479.0751 ThI
5480.8907 ThII
5481.4813 ThII
5481.9629 ThII
5483.1445 ThI
5484.1467 ThII
5484.5596 ThI
5487.8414 ThI
5488.6287 ThII
5490.1194 ArI
5492.6435 ThI
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5494.3306 ThI
5495.8738 ArI
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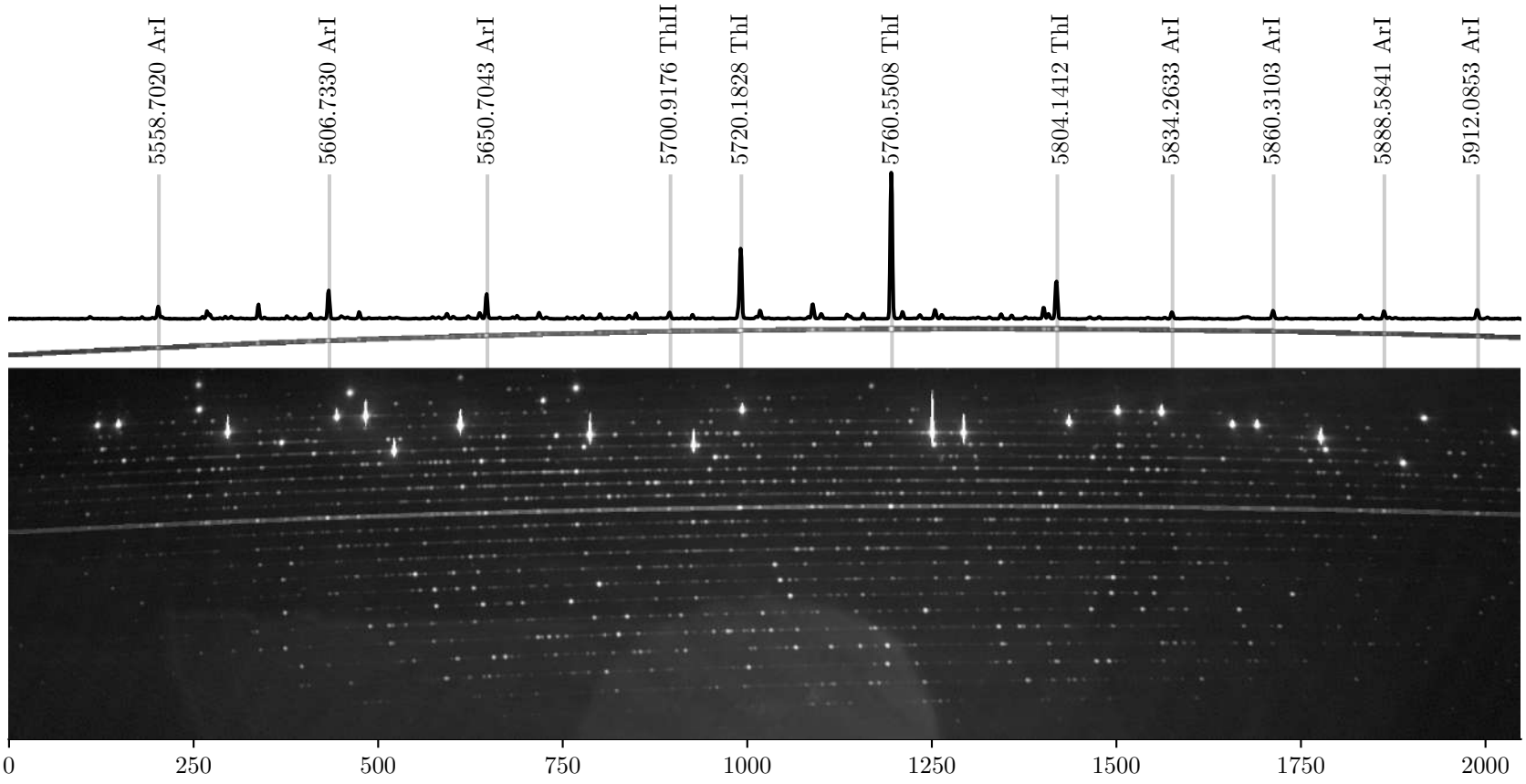

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5504.3018 ThI
5506.1128 ArI
5506.9196 ThI
5507.5385 ThI
5508.5585 ThI
5509.9938 ThI
5510.6832 ThII
5511.0608 Th
5512.7002 ThI
5512.9793 Th
5514.3760 ArII
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5524.5824 ThI
5524.9570 ArI
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5530.0763 ThI
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5535.9710 ThI
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5537.1307 ThII
5537.5563 ThI
5538.6087 ThI
5539.2618 ThI
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5541.1466 ThI
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5547.1339 ThI
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5551.3719 ThII
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5556.1487 ThII
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5545
5550
5555


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5555.5312 ThI
5556.1487 ThII
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5558.3422 ThI
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5572.4649 ThI
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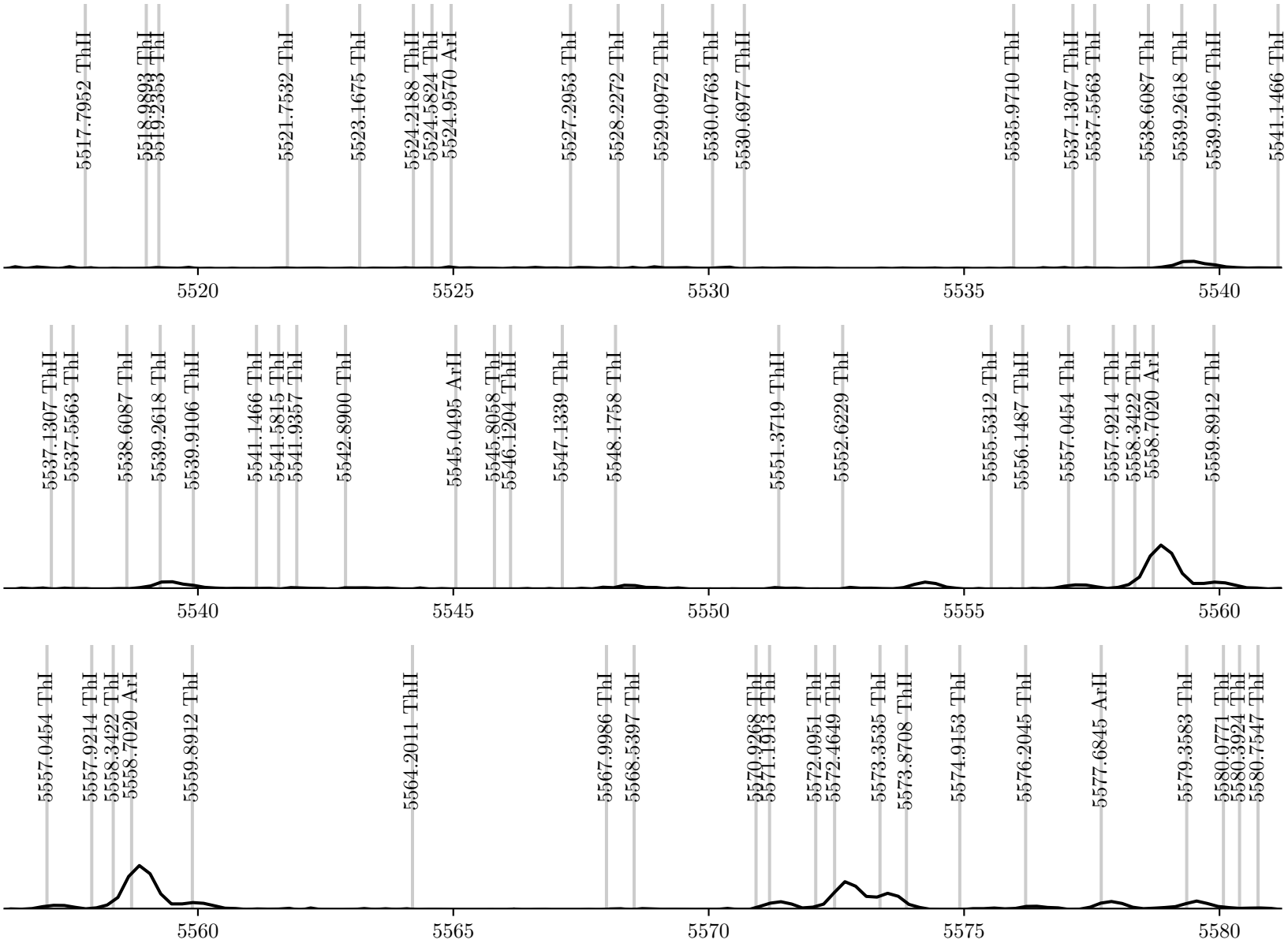

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5619.9755 ThI
5620.9213 ArI
5621.7884 ThI
5624.9129 ThI
5625.6782 ArII
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5650.7043 ArI
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5654.5185 ThII
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5657.9255 ThI
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5663.0420 ThI
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5665.1799 ThI
5665.6283 ThII
5666.4185 ThI
5667.1281 ThI
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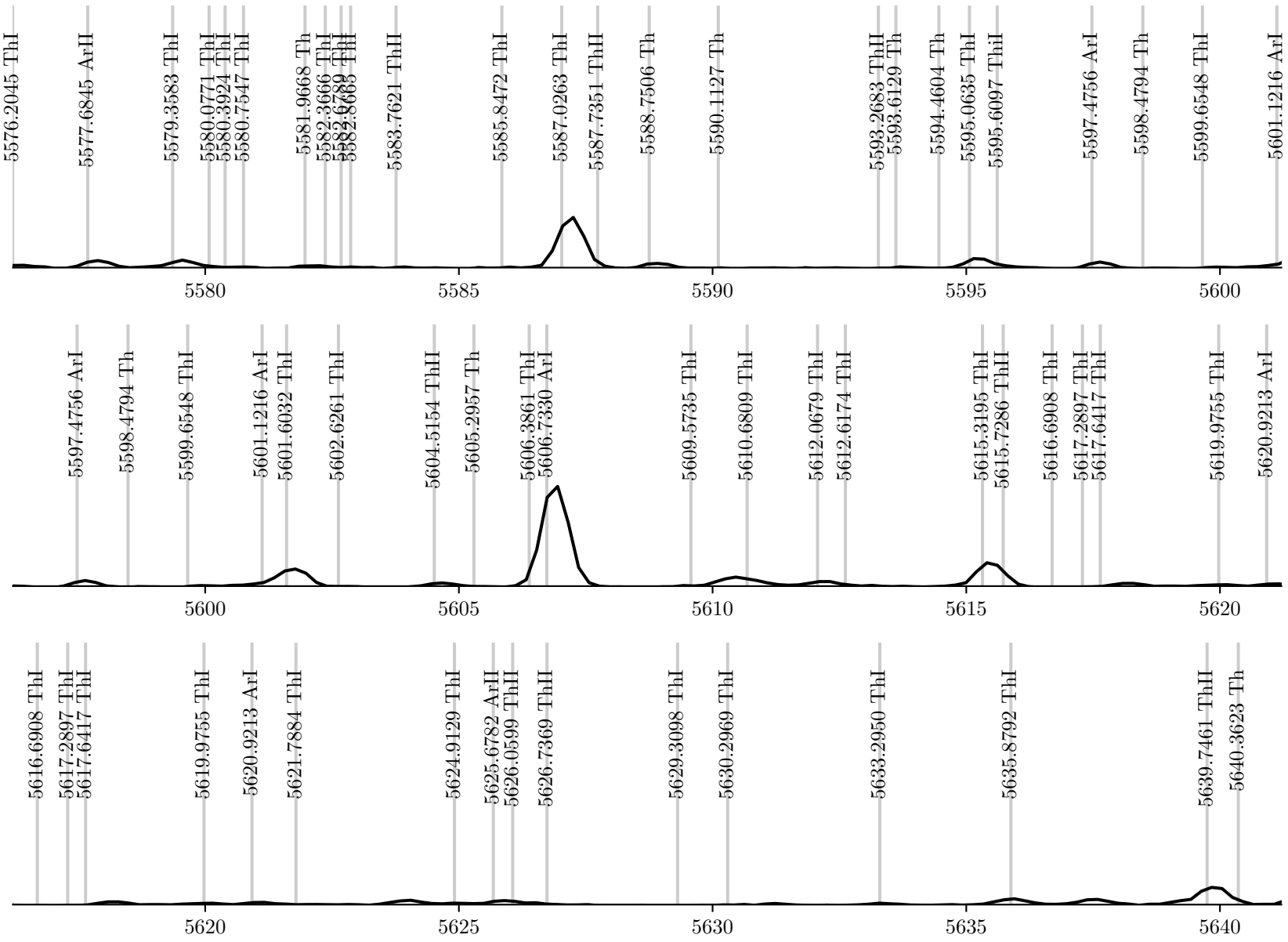

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5681.9001 ArI
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5687.3489 ThI
5690.6934 ThI
5691.6612 ArII
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5685
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5691.6612 ArII
5696.3904 ThI
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5715

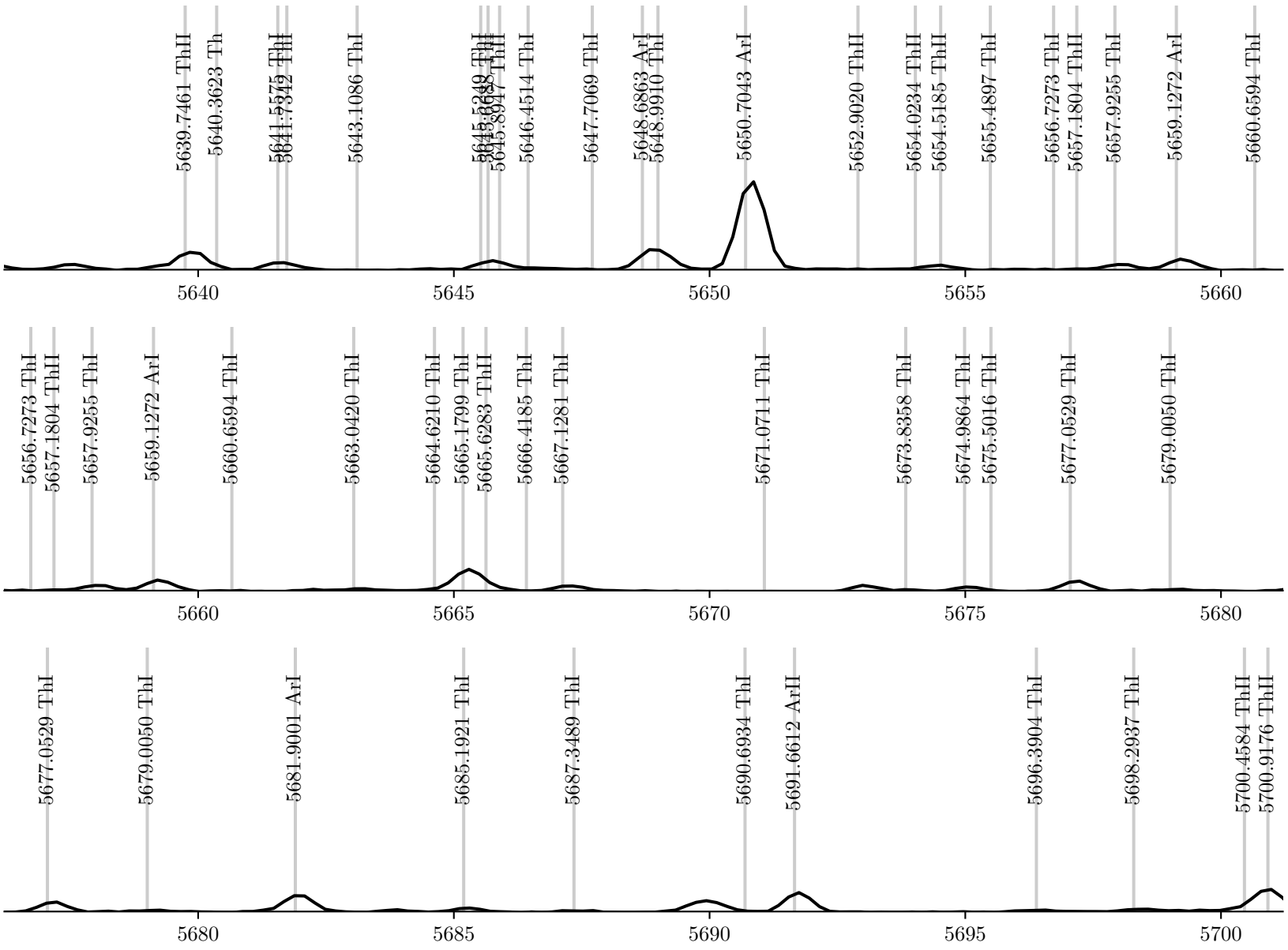
Order 26

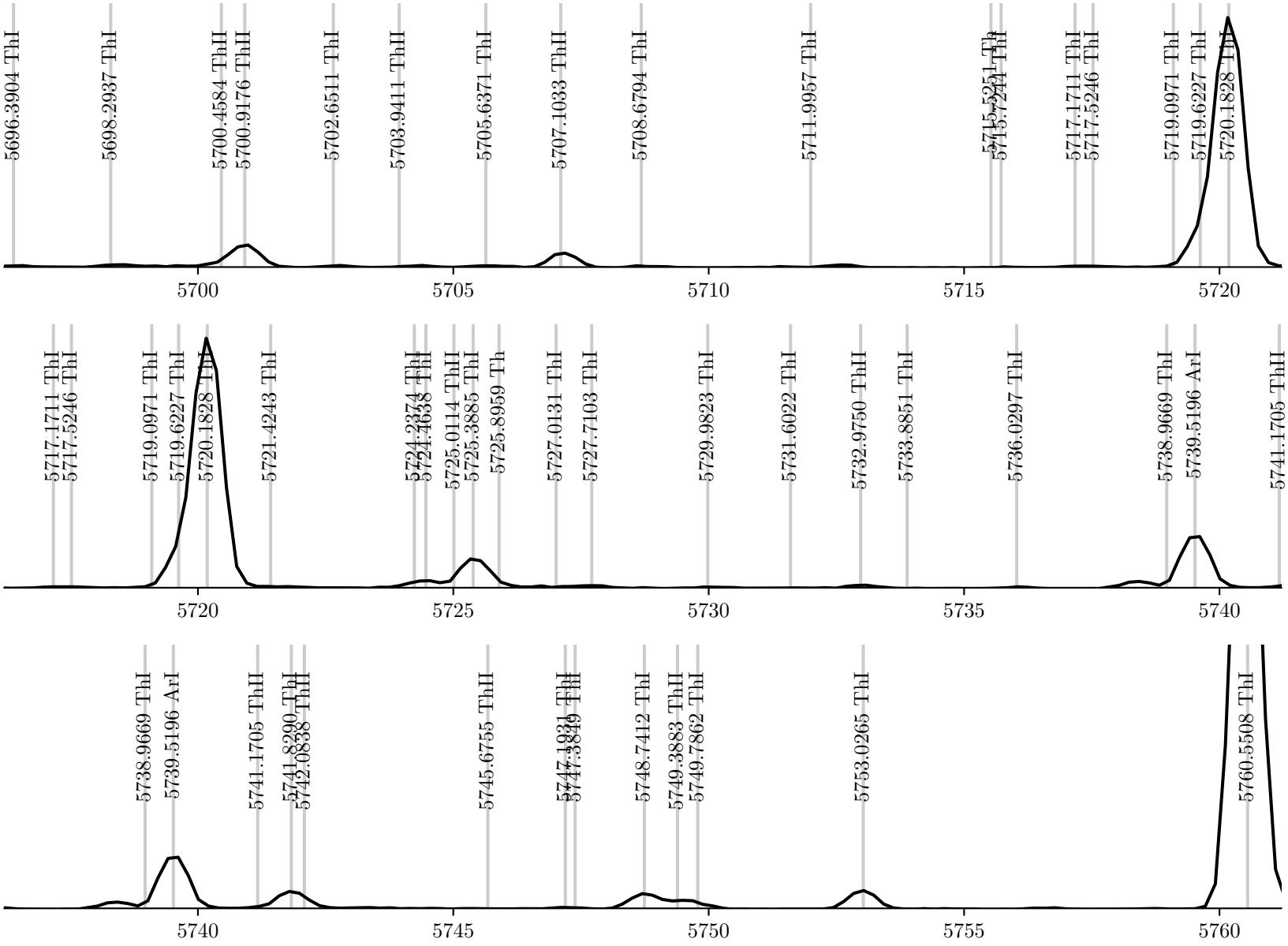


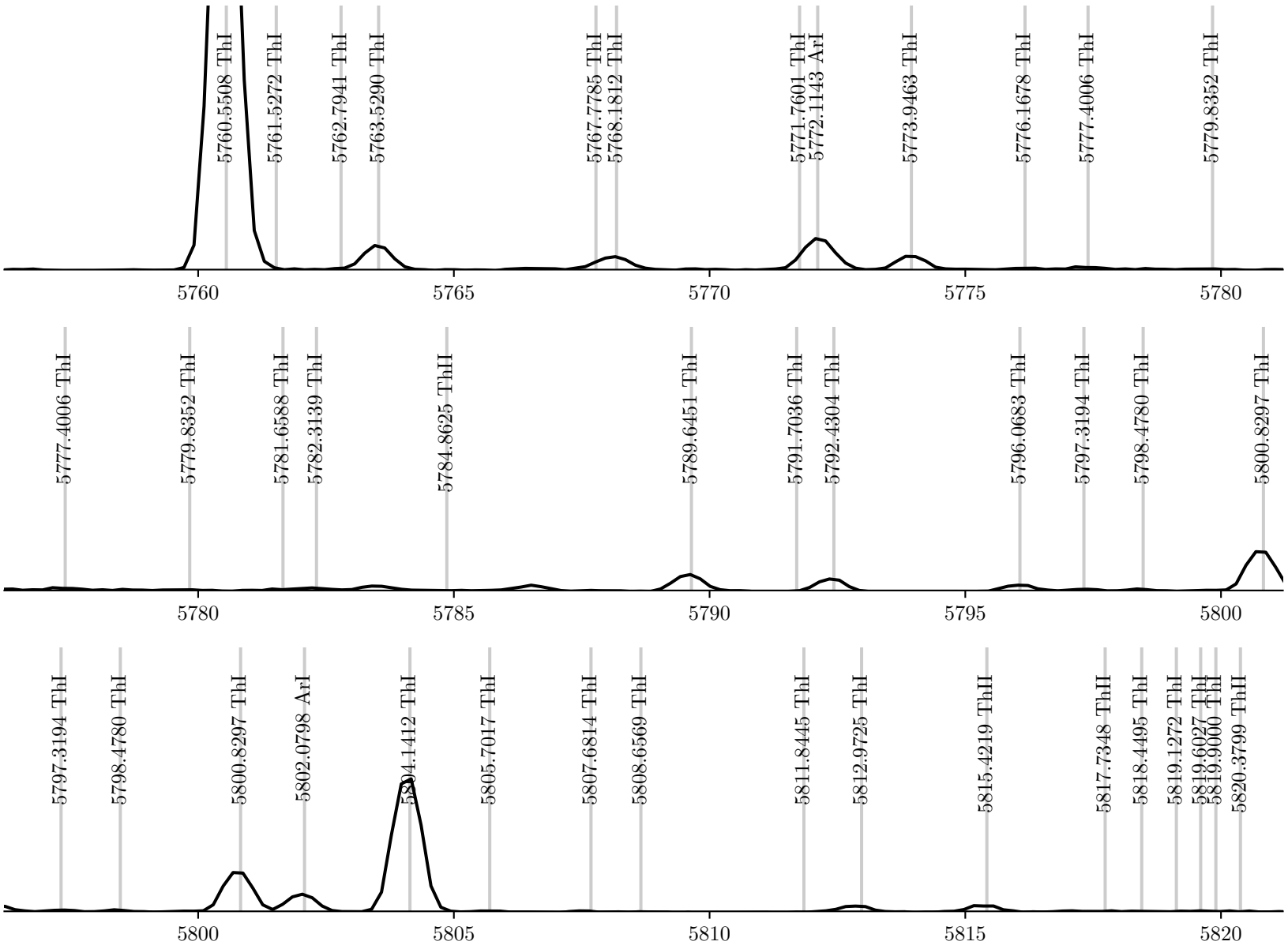
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Central wavelength:	5727 Å
Wavelength range:	5516 Å – 5923 Å

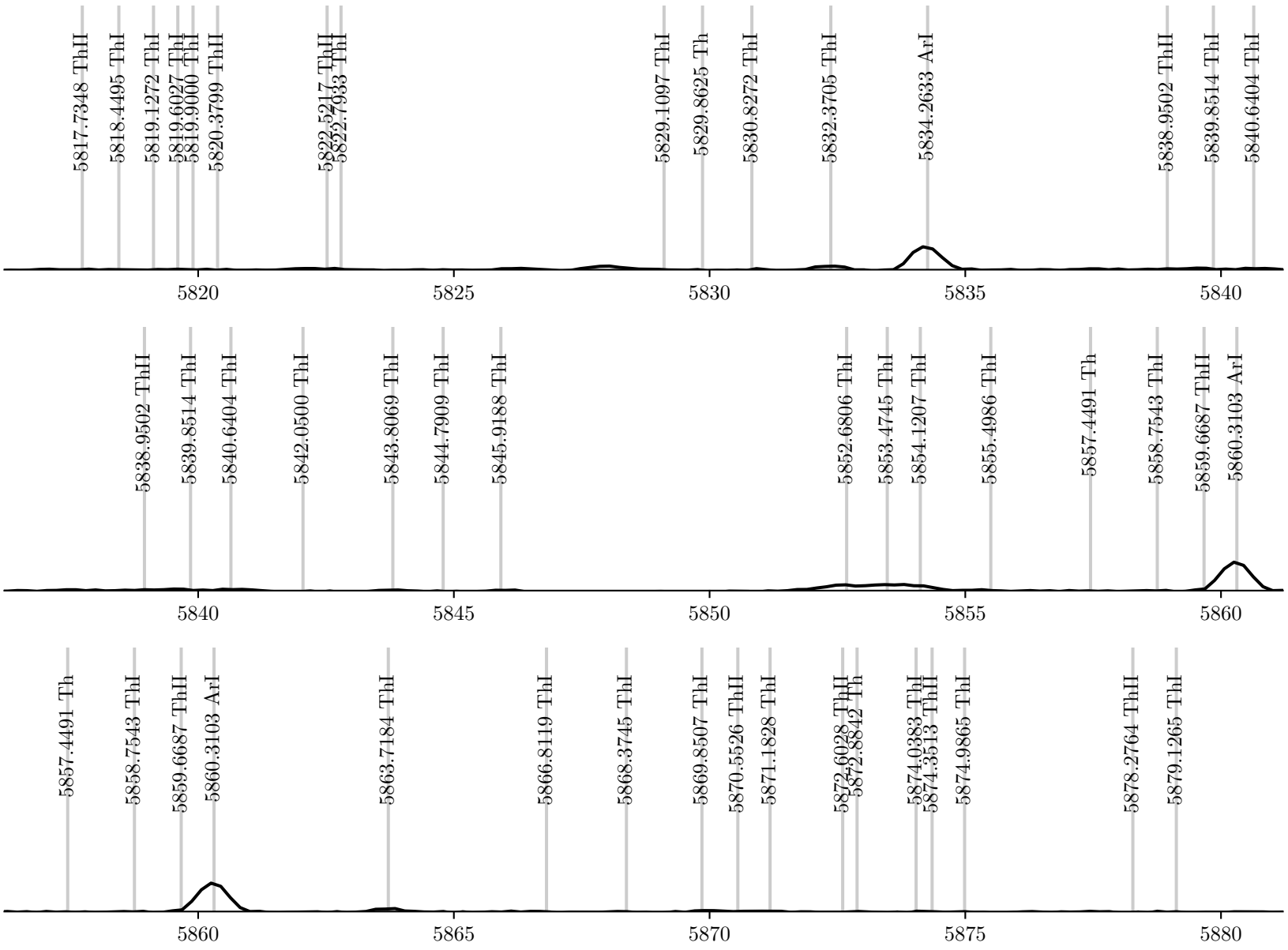

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5530.6977 ThII
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5537.1307 ThII
5537.5563 ThI
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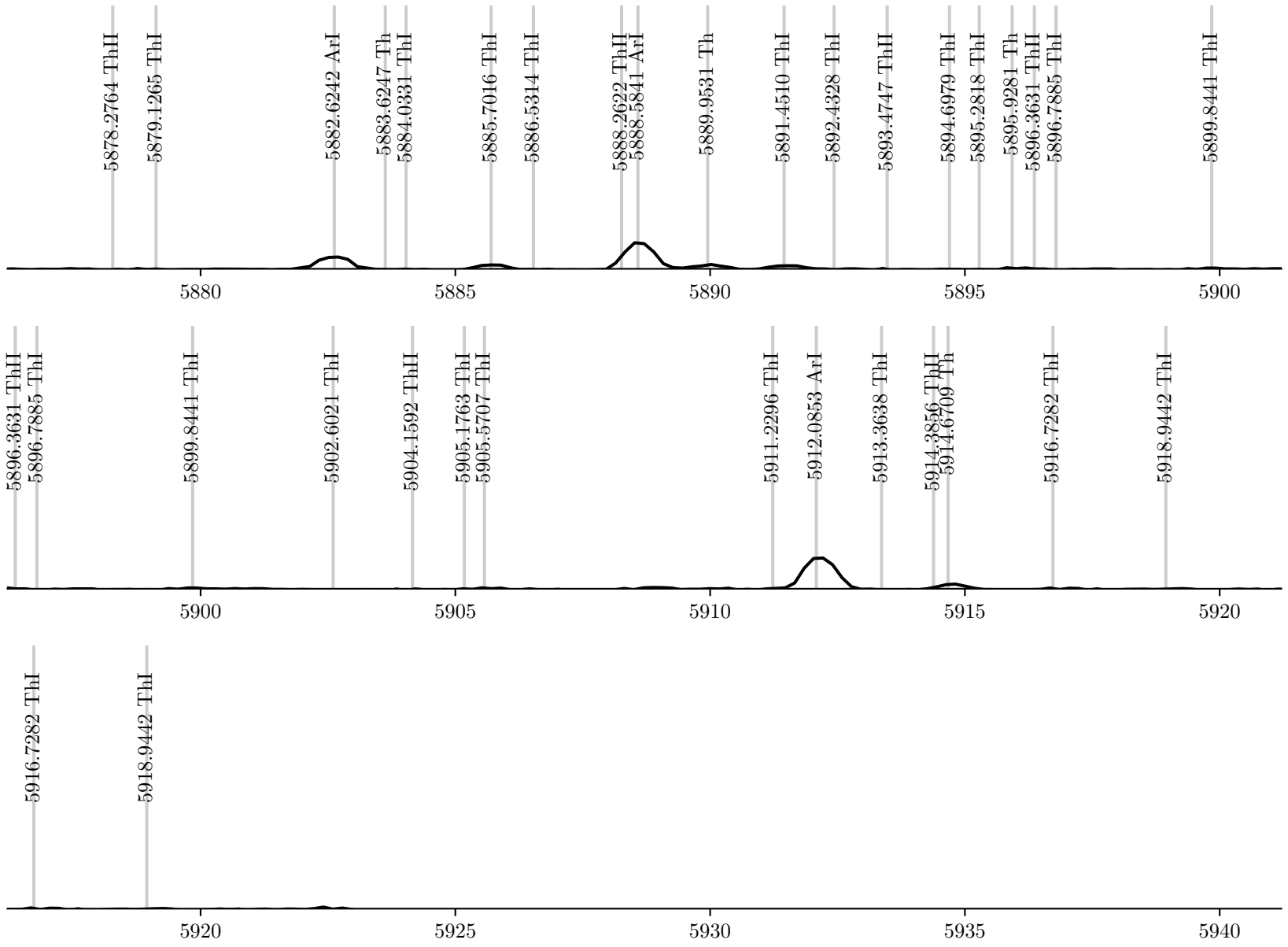

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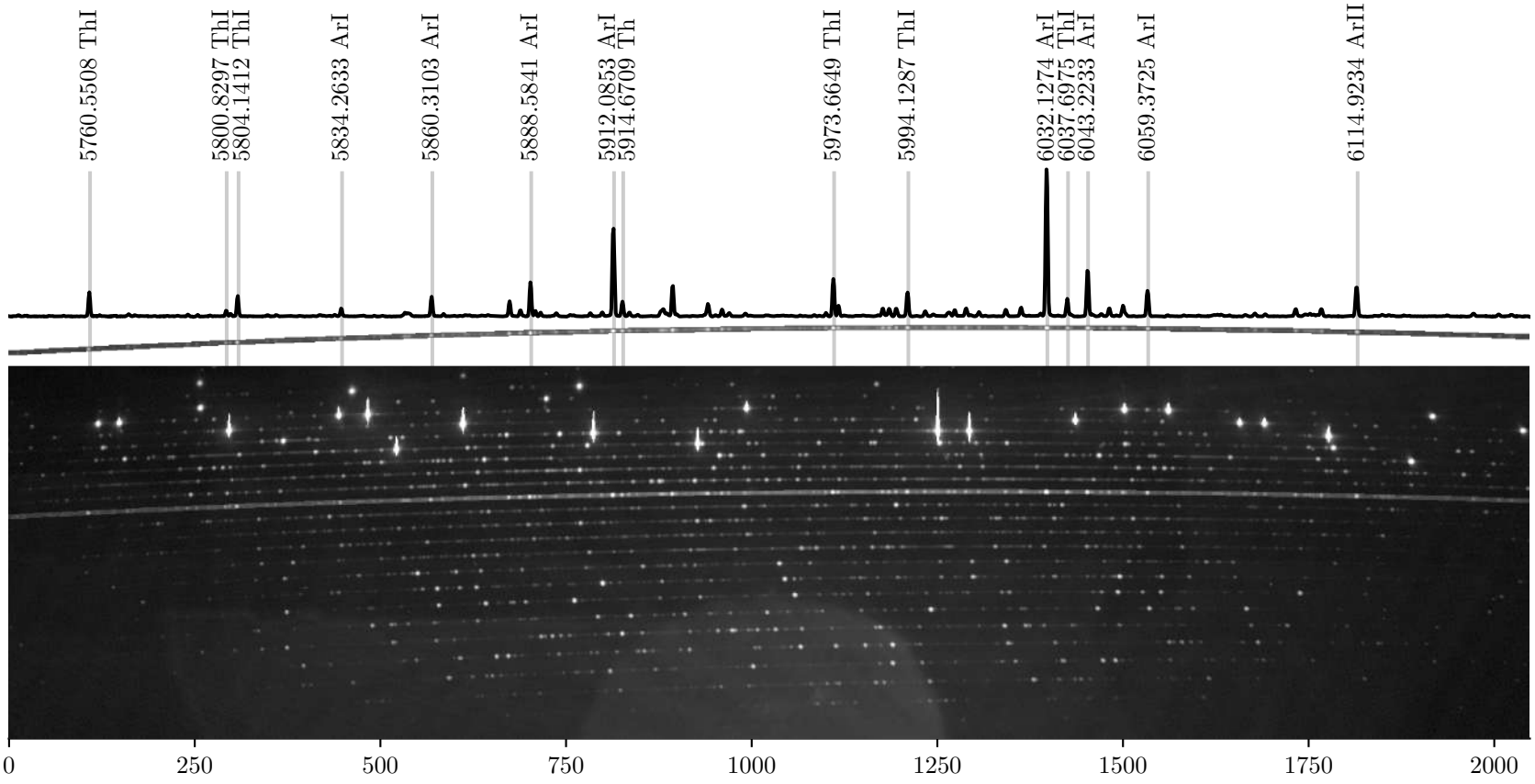

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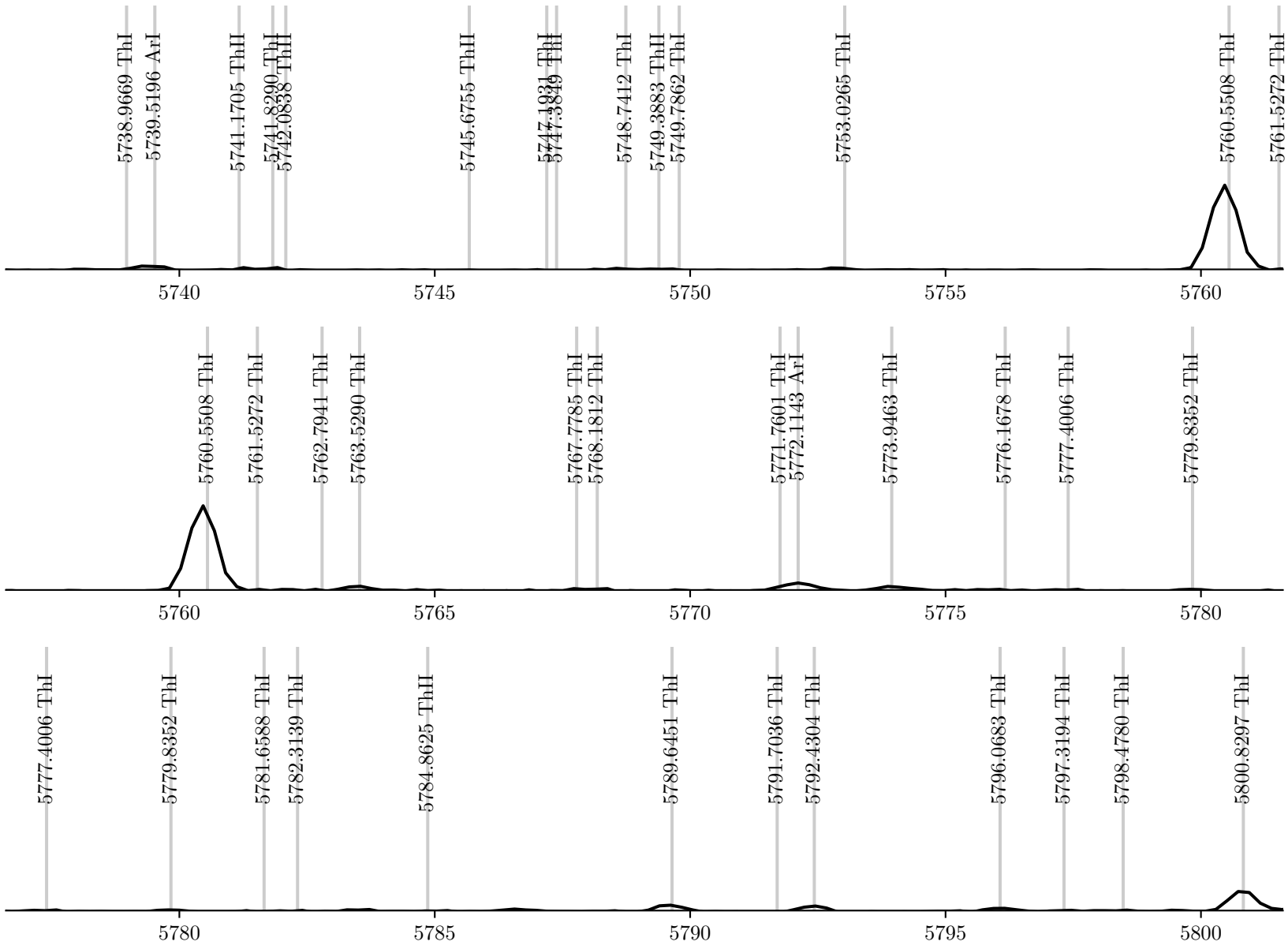

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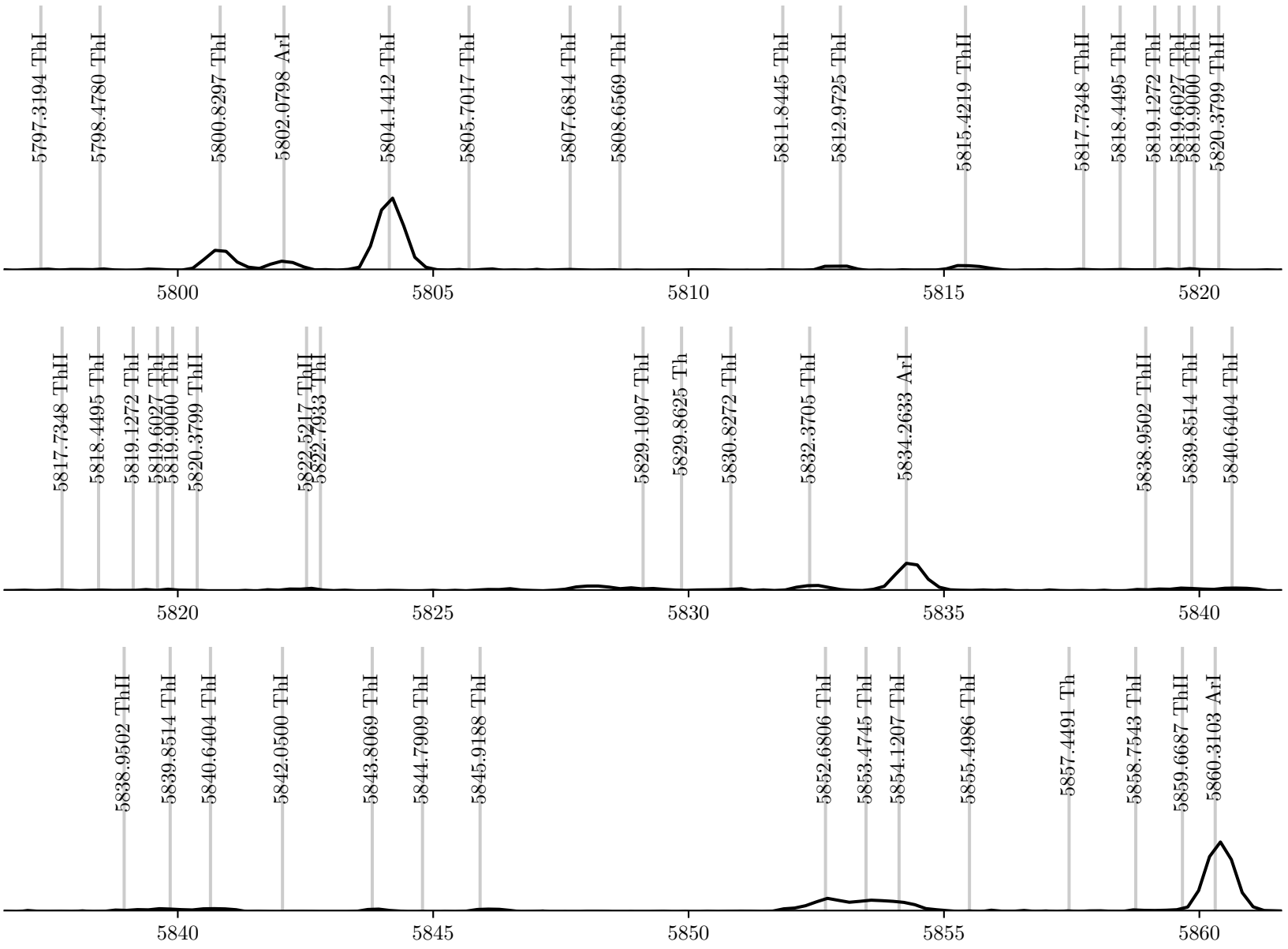

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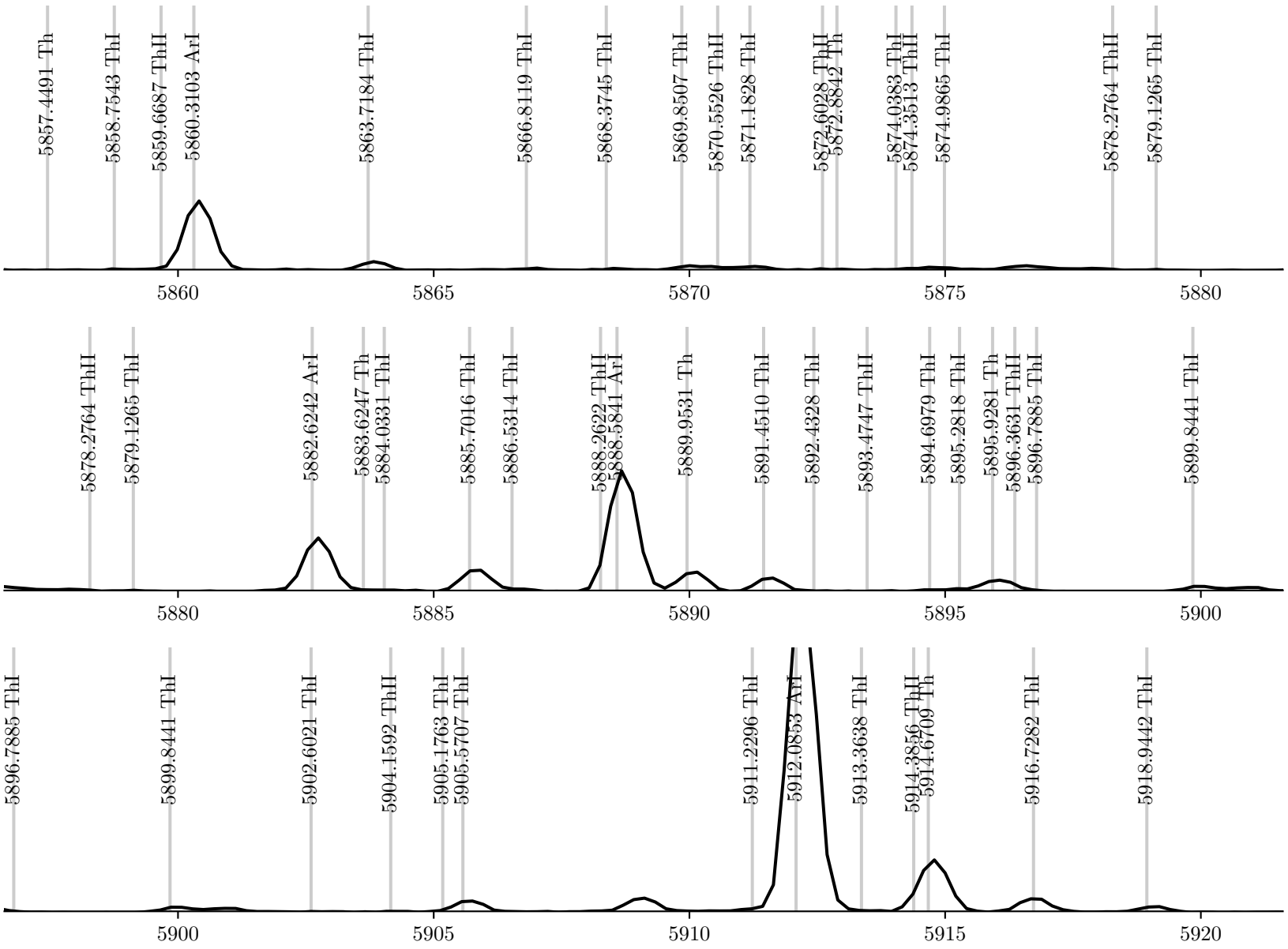
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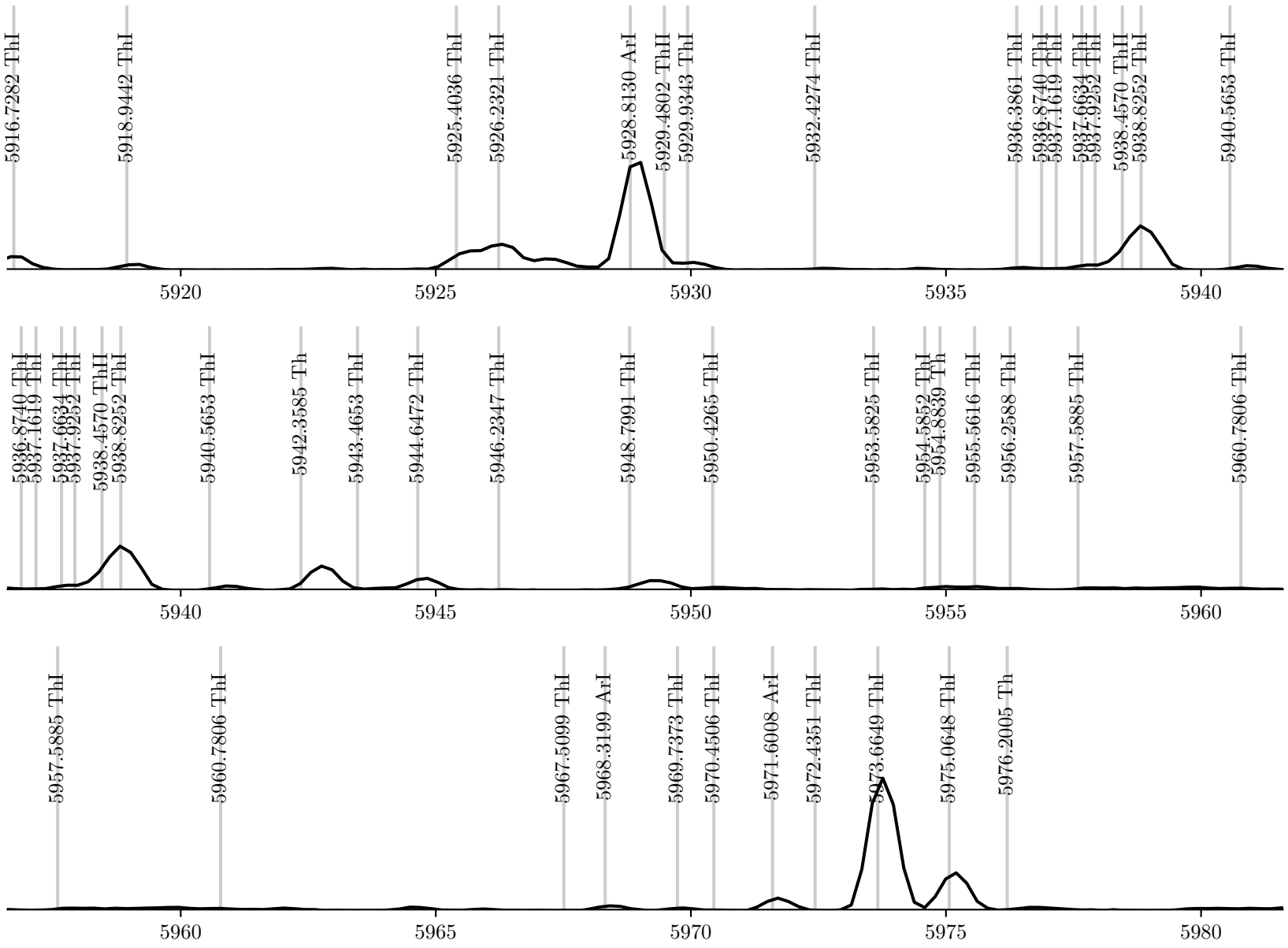


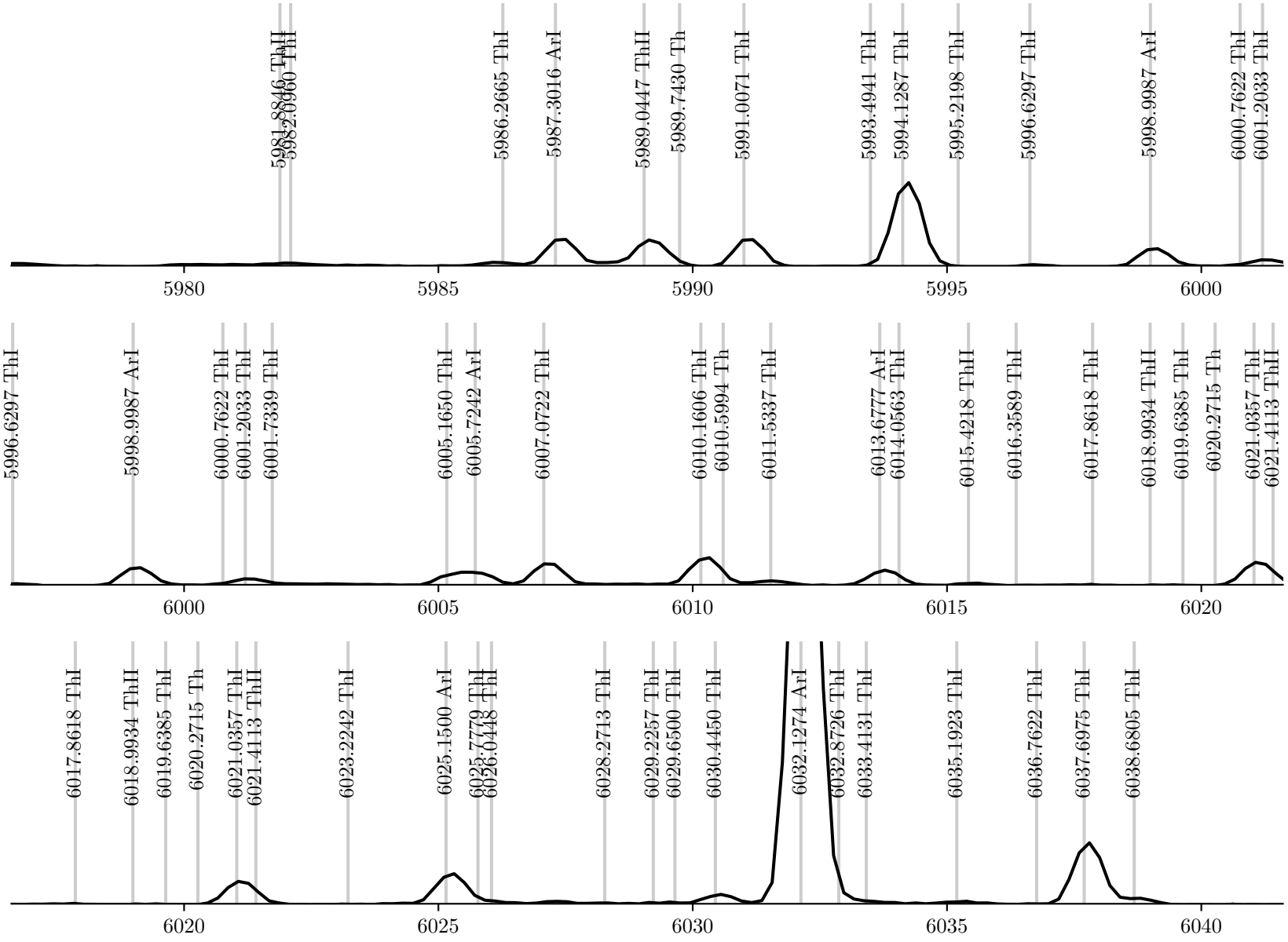
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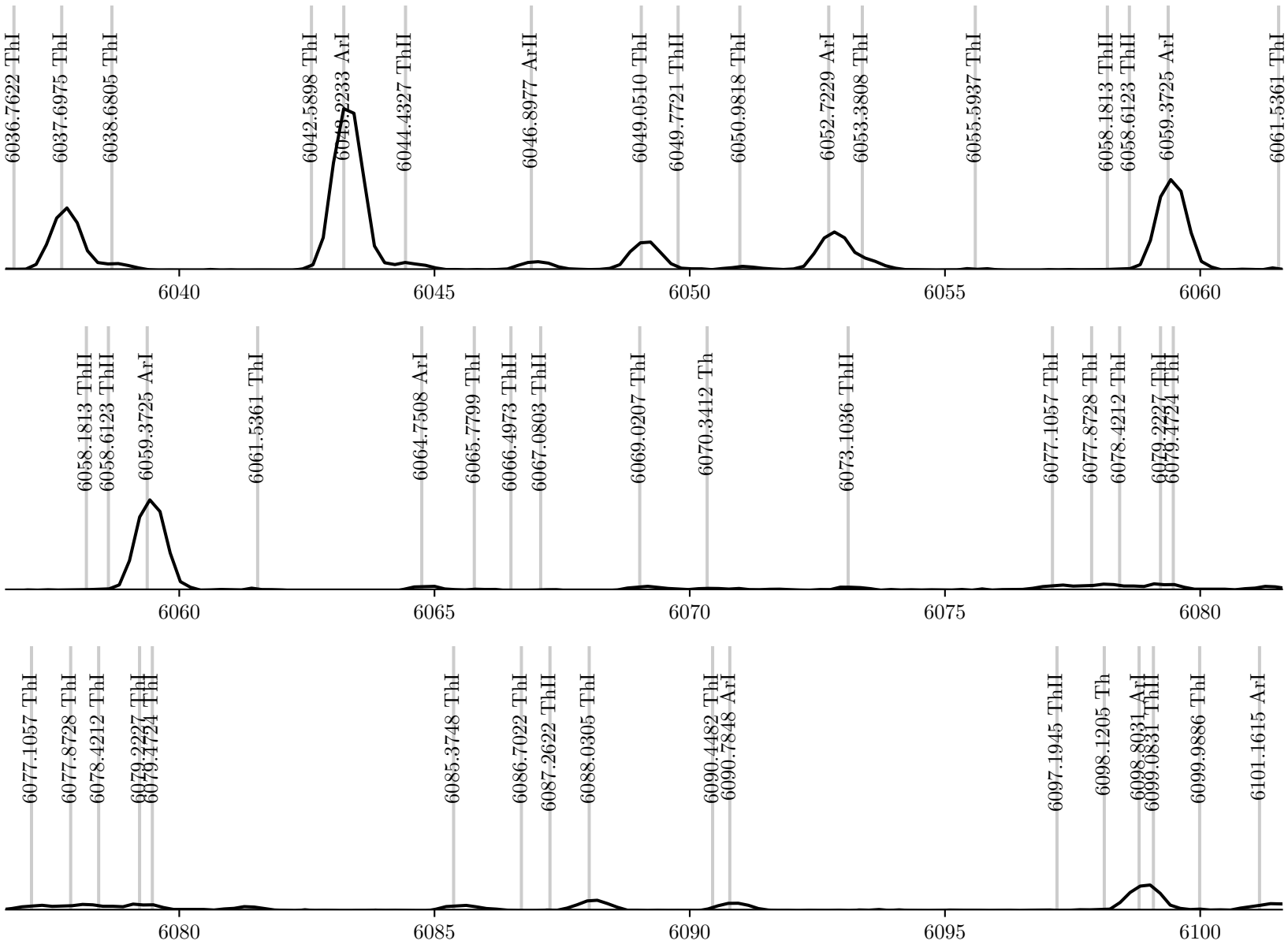

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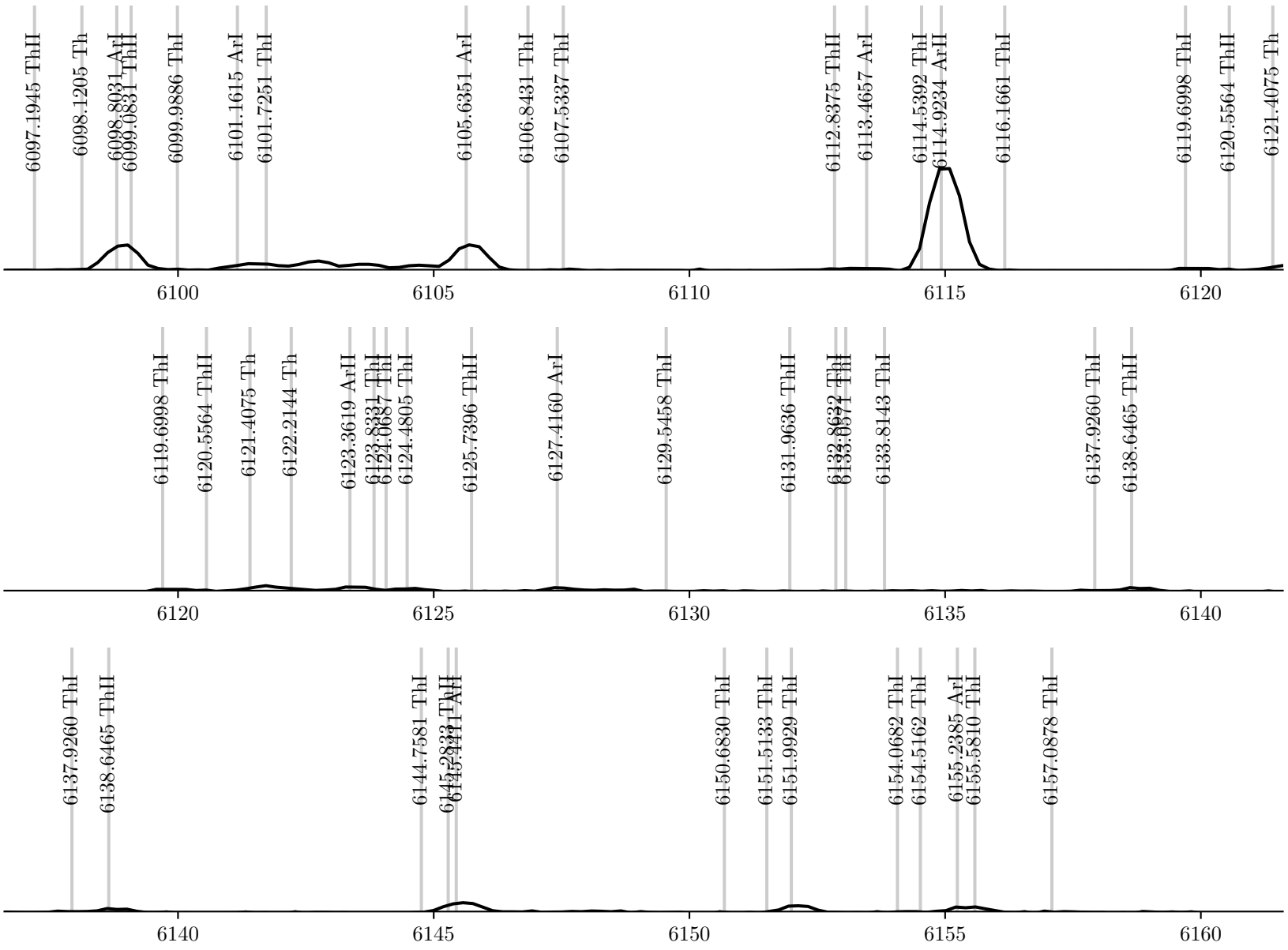

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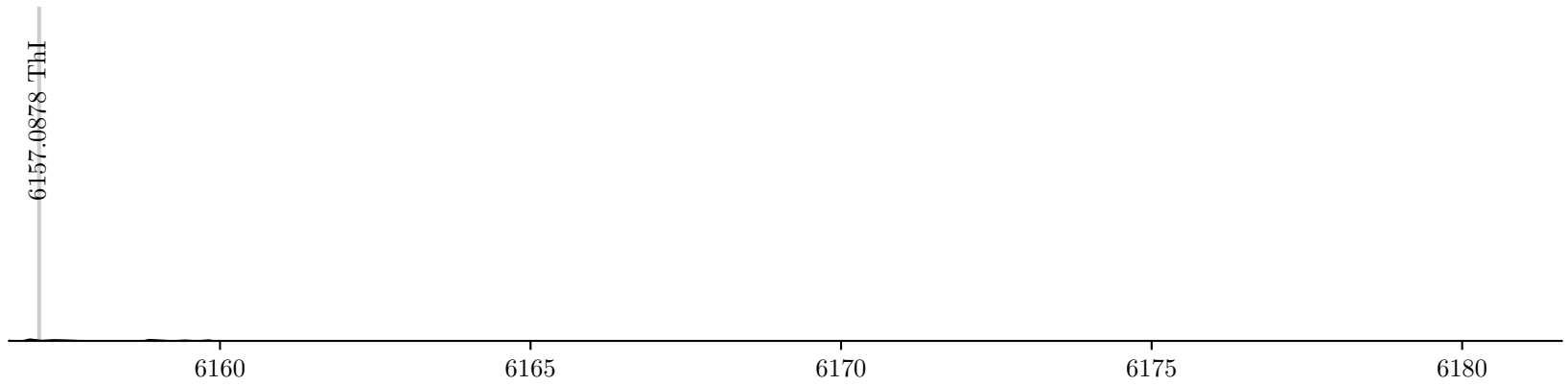

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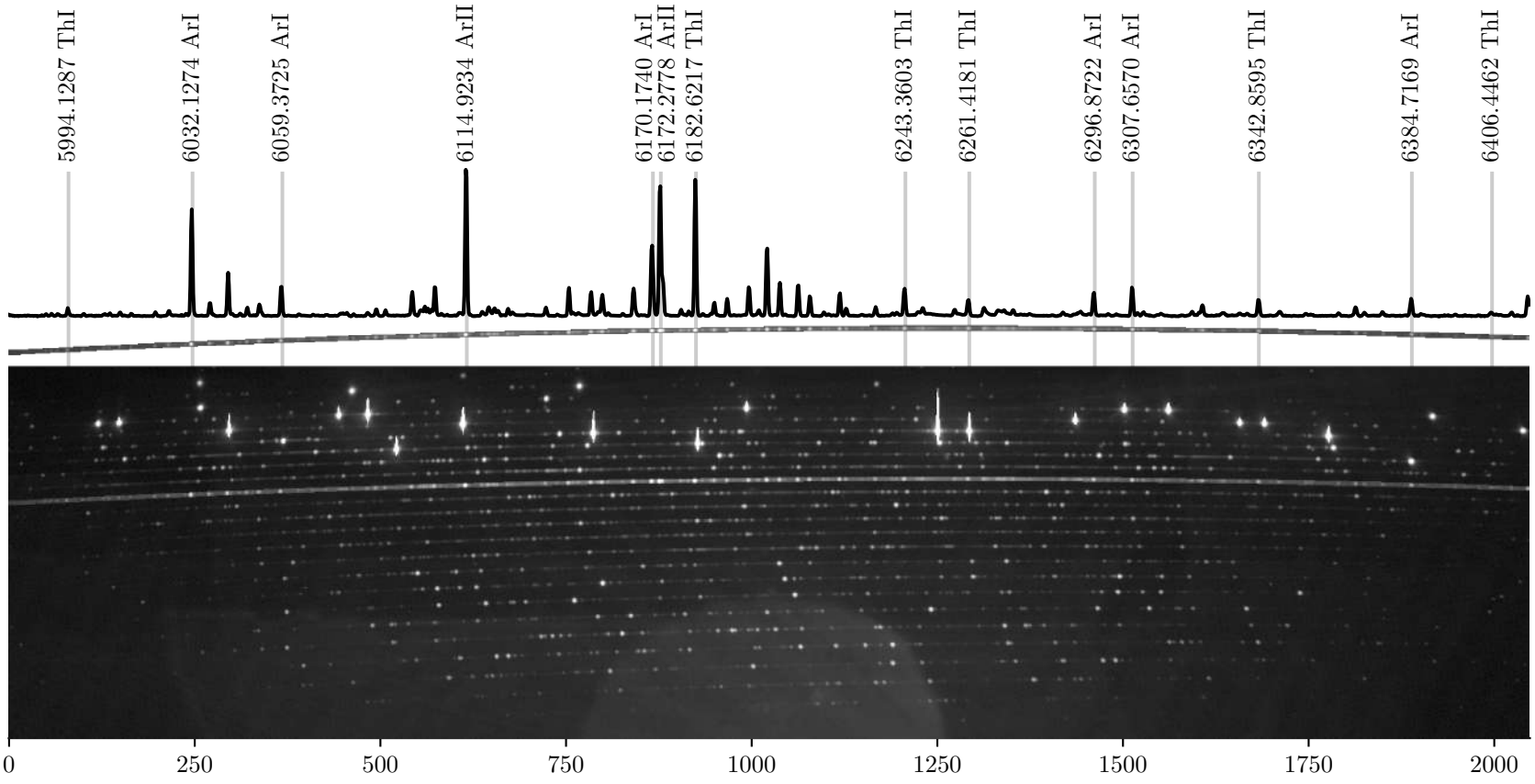

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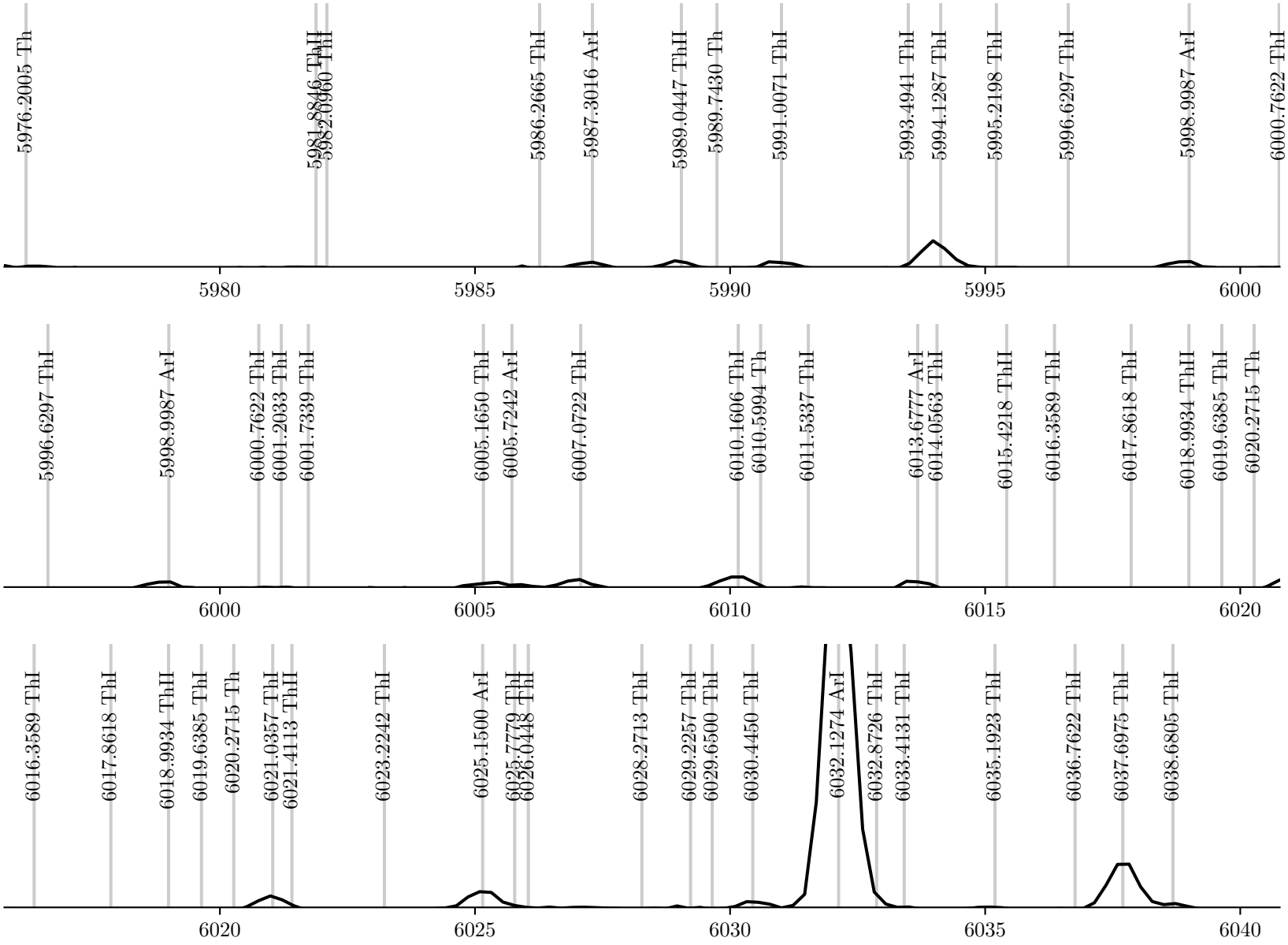

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6154.0682 ThI
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6155.2385 ArI
6155.5810 ThI
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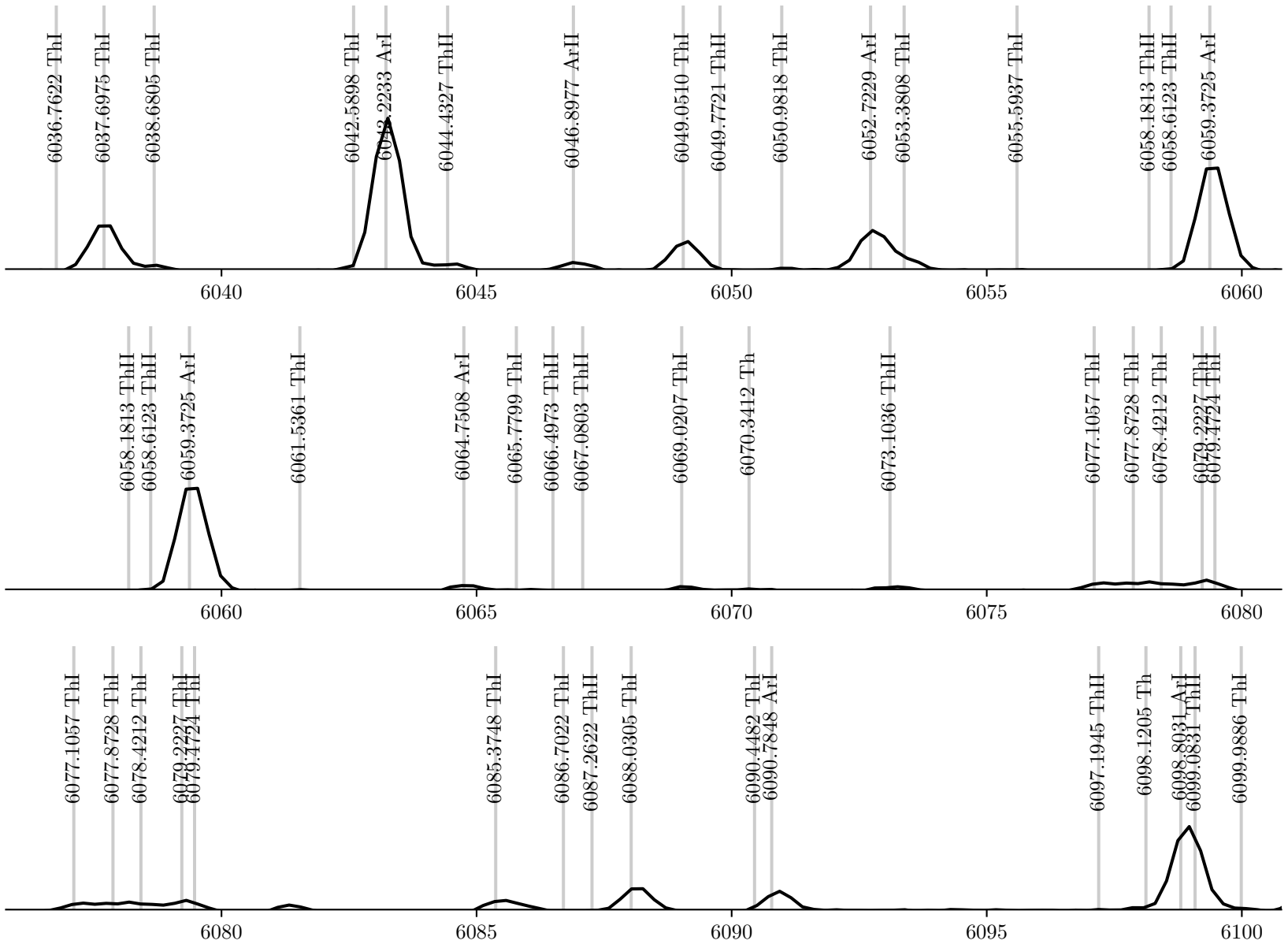

6157.0878 ThI
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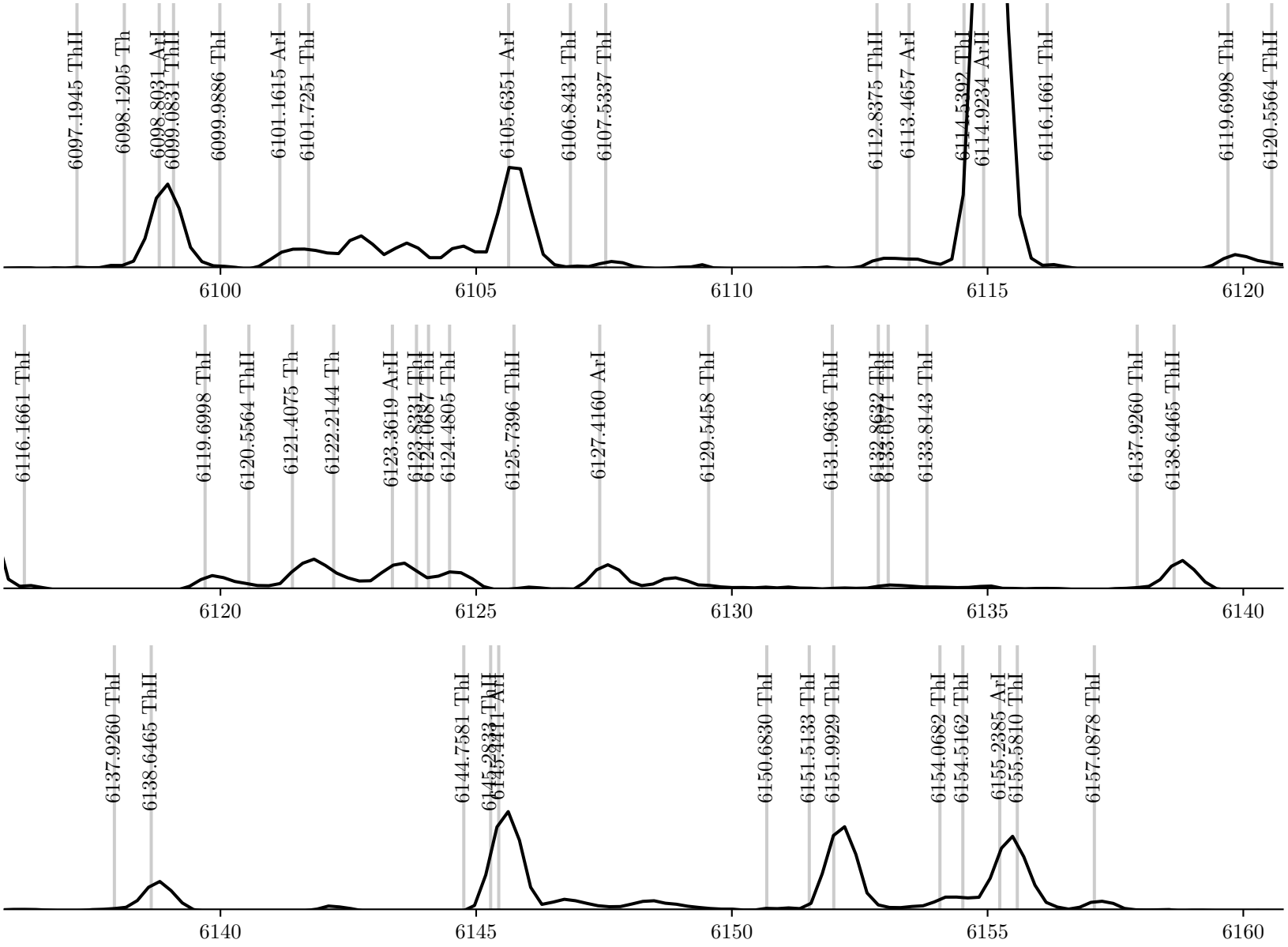
Order 24

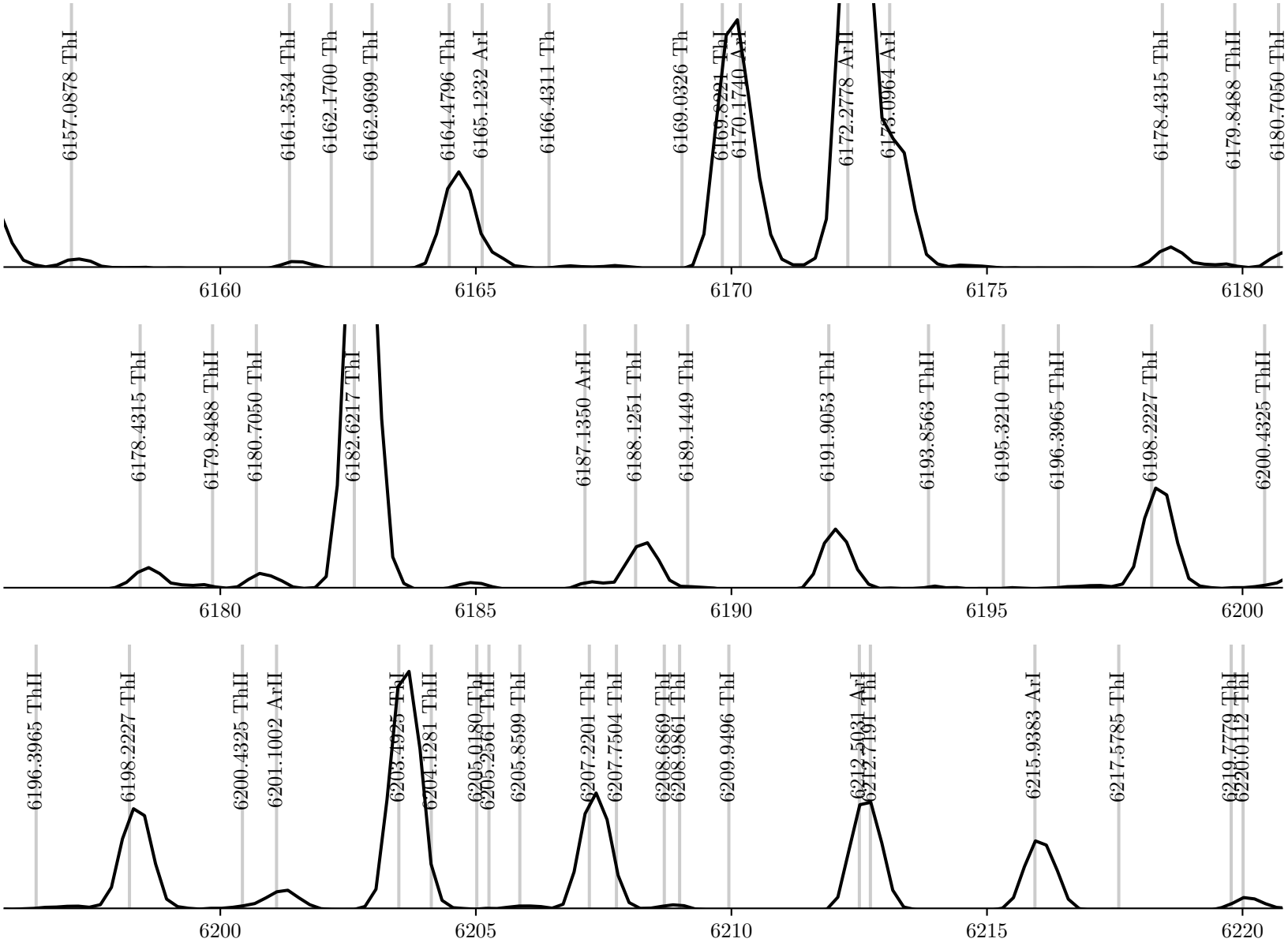


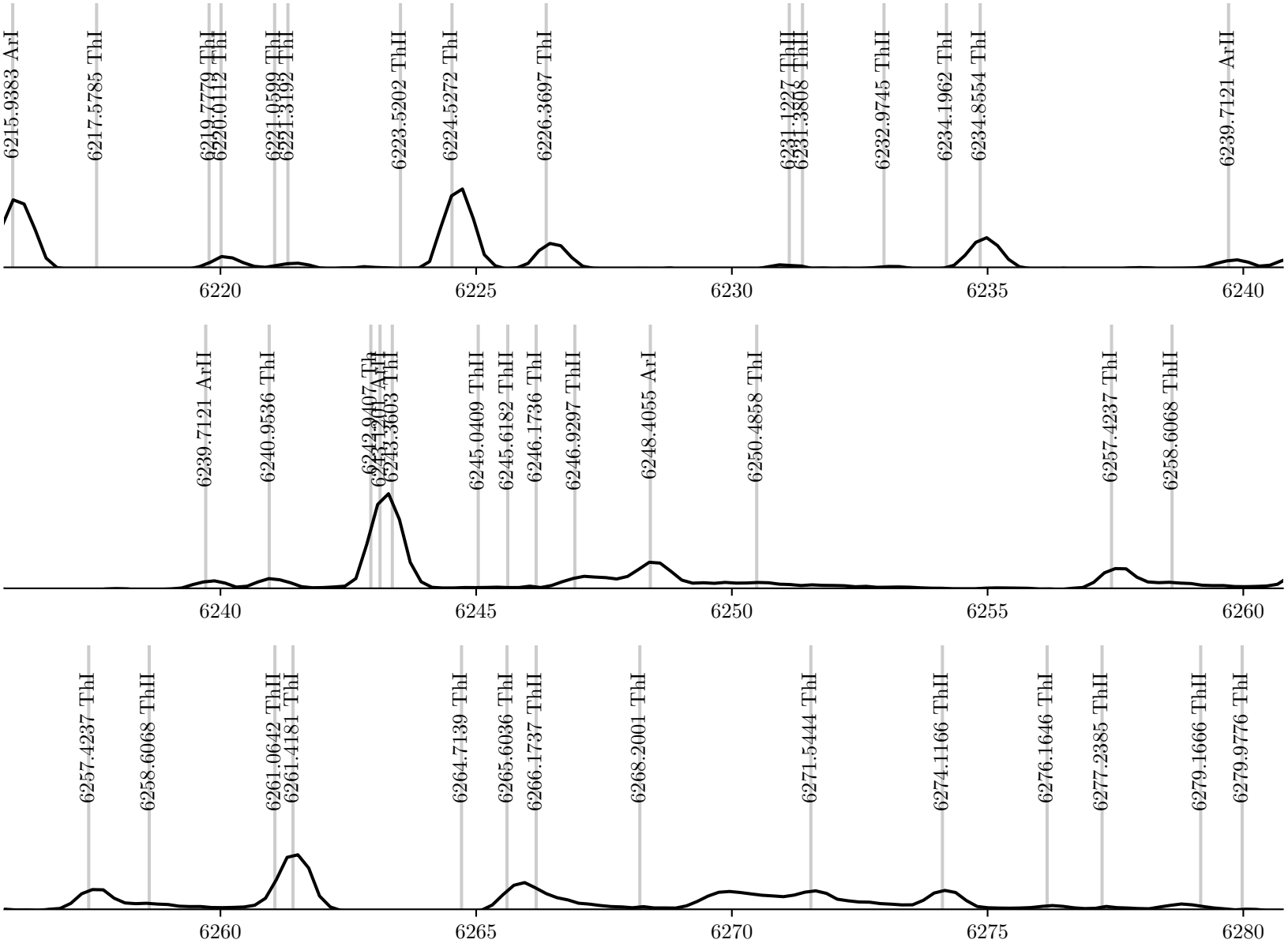
Dispersion:	0.215 Å pix^{-1}
Central wavelength:	6204 Å
Wavelength range:	5976 Å – 6417 Å

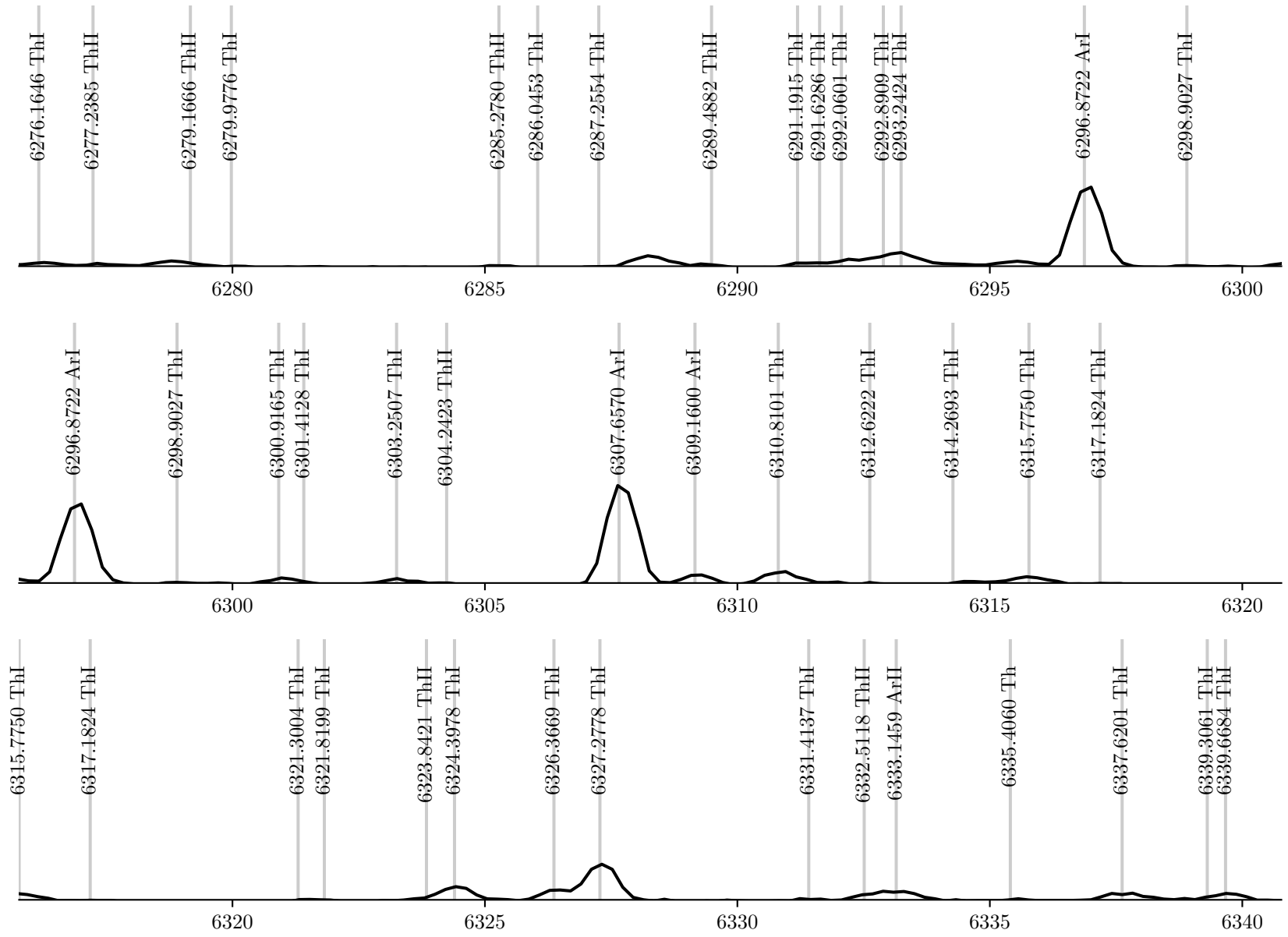

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5986.2665 ThI
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5989.0447 ThII
5989.7430 Th
5991.0071 ThI
5993.4941 ThI
5994.1287 ThI
5995.2198 ThI
5996.6297 ThI
5998.9987 ArI
6000.7622 ThI
5980
5985
5990
5995
6000
5996.6297 ThI
5998.9987 ArI
6000.7622 ThI
6001.2033 ThI
6001.7339 ThI
6005.1650 ThI
6005.7242 ArI
6007.0722 ThI
6010.1606 ThI
6010.5994 Th
6011.5337 ThI
6013.6777 ArI
6014.0563 ThI
6015.4218 ThII
6016.3589 ThI
6017.8618 ThI
6018.9934 ThII
6019.6385 ThI
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6000
6005
6010
6015
6020
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6017.8618 ThI
6018.9934 ThII
6019.6385 ThI
6020.2715 Th
6021.0357 ThI
6021.4113 ThII
6023.2242 ThI
6025.1500 ArI
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6028.2713 ThI
6029.2257 ThI
6029.6500 ThI
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6033.4131 ThI
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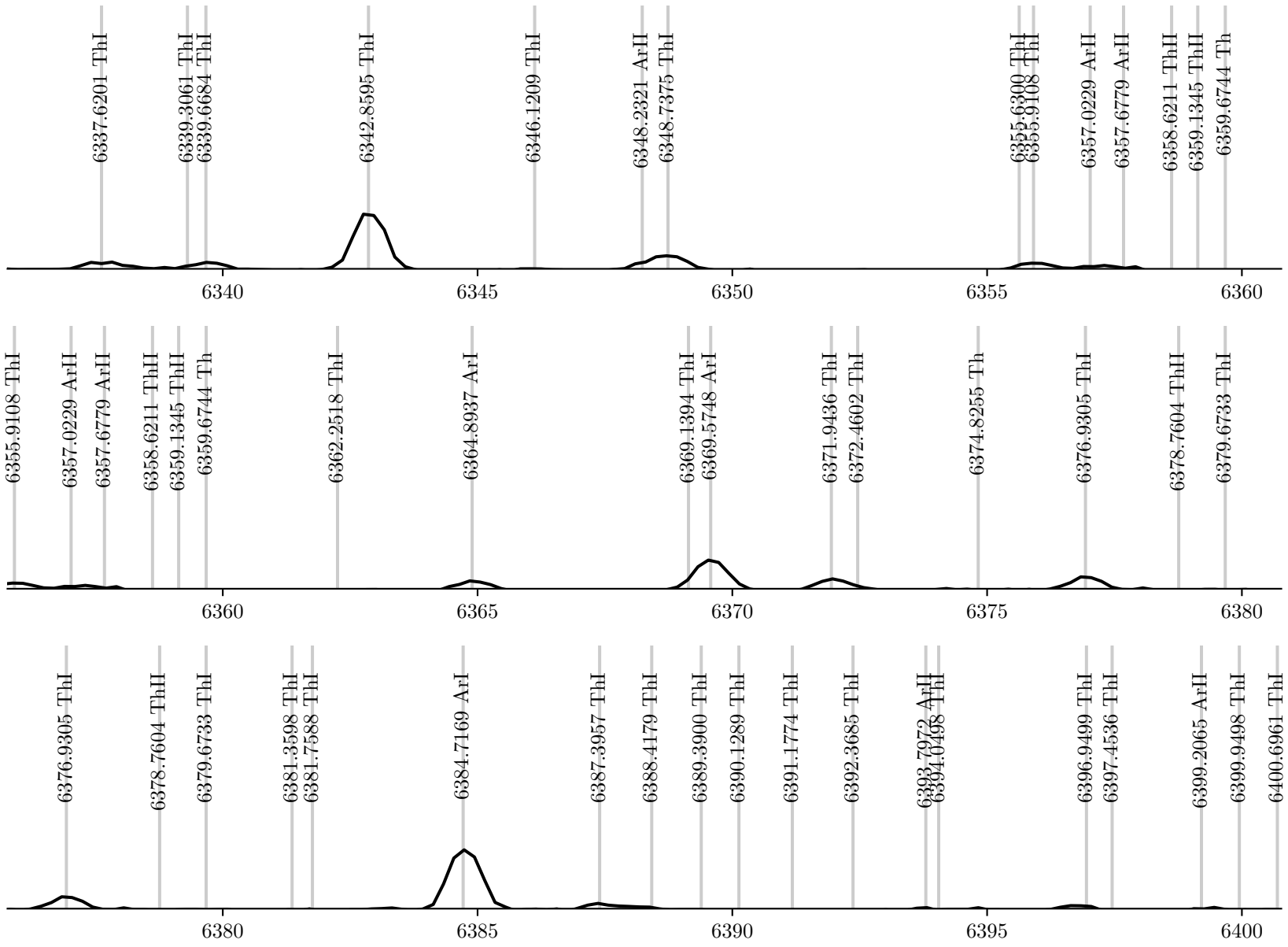

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6042.5898 ThI
6043.2233 ArI
6044.4327 ThII
6046.8977 ArII
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6049.7721 ThII
6050.9818 ThI
6052.7229 ArI
6053.3808 ThI
6055.5937 ThI
6058.1813 ThII
6058.6123 ThII
6059.3725 ArI
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6055
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6058.6123 ThII
6059.3725 ArI
6061.5361 ThI
6064.7508 ArI
6065.7799 ThI
6066.4973 ThII
6067.0803 ThII
6069.0207 ThI
6070.3412 Th
6073.1036 ThII
6077.1057 ThI
6077.8728 ThI
6078.4212 ThI
6079.2227 ThI
6079.4724 ThI
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6077.1057 ThI
6077.8728 ThI
6078.4212 ThI
6079.2227 ThI
6079.4724 ThI
6085.3748 ThI
6086.7022 ThI
6087.2622 ThII
6088.0305 ThI
6090.4482 ThI
6090.7848 ArI
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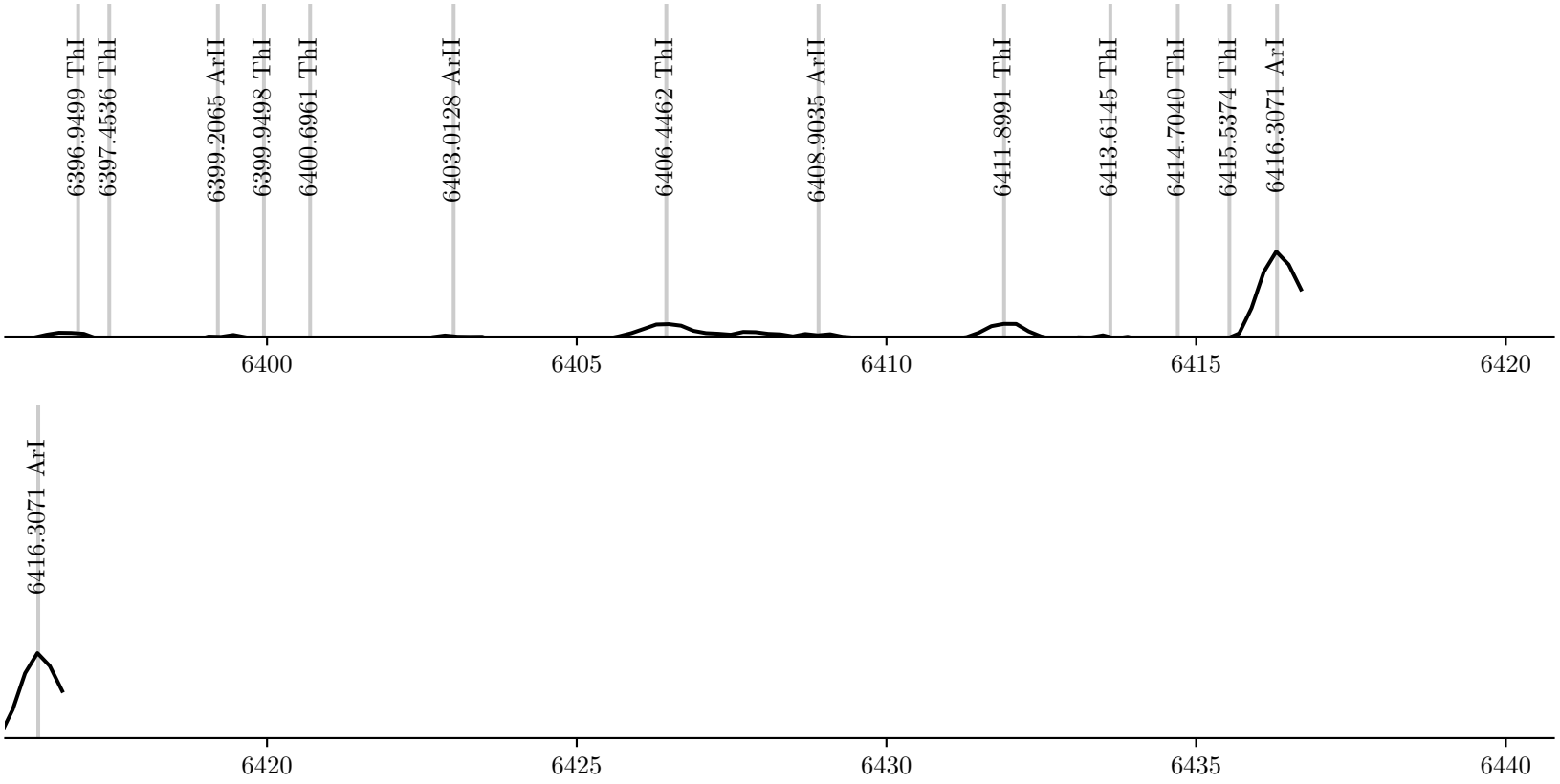

6097.1945 ThII
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6112.8375 ThII
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6114.5392 ThI
6114.9234 ArII
6116.1661 ThI
6119.6998 ThI
6120.5564 ThII
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6115
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6116.1661 ThI
6119.6998 ThI
6120.5564 ThII
6121.4075 Th
6122.2144 Th
6123.3619 ArII
6123.8331 ThI
6124.0687 ThI
6124.4805 ThI
6125.7396 ThII
6127.4160 ArI
6129.5458 ThI
6131.9636 ThII
6132.8632 ThI
6133.0871 ThI
6133.8143 ThI
6137.9260 ThI
6138.6465 ThII
6120
6125
6130
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6140
6137.9260 ThI
6138.6465 ThII
6144.7581 ThI
6145.2833 ThII
6145.4411 ArII
6150.6830 ThI
6151.5133 ThI
6151.9929 ThI
6154.0682 ThI
6154.5162 ThI
6155.2385 ArI
6155.5810 ThI
6157.0878 ThI
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6150
6155
6160


6157.0878 ThI
6161.3534 ThI
6162.1700 Th
6162.9699 ThI
6164.4796 ThI
6165.1232 ArI
6166.4311 Th
6169.0326 Th
6169.8221 ThI
6170.1740 ArI
6172.2778 ArII
6173.0964 ArI
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6180.7050 ThI
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6178.4315 ThI
6179.8488 ThII
6180.7050 ThI
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6188.1251 ThI
6189.1449 ThI
6191.9053 ThI
6193.8563 ThII
6195.3210 ThI
6196.3965 ThII
6198.2227 ThI
6200.4325 ThII
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6196.3965 ThII
6198.2227 ThI
6200.4325 ThII
6201.1002 ArII
6203.4925 ThI
6204.1281 ThII
6205.0180 ThI
6205.2561 ThII
6205.8599 ThI
6207.2201 ThI
6207.7504 ThI
6208.6869 ThI
6208.9861 ThI
6209.9496 ThI
6212.5031 ArI
6212.7191 ThI
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6220

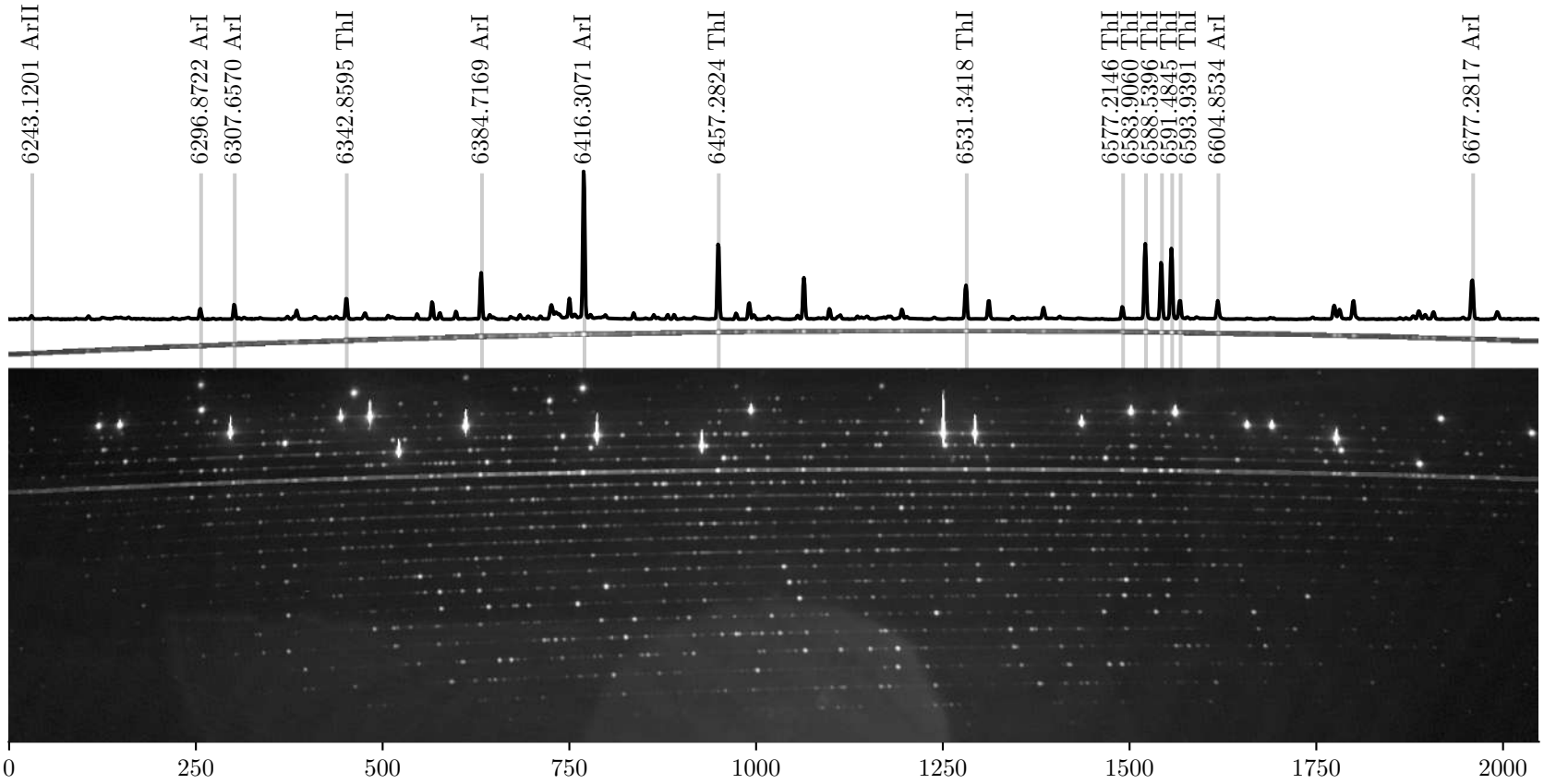

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6221.0599 ThI
6221.3192 ThI
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6224.5272 ThI
6226.3697 ThI
6231.1227 ThII
6231.3808 ThII
6232.9745 ThII
6234.1962 ThI
6234.8554 ThI
6239.7121 ArII
6220
6225
6230
6235
6240
6239.7121 ArII
6240.9536 ThI
6242.9407 ThI
6243.3603 ThI
6245.0409 ThII
6245.6182 ThII
6246.1736 ThI
6246.9297 ThII
6248.4055 ArI
6250.4858 ThI
6257.4237 ThI
6258.6068 ThII
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6250
6255
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6257.4237 ThI
6258.6068 ThII
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6261.4181 ThI
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6280


6276.1646 ThI
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6279.9776 ThI
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6287.2554 ThI
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6291.6286 ThI
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6301.4128 ThI
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6304.2423 ThII
6307.6570 ArI
6309.1600 ArI
6310.8101 ThI
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6314.2693 ThI
6315.7750 ThI
6317.1824 ThI
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6326.3669 ThI
6327.2778 ThI
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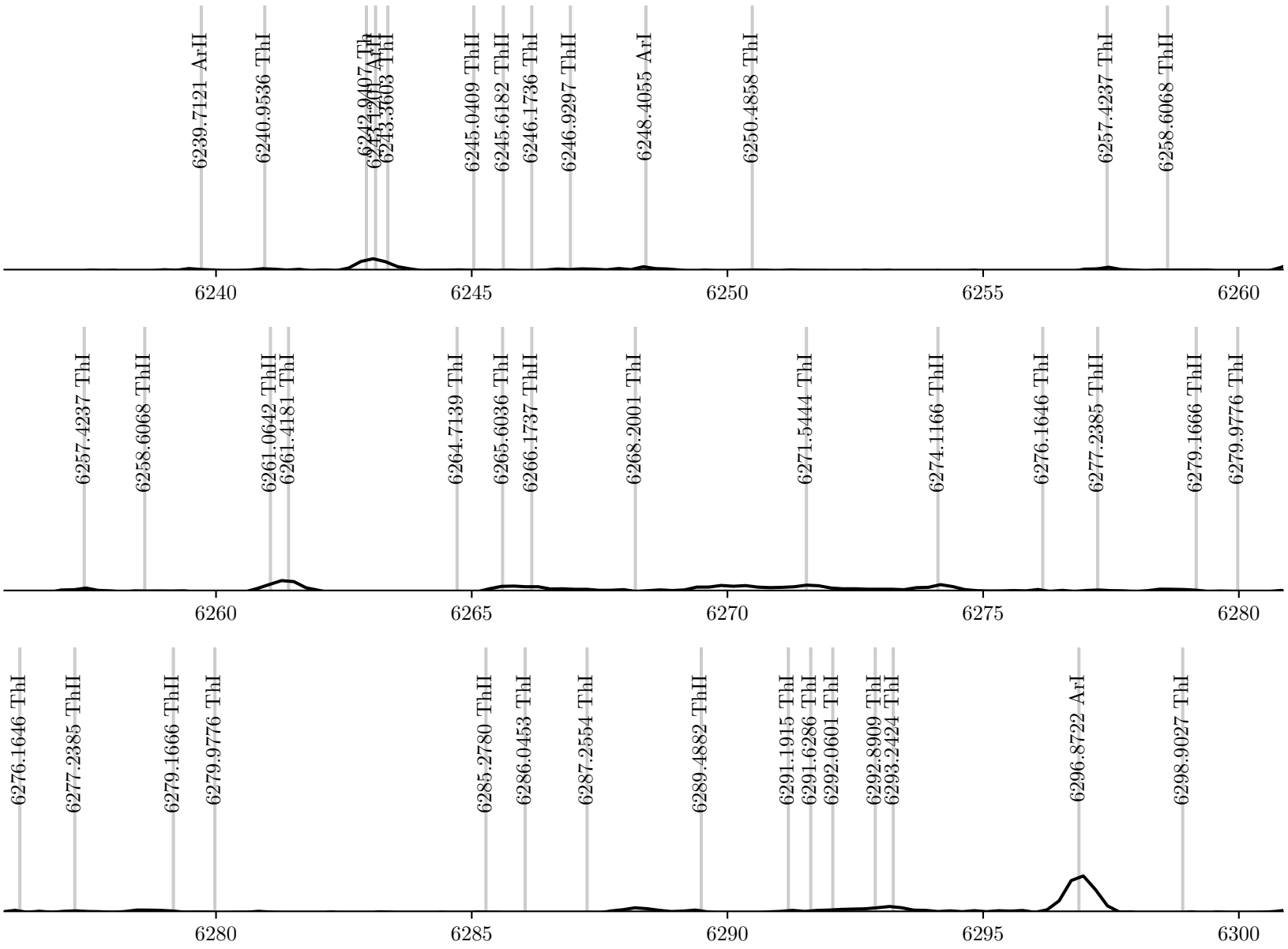

6337.6201 ThI
6339.3061 ThI
6339.6684 ThI
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6346.1209 ThI
6348.2321 ArII
6348.7375 ThI
6355.6300 ThI
6355.9108 ThI
6357.0229 ArII
6357.6779 ArII
6358.6211 ThII
6359.1345 ThII
6359.6744 Th
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6345
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6357.6779 ArII
6358.6211 ThII
6359.1345 ThII
6359.6744 Th
6362.2518 ThI
6364.8937 ArI
6369.1394 ThI
6369.5748 ArI
6371.9436 ThI
6372.4602 ThI
6374.8255 Th
6376.9305 ThI
6378.7604 ThII
6379.6733 ThI
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6370
6375
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6376.9305 ThI
6378.7604 ThII
6379.6733 ThI
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6381.7588 ThI
6384.7169 ArI
6387.3957 ThI
6388.4179 ThI
6389.3900 ThI
6390.1289 ThI
6391.1774 ThI
6392.3685 ThI
6393.7972 ArII
6394.0498 ThI
6396.9499 ThI
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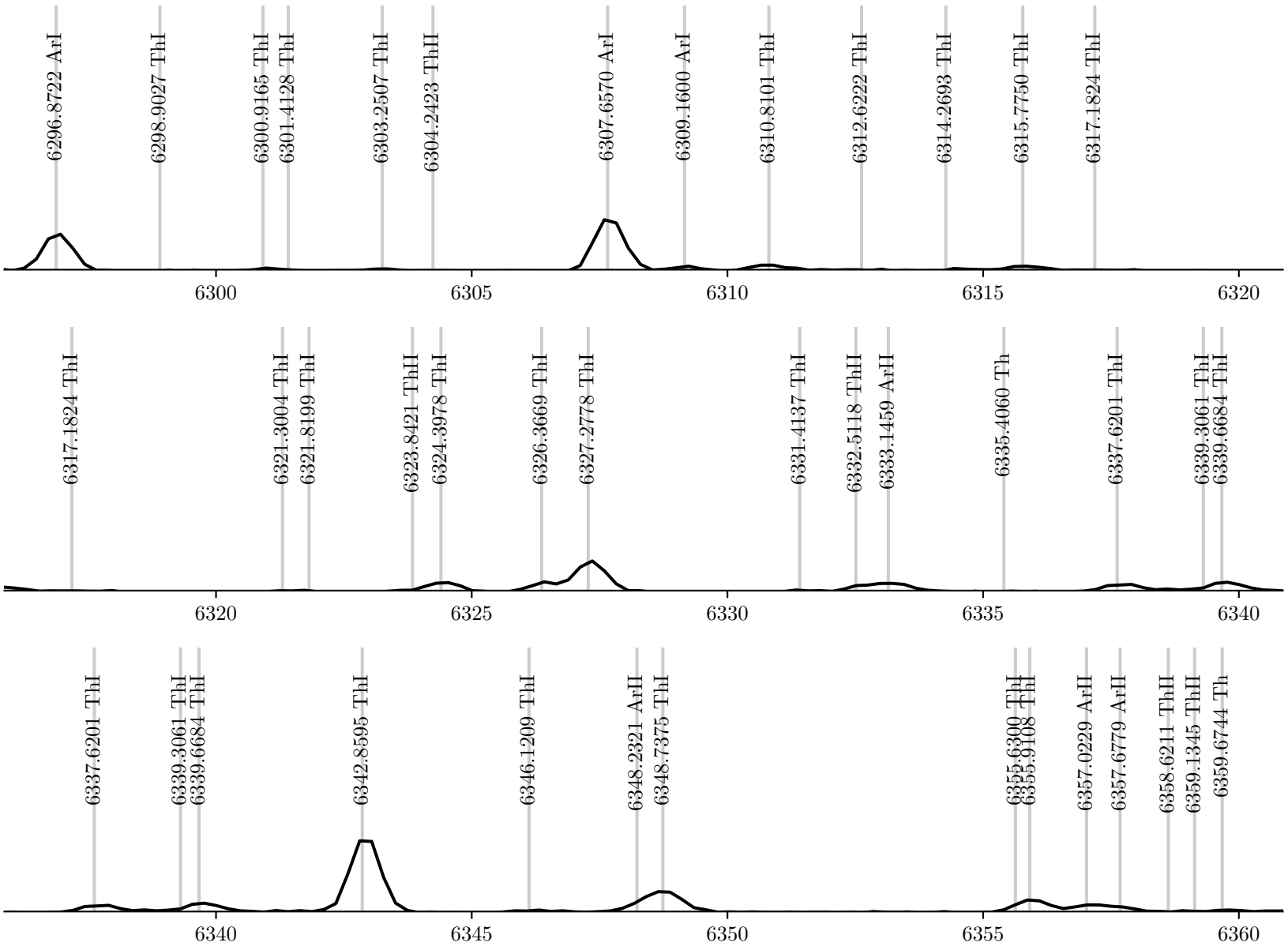

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6411.8991 ThI
6413.6145 ThI
6414.7040 ThI
6415.5374 ThI
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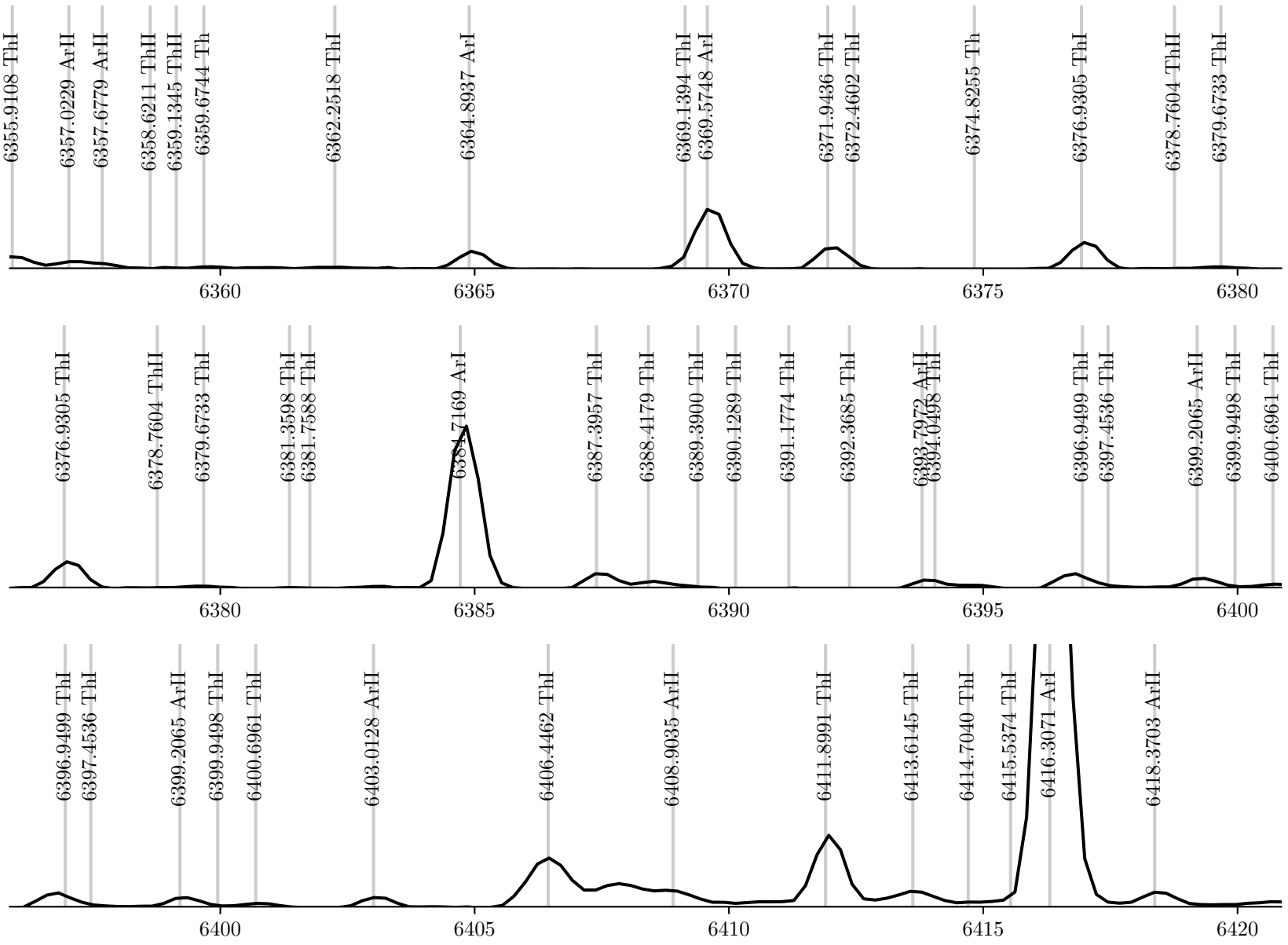
Order 23

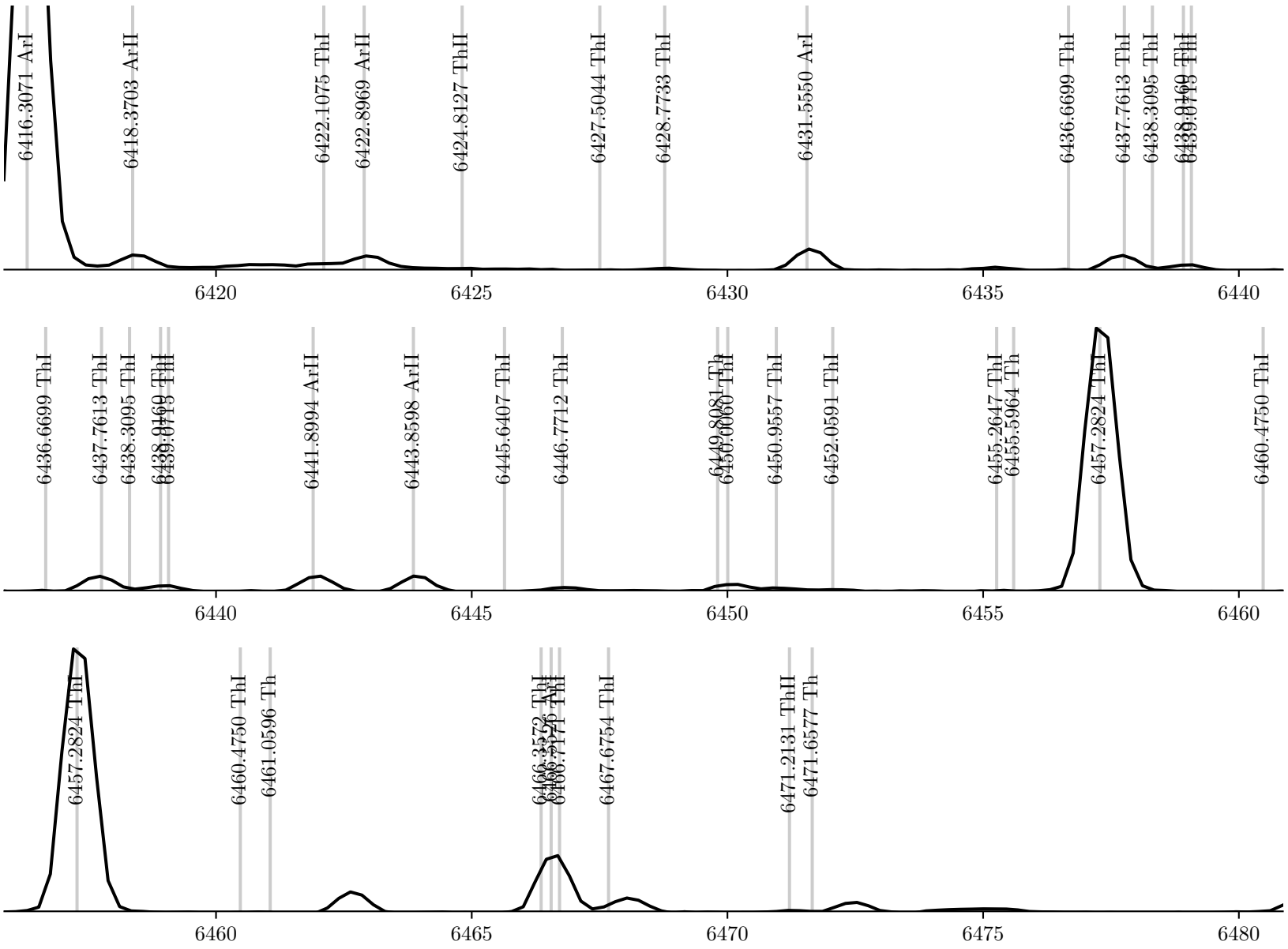


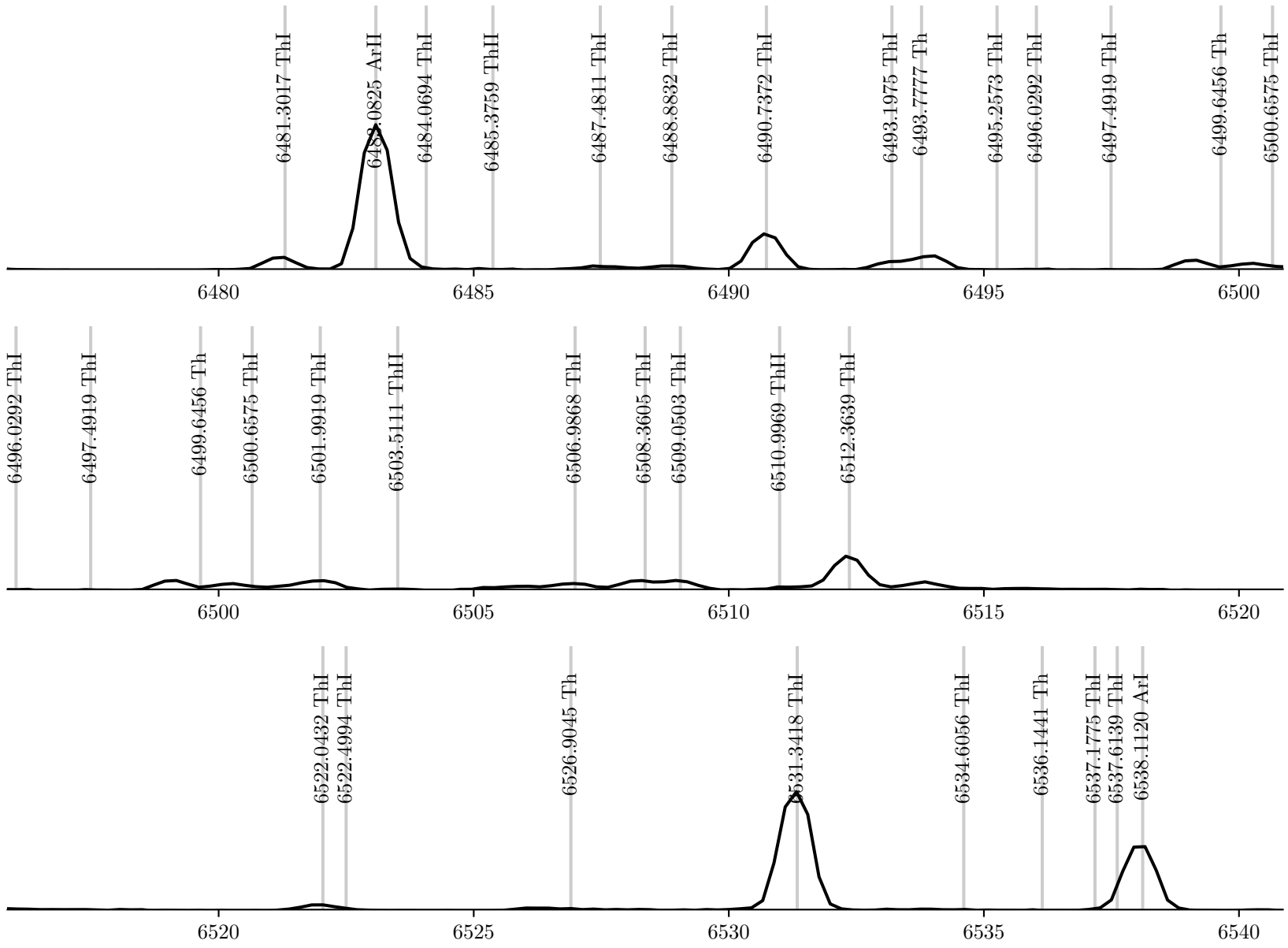
Dispersion:	0.225 Å pix^{-1}
Central wavelength:	6474 Å
Wavelength range:	6236 Å – 6696 Å

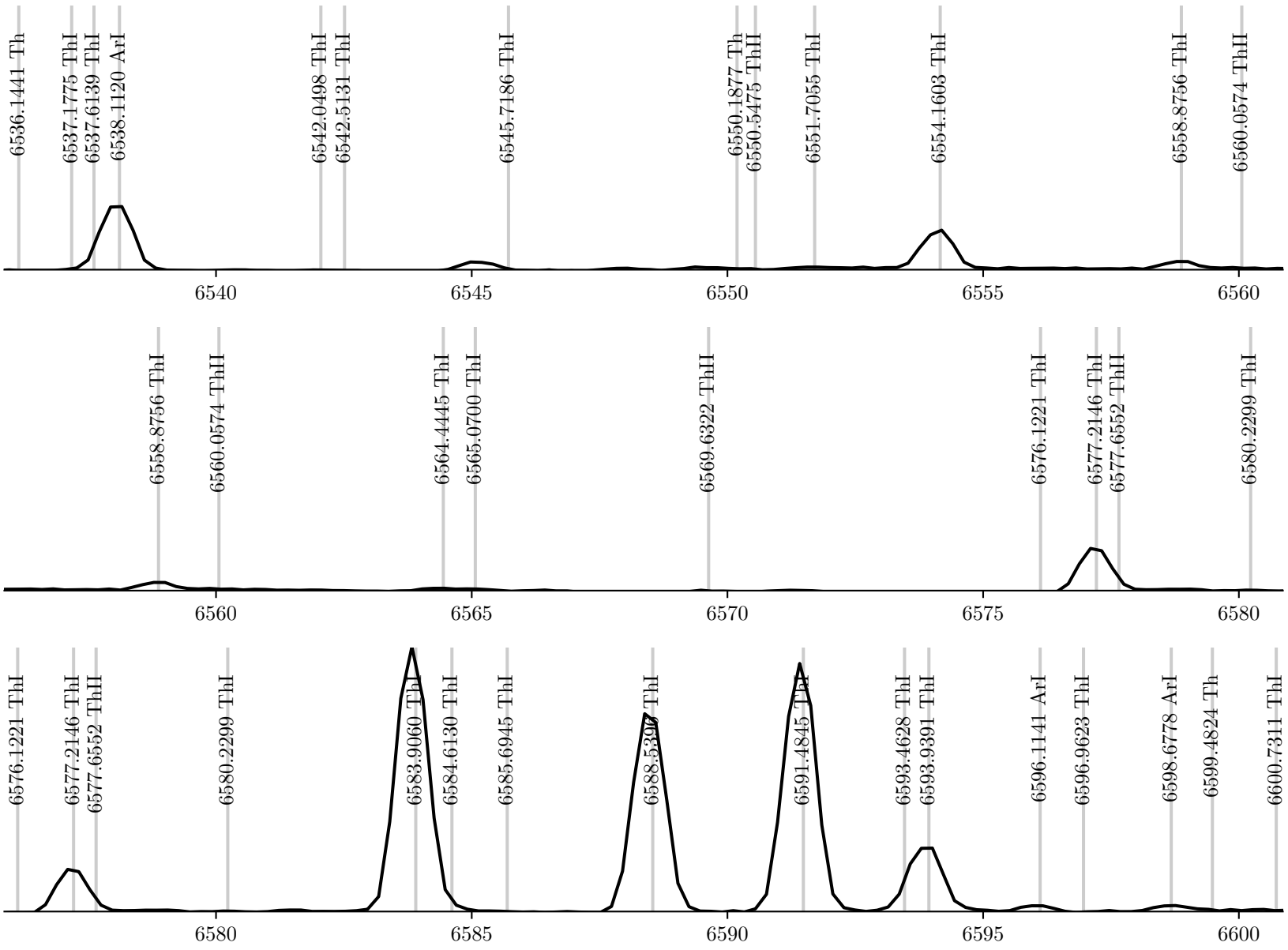

6239.7121 ArII
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6245.0409 ThII
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6246.1736 ThI
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6248.4055 ArI
6250.4858 ThI
6257.4237 ThI
6258.6068 ThII
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6245
6250
6255
6260
6257.4237 ThI
6258.6068 ThII
6261.0642 ThII
6261.4181 ThI
6264.7139 ThI
6265.6036 ThI
6266.1737 ThII
6268.2001 ThI
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6274.1166 ThII
6276.1646 ThI
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6277.2385 ThII
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6279.9776 ThI
6285.2780 ThII
6286.0453 ThI
6287.2554 ThI
6289.4882 ThII
6291.1915 ThI
6291.6286 ThI
6292.0601 ThI
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6293.2424 ThI
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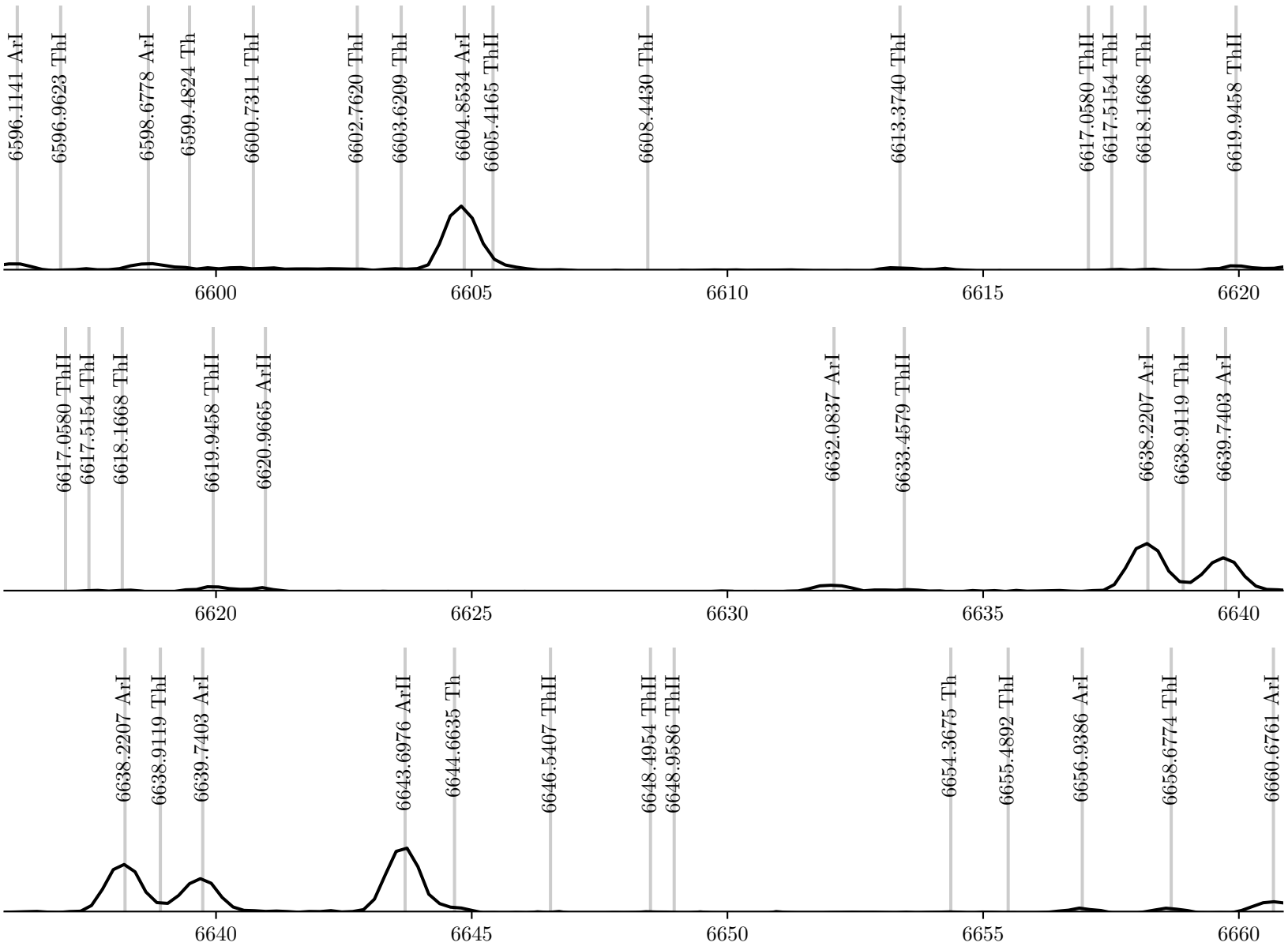

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6327.2778 ThI
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6332.5118 ThII
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6335.4060 Th
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6337.6201 ThI
6339.3061 ThI
6339.6684 ThI
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6348.7375 ThI
6355.6300 ThI
6355.9108 ThI
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6355
6360

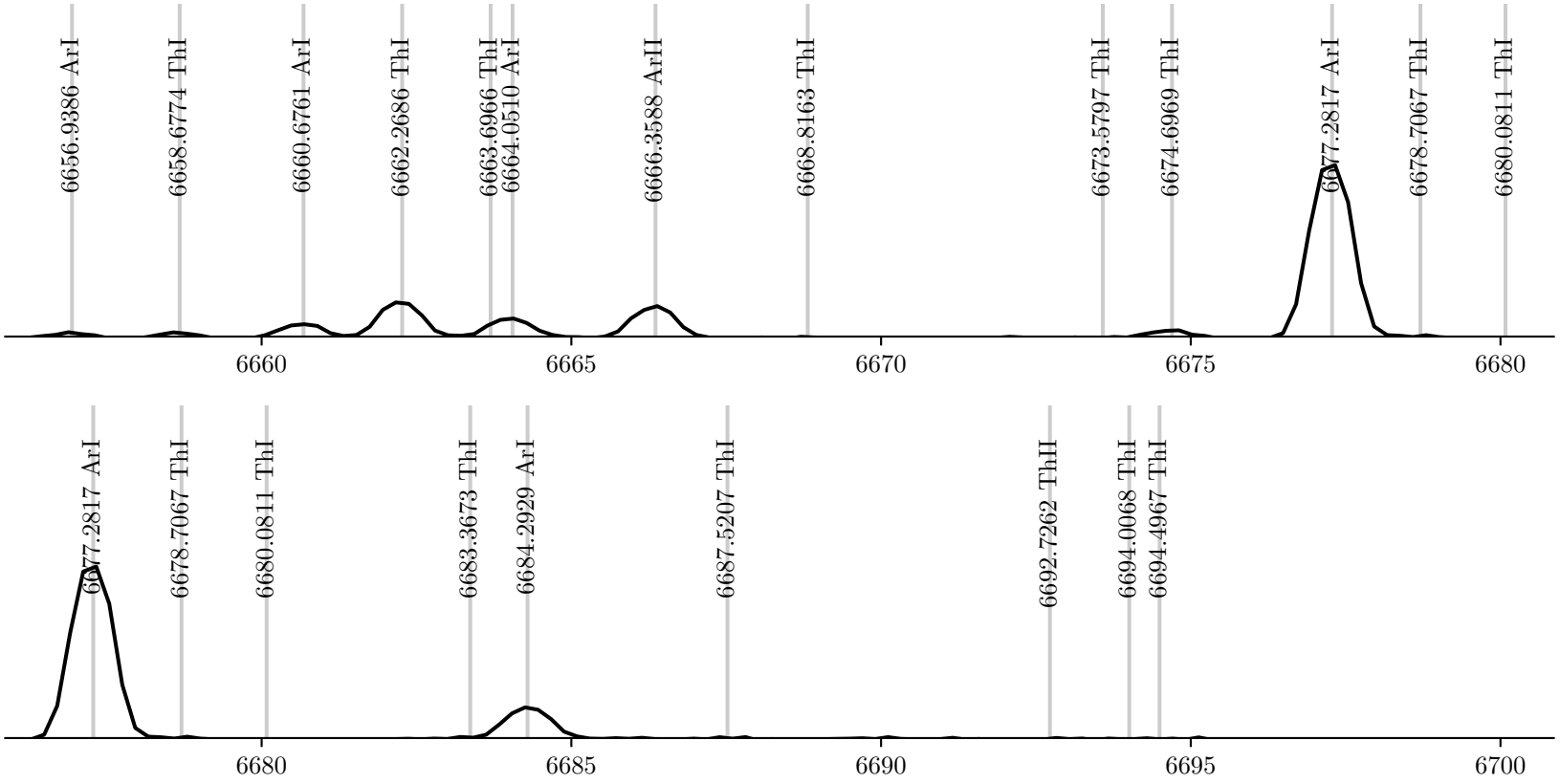

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6359.6744 Th
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6364.8937 ArI
6369.1394 ThI
6369.5748 ArI
6371.9436 ThI
6372.4602 ThI
6374.8255 Th
6376.9305 ThI
6378.7604 ThII
6379.6733 ThI
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6376.9305 ThI
6378.7604 ThII
6379.6733 ThI
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6381.7588 ThI
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6387.3957 ThI
6388.4179 ThI
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6392.3685 ThI
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6408.9035 ArII
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6418.3703 ArII
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6415
6420


6416.3071 ArI
6418.3703 ArII
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6437.7613 ThI
6438.3095 ThI
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6457.2824 ThI
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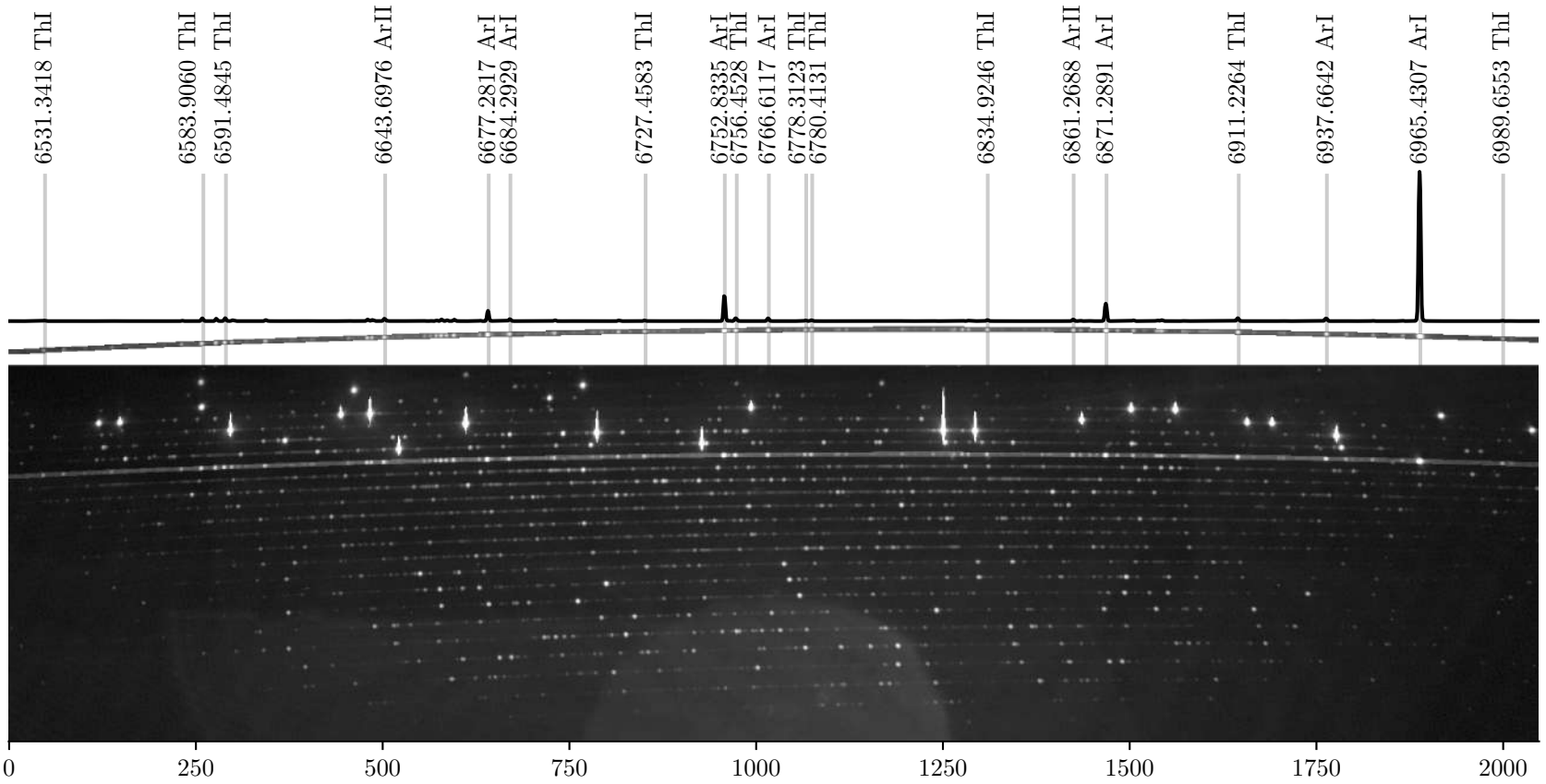

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6506.9868 ThI
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6509.0503 ThI
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6512.3639 ThI
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6505
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6520
6522.0432 ThI
6522.4994 ThI
6526.9045 Th
6531.3418 ThI
6534.6056 ThI
6536.1441 Th
6537.1775 ThI
6537.6139 ThI
6538.1120 ArI
6520
6525
6530
6535
6540


6536.1441 Th
6537.1775 ThI
6537.6139 ThI
6538.1120 ArI
6542.0498 ThI
6542.5131 ThI
6545.7186 ThI
6550.1877 Th
6550.5475 ThII
6551.7055 ThI
6554.1603 ThI
6558.8756 ThI
6560.0574 ThII
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6555
6560
6558.8756 ThI
6560.0574 ThII
6564.4445 ThI
6565.0700 ThI
6569.6322 ThII
6576.1221 ThI
6577.2146 ThI
6577.6552 ThII
6580.2299 ThI
6560
6565
6570
6575
6580
6576.1221 ThI
6577.2146 ThI
6577.6552 ThII
6580.2299 ThI
6583.9060 Th
6584.6130 ThI
6585.6945 ThI
6588.5300 ThI
6591.4845 ThI
6593.4628 ThI
6593.9391 ThI
6596.1141 ArI
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6600

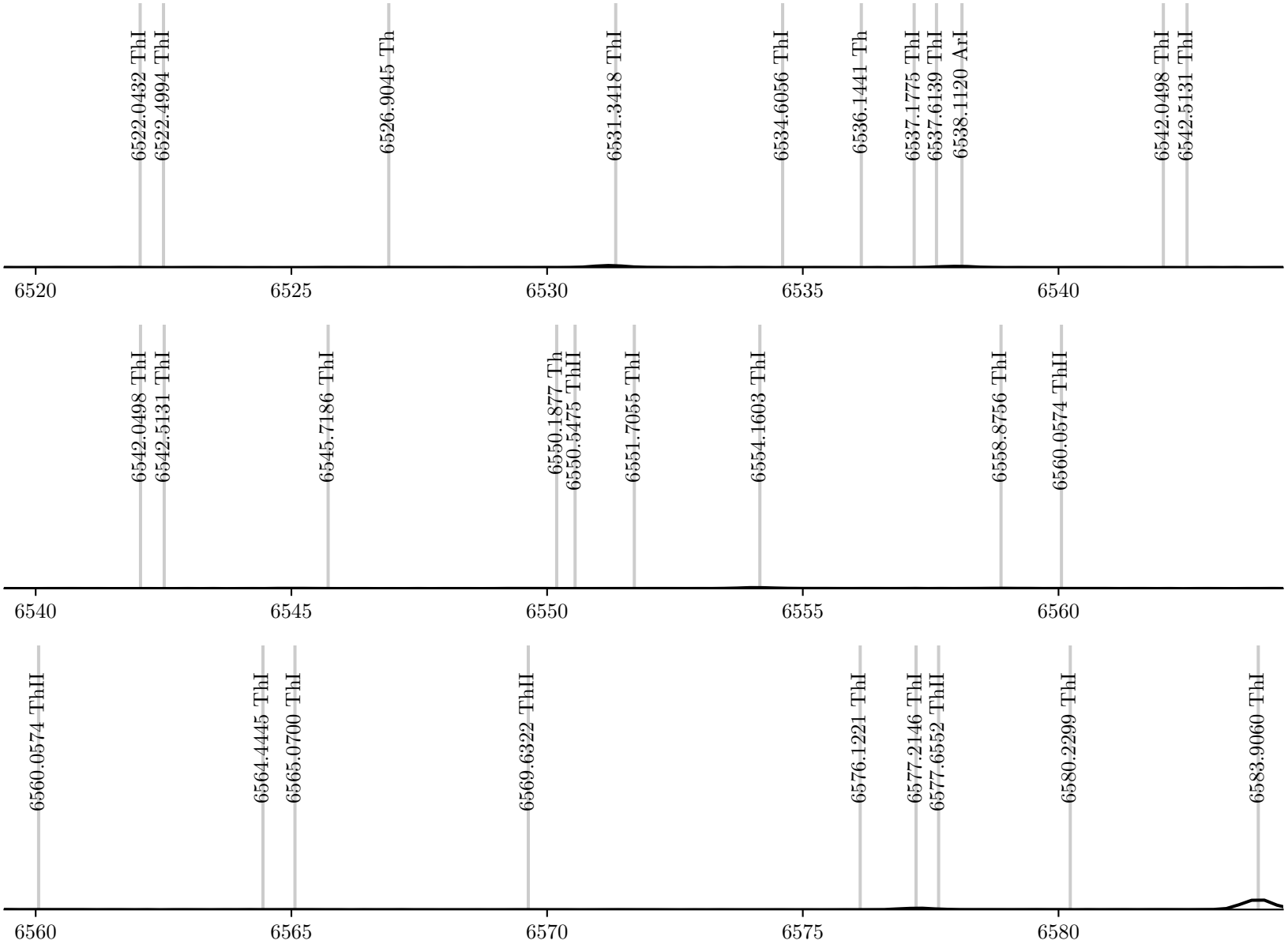

6596.1141 ArI
6596.9623 ThI
6598.6778 ArI
6599.4824 Th
6600.7311 ThI
6602.7620 ThI
6603.6209 ThI
6604.8534 ArI
6605.4165 ThII
6608.4430 ThI
6613.3740 ThI
6617.0580 ThII
6617.5154 ThI
6618.1668 ThI
6619.9458 ThII
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6605
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6615
6620
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6617.5154 ThI
6618.1668 ThI
6619.9458 ThII
6620.9665 ArII
6632.0837 ArI
6633.4579 ThII
6638.2207 ArI
6638.9119 ThI
6639.7403 ArI
6620
6625
6630
6635
6640
6638.2207 ArI
6638.9119 ThI
6639.7403 ArI
6643.6976 ArII
6644.6635 Th
6646.5407 ThII
6648.4954 ThII
6648.9586 ThII
6654.3675 Th
6655.4892 ThI
6656.9386 ArI
6658.6774 ThI
6660.6761 ArI
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6655
6660

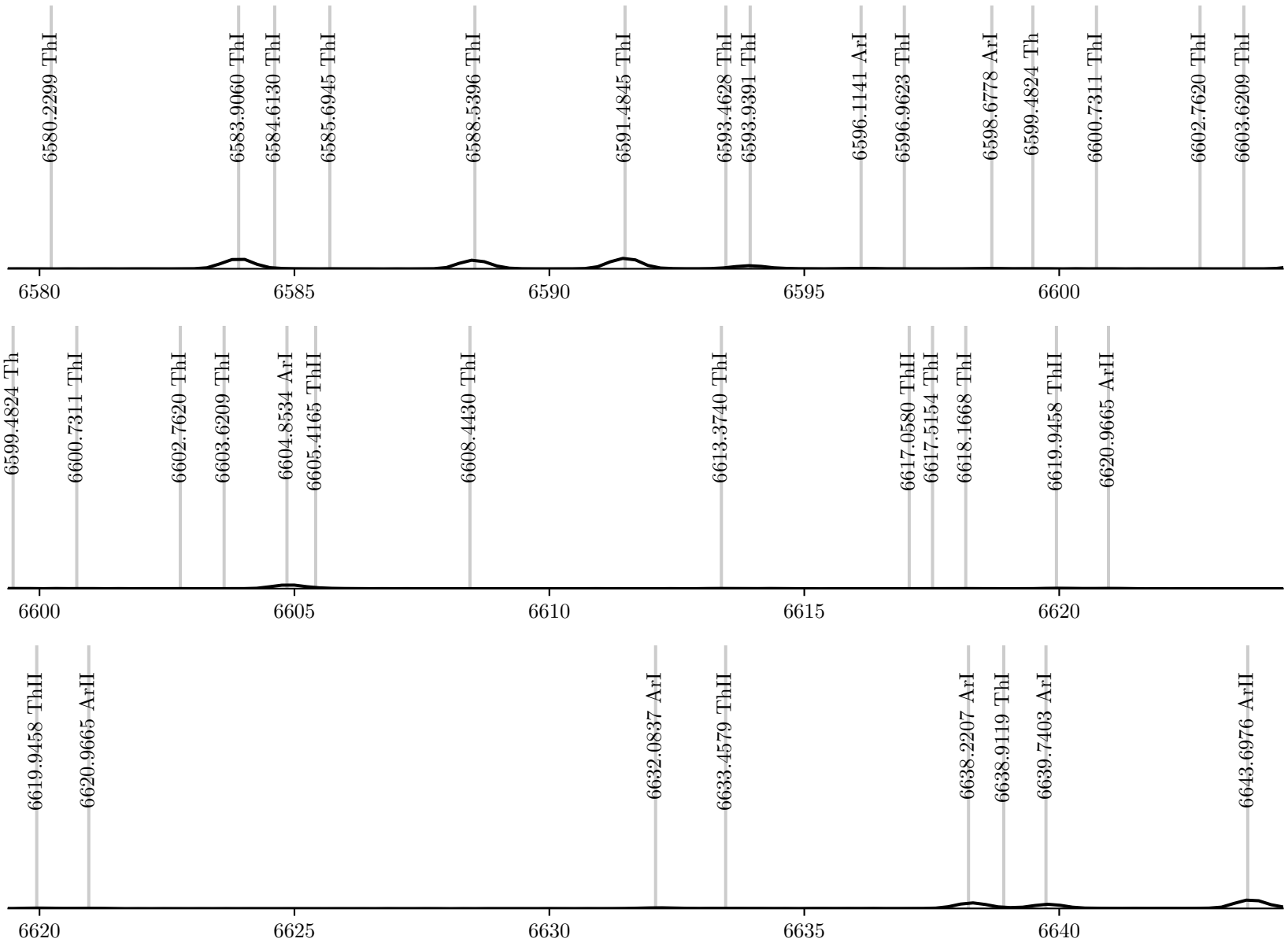

6656.9386 ArI
6658.6774 ThI
6660.6761 ArI
6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
6666.3588 ArII
6668.8163 ThI
6673.5797 ThI
6674.6969 ThI
6677.2817 ArI
6678.7067 ThI
6680.0811 ThI
6660
6665
6670
6675
6680
6677.2817 ArI
6678.7067 ThI
6680.0811 ThI
6683.3673 ThI
6684.2929 ArI
6687.5207 ThI
6692.7262 ThII
6694.0068 ThI
6694.4967 ThI
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6685
6690
6695
6700

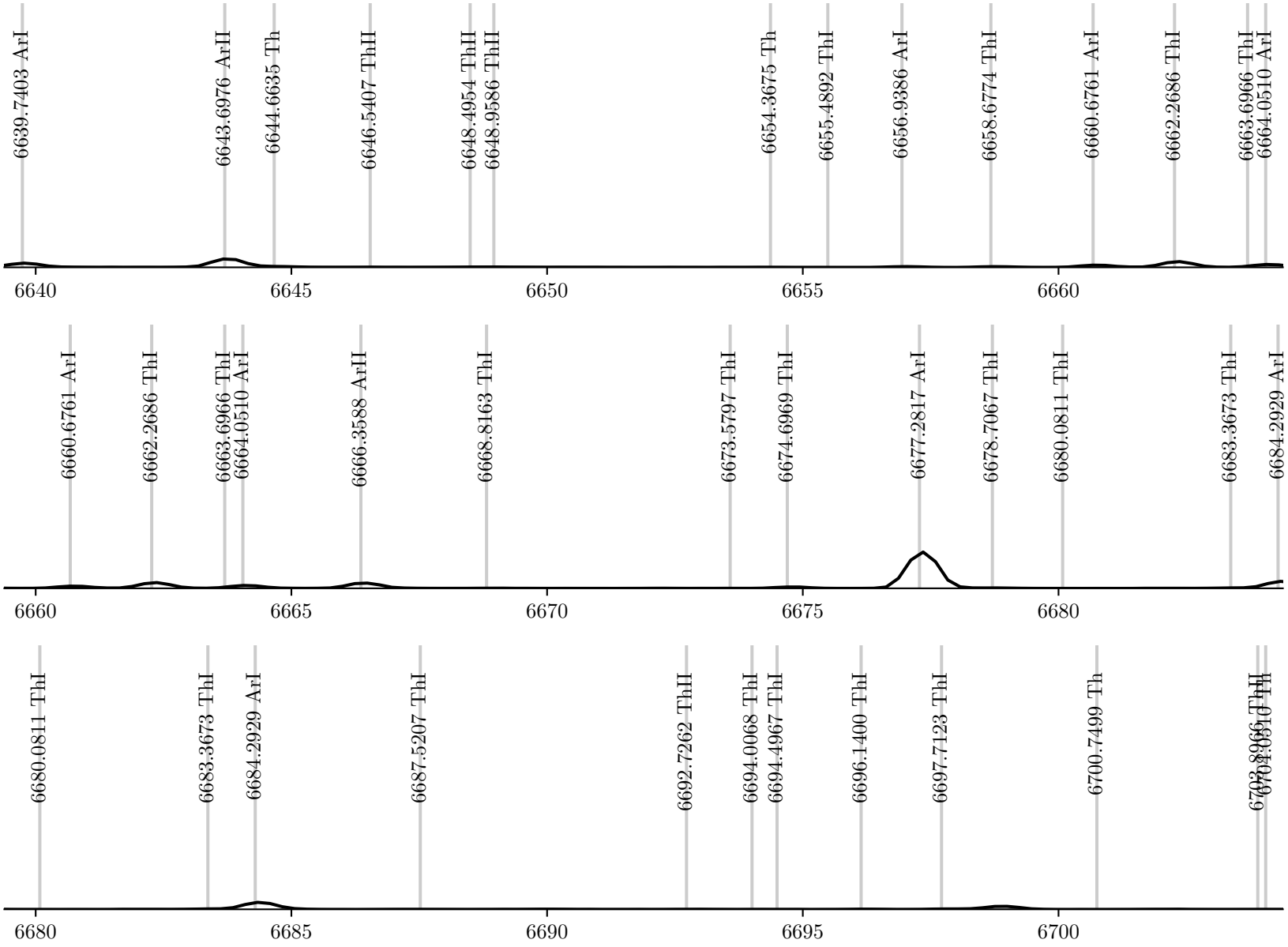
Order 22

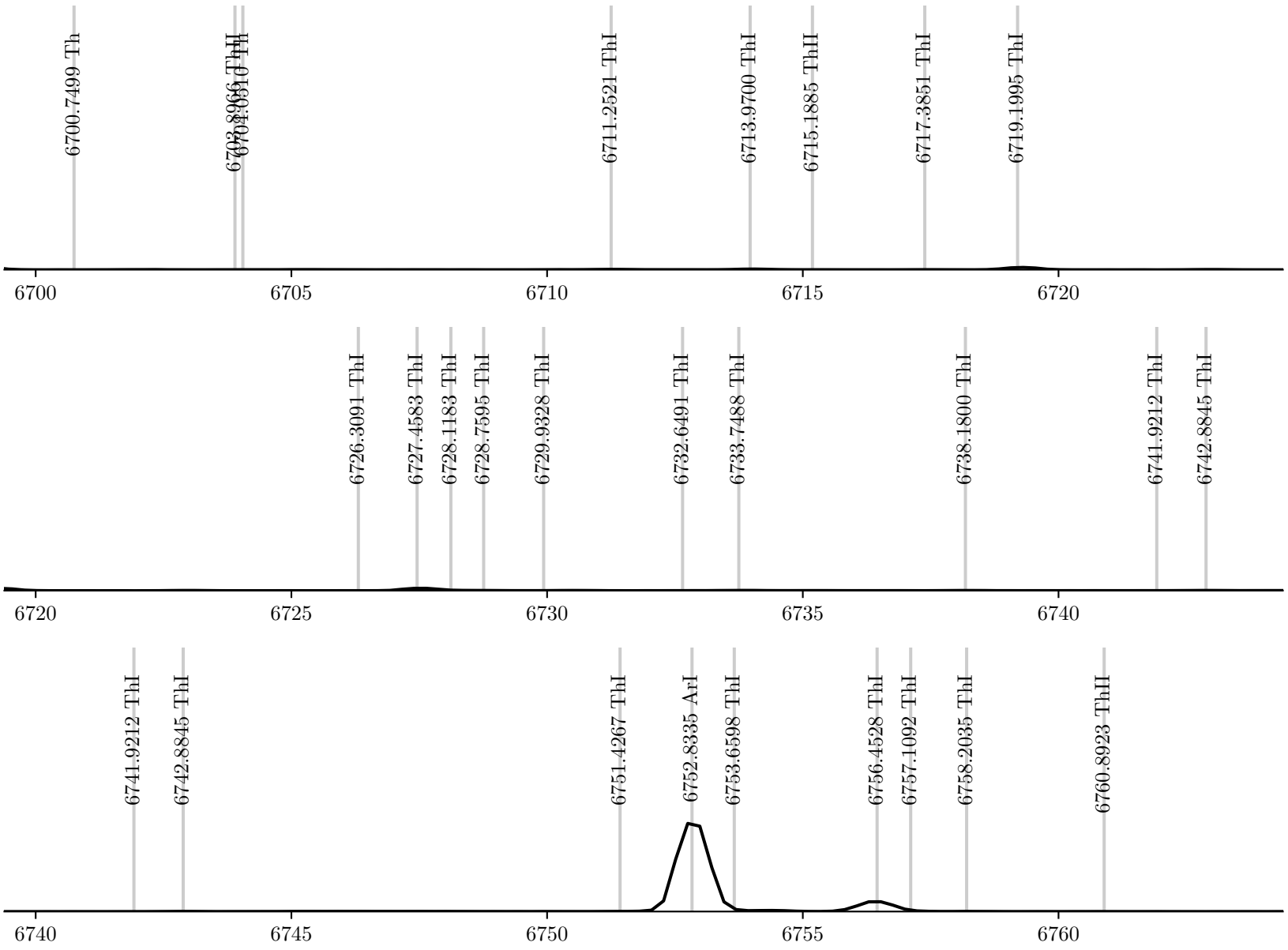


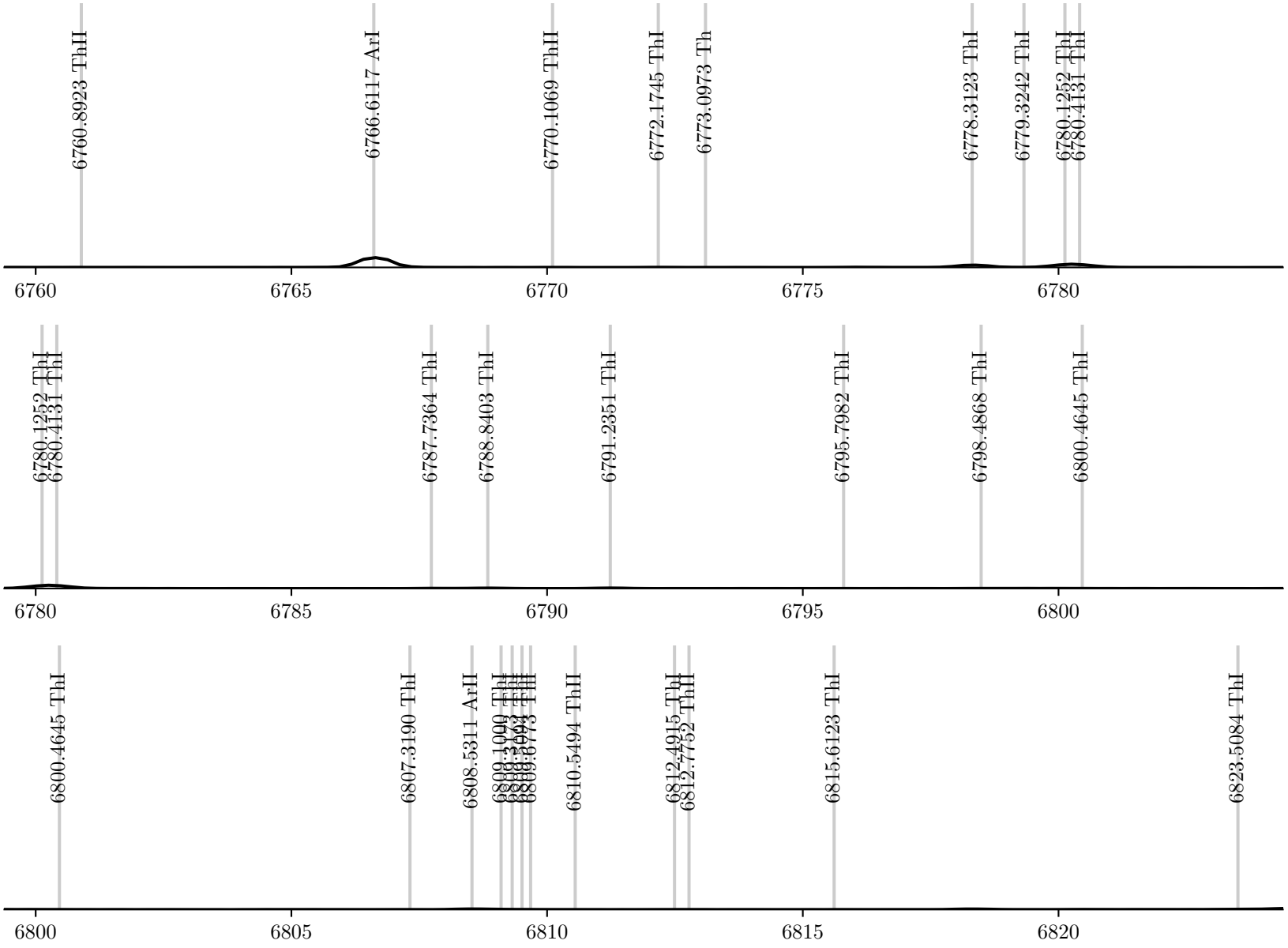
Dispersion:	0.235 Å pix^{-1}
Central wavelength:	6768 Å
Wavelength range:	6519 Å – 7000 Å

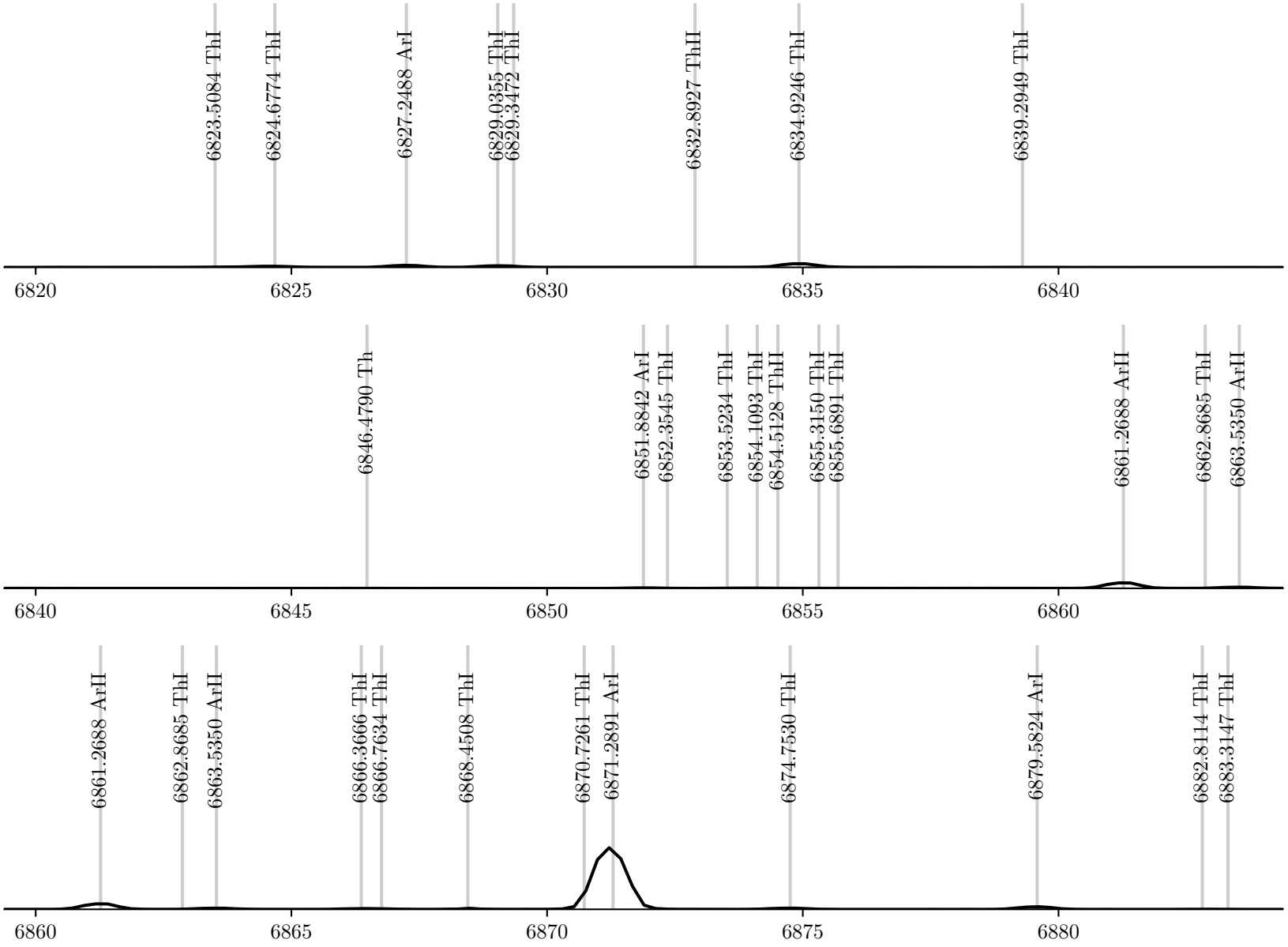

6522.0432 ThI
6522.4994 ThI
6526.9045 Th
6531.3418 ThI
6534.6056 ThI
6536.1441 Th
6537.1775 ThI
6537.6139 ThI
6538.1120 ArI
6542.0498 ThI
6542.5131 ThI
6520
6525
6530
6535
6540
6542.0498 ThI
6542.5131 ThI
6545.7186 ThI
6550.1877 Th
6550.5475 ThII
6551.7055 ThI
6554.1603 ThI
6558.8756 ThI
6560.0574 ThII
6540
6545
6550
6555
6560
6560.0574 ThII
6564.4445 ThI
6565.0700 ThI
6569.6322 ThII
6576.1221 ThI
6577.2146 ThI
6577.6552 ThII
6580.2299 ThI
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6575
6580

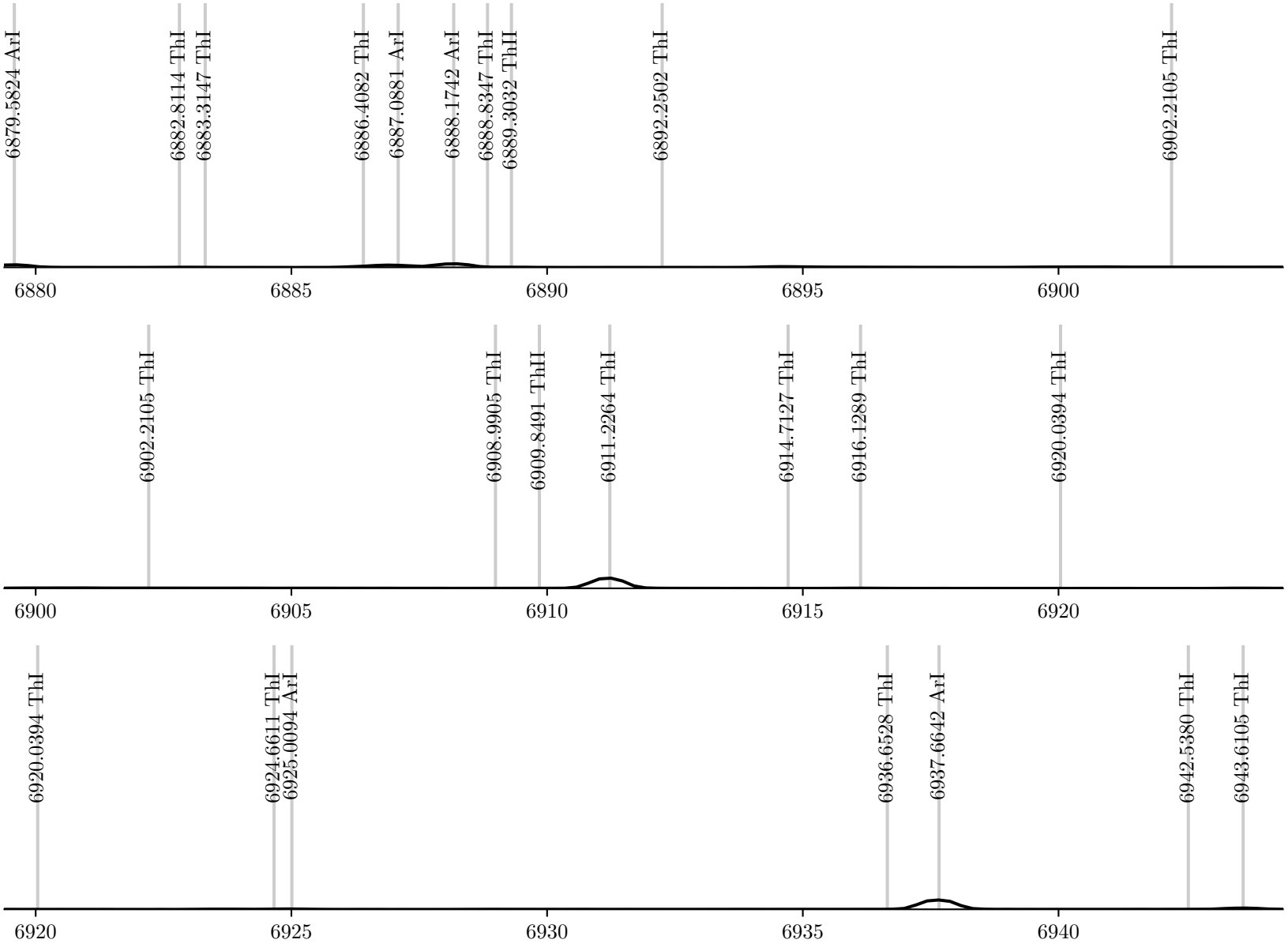

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6584.6130 ThI
6585.6945 ThI
6588.5396 ThI
6591.4845 ThI
6593.4628 ThI
6593.9391 ThI
6596.1141 ArI
6596.9623 ThI
6598.6778 ArI
6599.4824 Th
6600.7311 ThI
6602.7620 ThI
6603.6209 ThI
6580
6585
6590
6595
6600
6599.4824 Th
6600.7311 ThI
6602.7620 ThI
6603.6209 ThI
6604.8534 ArI
6605.4165 ThII
6608.4430 ThI
6613.3740 ThI
6617.0580 ThII
6617.5154 ThI
6618.1668 ThI
6619.9458 ThII
6620.9665 ArII
6600
6605
6610
6615
6620
6619.9458 ThII
6620.9665 ArII
6632.0837 ArI
6633.4579 ThII
6638.2207 ArI
6638.9119 ThI
6639.7403 ArI
6643.6976 ArII
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6640

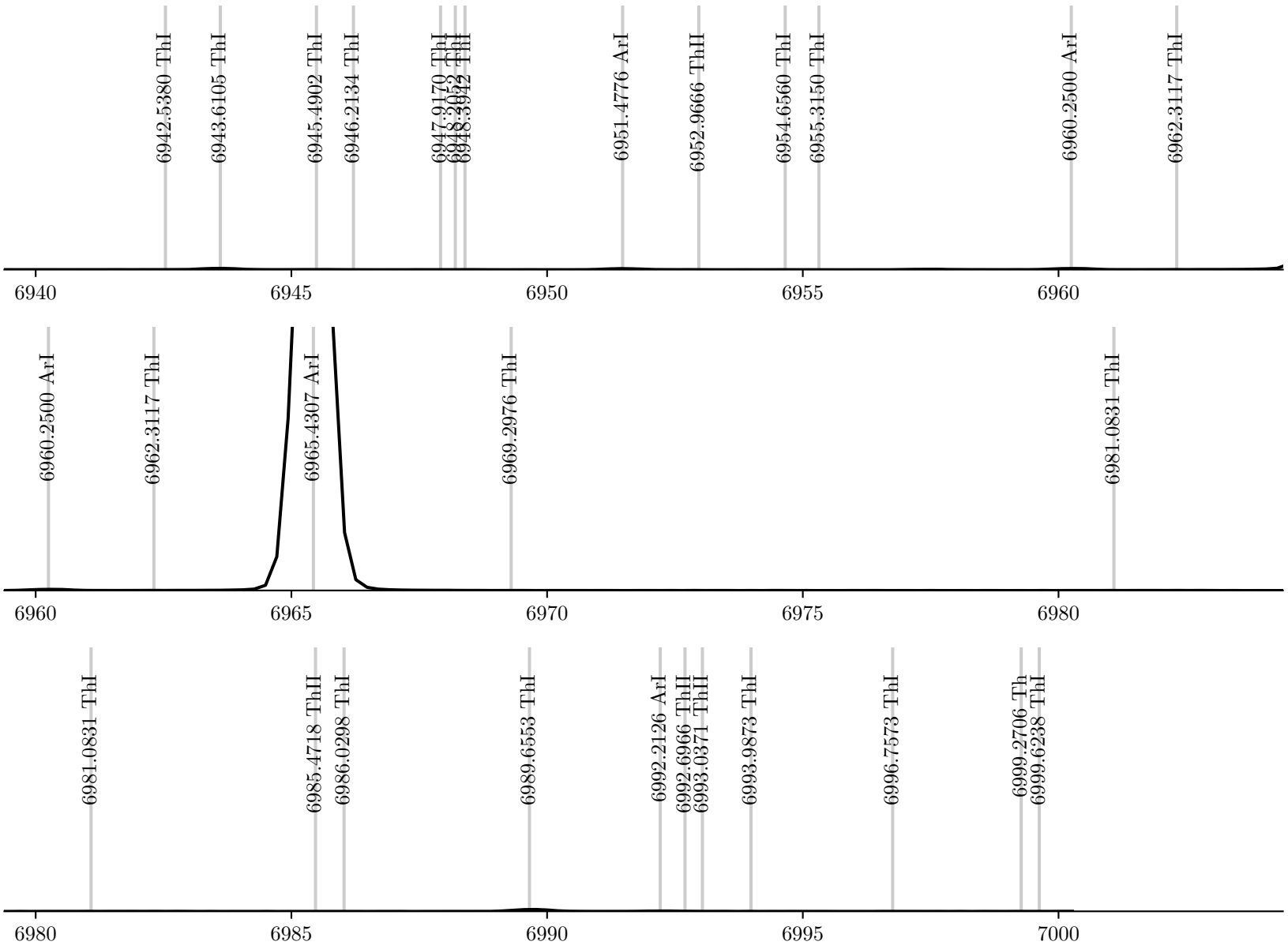

6639.7403 ArI
6643.6976 ArII
6644.6635 Th
6646.5407 ThII
6648.4954 ThII
6648.9586 ThII
6654.3675 Th
6655.4892 ThI
6656.9386 ArI
6658.6774 ThI
6660.6761 ArI
6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
6640
6645
6650
6655
6660
6660.6761 ArI
6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
6666.3588 ArII
6668.8163 ThI
6673.5797 ThI
6674.6969 ThI
6677.2817 ArI
6678.7067 ThI
6680.0811 ThI
6683.3673 ThI
6684.2929 ArI
6660
6665
6670
6675
6680
6680.0811 ThI
6683.3673 ThI
6684.2929 ArI
6687.5207 ThI
6692.7262 ThII
6694.0068 ThI
6694.4967 ThI
6696.1400 ThI
6697.7123 ThI
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6685
6690
6695
6700

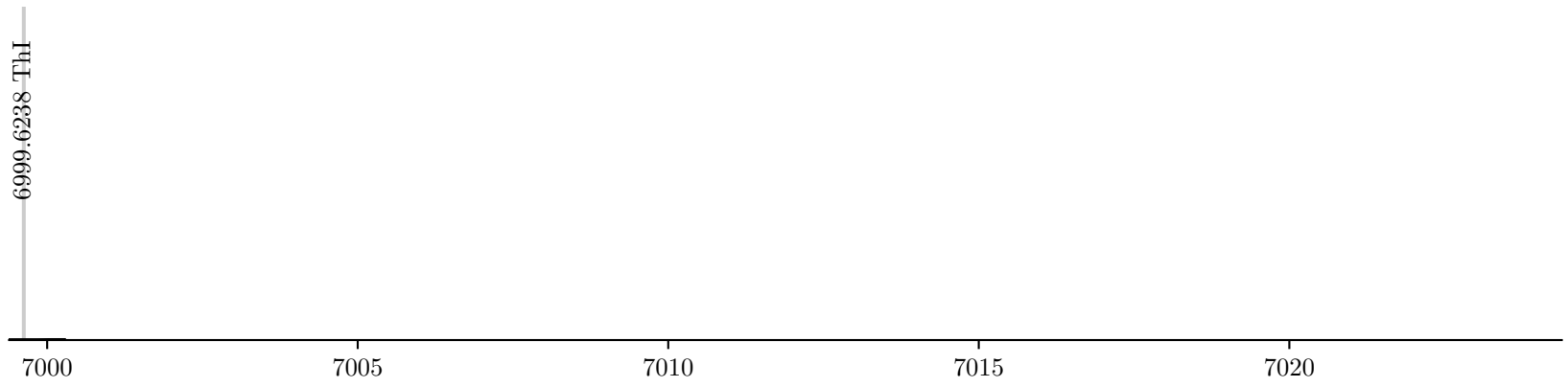

6700.7499 Th
6711.2521 ThI
6713.9700 ThI
6715.1885 ThII
6717.3851 ThI
6719.1995 ThI
6700
6705
6710
6715
6720
6726.3091 ThI
6727.4583 ThI
6728.1183 ThI
6728.7595 ThI
6729.9328 ThI
6732.6491 ThI
6733.7488 ThI
6738.1800 ThI
6741.9212 ThI
6742.8845 ThI
6720
6725
6730
6735
6740
6741.9212 ThI
6742.8845 ThI
6751.4267 ThI
6752.8335 ArI
6753.6598 ThI
6756.4528 ThI
6757.1092 ThI
6758.2035 ThI
6760.8923 ThII
6740
6745
6750
6755
6760


6760.8923 ThII
6766.6117 ArI
6770.1069 ThII
6772.1745 ThI
6773.0973 Th
6778.3123 ThI
6779.3242 ThI
6780.1252 ThI
6780.4131 ThI
6760
6765
6770
6775
6780
6780.1252 ThI
6780.4131 ThI
6787.7364 ThI
6788.8403 ThI
6791.2351 ThI
6795.7982 ThI
6798.4868 ThI
6800.4645 ThI
6780
6785
6790
6795
6800
6800.4645 ThI
6807.3190 ThI
6808.5311 ArII
6810.5494 ThII
6812.4915 ThI
6812.7752 ThII
6815.6123 ThI
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6820

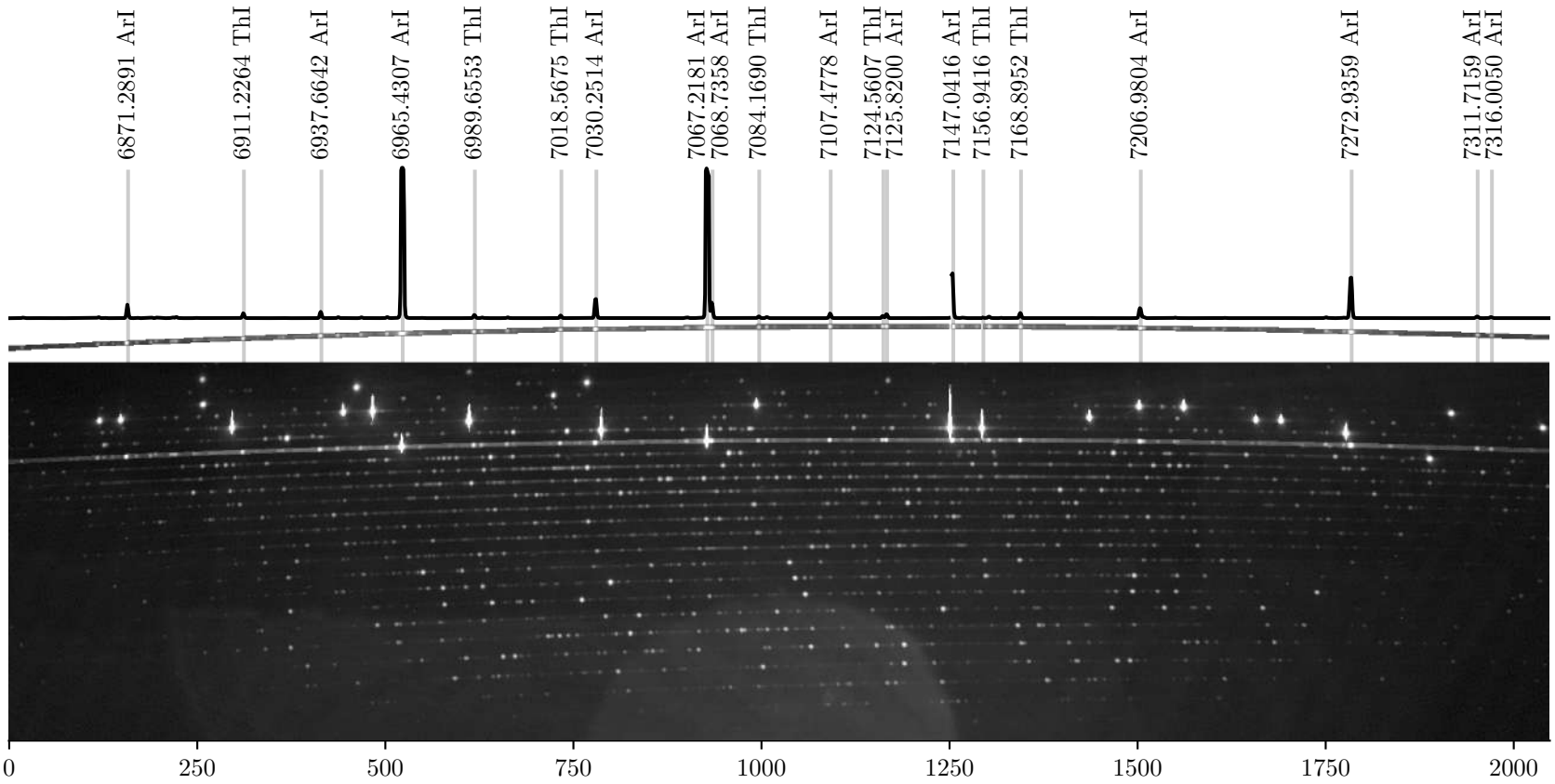

6823.5084 ThI
6824.6774 ThI
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6829.0355 ThI
6829.3472 ThI
6832.8927 ThII
6834.9246 ThI
6839.2949 ThI
6820
6825
6830
6835
6840
6846.4790 Th
6851.8842 ArI
6852.3545 ThI
6853.5234 ThI
6854.1093 ThI
6854.5128 ThII
6855.3150 ThI
6855.6891 ThI
6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6840
6845
6850
6855
6860
6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6866.3666 ThI
6866.7634 ThI
6868.4508 ThI
6870.7261 ThI
6871.2891 ArI
6874.7530 ThI
6879.5824 ArI
6882.8114 ThI
6883.3147 ThI
6860
6865
6870
6875
6880


6879.5824 ArI
6882.8114 ThI
6883.3147 ThI
6886.4082 ThI
6887.0881 ArI
6888.1742 ArI
6888.8347 ThI
6889.3032 ThII
6892.2502 ThI
6902.2105 ThI
6880
6885
6890
6895
6900
6902.2105 ThI
6908.9905 ThI
6909.8491 ThII
6911.2264 ThI
6914.7127 ThI
6916.1289 ThI
6920.0394 ThI
6900
6905
6910
6915
6920
6920.0394 ThI
6924.6611 ThI
6925.0094 ArI
6936.6528 ThI
6937.6642 ArI
6942.5380 ThI
6943.6105 ThI
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6940

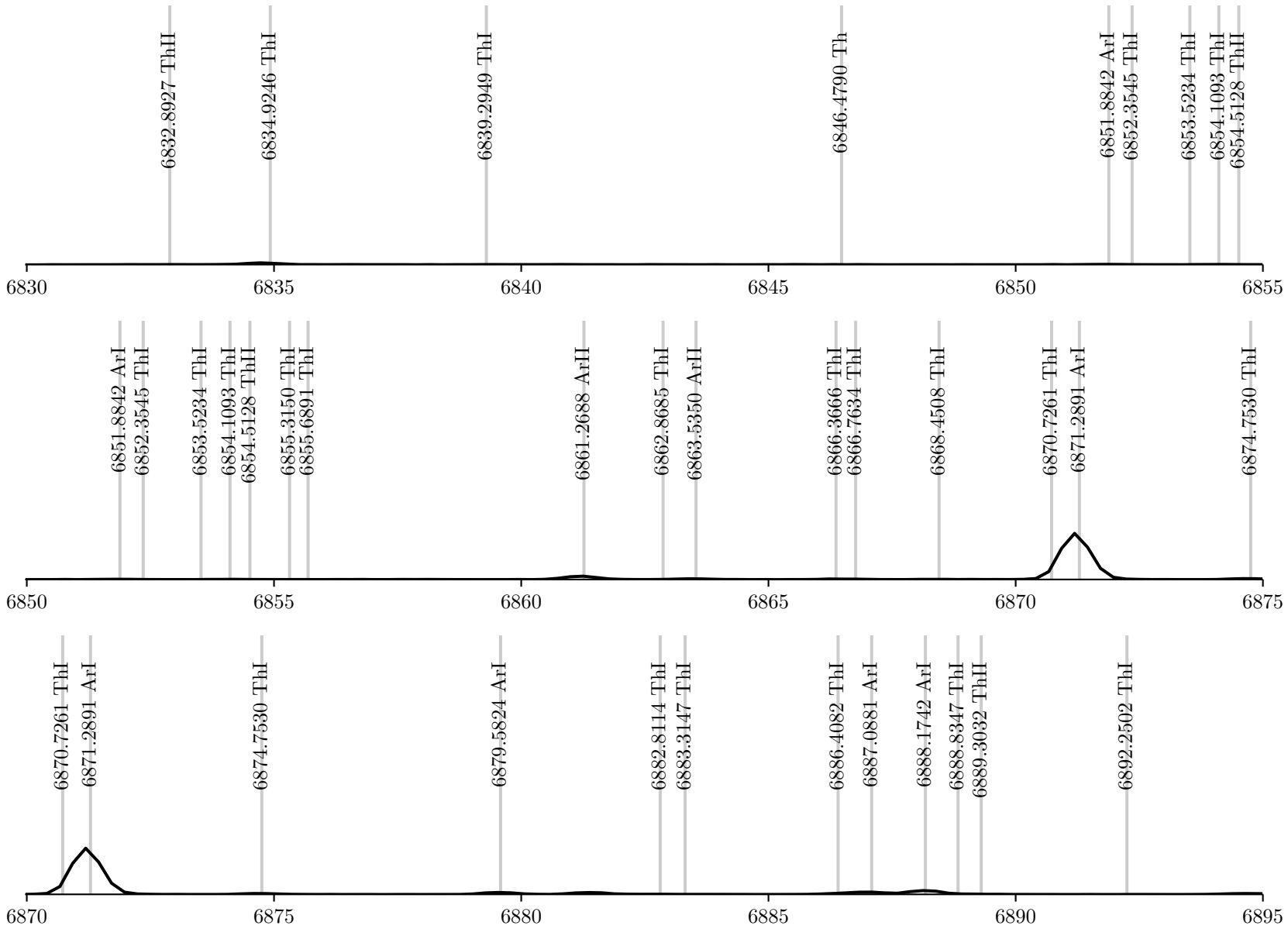

6942.5380 ThI
6943.6105 ThI
6945.4902 ThI
6946.2134 ThI
6947.9170 ThI
6948.3052 ThI
6948.3942 ThI
6951.4776 ArI
6952.9666 ThII
6954.6560 ThI
6955.3150 ThI
6960.2500 ArI
6962.3117 ThI
6940
6945
6950
6955
6960
6960.2500 ArI
6962.3117 ThI
6965.4307 ArI
6969.2976 ThI
6981.0831 ThI
6960
6965
6970
6975
6980
6981.0831 ThI
6985.4718 ThII
6986.0298 ThI
6989.6553 ThI
6992.2126 ArI
6992.6966 ThII
6993.0371 ThII
6993.9873 ThI
6996.7573 ThI
6999.2706 Th
6999.6238 ThI
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6995
7000

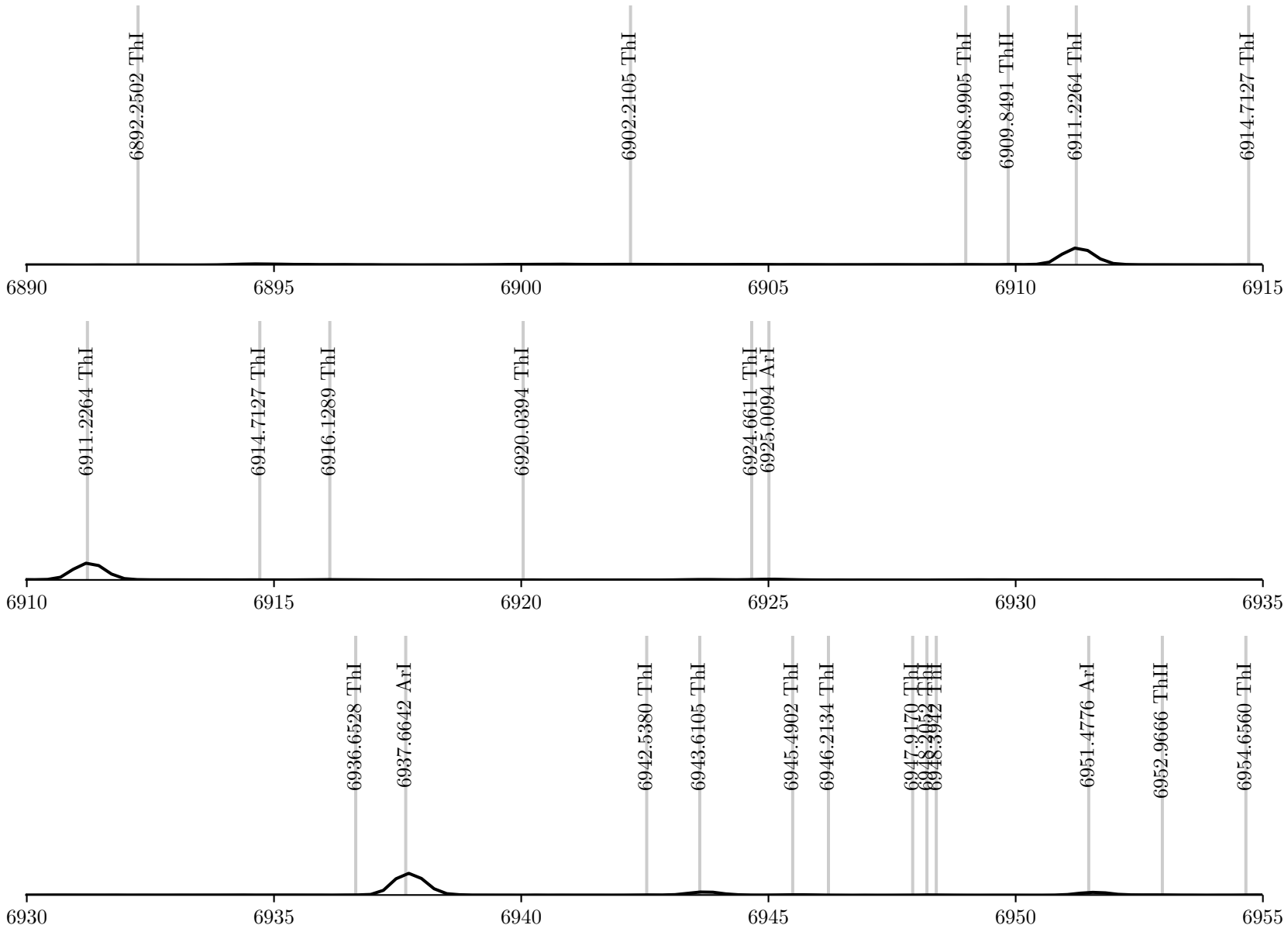

6999.6238 ThI
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7010
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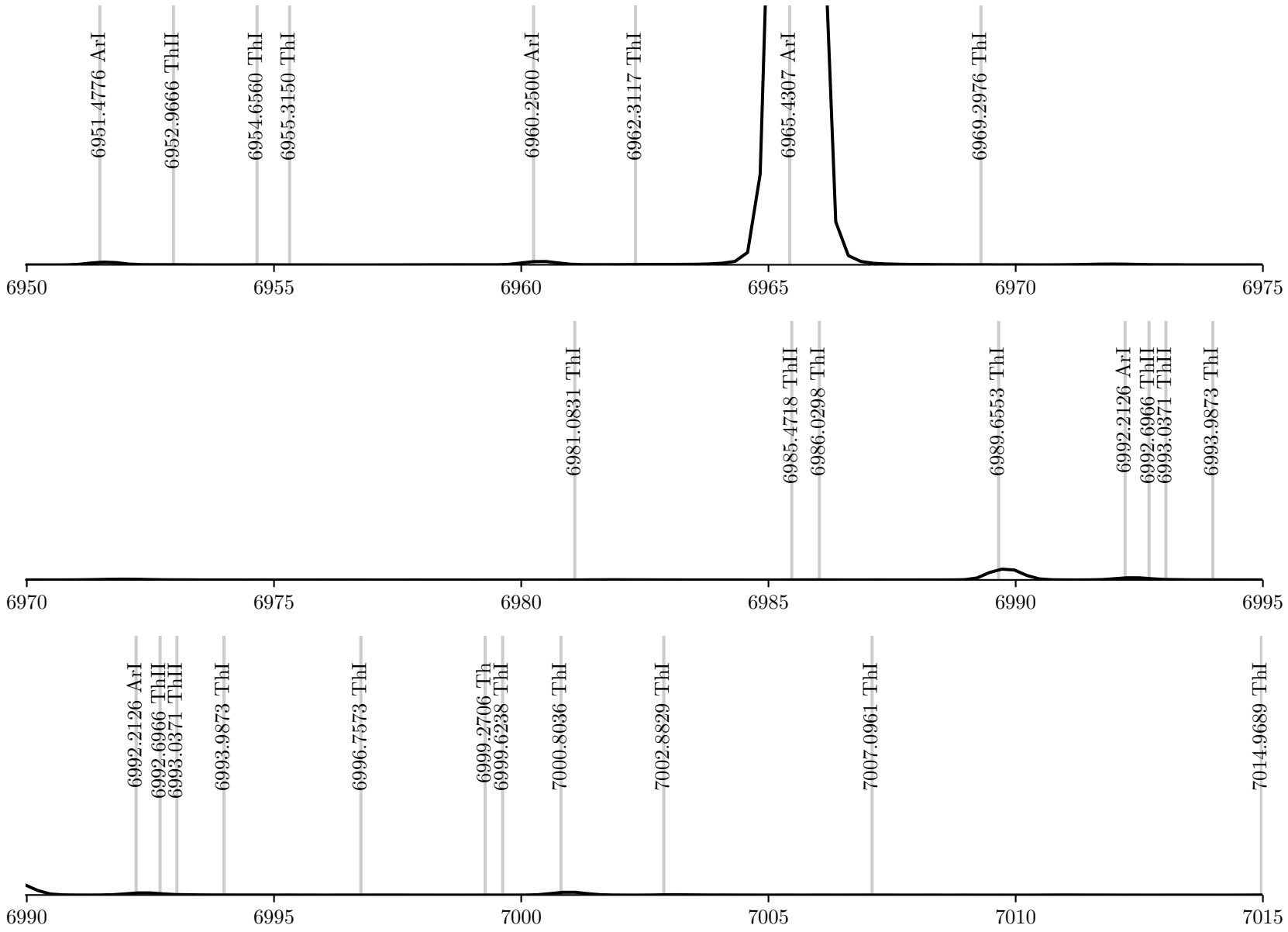
Order 21

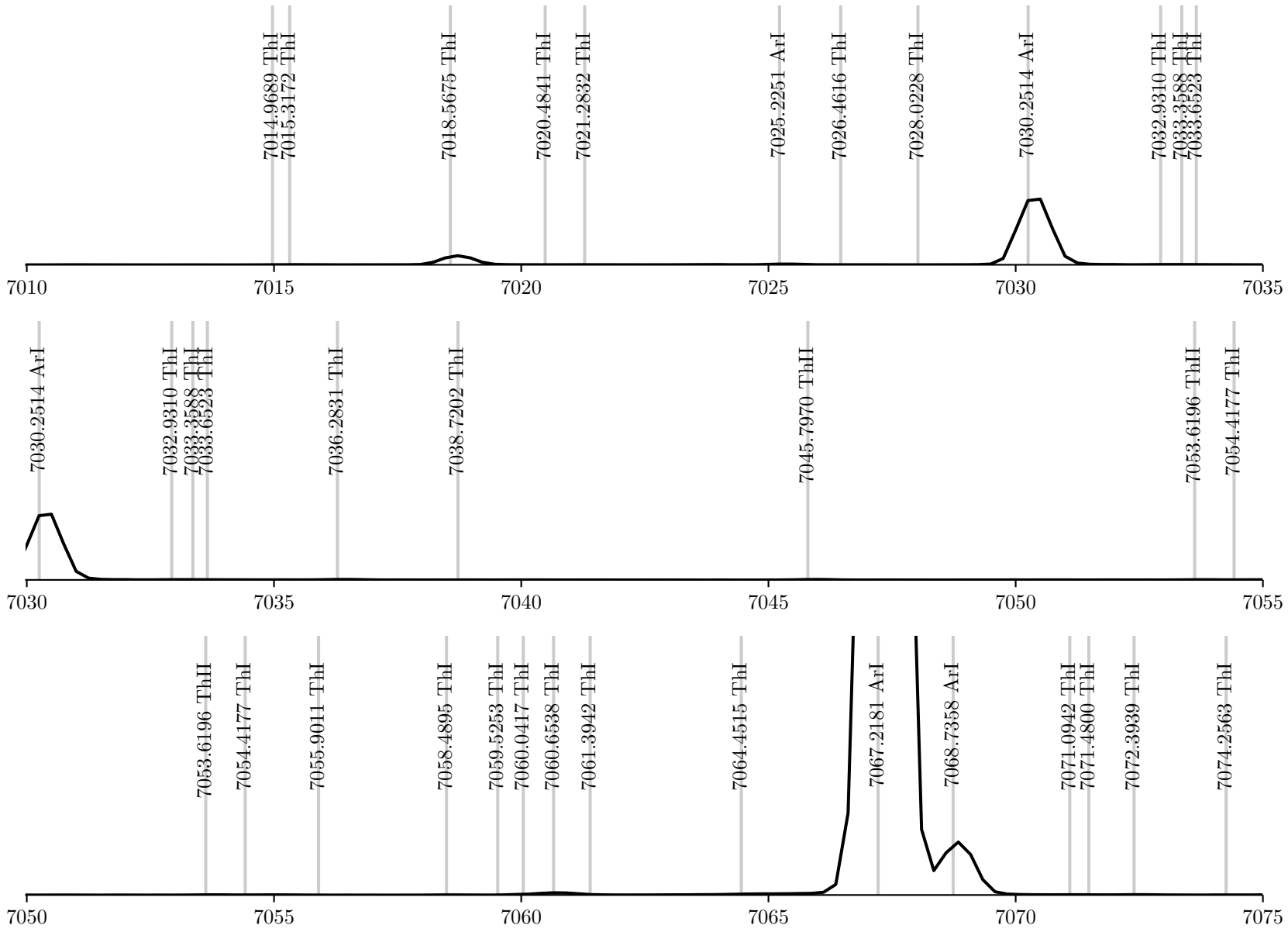


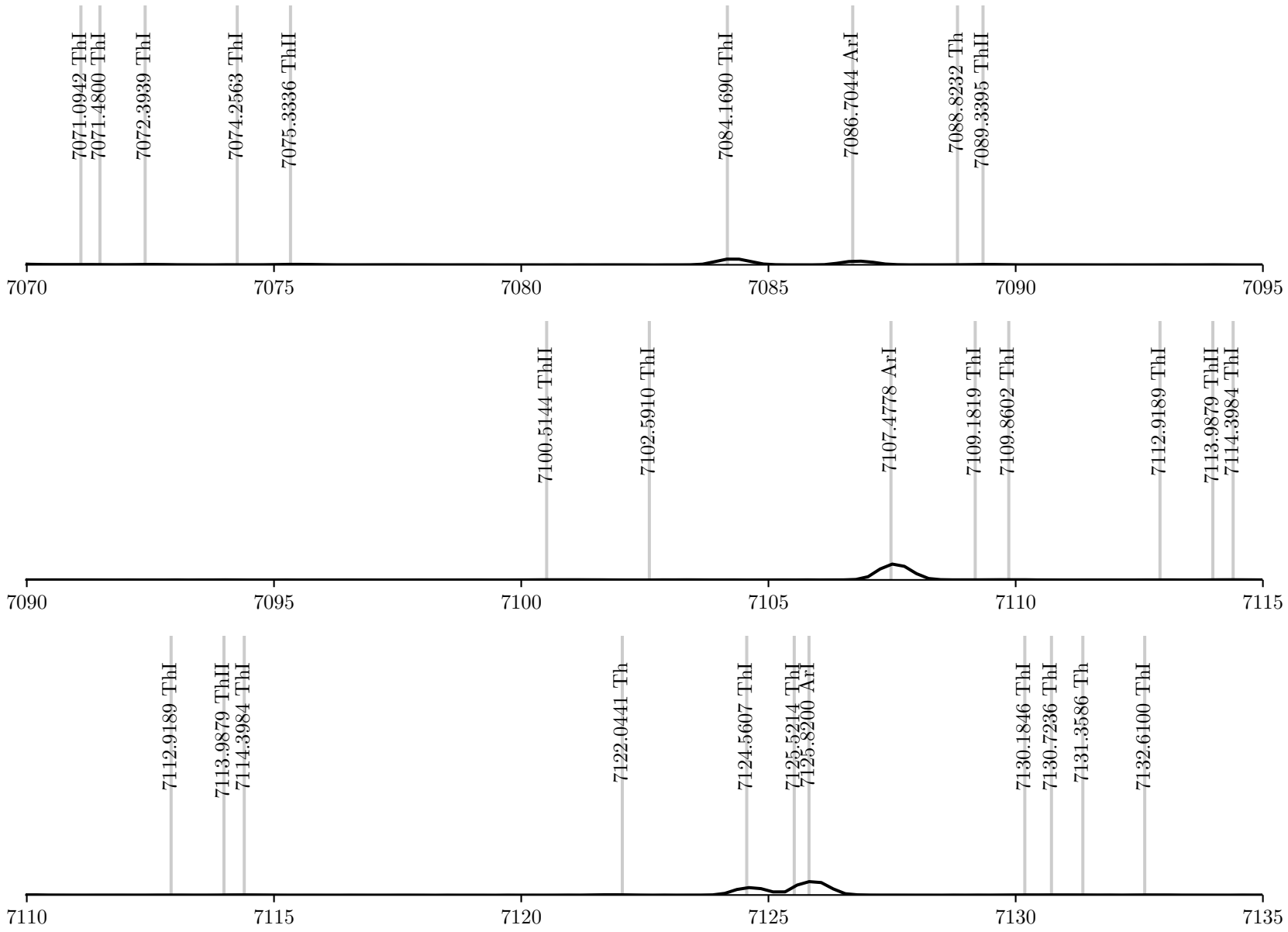
Dispersion:	0.246 Å pix^{-1}
Central wavelength:	7091 Å
Wavelength range:	6830 Å – 7334 Å

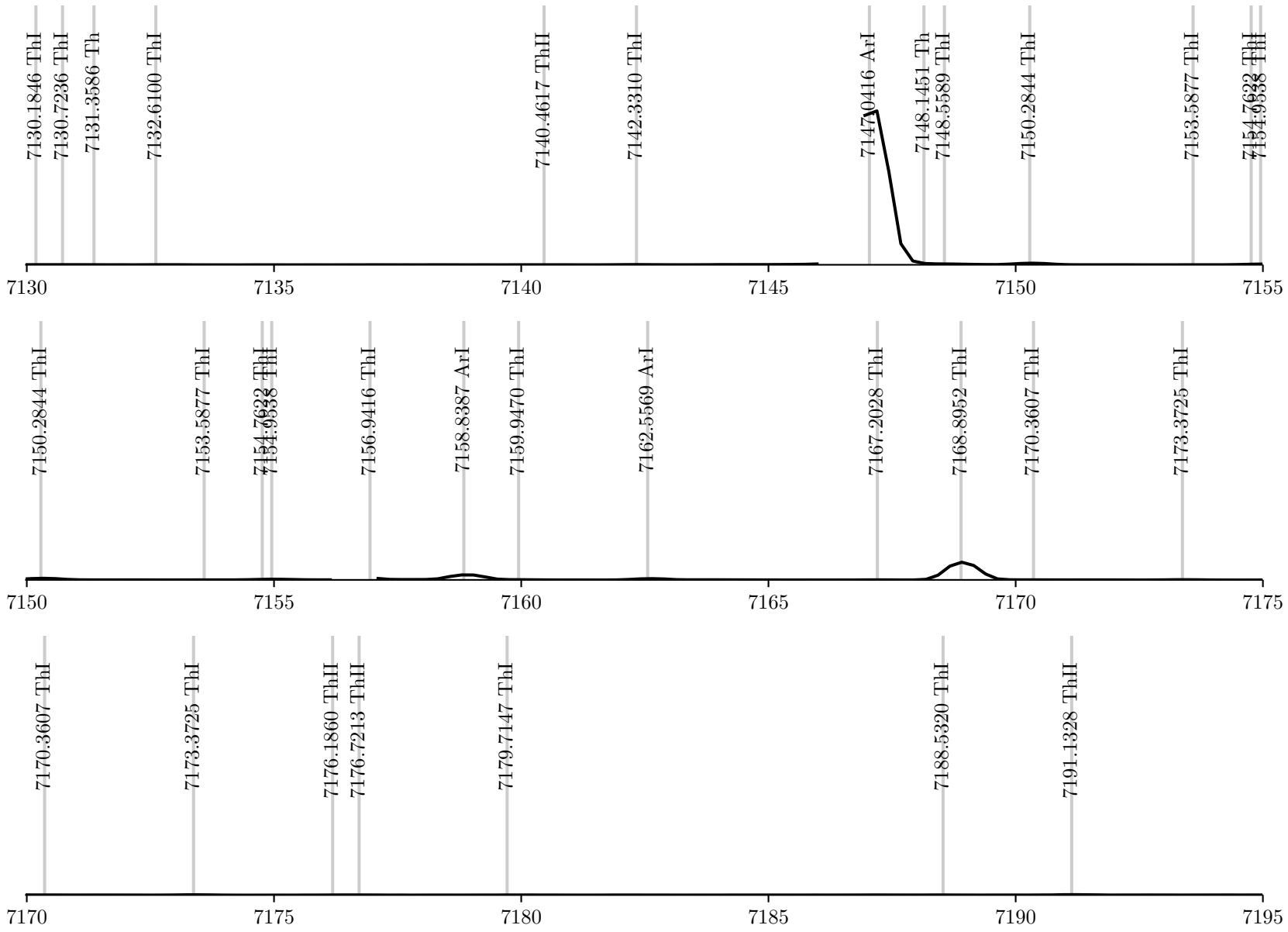

6832.8927 ThII
6834.9246 ThI
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6846.4790 Th
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6852.3545 ThI
6853.5234 ThI
6854.1093 ThI
6854.5128 ThII
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6835
6840
6845
6850
6855
6851.8842 ArI
6852.3545 ThI
6853.5234 ThI
6854.1093 ThI
6854.5128 ThII
6855.3150 ThI
6855.6891 ThI
6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6866.3666 ThI
6866.7634 ThI
6868.4508 ThI
6870.7261 ThI
6871.2891 ArI
6874.7530 ThI
6850
6855
6860
6865
6870
6875
6870.7261 ThI
6871.2891 ArI
6874.7530 ThI
6879.5824 ArI
6882.8114 ThI
6883.3147 ThI
6886.4082 ThI
6887.0881 ArI
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6888.8347 ThI
6889.3032 ThII
6892.2502 ThI
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6885
6890
6895

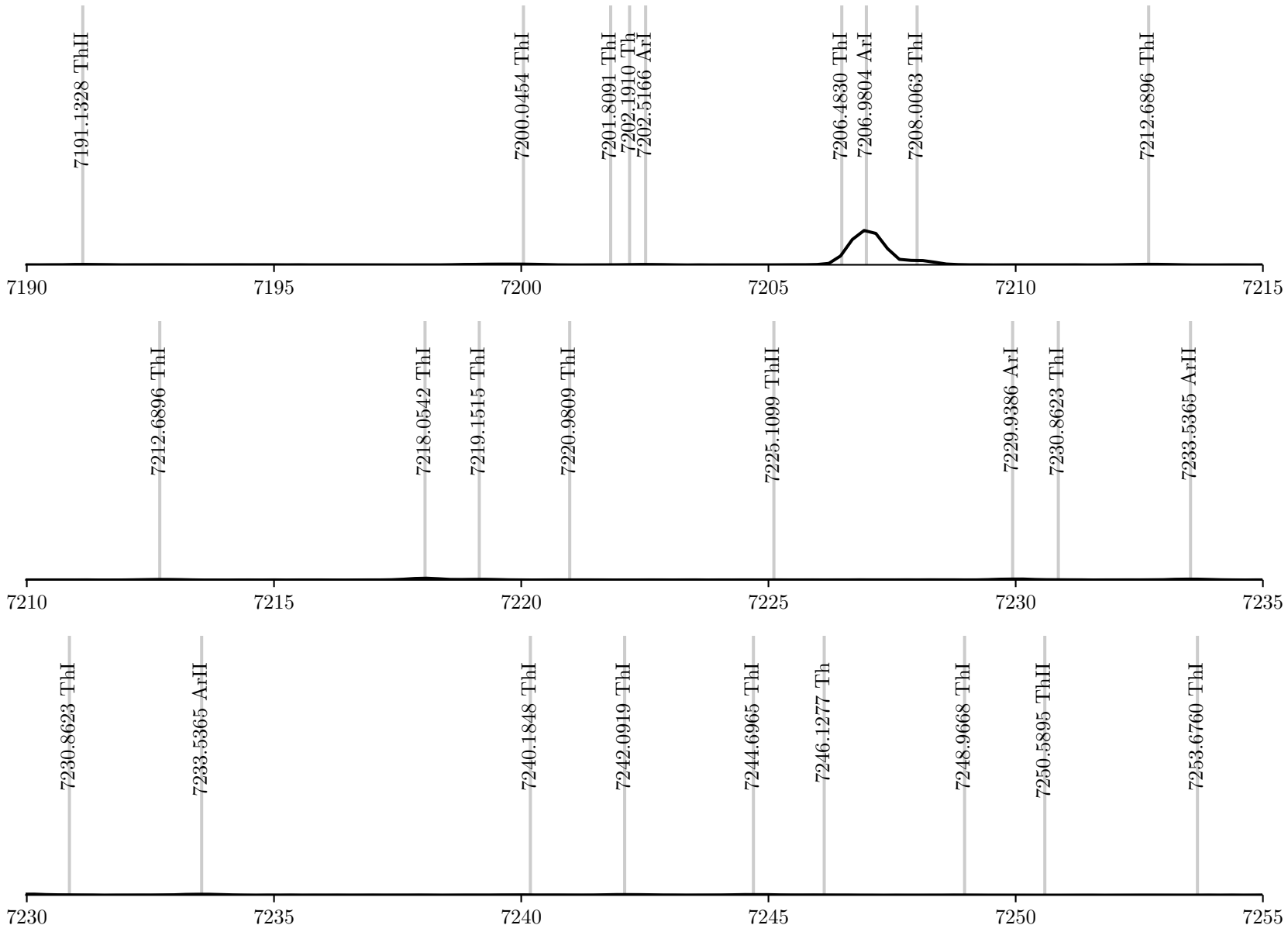

6892.2502 ThI
6902.2105 ThI
6908.9905 ThI
6909.8491 ThII
6911.2264 ThI
6914.7127 ThI
6890
6895
6900
6905
6910
6915
6911.2264 ThI
6914.7127 ThI
6916.1289 ThI
6920.0394 ThI
6924.6611 ThI
6925.0094 ArI
6910
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6920
6925
6930
6935
6936.6528 ThI
6937.6642 ArI
6942.5380 ThI
6943.6105 ThI
6945.4902 ThI
6946.2134 ThI
6947.9170 ThI
6948.3052 ThI
6948.3942 ThI
6951.4776 ArI
6952.9666 ThII
6954.6560 ThI
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6945
6950
6955

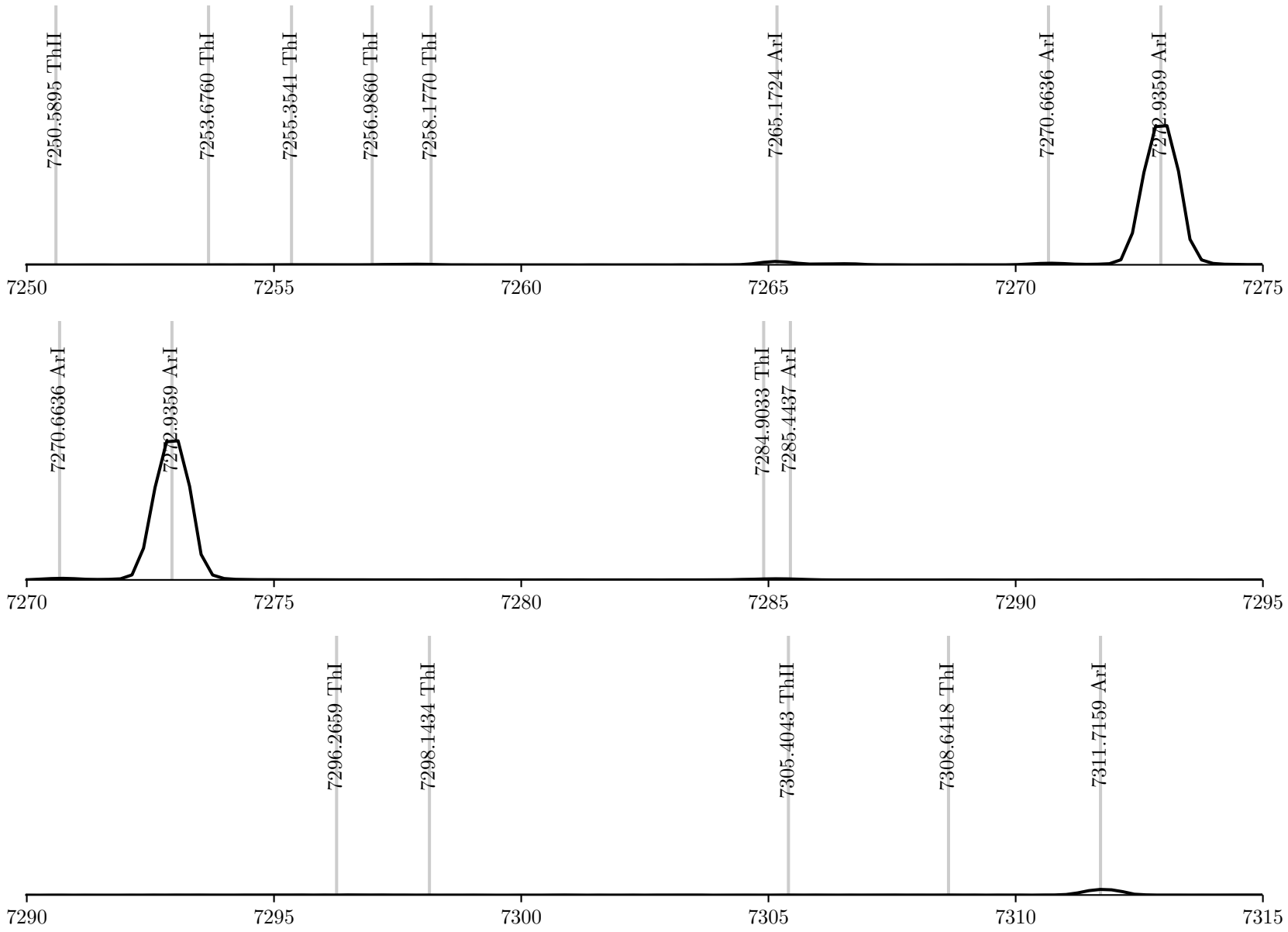

6951.4776 ArI
6952.9666 ThII
6954.6560 ThI
6955.3150 ThI
6960.2500 ArI
6962.3117 ThI
6965.4307 ArI
6969.2976 ThI
6950
6955
6960
6965
6970
6975
6981.0831 ThI
6985.4718 ThII
6986.0298 ThI
6989.6553 ThI
6992.2126 ArI
6992.6966 ThII
6993.0371 ThII
6993.9873 ThI
6970
6975
6980
6985
6990
6995
6992.2126 ArI
6992.6966 ThII
6993.0371 ThII
6993.9873 ThI
6996.7573 ThI
6999.2706 Th
6999.6238 ThI
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7007.0961 ThI
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7005
7010
7015

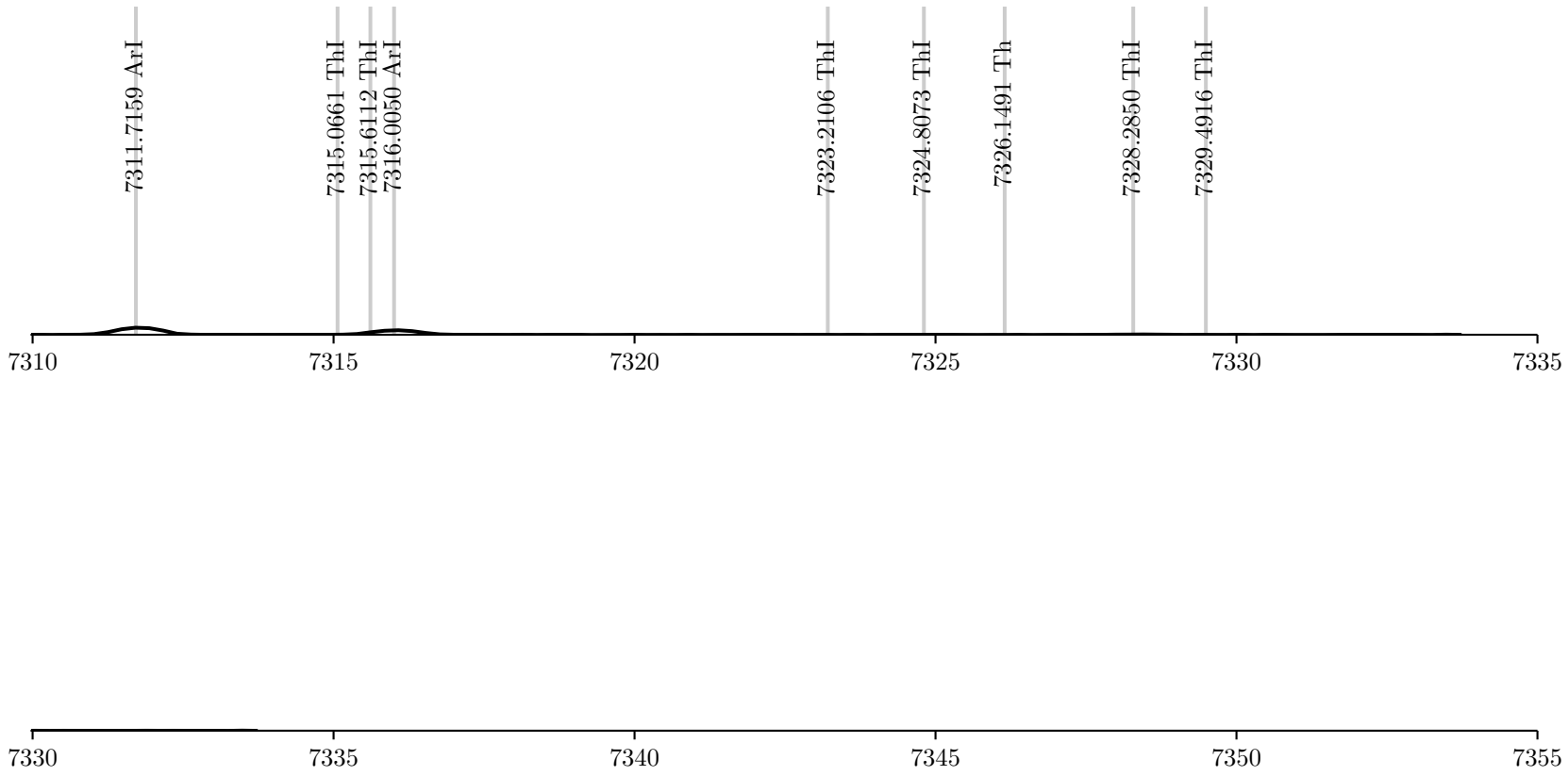

7014.9689 ThI
7015.3172 ThI
7018.5675 ThI
7020.4841 ThI
7021.2832 ThI
7025.2251 ArI
7026.4616 ThI
7028.0228 ThI
7030.2514 ArI
7032.9310 ThI
7033.3588 ThI
7033.6523 ThI
7010
7015
7020
7025
7030
7035
7030.2514 ArI
7032.9310 ThI
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7033.6523 ThI
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7038.7202 ThI
7045.7970 ThII
7053.6196 ThII
7054.4177 ThI
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7040
7045
7050
7055
7053.6196 ThII
7054.4177 ThI
7055.9011 ThI
7058.4895 ThI
7059.5253 ThI
7060.0417 ThI
7060.6538 ThI
7061.3942 ThI
7064.4515 ThI
7067.2181 ArI
7068.7358 ArI
7071.0942 ThI
7071.4800 ThI
7072.3939 ThI
7074.2563 ThI
7050
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7060
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7070
7075


7071.0942 ThI
7071.4800 ThI
7072.3939 ThI
7074.2563 ThI
7075.3336 ThII
7084.1690 ThI
7086.7044 ArI
7088.8232 Th
7089.3395 ThII
7070
7075
7080
7085
7090
7095
7100.5144 ThII
7102.5910 ThI
7107.4778 ArI
7109.1819 ThI
7109.8602 ThI
7112.9189 ThI
7113.9879 ThII
7114.3984 ThI
7090
7095
7100
7105
7110
7115
7112.9189 ThI
7113.9879 ThII
7114.3984 ThI
7122.0441 Th
7124.5607 ThI
7125.5214 ThI
7125.8200 ArI
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7130.7236 ThI
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7130
7135

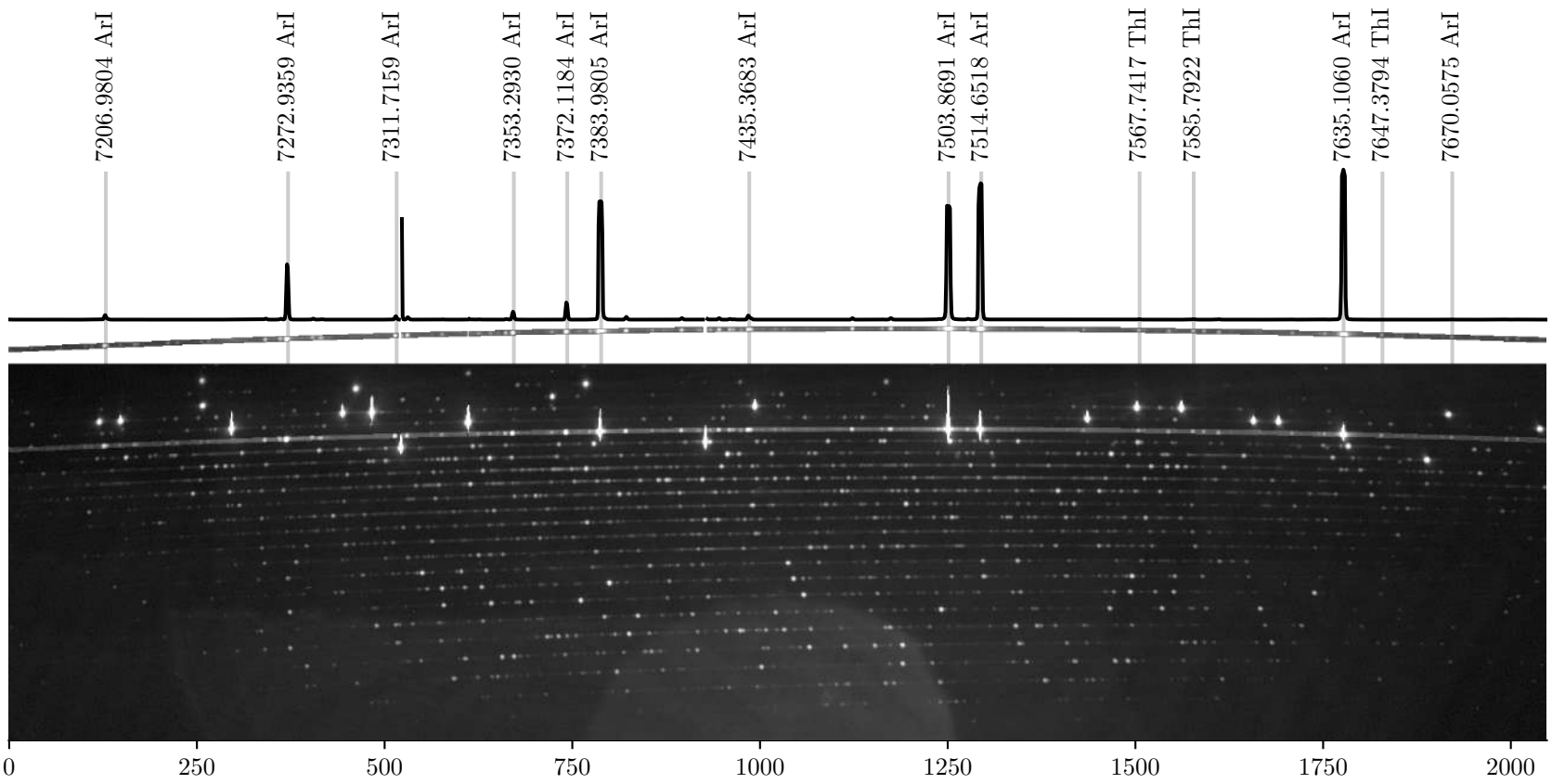

7130.1846 ThI
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7140.4617 ThII
7142.3310 ThI
7147.0416 ArI
7148.1451 Th
7148.5589 ThI
7150.2844 ThI
7153.5877 ThI
7154.7632 ThI
7154.9338 ThI
7130
7135
7140
7145
7150
7155
7150.2844 ThI
7153.5877 ThI
7154.7632 ThI
7154.9338 ThI
7156.9416 ThI
7158.8387 ArI
7159.9470 ThI
7162.5569 ArI
7167.2028 ThI
7168.8952 ThI
7170.3607 ThI
7173.3725 ThI
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7170.3607 ThI
7173.3725 ThI
7176.1860 ThII
7176.7213 ThII
7179.7147 ThI
7188.5320 ThI
7191.1328 ThII
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7180
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7190
7195


7191.1328 ThII
7200.0454 ThI
7201.8091 ThI
7202.1910 Th
7202.5166 ArI
7206.4830 ThI
7206.9804 ArI
7208.0063 ThI
7212.6896 ThI
7190
7195
7200
7205
7210
7215
7212.6896 ThI
7218.0542 ThI
7219.1515 ThI
7220.9809 ThI
7225.1099 ThII
7229.9386 ArI
7230.8623 ThI
7233.5365 ArII
7210
7215
7220
7225
7230
7235
7230.8623 ThI
7233.5365 ArII
7240.1848 ThI
7242.0919 ThI
7244.6965 ThI
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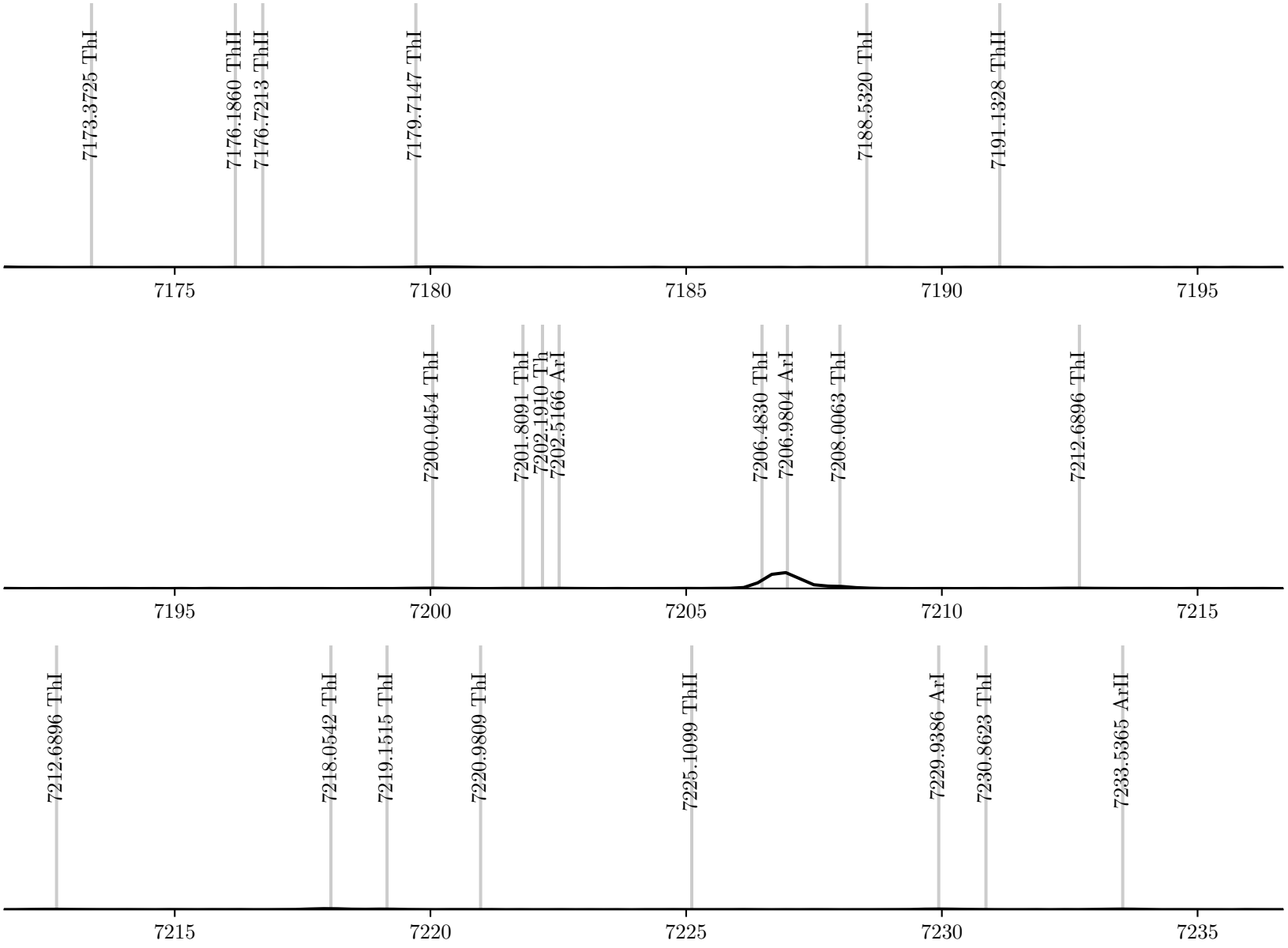

7250.5895 ThII
7253.6760 ThI
7255.3541 ThI
7256.9860 ThI
7258.1770 ThI
7265.1724 ArI
7270.6636 ArI
7272.9359 ArI
7250
7255
7260
7265
7270
7275
7270.6636 ArI
7272.9359 ArI
7284.9033 ThI
7285.4437 ArI
7270
7275
7280
7285
7290
7295
7296.2659 ThI
7298.1434 ThI
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7308.6418 ThI
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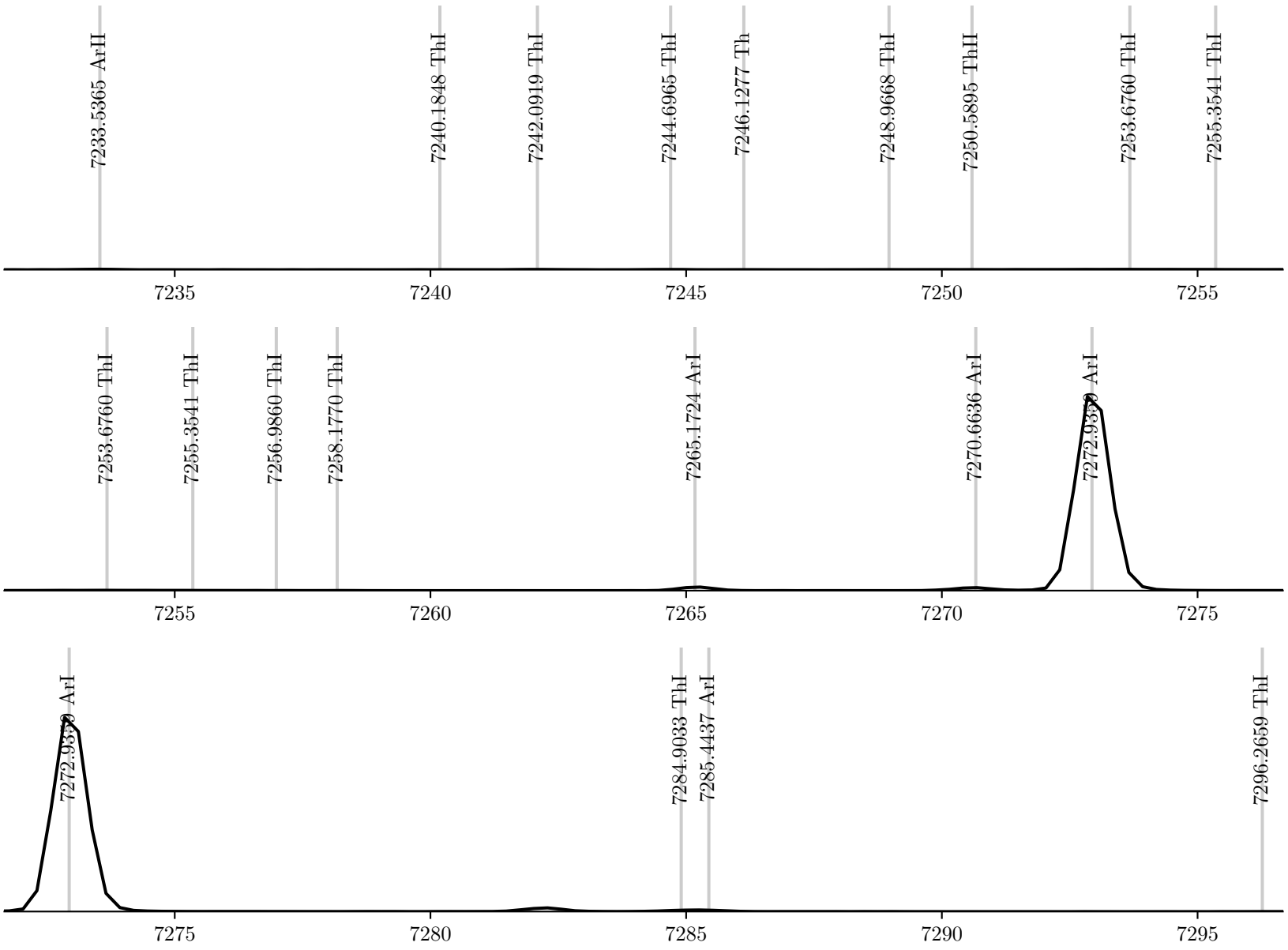

7311.7159 ArI
7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7323.2106 ThI
7324.8073 ThI
7326.1491 Th
7328.2850 ThI
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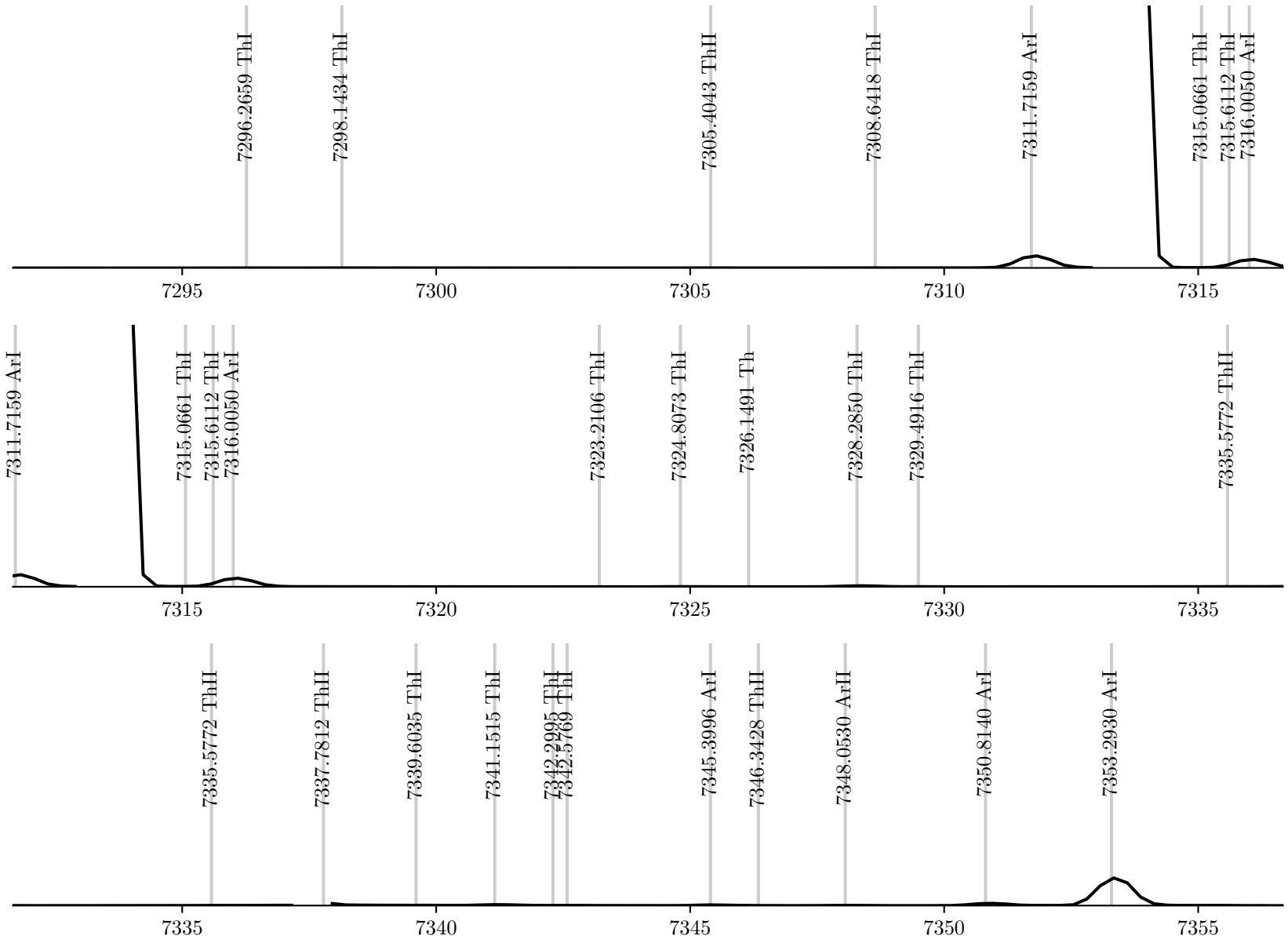
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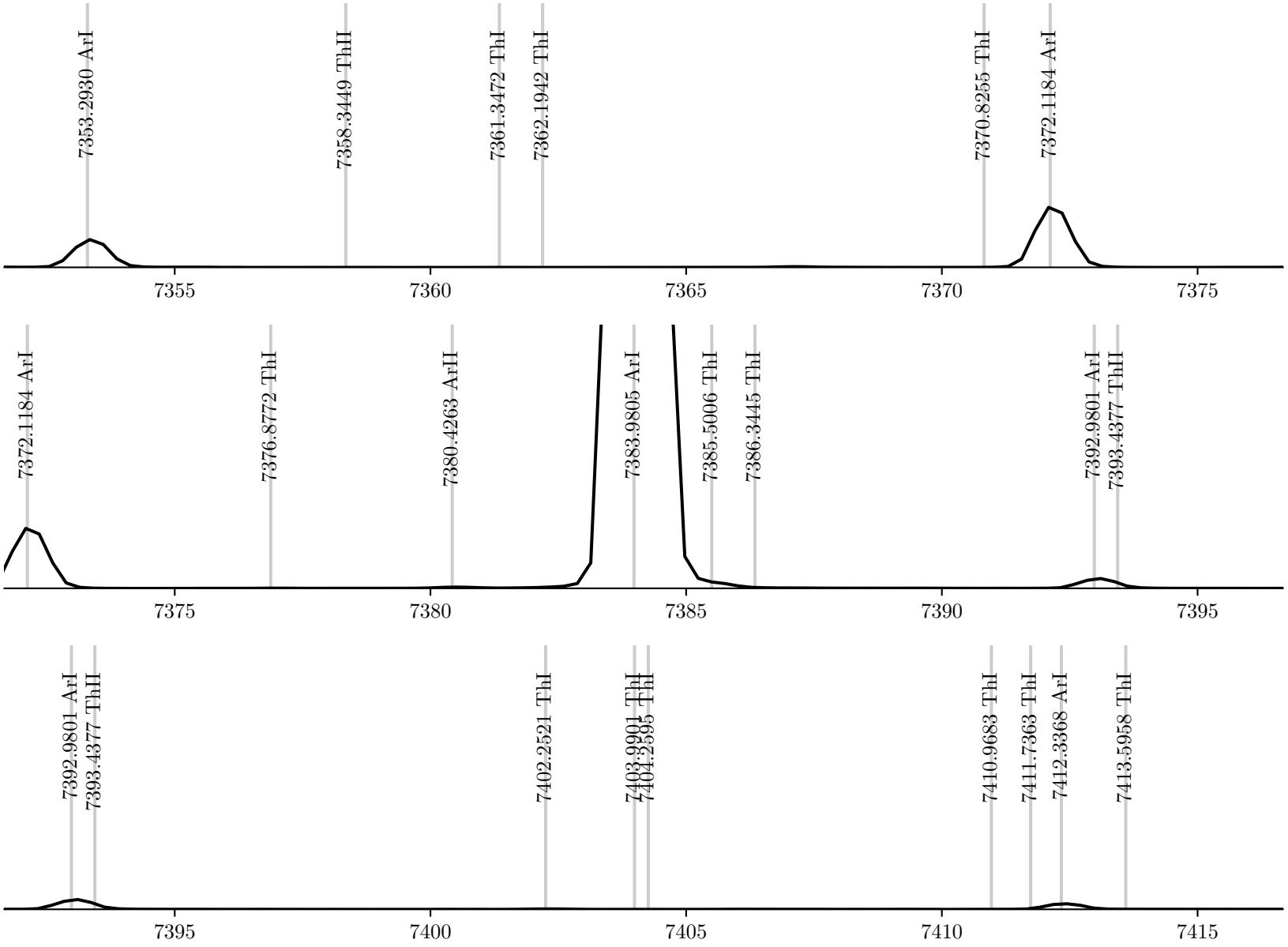


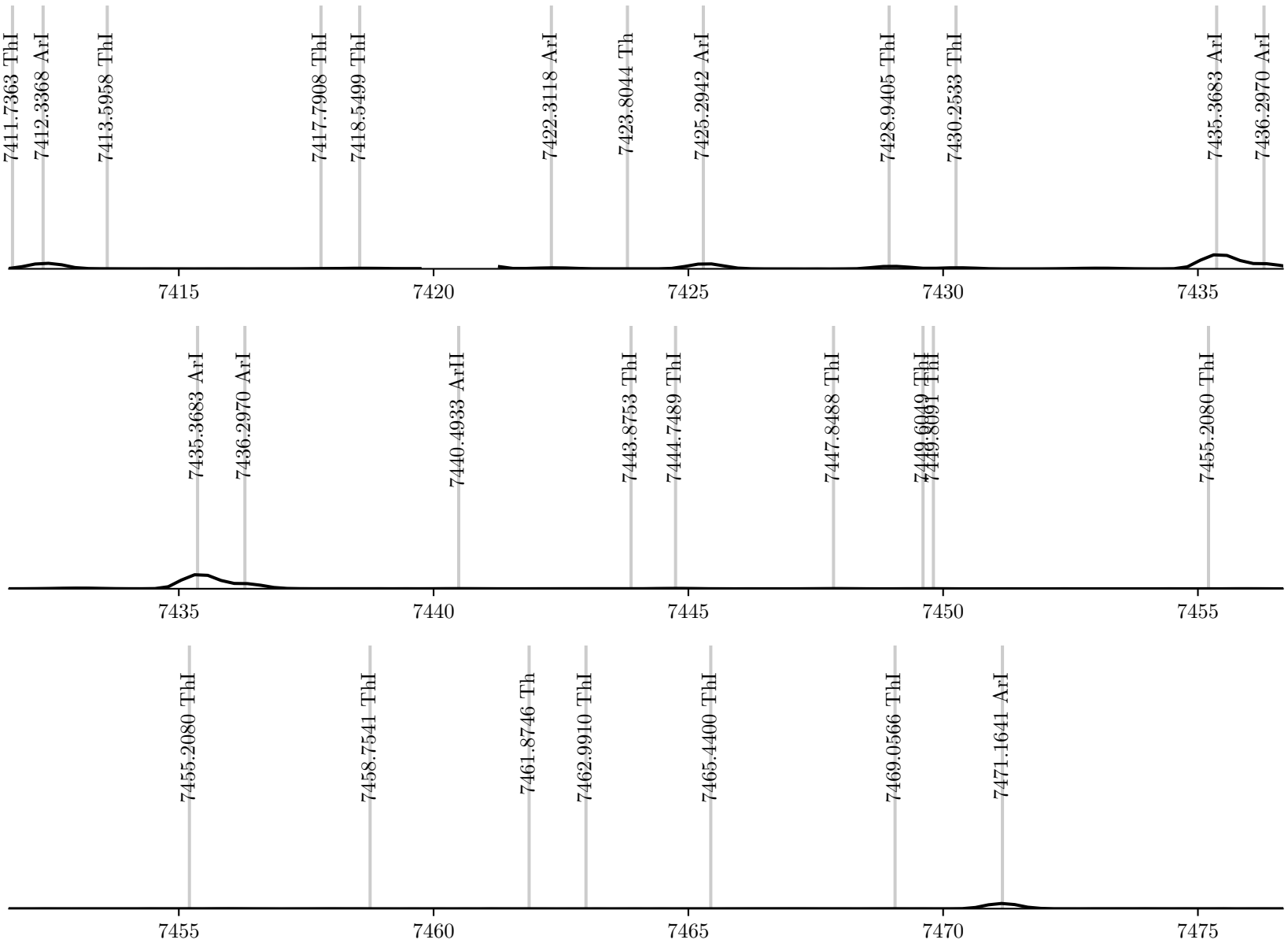
Dispersion:	0.258 Å pix^{-1}
Central wavelength:	7445 Å
Wavelength range:	7172 Å – 7701 Å

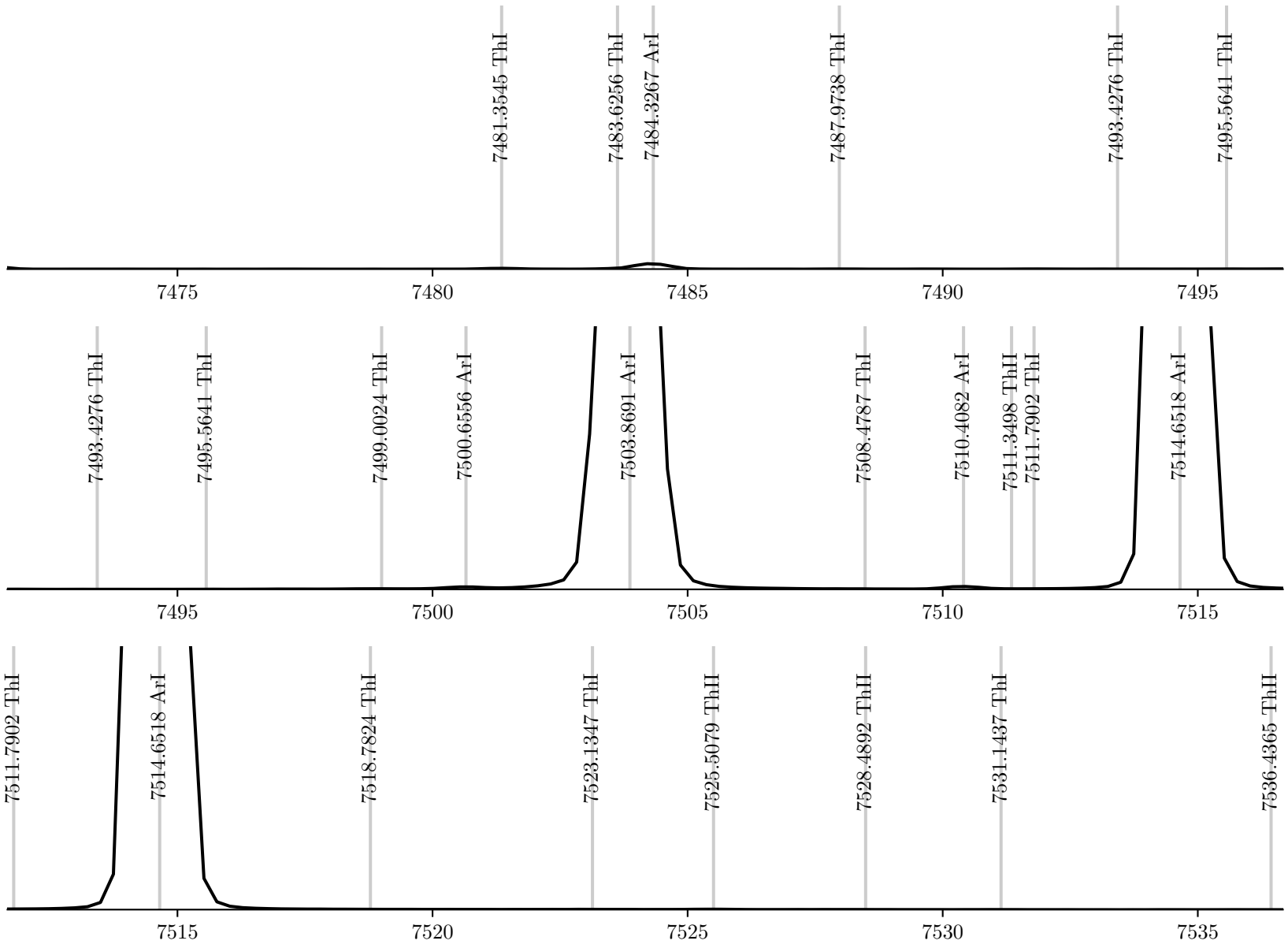

7173.3725 ThI
7176.1860 ThII
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7191.1328 ThII
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7185
7190
7195
7200.0454 ThI
7201.8091 ThI
7202.1910 Th
7202.5166 ArI
7206.4830 ThI
7206.9804 ArI
7208.0063 ThI
7212.6896 ThI
7195
7200
7205
7210
7215
7212.6896 ThI
7218.0542 ThI
7219.1515 ThI
7220.9809 ThI
7225.1099 ThII
7229.9386 ArI
7230.8623 ThI
7233.5365 ArII
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7230
7235

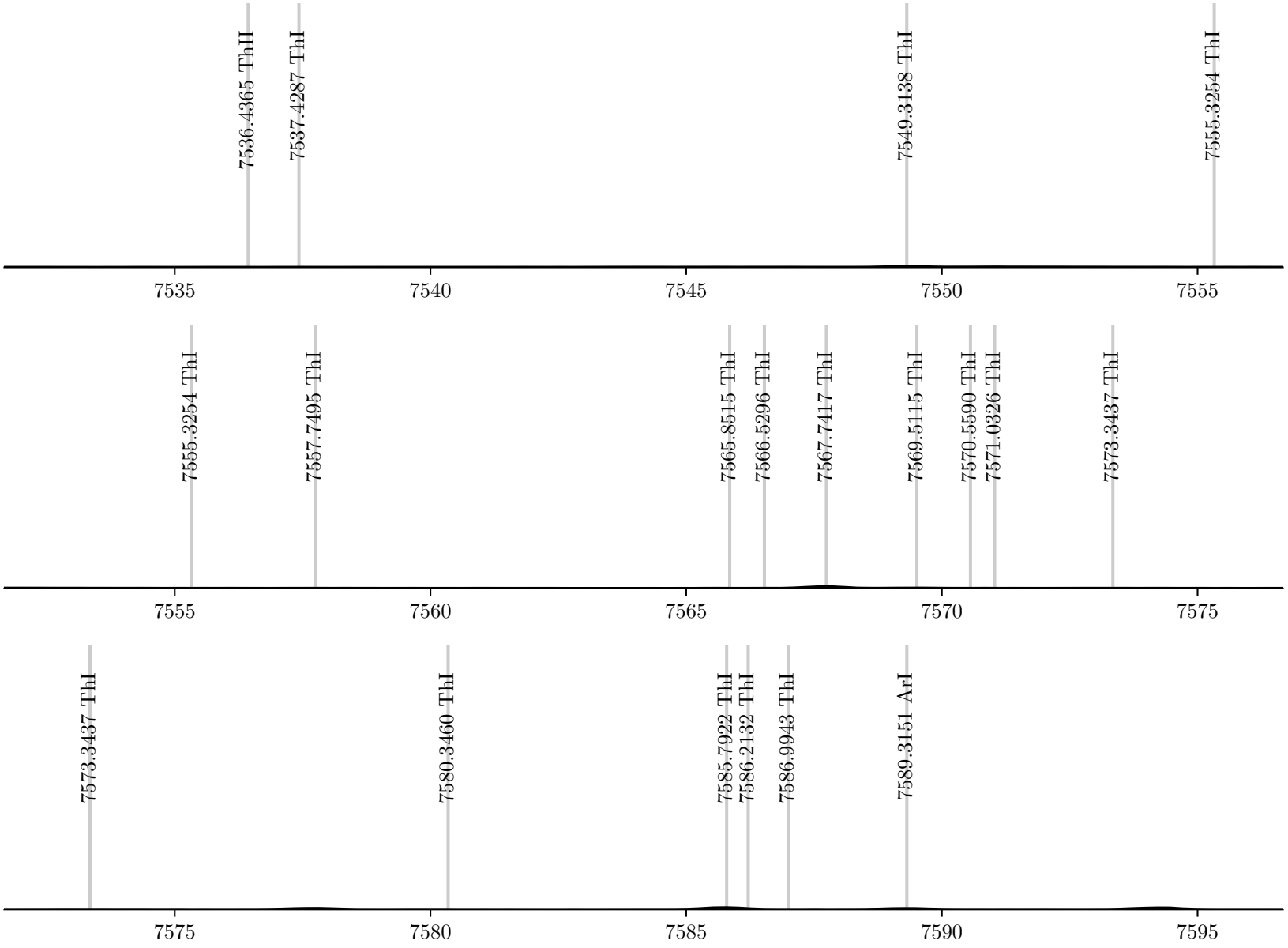

7233.5365 ArII
7240.1848 ThI
7242.0919 ThI
7244.6965 ThI
7246.1277 Th
7248.9668 ThI
7250.5895 ThII
7253.6760 ThI
7255.3541 ThI
7235
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7250
7255
7253.6760 ThI
7255.3541 ThI
7256.9860 ThI
7258.1770 ThI
7265.1724 ArI
7270.6636 ArI
7272.9359 ArI
7255
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7270
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7272.9359 ArI
7284.9033 ThI
7285.4437 ArI
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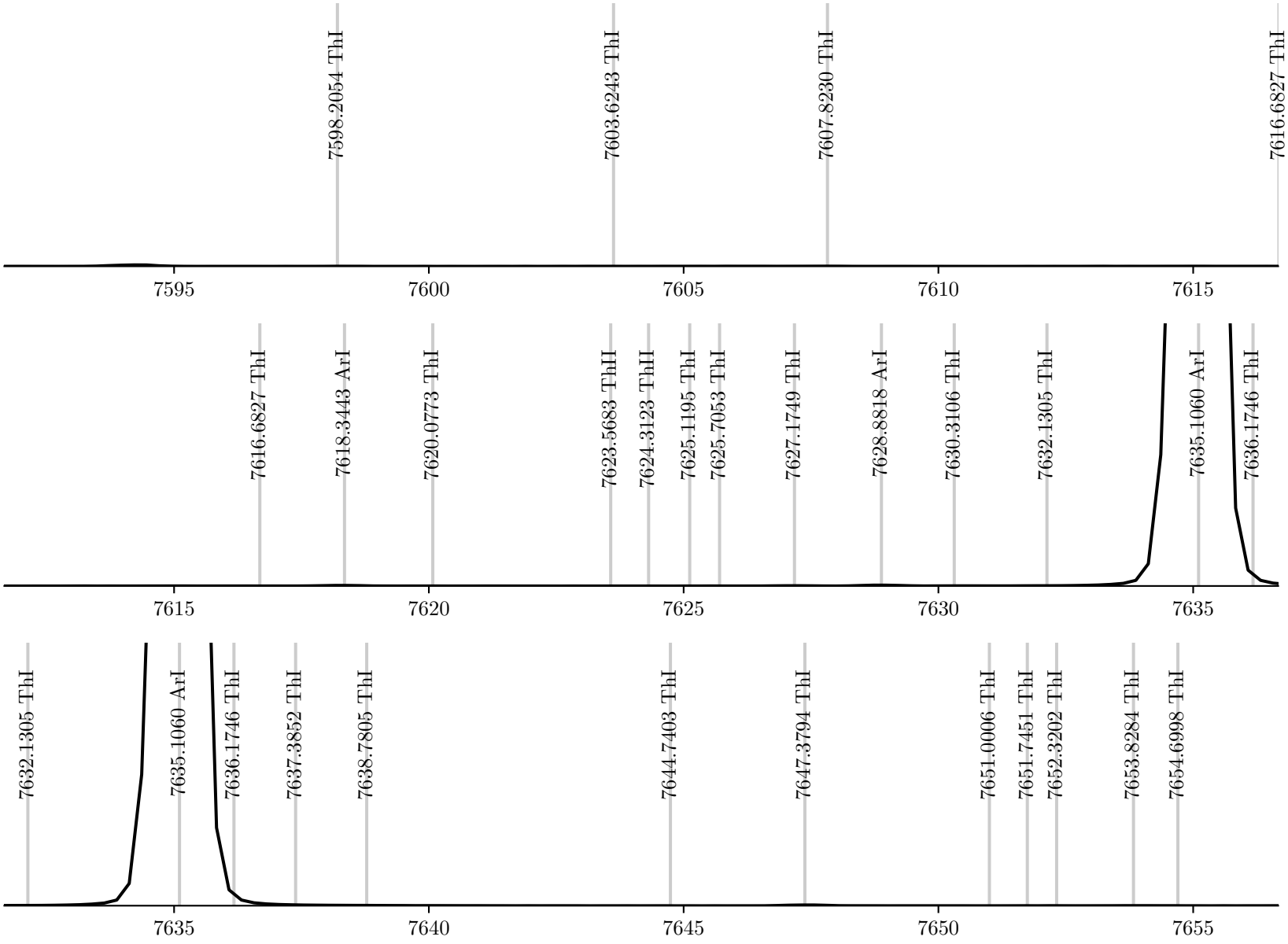

7296.2659 ThI
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7305.4043 ThII
7308.6418 ThI
7311.7159 ArI
7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7295
7300
7305
7310
7315
7311.7159 ArI
7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7323.2106 ThI
7324.8073 ThI
7326.1491 Th
7328.2850 ThI
7329.4916 ThI
7335.5772 ThII
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7325
7330
7335
7335.5772 ThII
7337.7812 ThII
7339.6035 ThI
7341.1515 ThI
7345.3996 ArI
7346.3428 ThII
7348.0530 ArII
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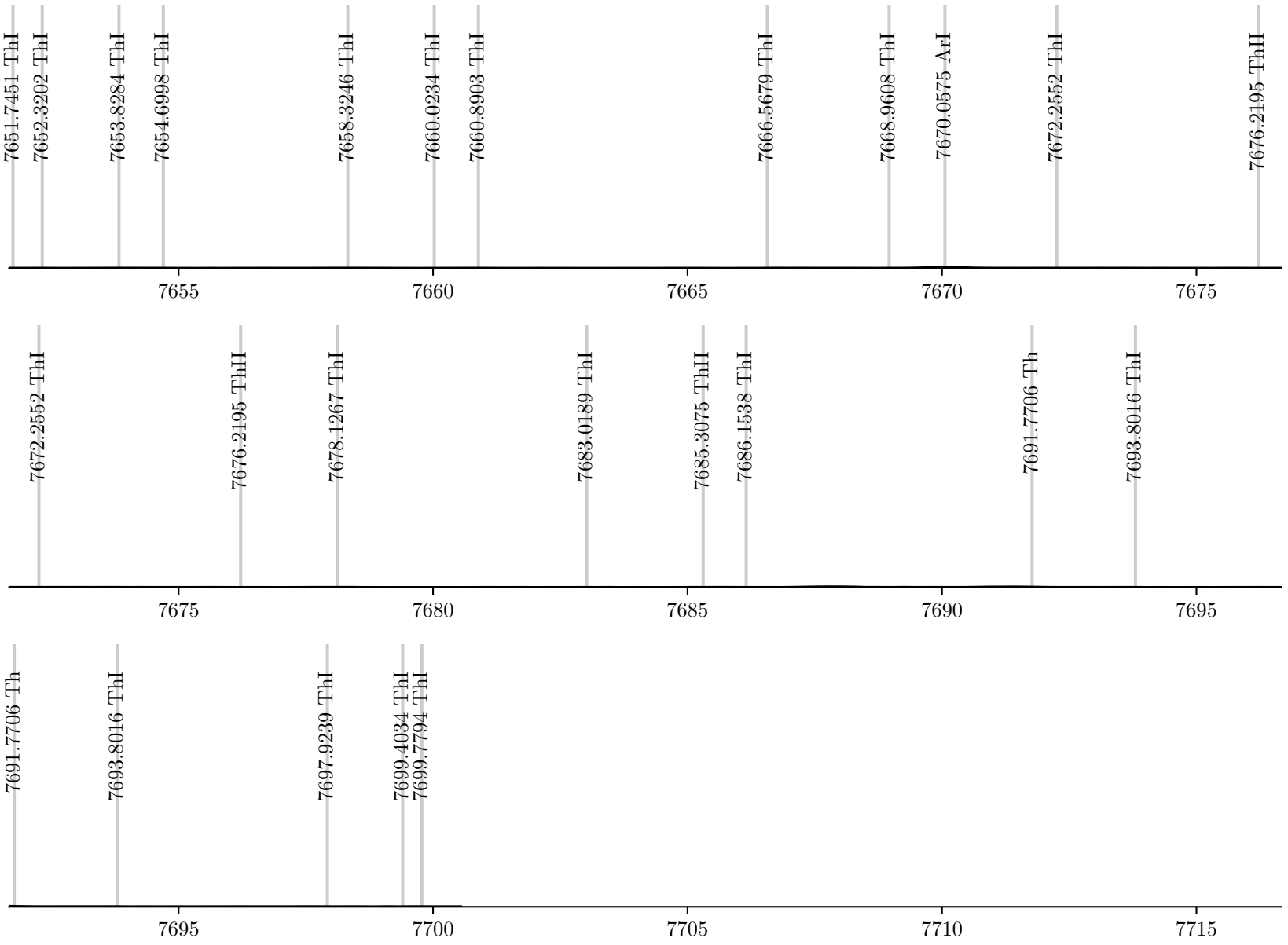

7353.2930 ArI
7358.3449 ThII
7361.3472 ThI
7362.1942 ThI
7370.8255 ThI
7372.1184 ArI
7355
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7370
7375
7372.1184 ArI
7376.8772 ThI
7380.4263 ArII
7383.9805 ArI
7385.5006 ThI
7386.3445 ThI
7392.9801 ArI
7393.4377 ThII
7375
7380
7385
7390
7395
7392.9801 ArI
7393.4377 ThII
7402.2521 ThI
7403.9901 ThI
7404.2395 ThI
7410.9683 ThI
7411.7363 ThI
7412.3368 ArI
7413.5958 ThI
7395
7400
7405
7410
7415


7411.7363 ThI
7412.3368 ArI
7413.5958 ThI
7417.7908 ThI
7418.5499 ThI
7422.3118 ArI
7423.8044 Th
7425.2942 ArI
7428.9405 ThI
7430.2533 ThI
7435.3683 ArI
7436.2970 ArI
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7420
7425
7430
7435
7435.3683 ArI
7436.2970 ArI
7440.4933 ArII
7443.8753 ThI
7444.7489 ThI
7447.8488 ThI
7455.2080 ThI
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7455.2080 ThI
7458.7541 ThI
7461.8746 Th
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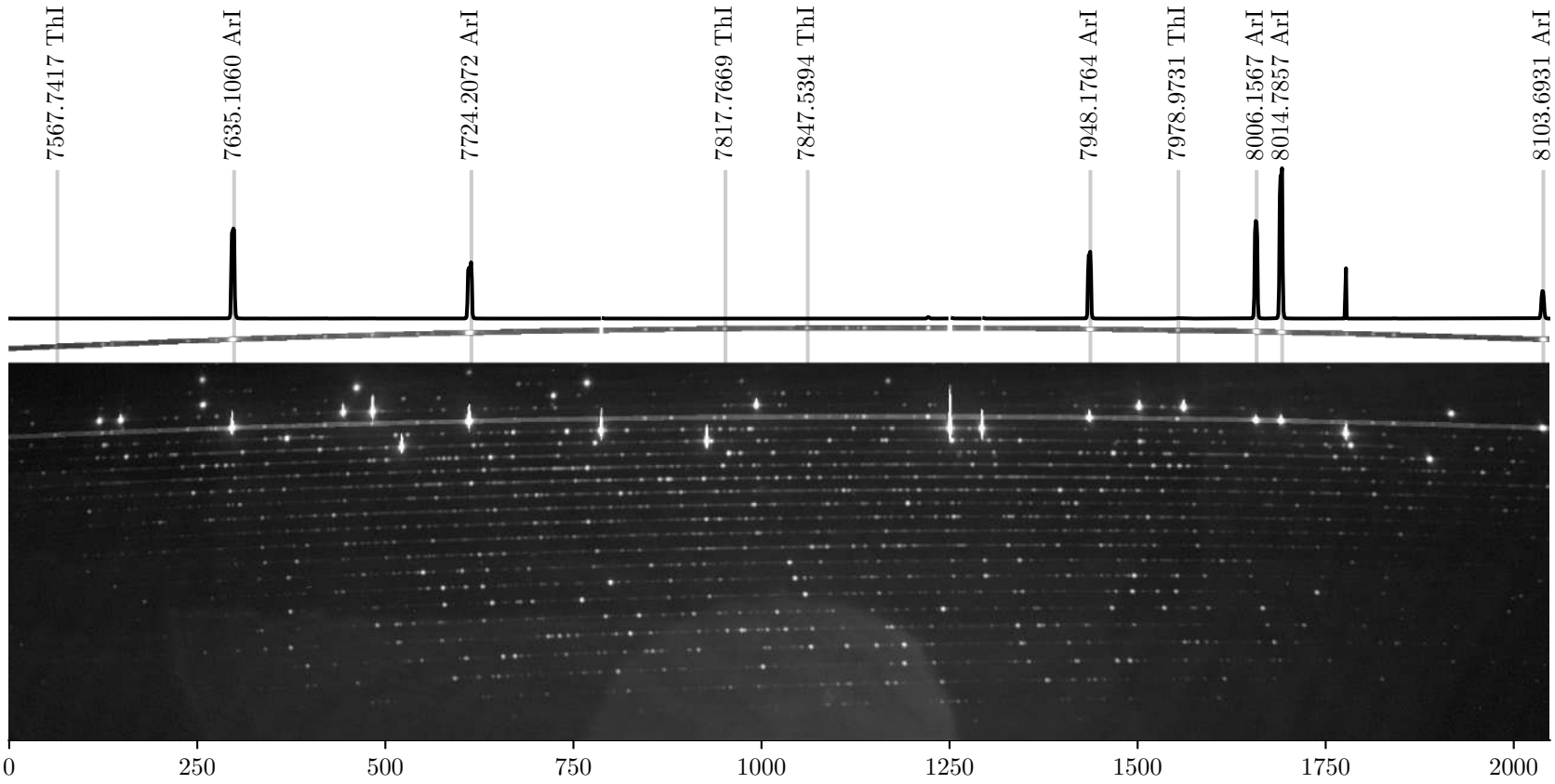

7481.3545 ThI
7483.6256 ThI
7484.3267 ArI
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7493.4276 ThI
7495.5641 ThI
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7485
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7495
7493.4276 ThI
7495.5641 ThI
7499.0024 ThI
7500.6556 ArI
7503.8691 ArI
7508.4787 ThI
7510.4082 ArI
7511.3498 ThII
7511.7902 ThI
7514.6518 ArI
7495
7500
7505
7510
7515
7511.7902 ThI
7514.6518 ArI
7518.7824 ThI
7523.1347 ThI
7525.5079 ThII
7528.4892 ThII
7531.1437 ThI
7536.4365 ThII
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7520
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7535


7536.4365 ThII
7537.4287 ThI
7549.3138 ThI
7555.3254 ThI
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7555.3254 ThI
7557.7495 ThI
7565.8515 ThI
7566.5296 ThI
7567.7417 ThI
7569.5115 ThI
7570.5590 ThI
7571.0326 ThI
7573.3437 ThI
7555
7560
7565
7570
7575
7573.3437 ThI
7580.3460 ThI
7585.7922 ThI
7586.2132 ThI
7586.9943 ThI
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7585
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7595

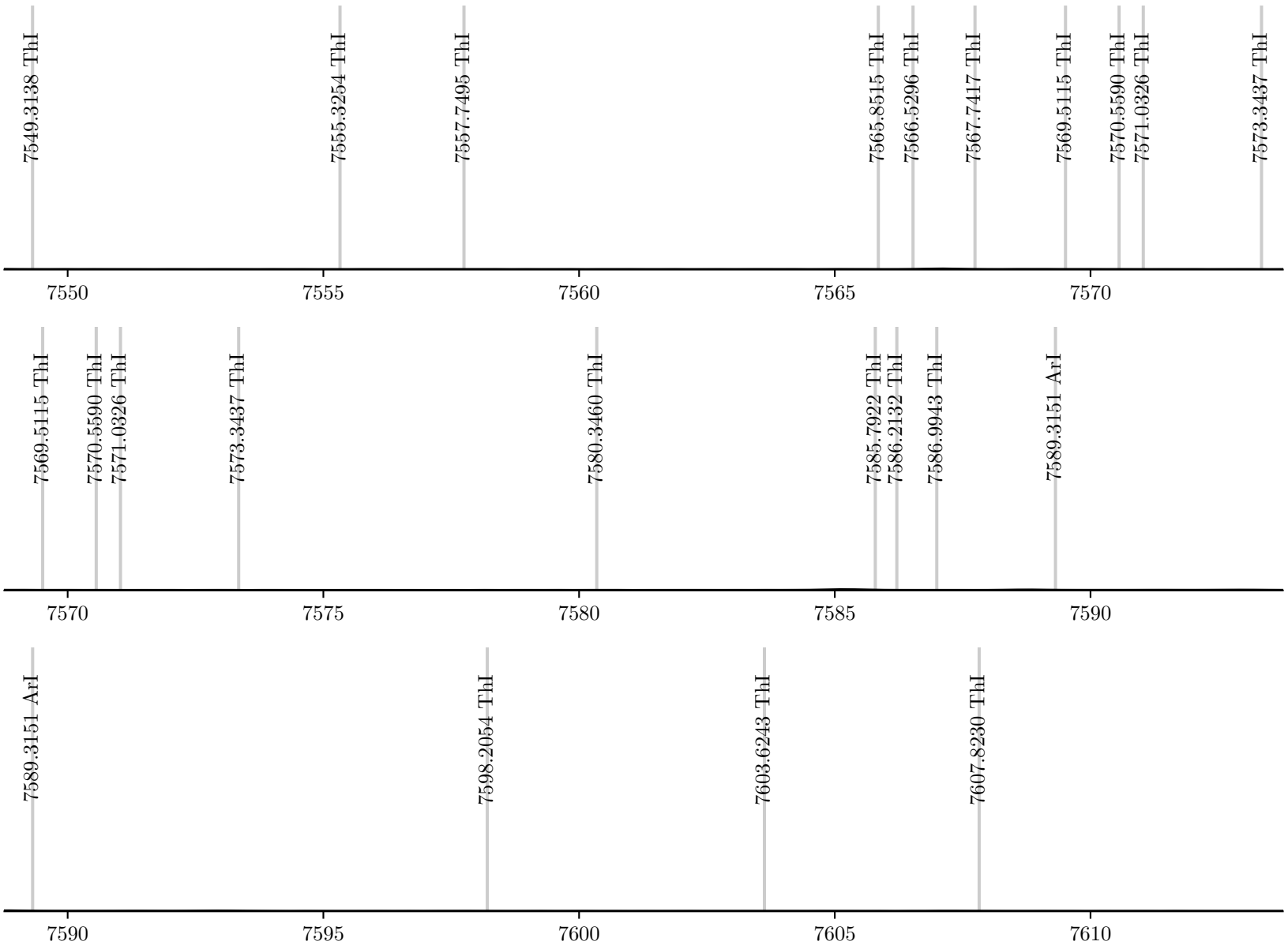

7598.2054 ThI
7603.6243 ThI
7607.8230 ThI
7616.6827 ThI
7595
7600
7605
7610
7615
7616.6827 ThI
7618.3443 ArI
7620.0773 ThI
7623.5683 ThII
7624.3123 ThII
7625.1195 ThI
7625.7053 ThI
7627.1749 ThI
7628.8818 ArI
7630.3106 ThI
7632.1305 ThI
7635.1060 ArI
7636.1746 ThI
7615
7620
7625
7630
7635
7632.1305 ThI
7635.1060 ArI
7636.1746 ThI
7637.3852 ThI
7638.7805 ThI
7644.7403 ThI
7647.3794 ThI
7651.0006 ThI
7651.7451 ThI
7652.3202 ThI
7653.8284 ThI
7654.6998 ThI
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7645
7650
7655

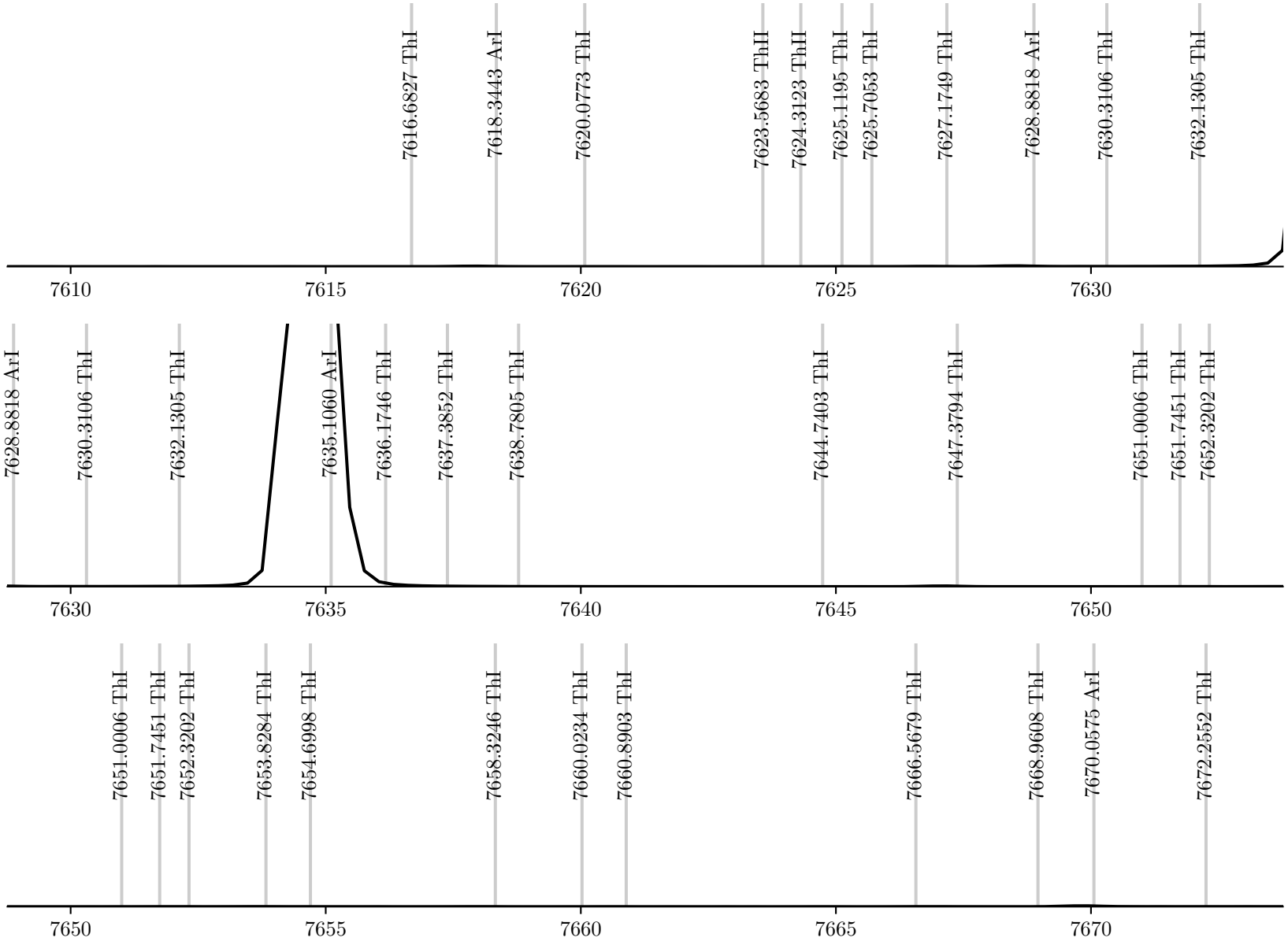

7651.7451 ThI
7652.3202 ThI
7653.8284 ThI
7654.6998 ThI
7658.3246 ThI
7660.0234 ThI
7660.8903 ThI
7666.5679 ThI
7668.9608 ThI
7670.0575 ArI
7672.2552 ThI
7676.2195 ThII
7655
7660
7665
7670
7675
7672.2552 ThI
7676.2195 ThII
7678.1267 ThI
7683.0189 ThI
7685.3075 ThII
7686.1538 ThI
7691.7706 Th
7693.8016 ThI
7675
7680
7685
7690
7695
7691.7706 Th
7693.8016 ThI
7697.9239 ThI
7699.4034 ThI
7699.7794 ThI
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7705
7710
7715

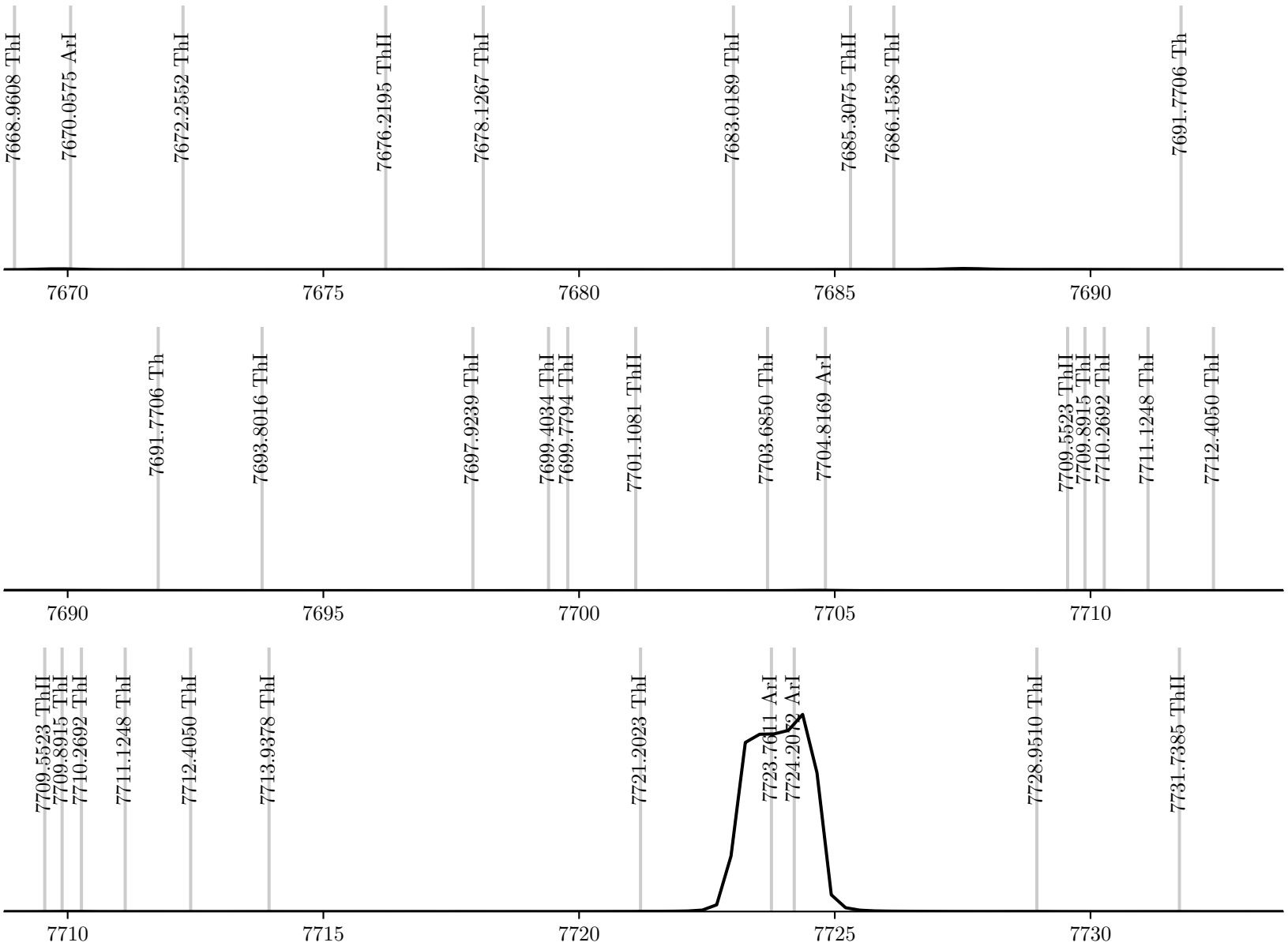
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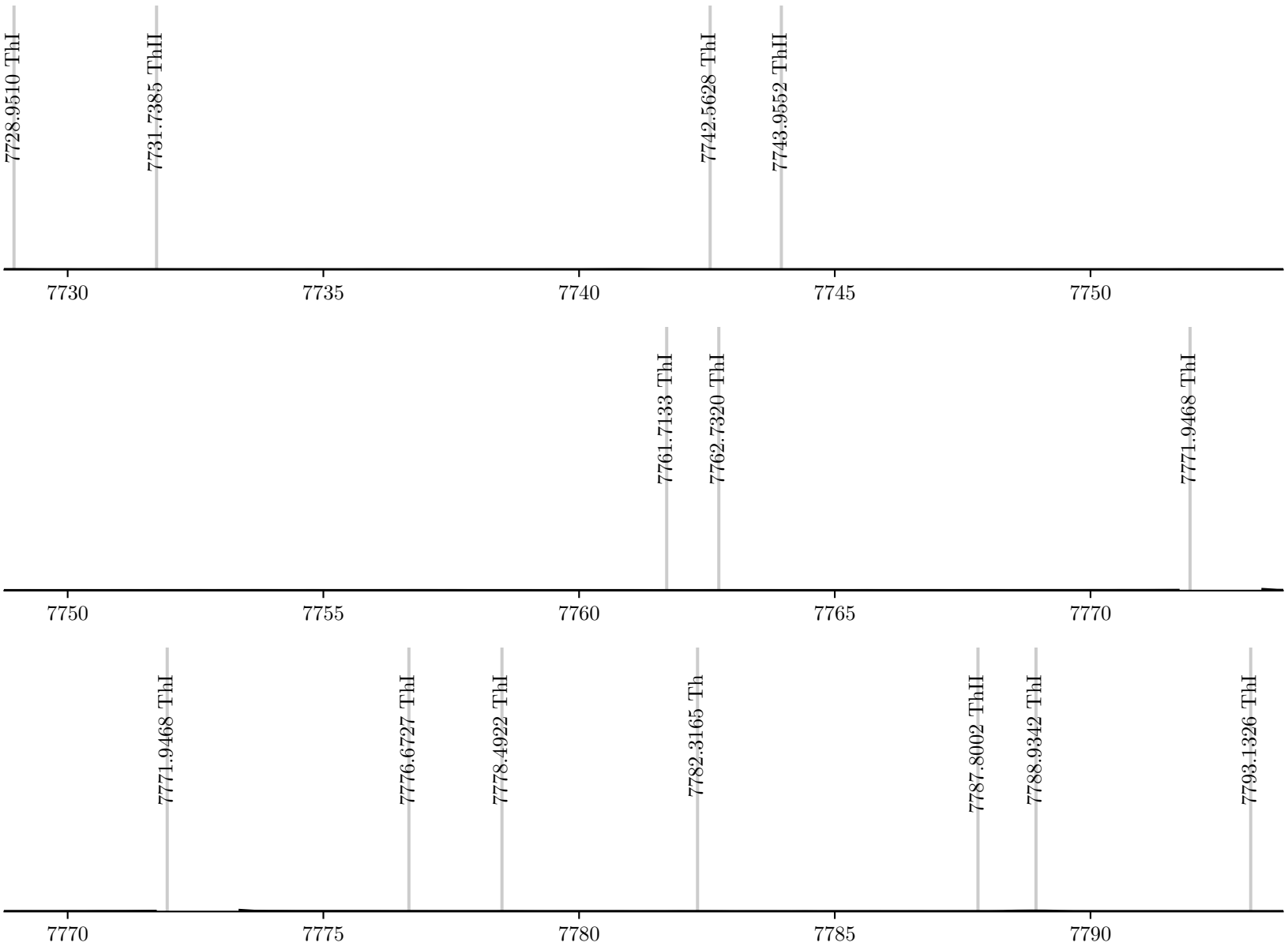


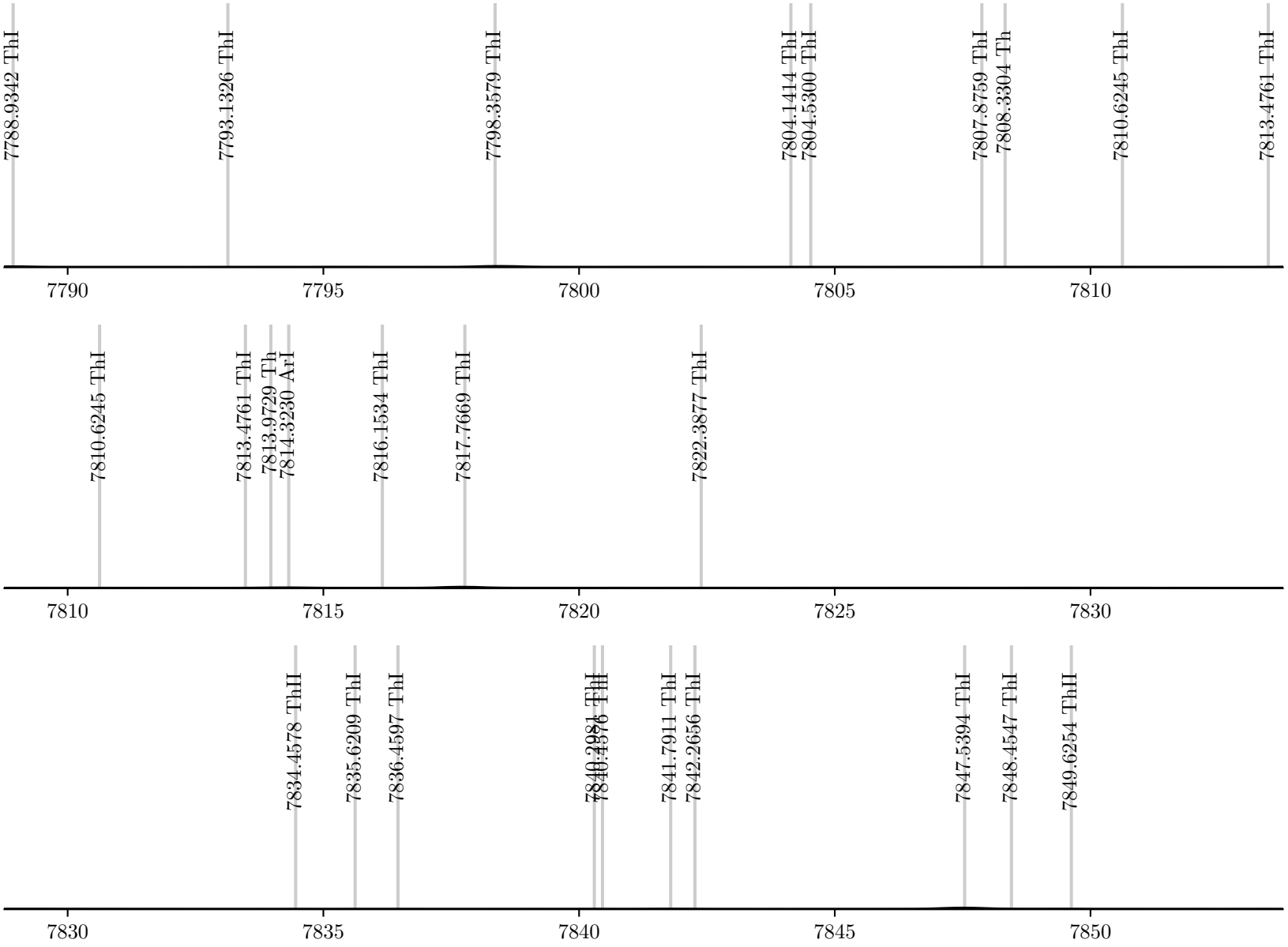
Dispersion:	0.272 Å pix^{-1}
Central wavelength:	7837 Å
Wavelength range:	7549 Å – 8106 Å

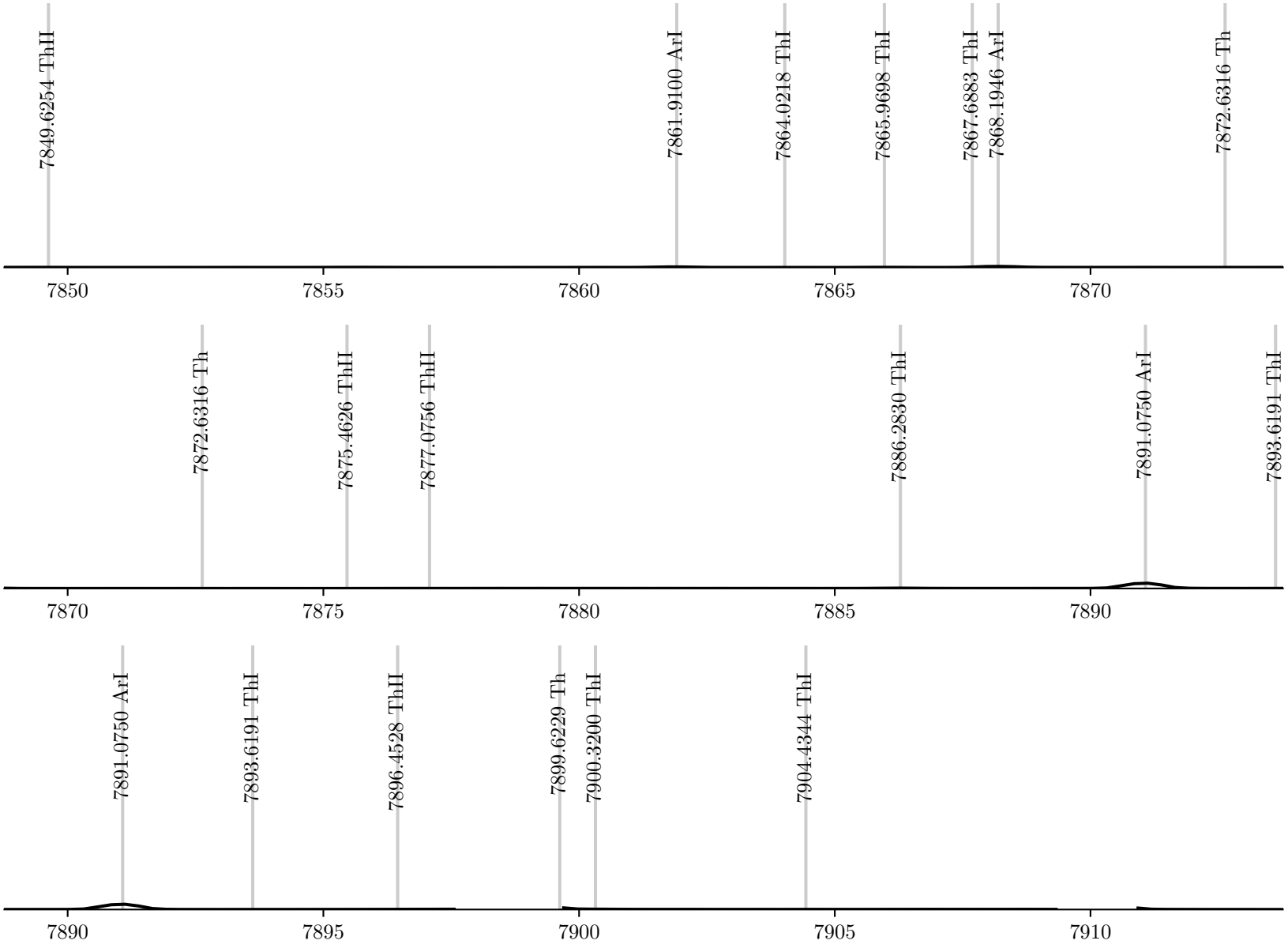

7549.3138 ThI
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7566.5296 ThI
7567.7417 ThI
7569.5115 ThI
7570.5590 ThI
7571.0326 ThI
7573.3437 ThI
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7560
7565
7570
7569.5115 ThI
7570.5590 ThI
7571.0326 ThI
7573.3437 ThI
7580.3460 ThI
7585.7922 ThI
7586.2132 ThI
7586.9943 ThI
7589.3151 ArI
7570
7575
7580
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7603.6243 ThI
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7610

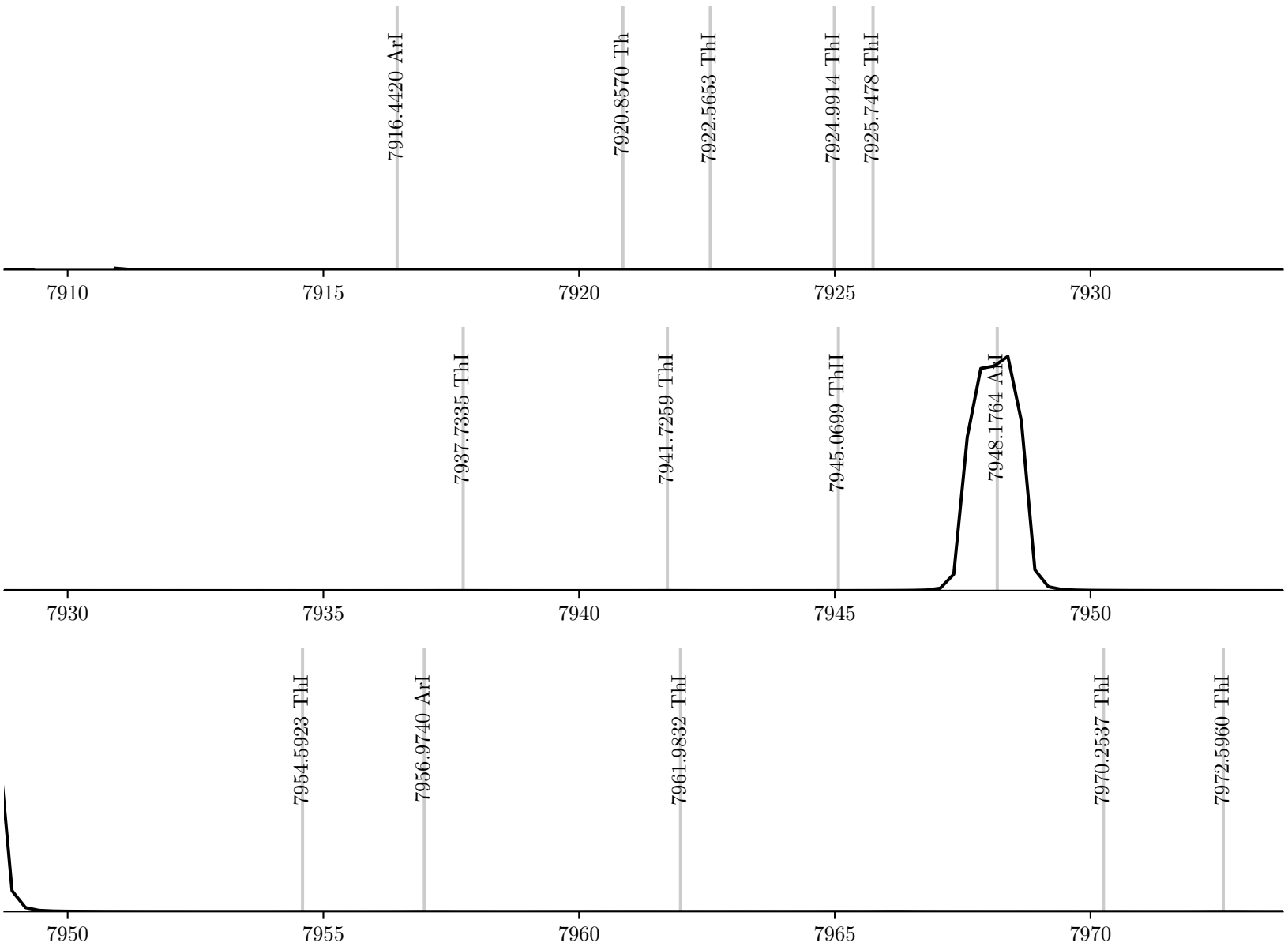

7616.6827 ThI
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7624.3123 ThII
7625.1195 ThI
7625.7053 ThI
7627.1749 ThI
7628.8818 ArI
7630.3106 ThI
7632.1305 ThI
7610
7615
7620
7625
7630
7628.8818 ArI
7630.3106 ThI
7632.1305 ThI
7635.1060 ArI
7636.1746 ThI
7637.3852 ThI
7638.7805 ThI
7644.7403 ThI
7647.3794 ThI
7651.0006 ThI
7651.7451 ThI
7652.3202 ThI
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7640
7645
7650
7651.0006 ThI
7651.7451 ThI
7652.3202 ThI
7653.8284 ThI
7654.6998 ThI
7658.3246 ThI
7660.0234 ThI
7660.8903 ThI
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7670.0575 ArI
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7670

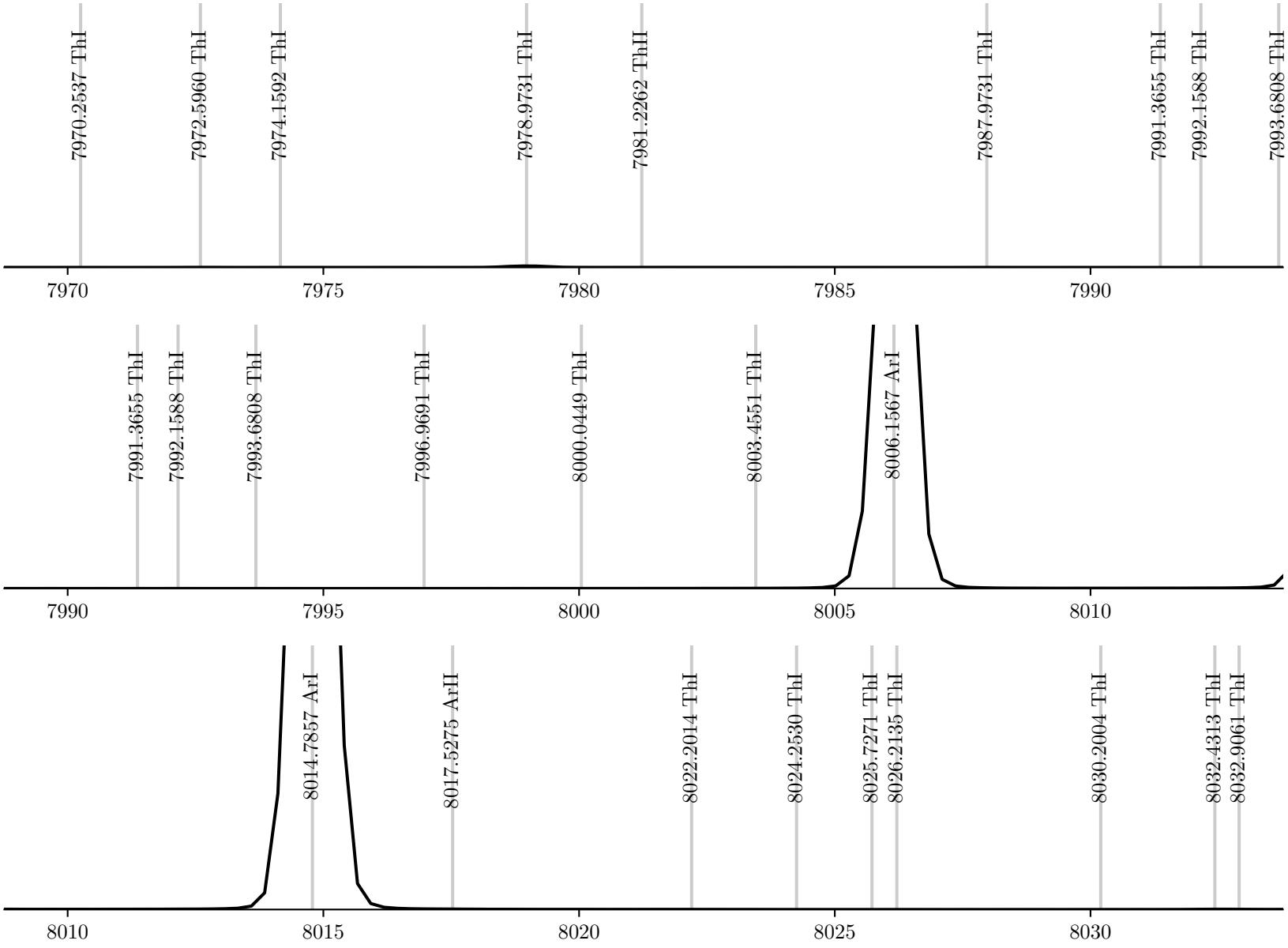

7668.9608 ThI
7670.0575 ArI
7672.2552 ThI
7676.2195 ThII
7678.1267 ThI
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7685.3075 ThII
7686.1538 ThI
7691.7706 Th
7670
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7680
7685
7690
7691.7706 Th
7693.8016 ThI
7697.9239 ThI
7699.4034 ThI
7699.7794 ThI
7701.1081 ThII
7703.6850 ThI
7704.8169 ArI
7709.5523 ThII
7709.8915 ThI
7710.2692 ThI
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7712.4050 ThI
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7709.5523 ThII
7709.8915 ThI
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7711.1248 ThI
7712.4050 ThI
7713.9378 ThI
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7724.2072 ArI
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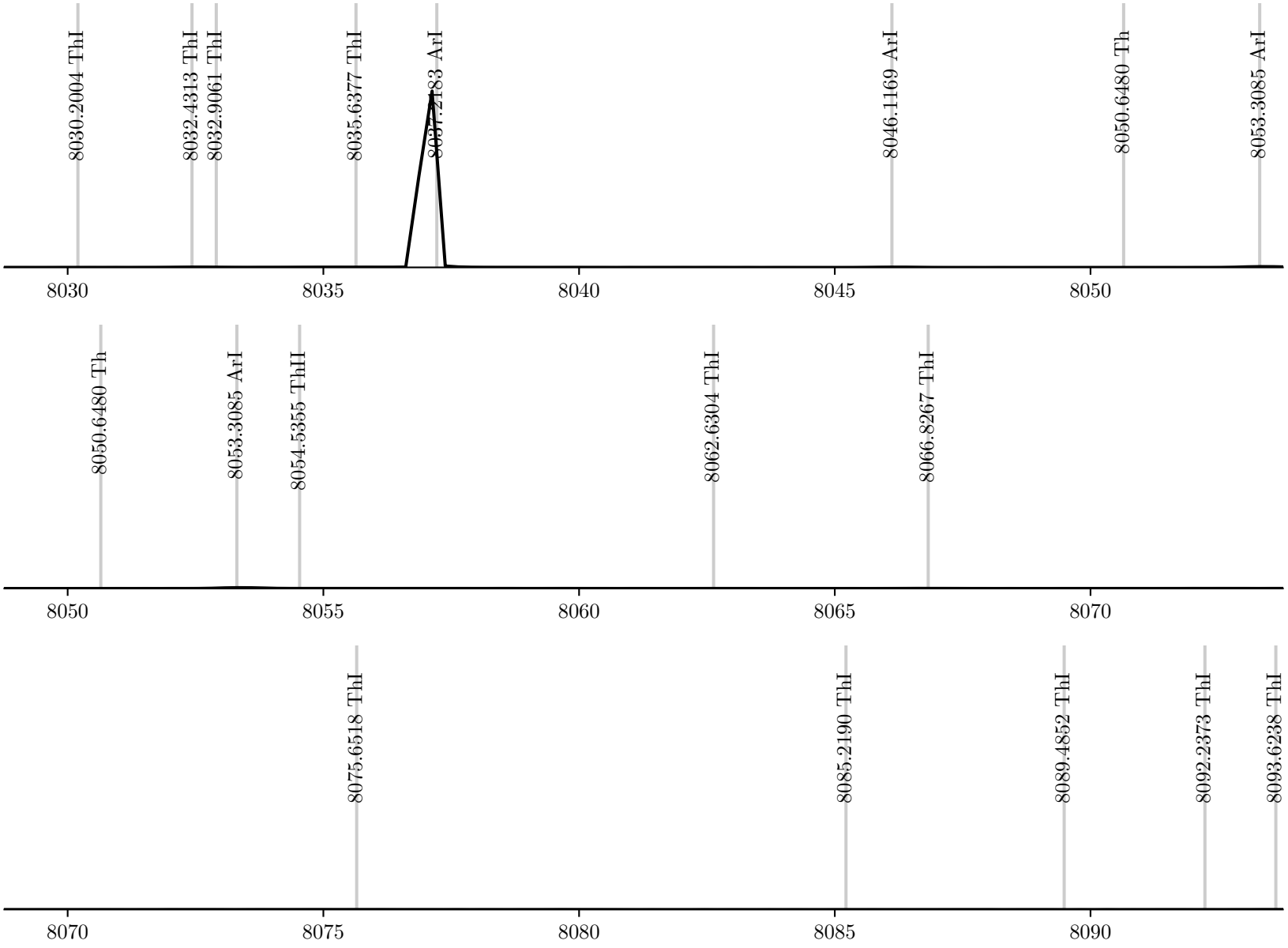

7728.9510 ThI
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7762.7320 ThI
7771.9468 ThI
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7771.9468 ThI
7776.6727 ThI
7778.4922 ThI
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7787.8002 ThII
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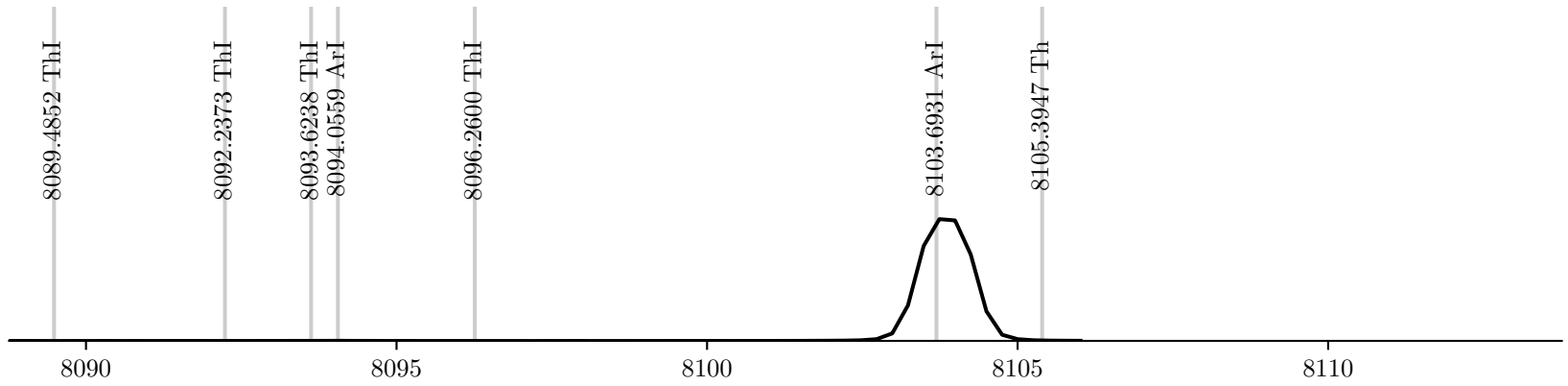

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7804.5300 ThI
7807.8759 ThI
7808.3304 Th
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7813.4761 ThI
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7841.7911 ThI
7842.2656 ThI
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7848.4547 ThI
7849.6254 ThII
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7850


7849.6254 ThII
7861.9100 ArI
7864.0218 ThI
7865.9698 ThI
7867.6883 ThI
7868.1946 ArI
7872.6316 Th
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7860
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7870
7872.6316 Th
7875.4626 ThII
7877.0756 ThII
7886.2830 ThI
7891.0750 ArI
7893.6191 ThI
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7890
7891.0750 ArI
7893.6191 ThI
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7910

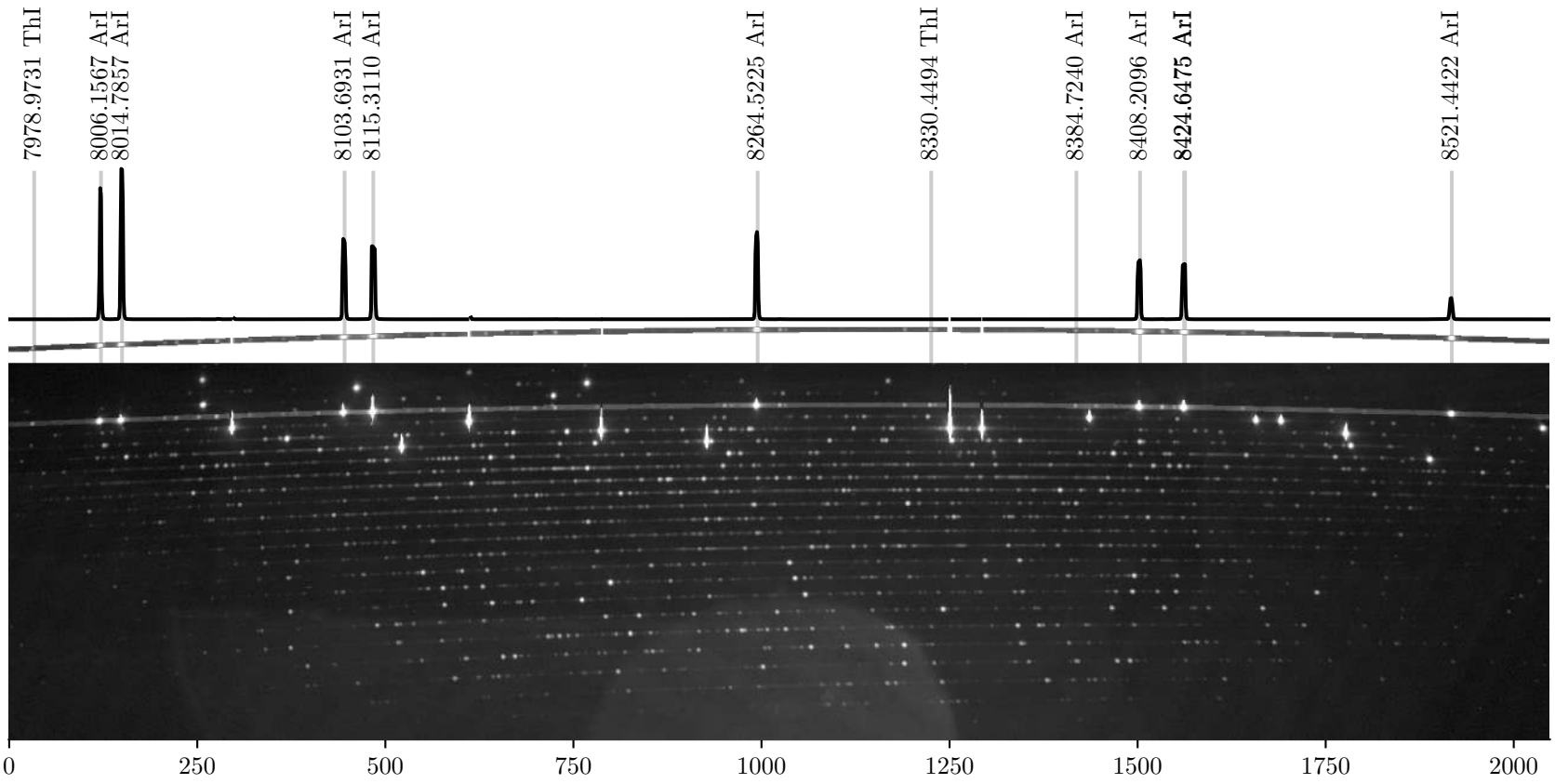

7916.4420 ArI
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7922.5653 ThI
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7925.7478 ThI
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7930
7937.7335 ThI
7941.7259 ThI
7945.0699 ThII
7948.1764 AlI
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7954.5923 ThI
7956.9740 ArI
7961.9832 ThI
7970.2537 ThI
7972.5960 ThI
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7955
7960
7965
7970


7970.2537 ThI
7972.5960 ThI
7974.1592 ThI
7978.9731 ThI
7981.2262 ThII
7987.9731 ThI
7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
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7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
7996.9691 ThI
8000.0449 ThI
8003.4551 ThI
8006.1567 ArI
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7995
8000
8005
8010
8014.7857 ArI
8017.5275 ArII
8022.2014 ThI
8024.2530 ThI
8025.7271 ThI
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8030.2004 ThI
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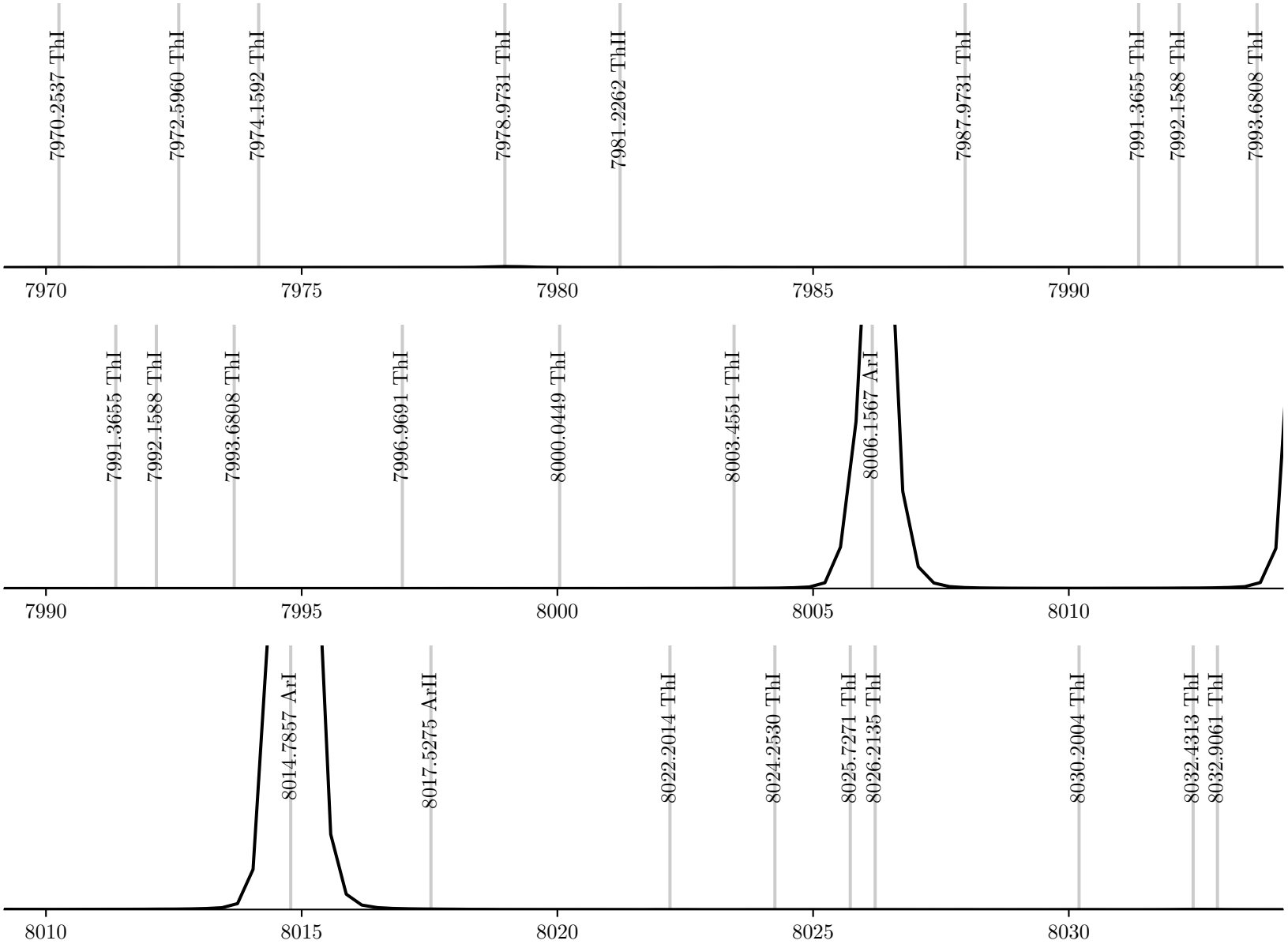

8030.2004 ThI
8032.4313 ThI
8032.9061 ThI
8035.6377 ThI
8037.2183 ArI
8046.1169 ArI
8050.6480 Th
8053.3085 ArI
8030
8035
8040
8045
8050
8050.6480 Th
8053.3085 ArI
8054.5355 ThII
8062.6304 ThI
8066.8267 ThI
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8060
8065
8070
8075.6518 ThI
8085.2190 ThI
8089.4852 ThI
8092.2373 ThI
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8080
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8090

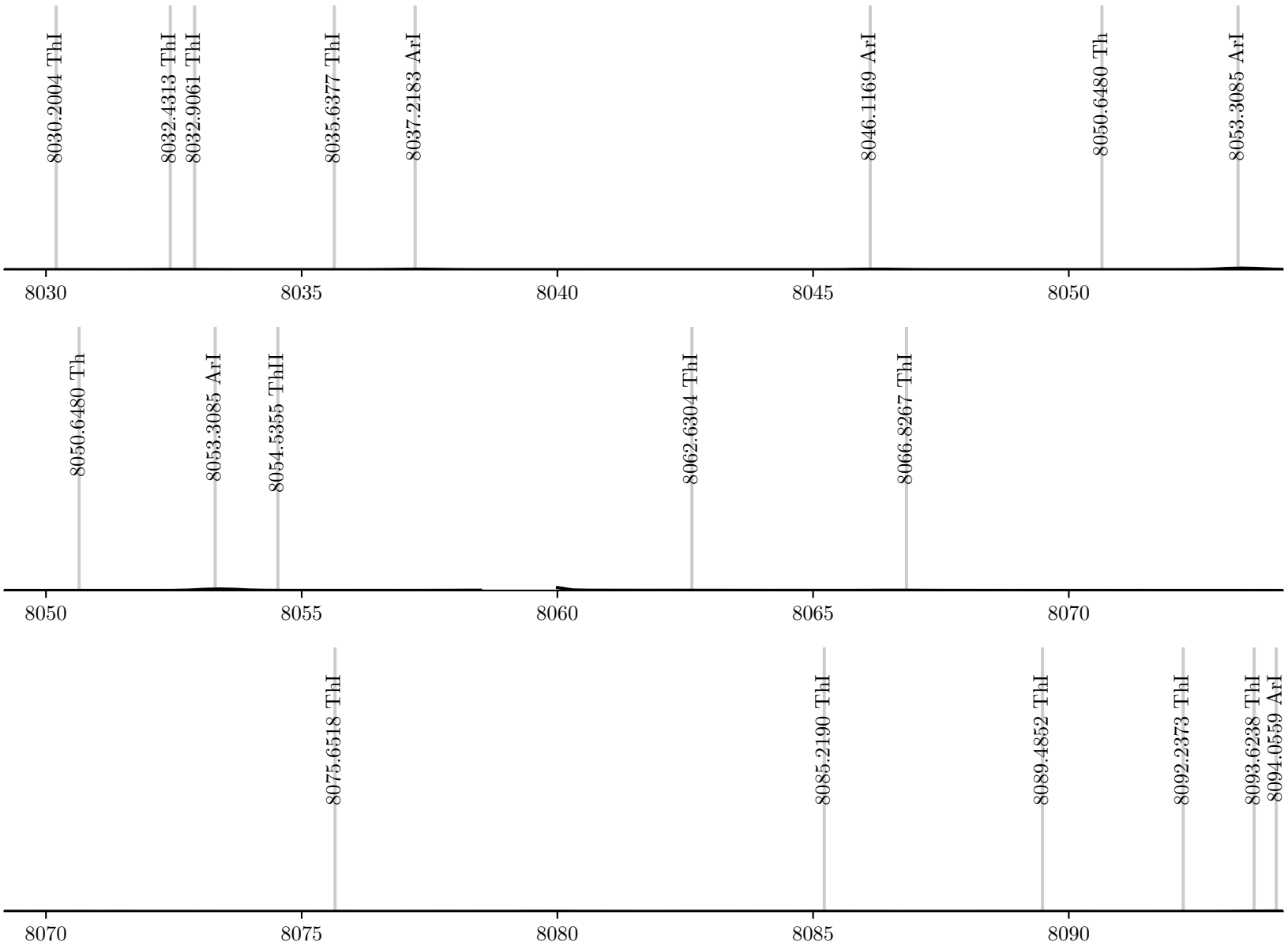

8089.4852 ThI
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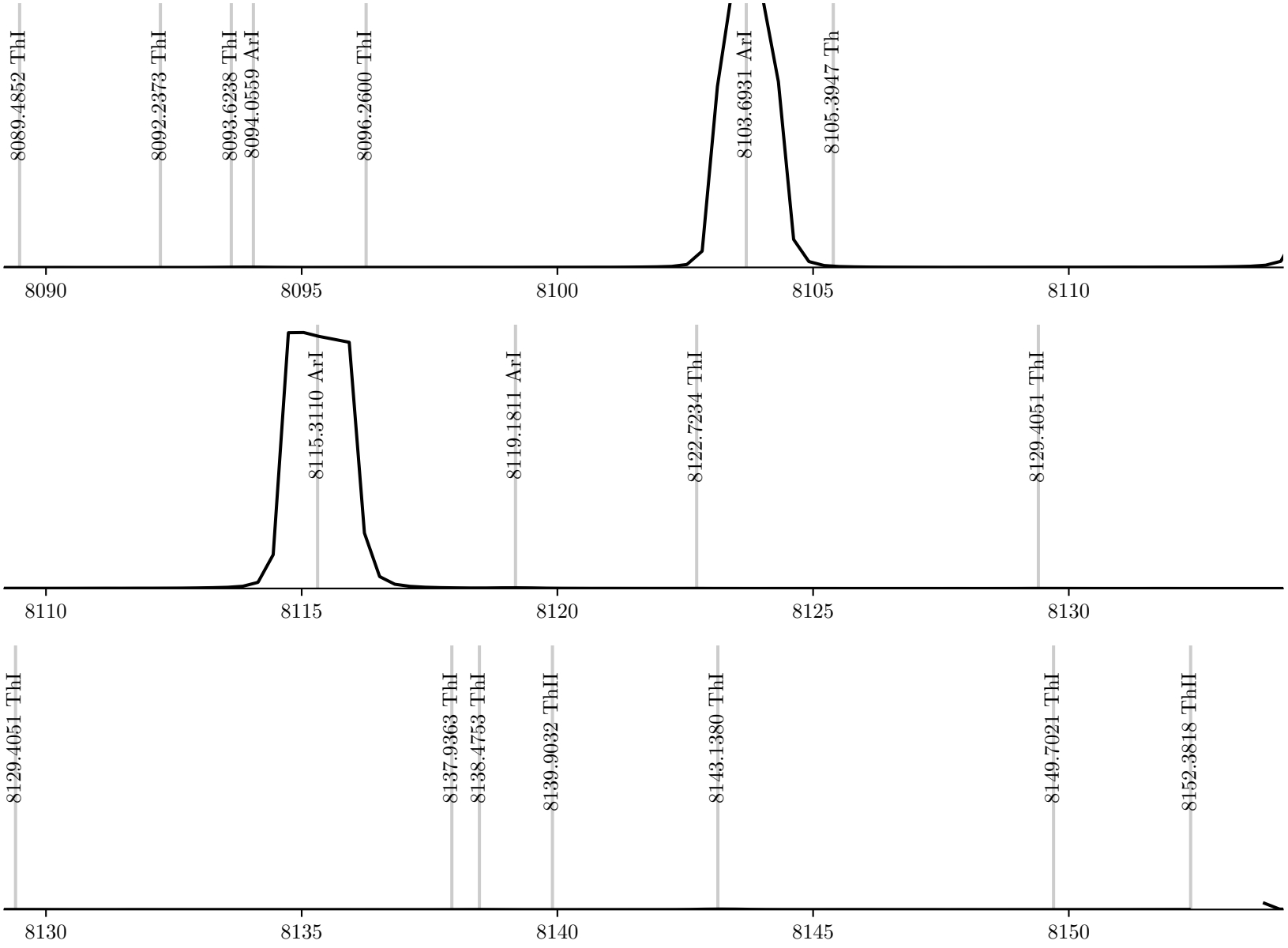
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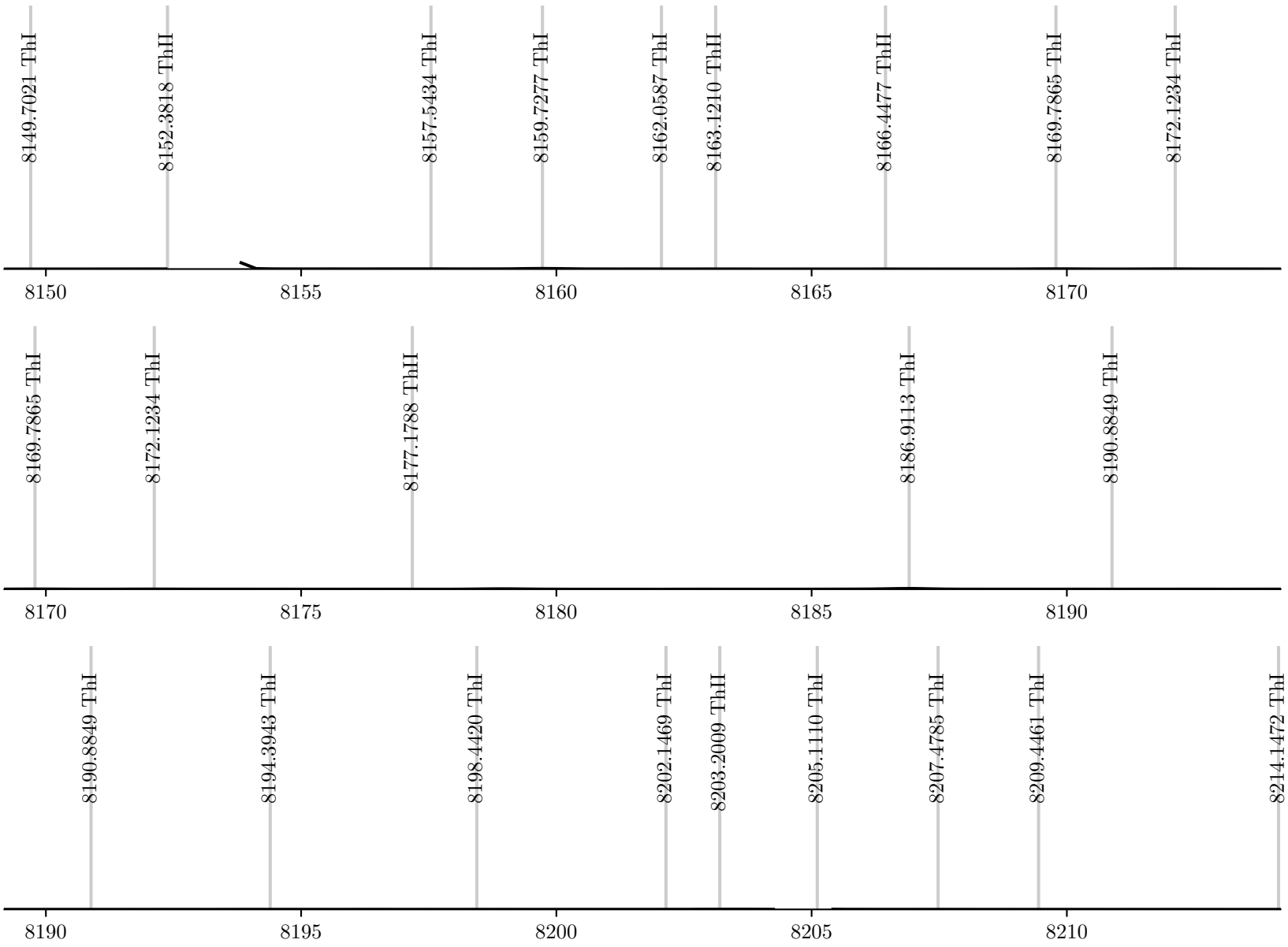


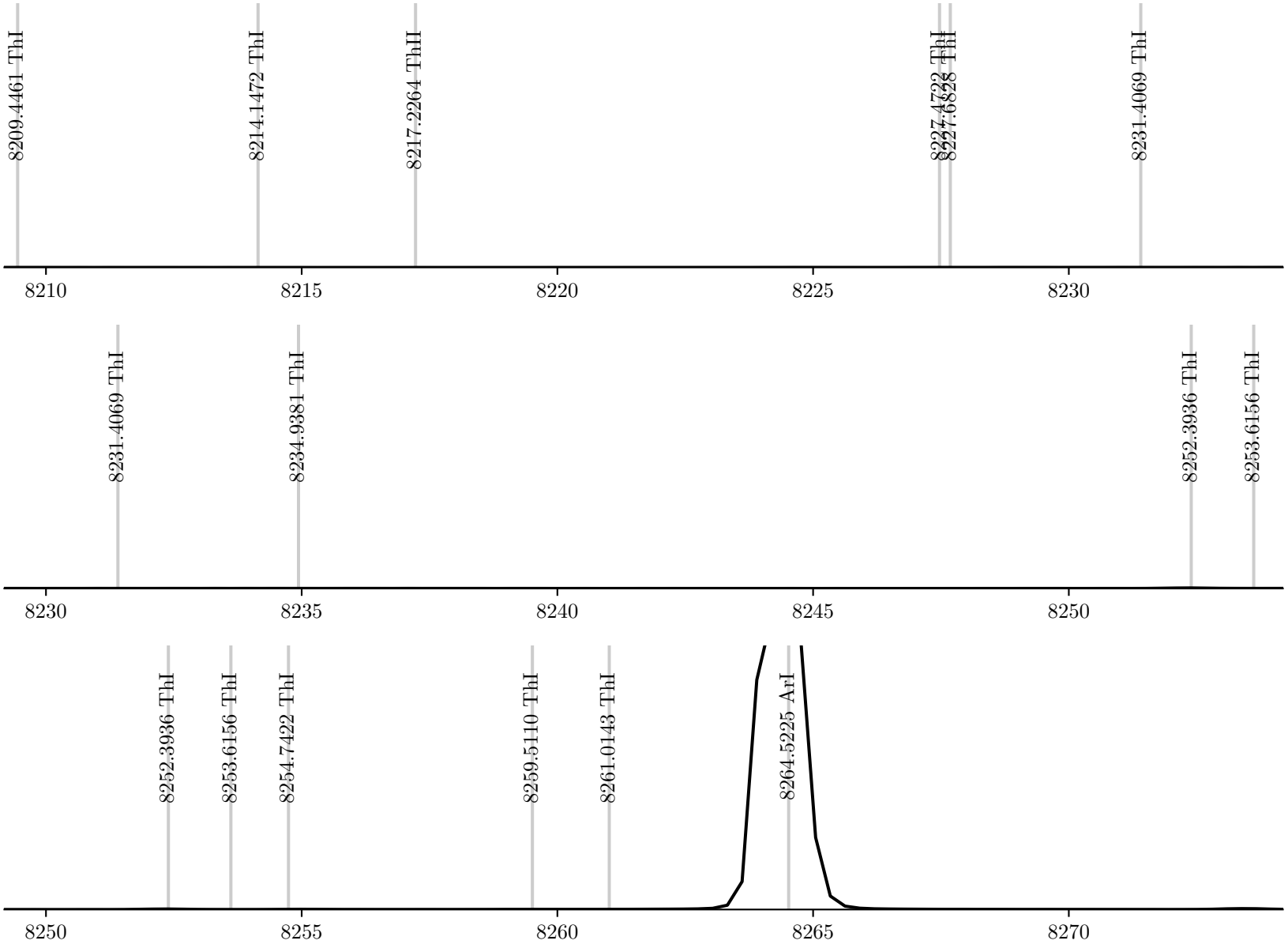
Dispersion:	0.287 Å pix^{-1}
Central wavelength:	8273 Å
Wavelength range:	7969 Å – 8556 Å

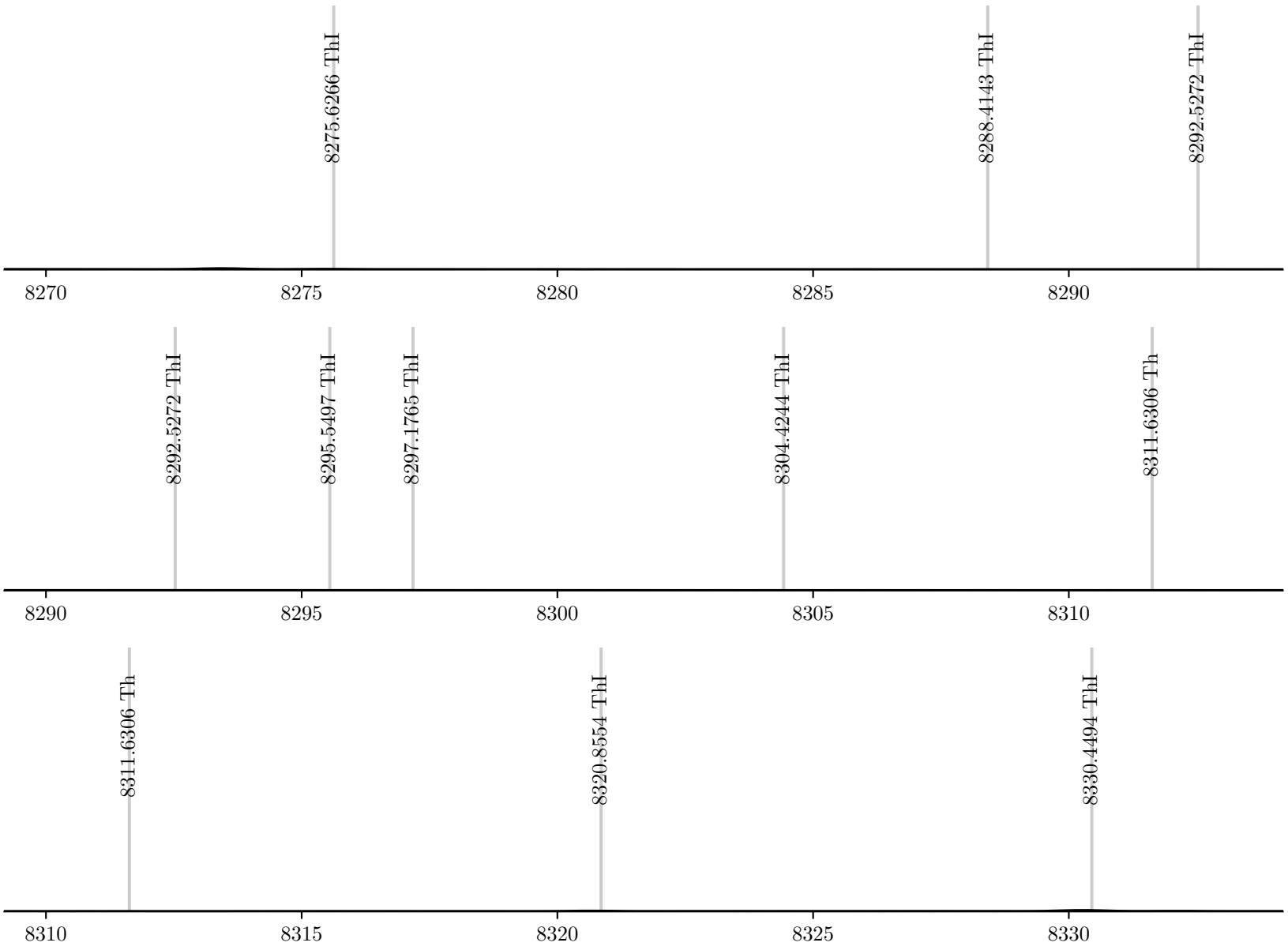

7970.2537 ThI
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7974.1592 ThI
7978.9731 ThI
7981.2262 ThII
7987.9731 ThI
7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
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7975
7980
7985
7990
7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
7996.9691 ThI
8000.0449 ThI
8003.4551 ThI
8006.1567 ArI
7990
7995
8000
8005
8010
8014.7857 ArI
8017.5275 ArII
8022.2014 ThI
8024.2530 ThI
8025.7271 ThI
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8032.4313 ThI
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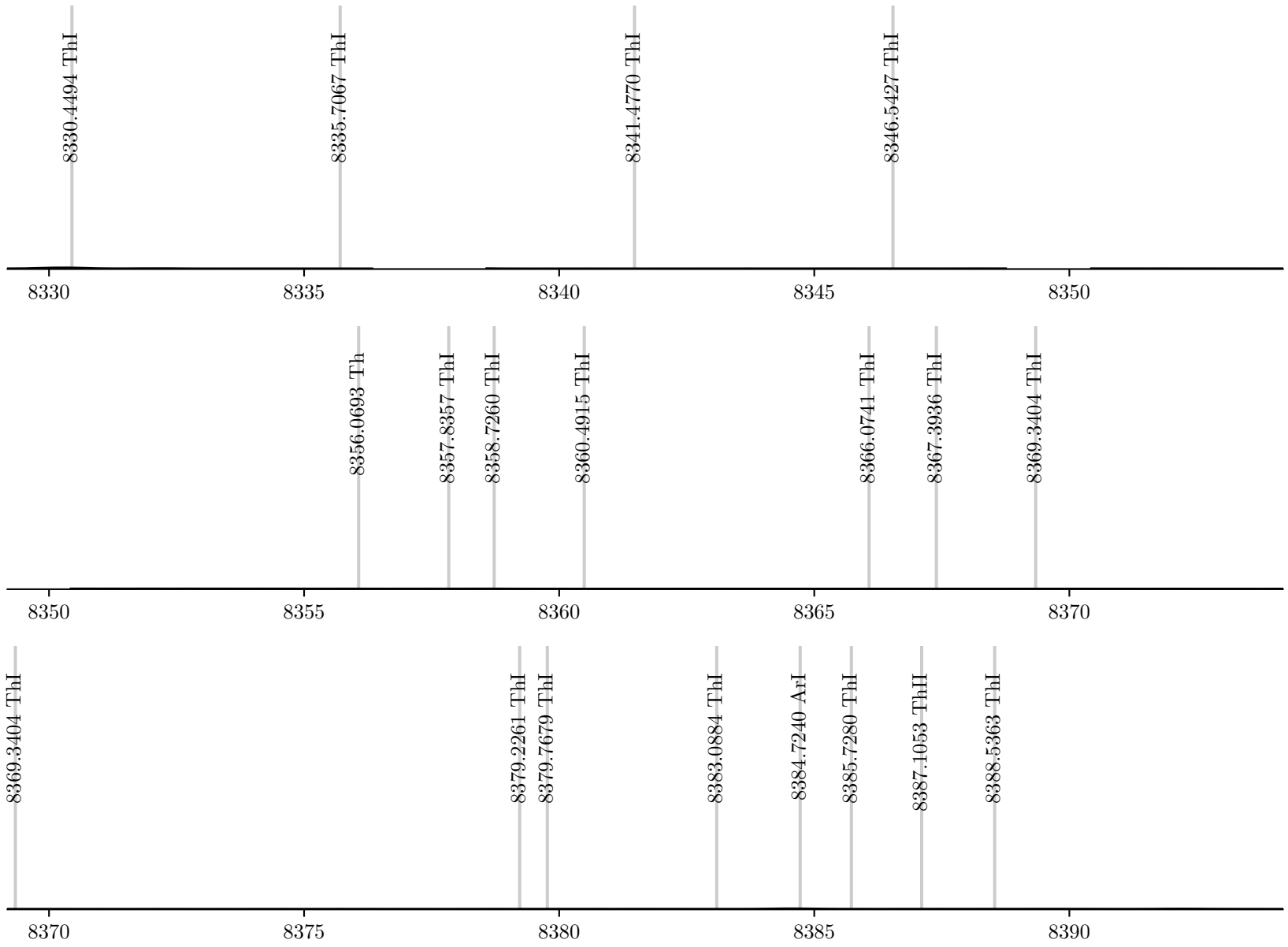

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8032.9061 ThI
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8046.1169 ArI
8050.6480 Th
8053.3085 ArI
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8045
8050
8050.6480 Th
8053.3085 ArI
8054.5355 ThII
8062.6304 ThI
8066.8267 ThI
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8055
8060
8065
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8075.6518 ThI
8085.2190 ThI
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8093.6238 ThI
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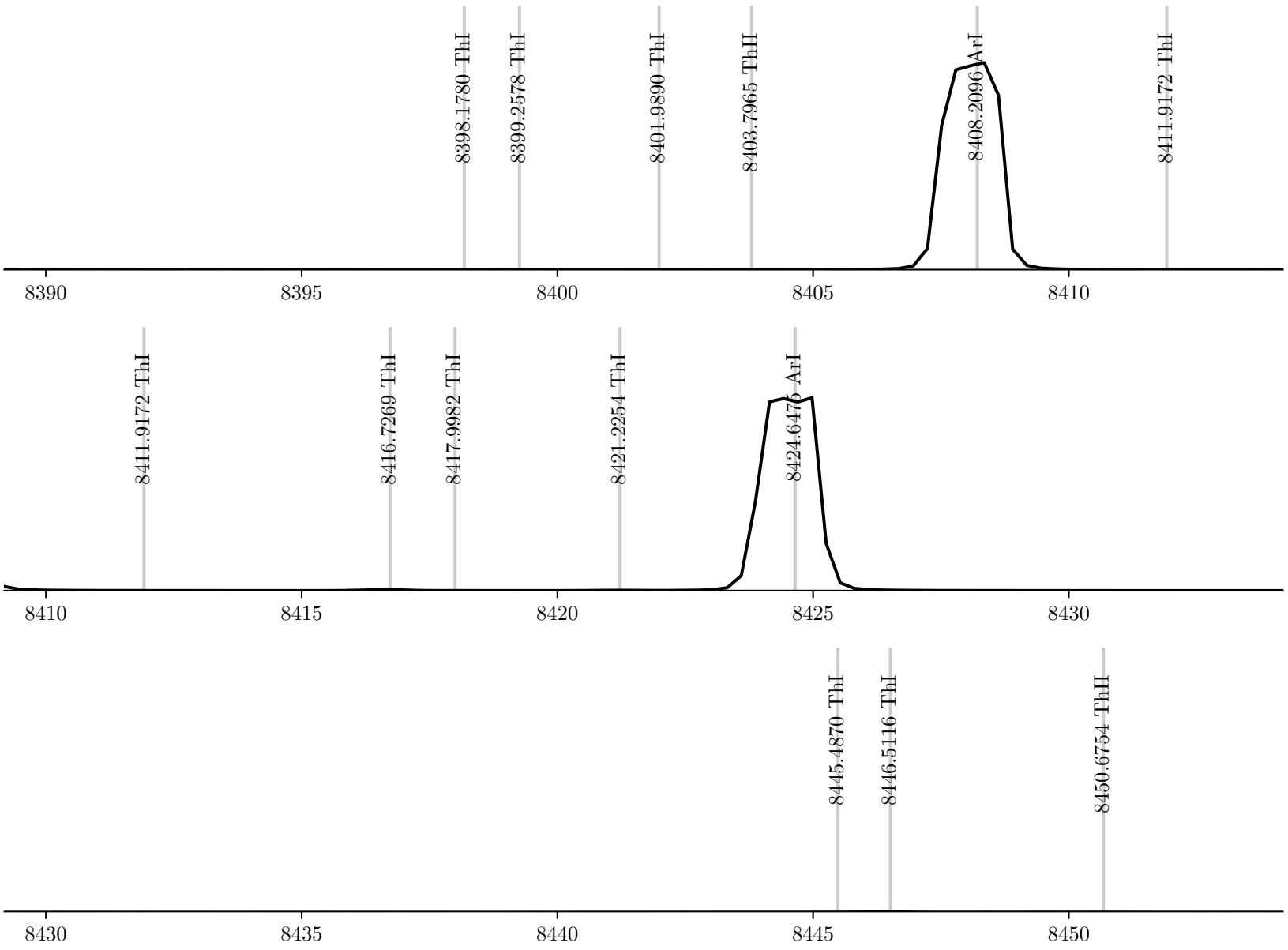

8089.4852 ThI
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8093.6238 ThI
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8105.3947 Th
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8115.3110 ArI
8119.1811 ArI
8122.7234 ThI
8129.4051 ThI
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8129.4051 ThI
8137.9363 ThI
8138.4753 ThI
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8150

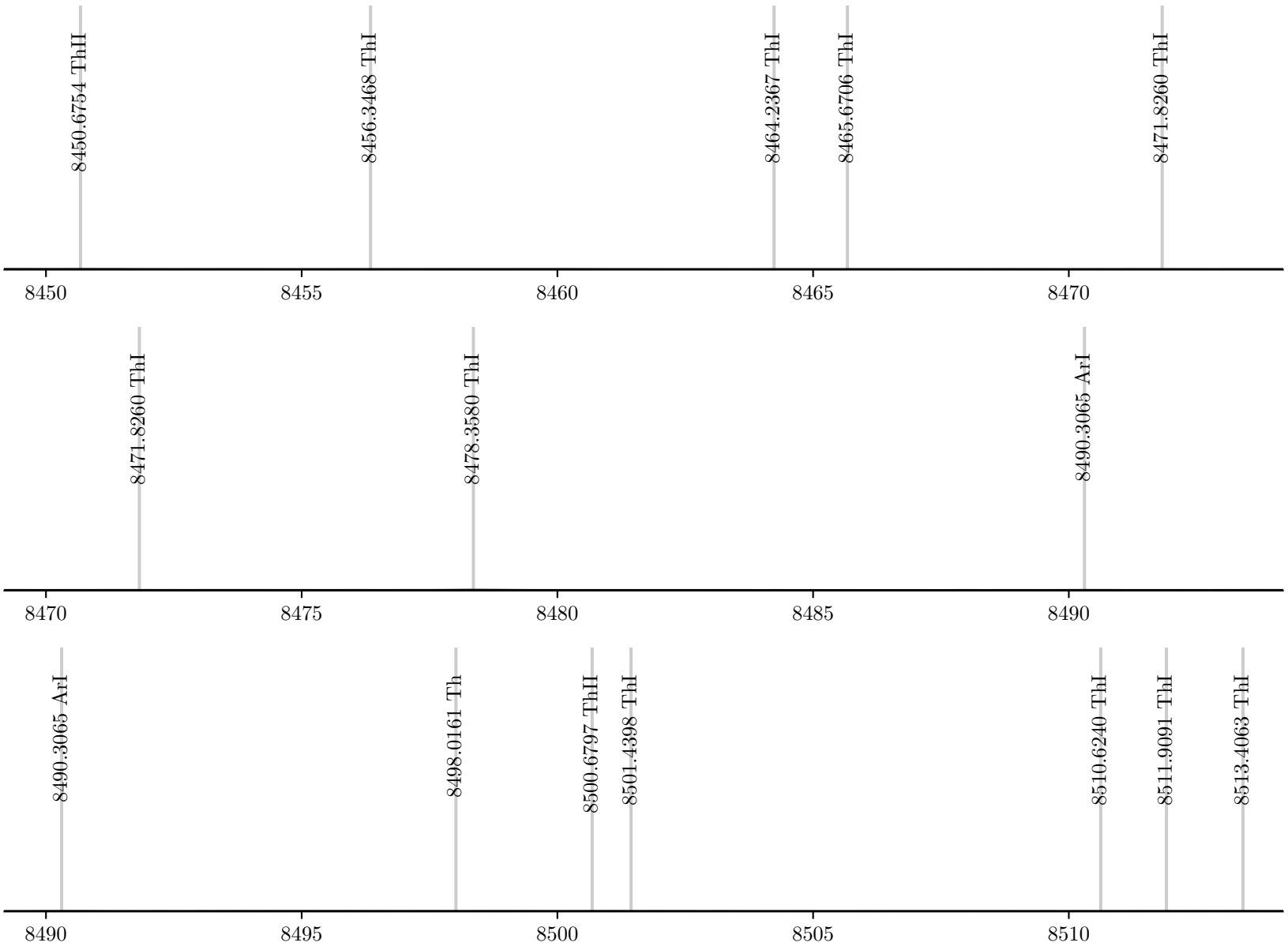

8149.7021 ThI
8152.3818 ThII
8157.5434 ThI
8159.7277 ThI
8162.0587 ThI
8163.1210 ThII
8166.4477 ThII
8169.7865 ThI
8172.1234 ThI
8150
8155
8160
8165
8170
8169.7865 ThI
8172.1234 ThI
8177.1788 ThII
8186.9113 ThI
8190.8849 ThI
8170
8175
8180
8185
8190
8190.8849 ThI
8194.3943 ThI
8198.4420 ThI
8202.1469 ThI
8203.2009 ThII
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8207.4785 ThI
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8205
8210

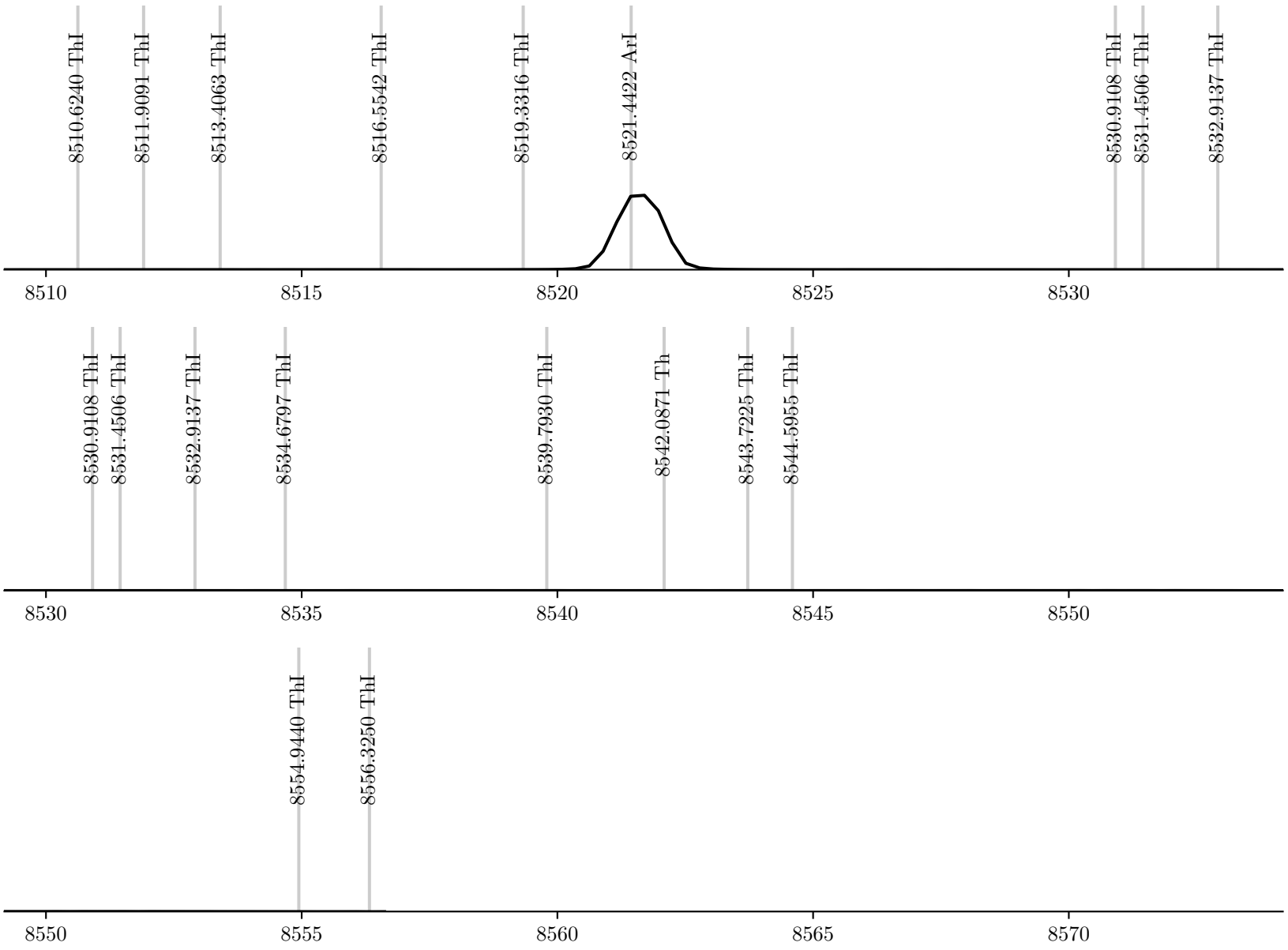

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8214.1472 ThI
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8227.6828 ThI
8231.4069 ThI
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8225
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8231.4069 ThI
8234.9381 ThI
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8253.6156 ThI
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8252.3936 ThI
8253.6156 ThI
8254.7422 ThI
8259.5110 ThI
8261.0143 ThI
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8275.6266 ThI
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8295.5497 ThI
8297.1765 ThI
8304.4244 ThI
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8330

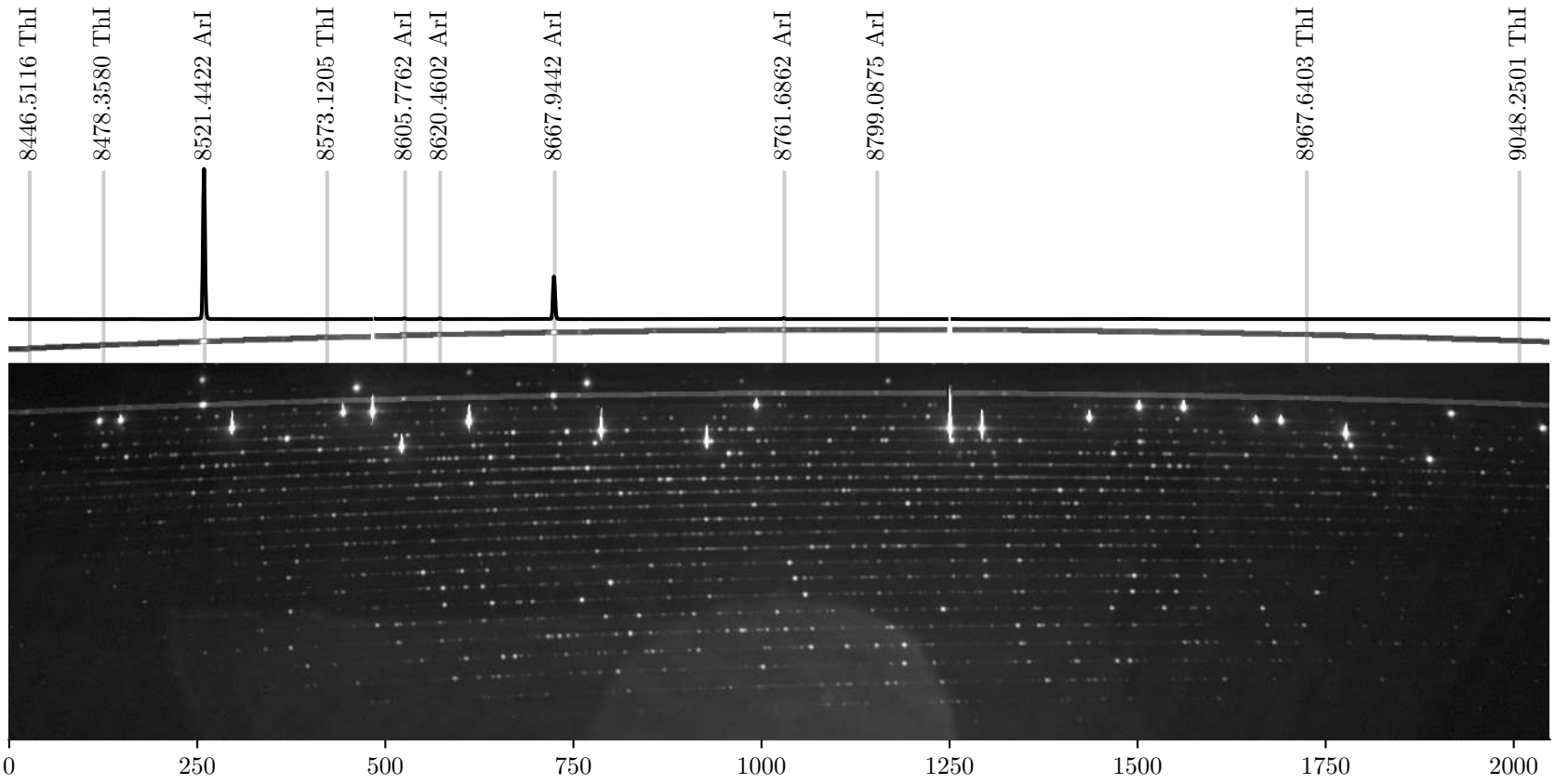

8330.4494 ThI
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8345
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8356.0693 Th
8357.8357 ThI
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8360.4915 ThI
8366.0741 ThI
8367.3936 ThI
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8369.3404 ThI
8379.2261 ThI
8379.7679 ThI
8383.0884 ThI
8384.7240 ArI
8385.7280 ThI
8387.1053 ThII
8388.5363 ThI
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8385
8390


8398.1780 ThI
8399.2578 ThI
8401.9890 ThI
8403.7965 ThII
8408.2096 ArI
8411.9172 ThI
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8395
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8405
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8411.9172 ThI
8416.7269 ThI
8417.9982 ThI
8421.2254 ThI
8424.6476 ArI
8410
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8445.4870 ThI
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8445
8450

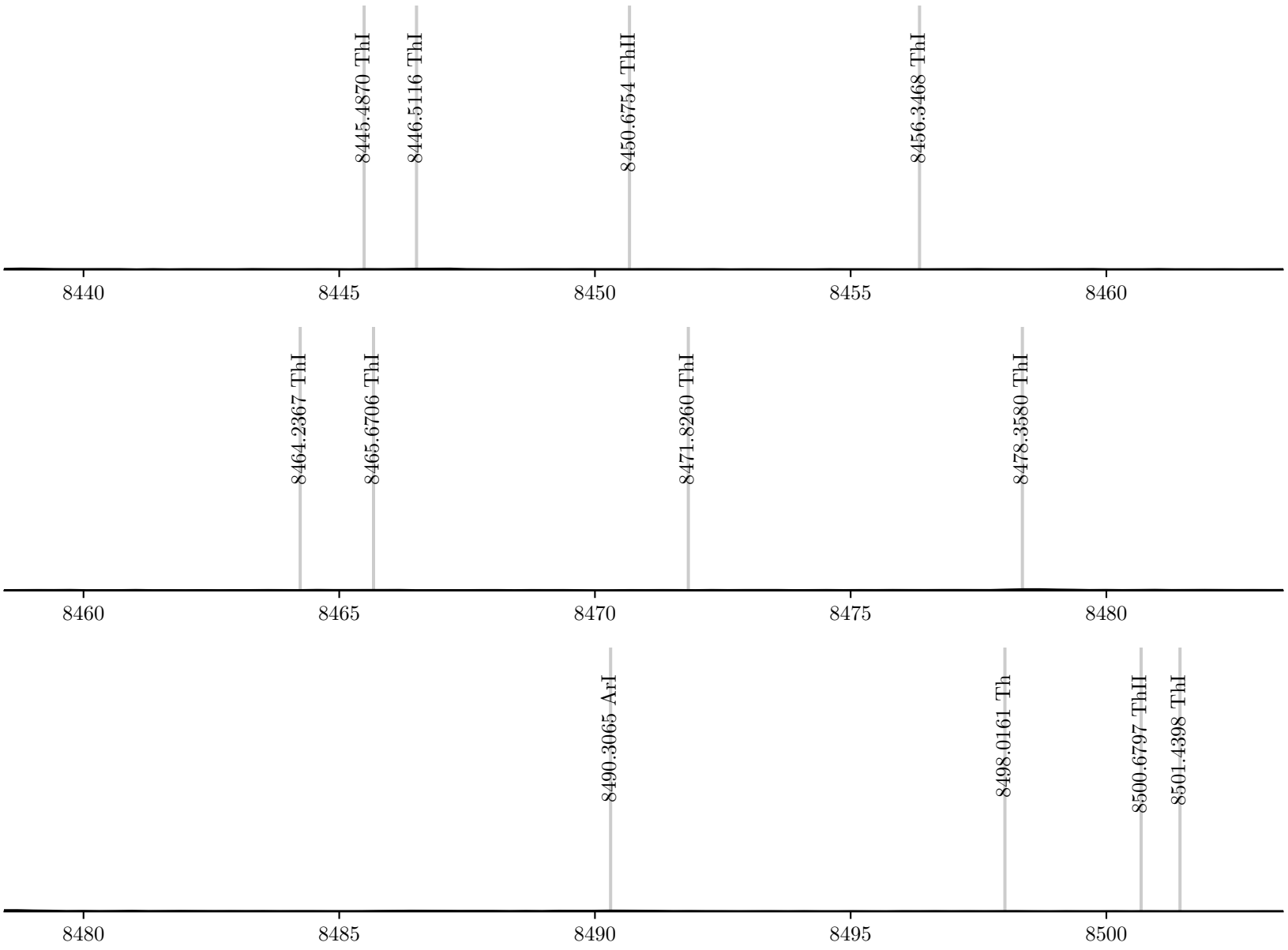

8450.6754 ThII
8456.3468 ThI
8464.2367 ThI
8465.6706 ThI
8471.8260 ThI
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8465
8470
8471.8260 ThI
8478.3580 ThI
8490.3065 ArI
8470
8475
8480
8485
8490
8490.3065 ArI
8498.0161 Th
8500.6797 ThII
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8510.6240 ThI
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8500
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8510

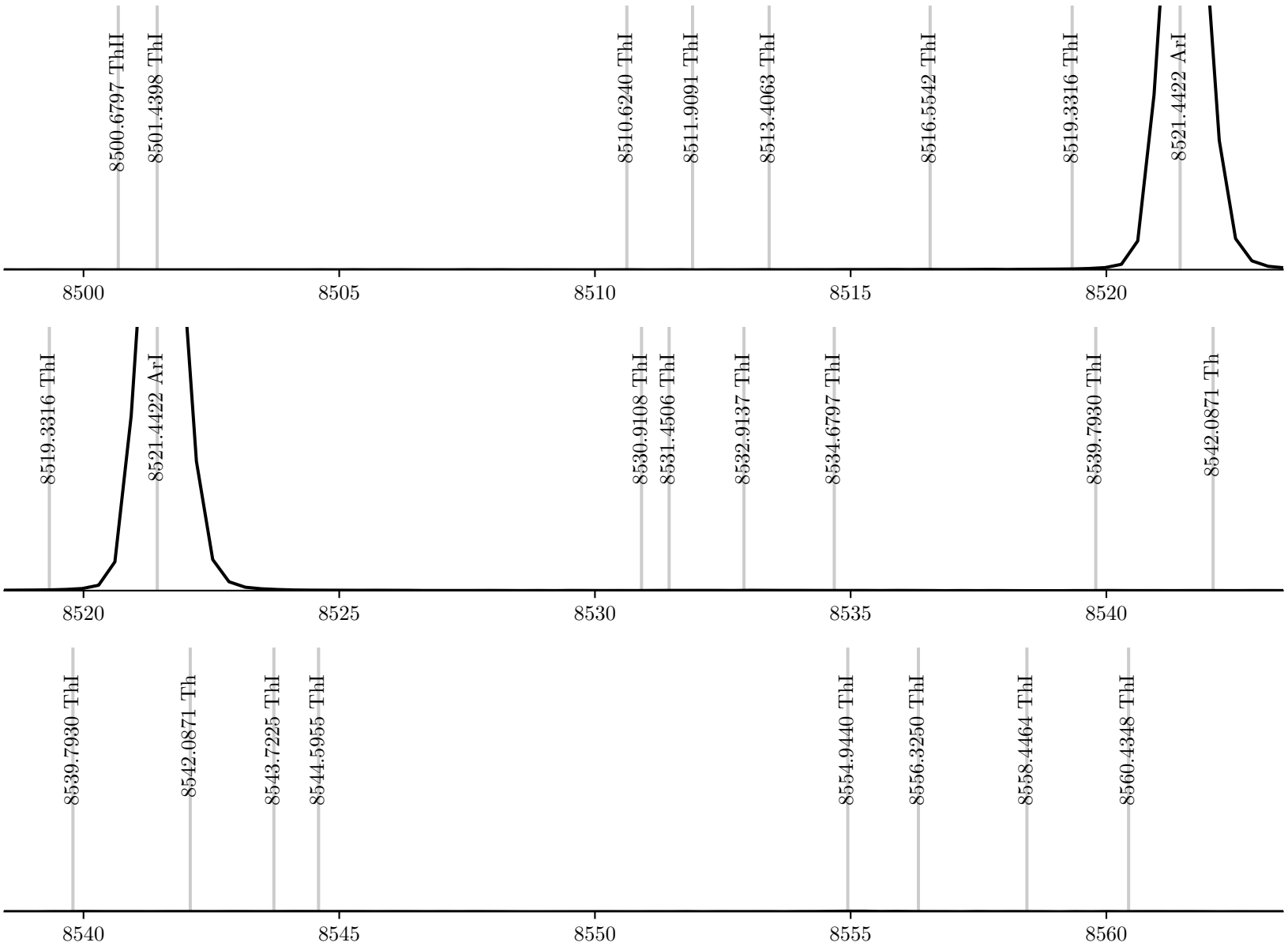

8510.6240 ThI
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8513.4063 ThI
8516.5542 ThI
8519.3316 ThI
8521.4422 ArI
8530.9108 ThI
8531.4506 ThI
8532.9137 ThI
8510
8515
8520
8525
8530
8530.9108 ThI
8531.4506 ThI
8532.9137 ThI
8534.6797 ThI
8539.7930 ThI
8542.0871 Th
8543.7225 ThI
8544.5955 ThI
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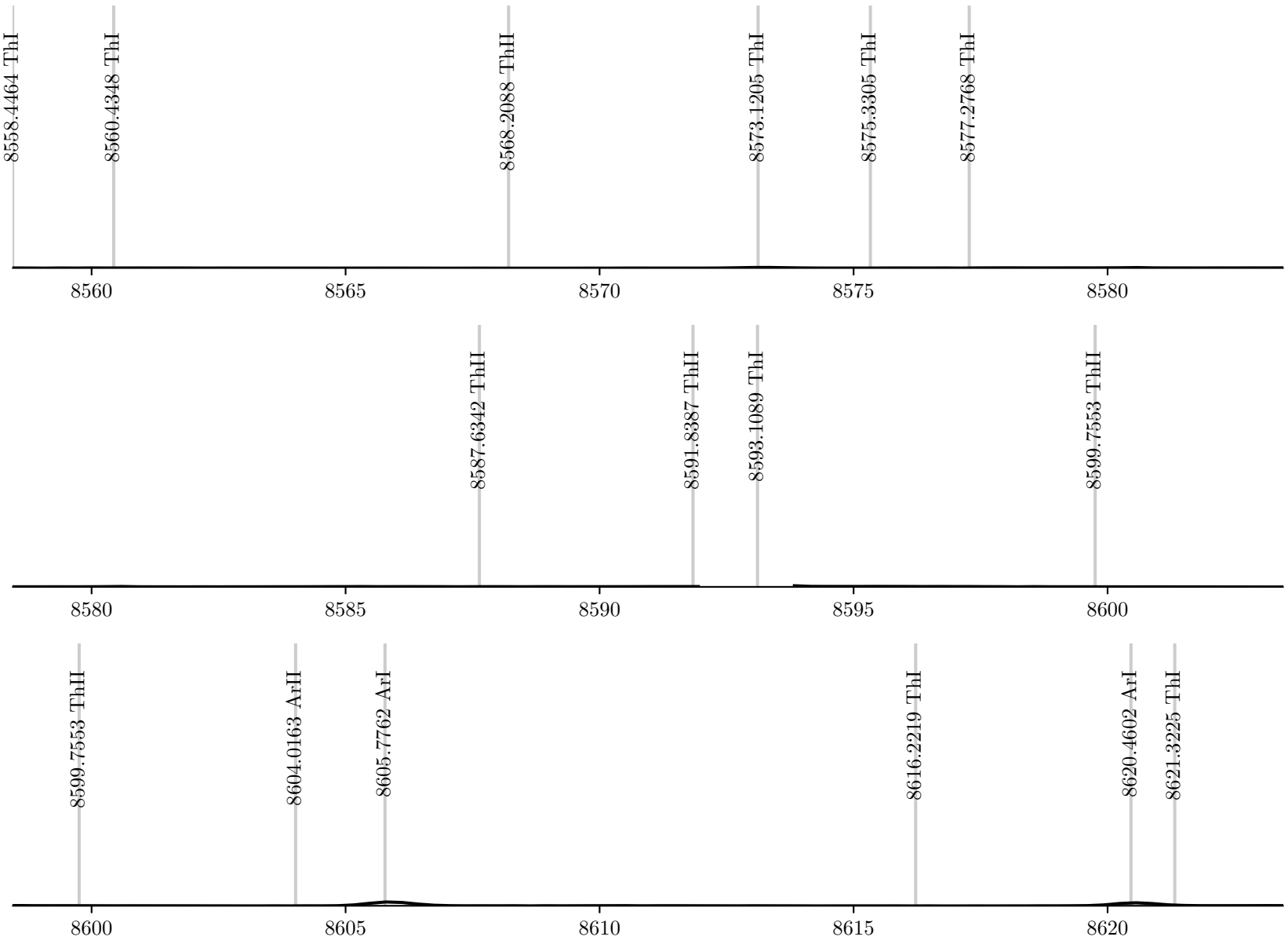
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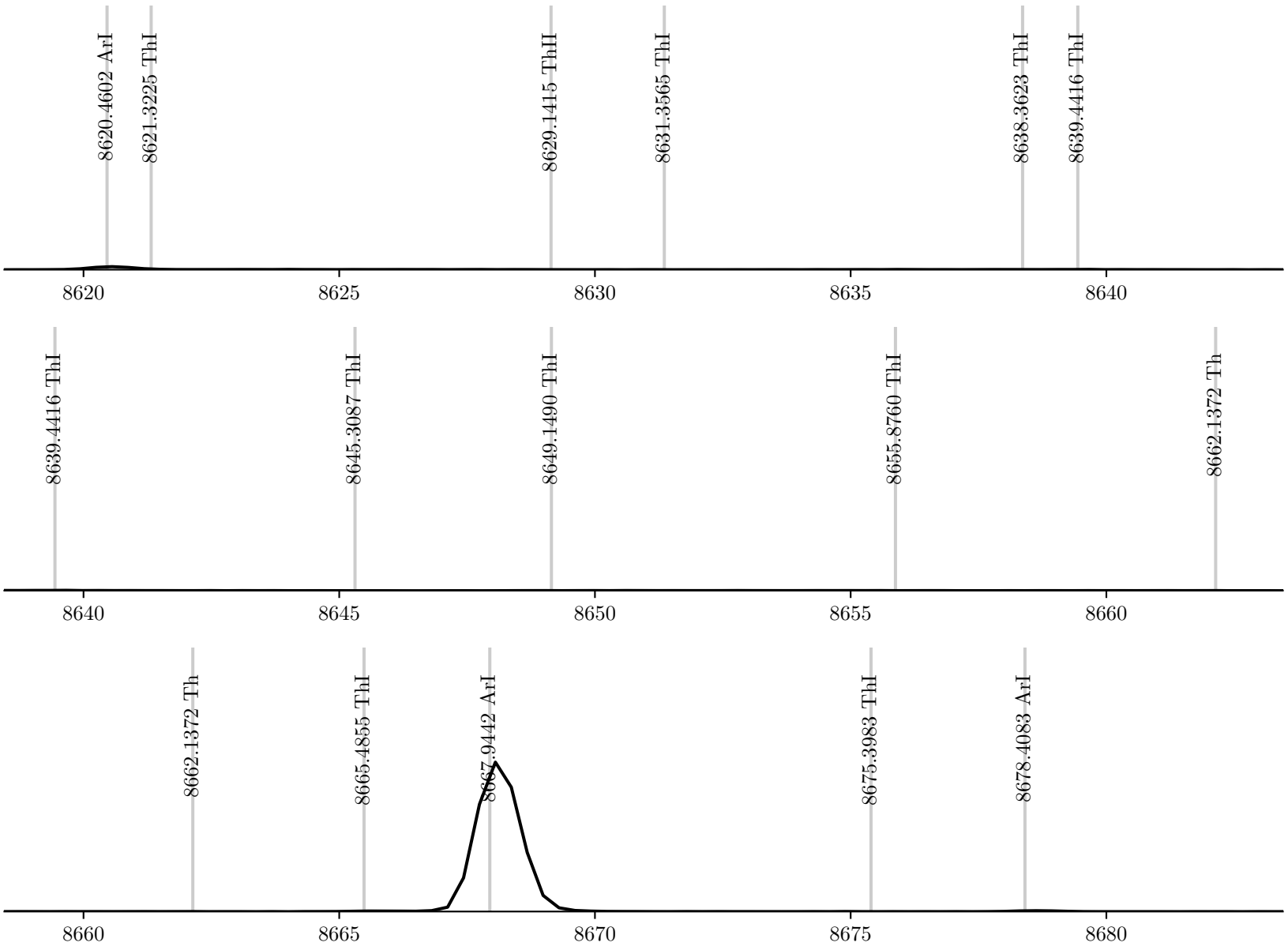


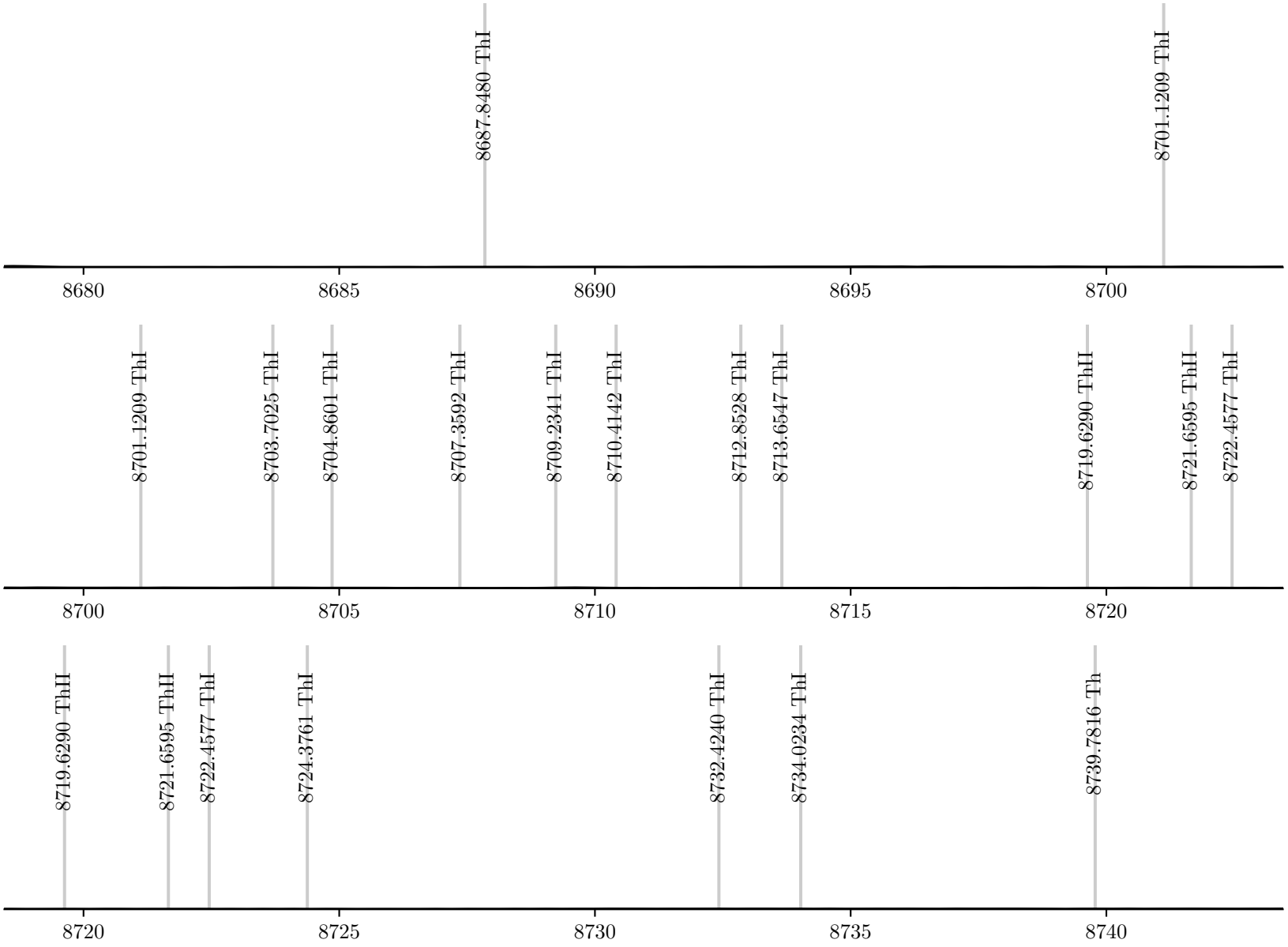
Dispersion:	0.304 Å pix^{-1}
Central wavelength:	8760 Å
Wavelength range:	8438 Å – 9060 Å

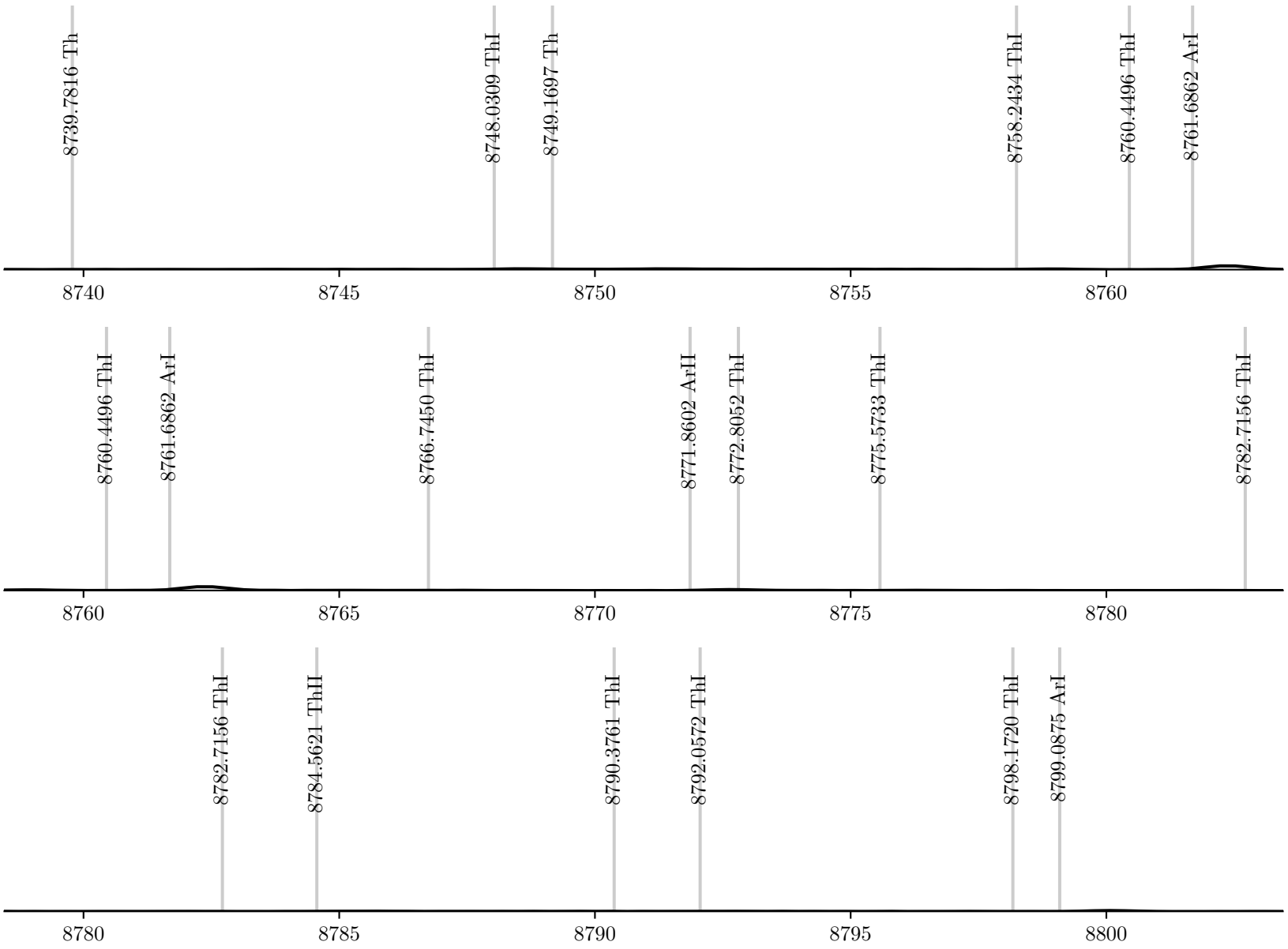

8445.4870 ThI
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8471.8260 ThI
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8470
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8500

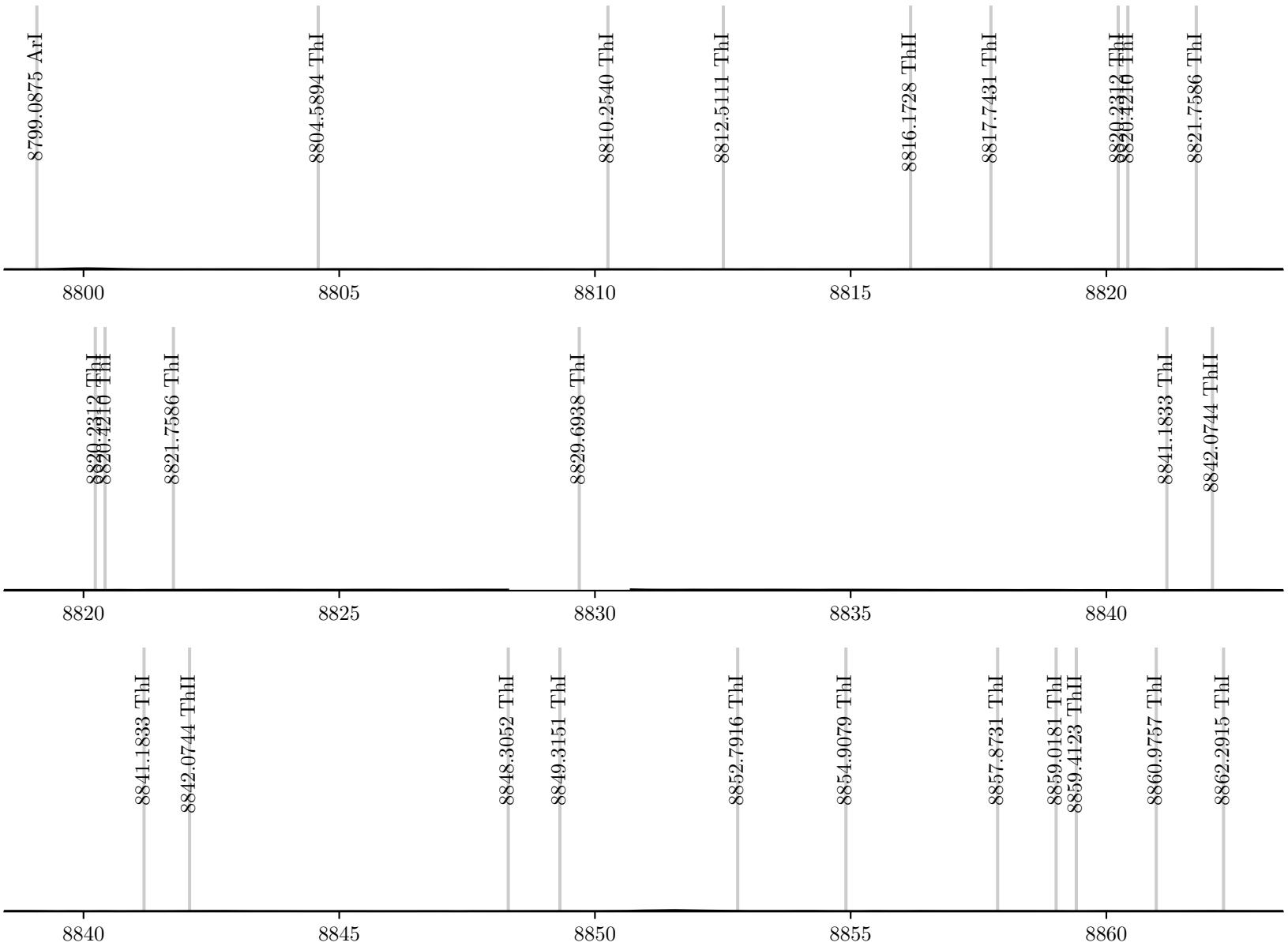

8500.6797 ThII
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8520
8519.3316 ThI
8521.4422 ArI
8530.9108 ThI
8531.4506 ThI
8532.9137 ThI
8534.6797 ThI
8539.7930 ThI
8542.0871 Th
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8525
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8539.7930 ThI
8542.0871 Th
8543.7225 ThI
8544.5955 ThI
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8560.4348 ThI
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8560

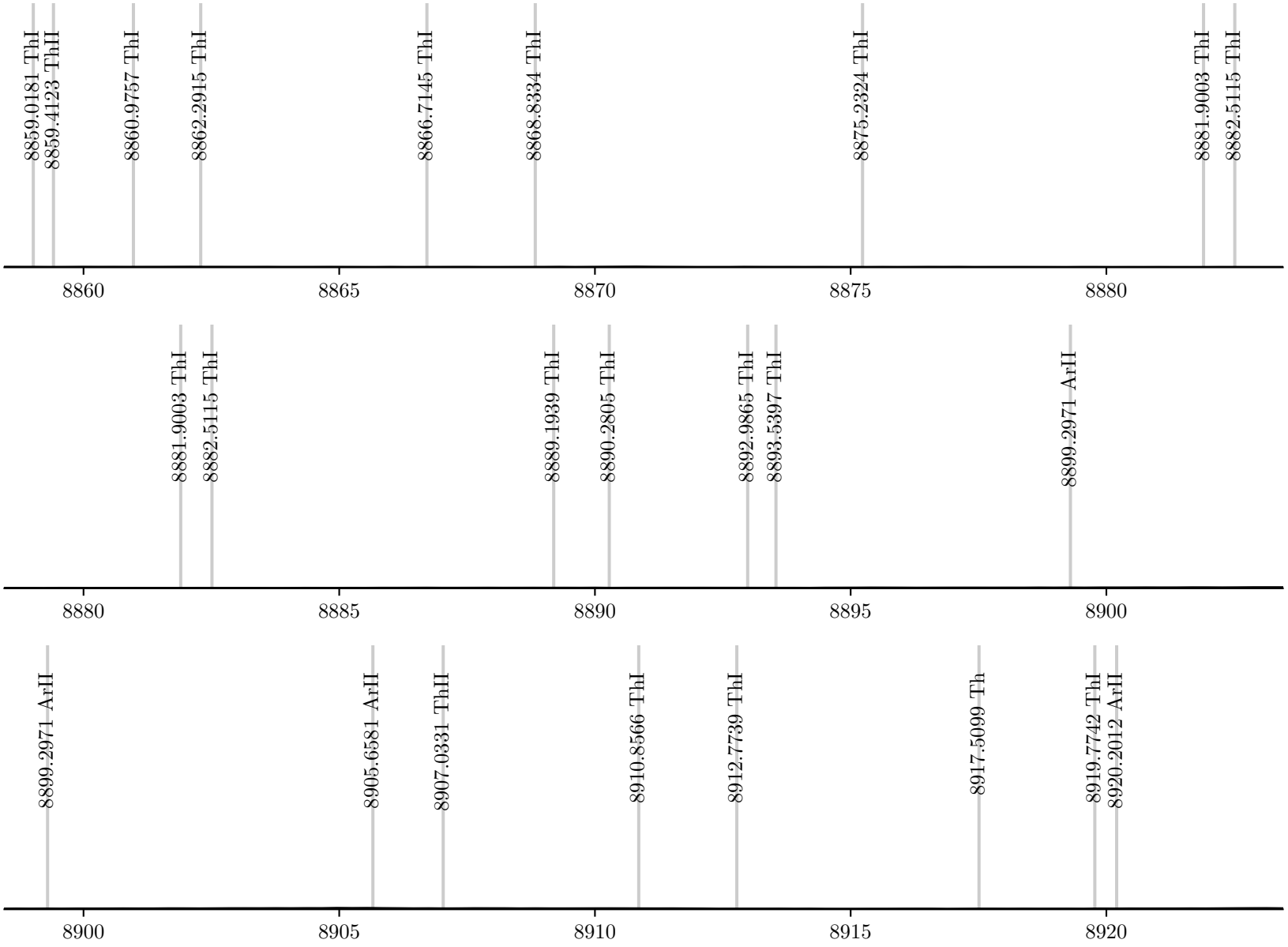

8558.4464 ThI
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8575.3305 ThI
8577.2768 ThI
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8580
8587.6342 ThII
8591.8387 ThII
8593.1089 ThI
8599.7553 ThII
8580
8585
8590
8595
8600
8599.7553 ThII
8604.0163 ArII
8605.7762 ArI
8616.2219 ThI
8620.4602 ArI
8621.3225 ThI
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8610
8615
8620

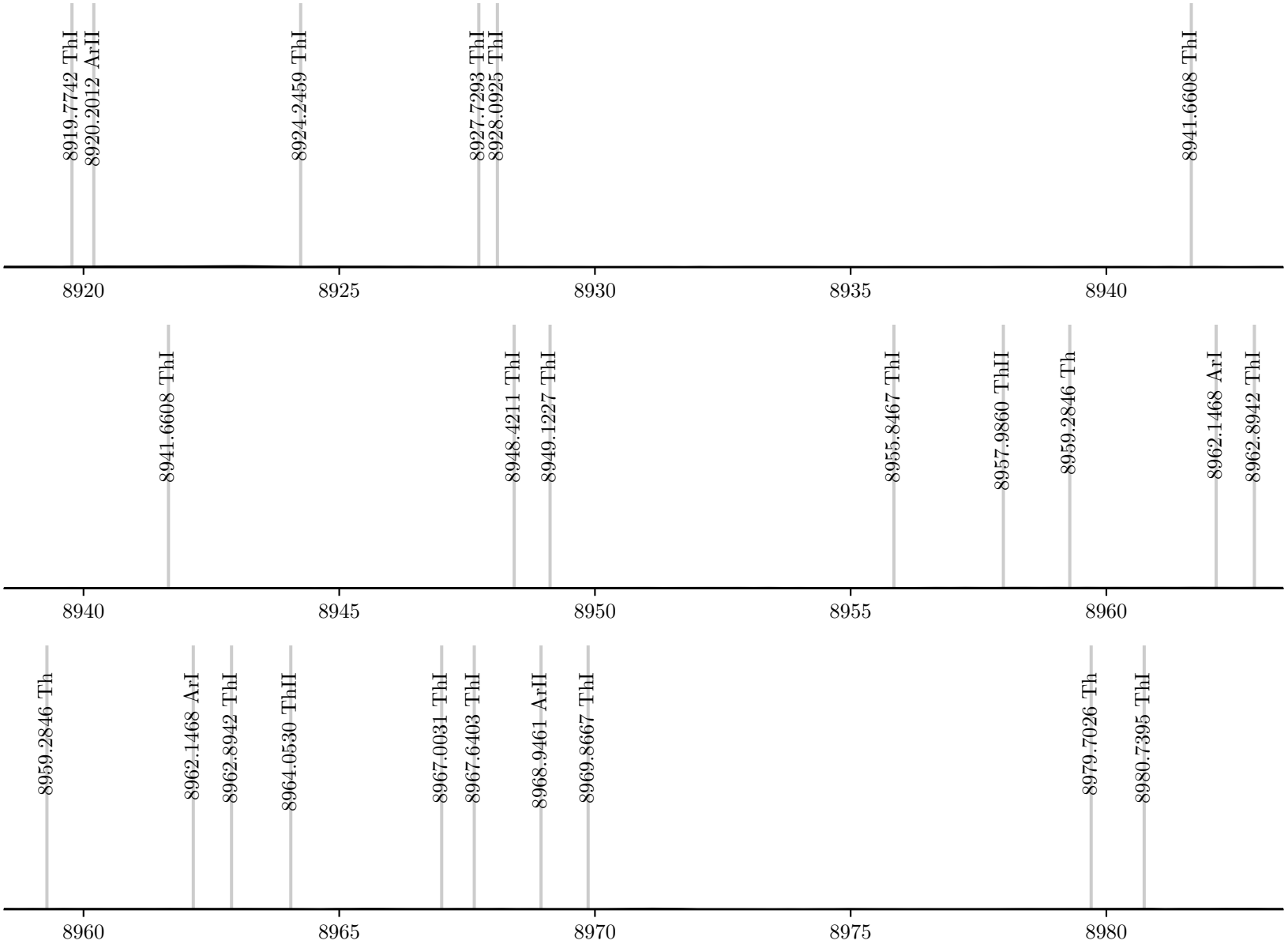

8620.4602 ArI
8621.3225 ThI
8629.1415 ThII
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8639.4416 ThI
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8639.4416 ThI
8645.3087 ThI
8649.1490 ThI
8655.8760 ThI
8662.1372 Th
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8655
8660
8662.1372 Th
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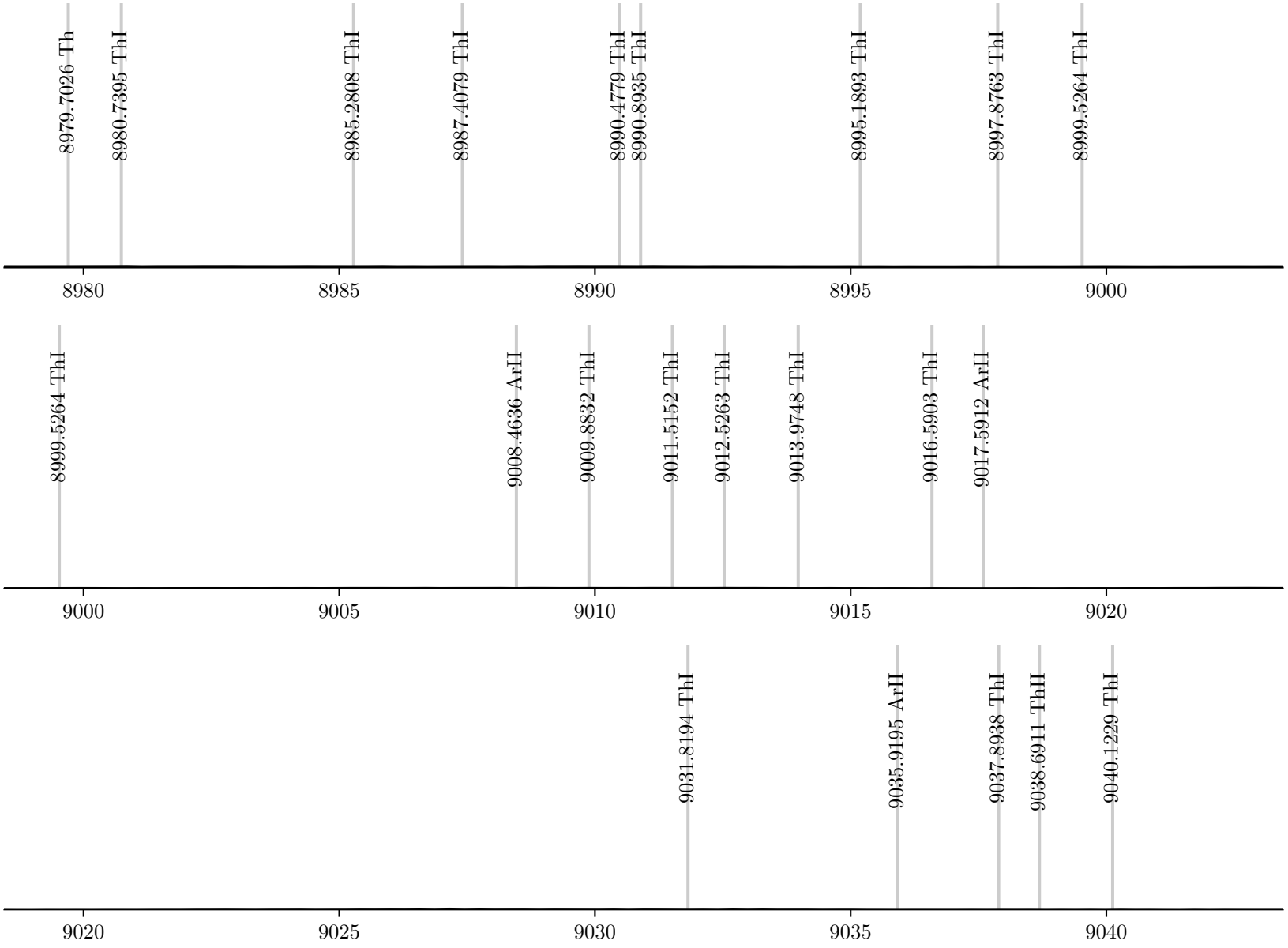

8687.8480 ThI
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8719.6290 ThII
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8722.4577 ThI
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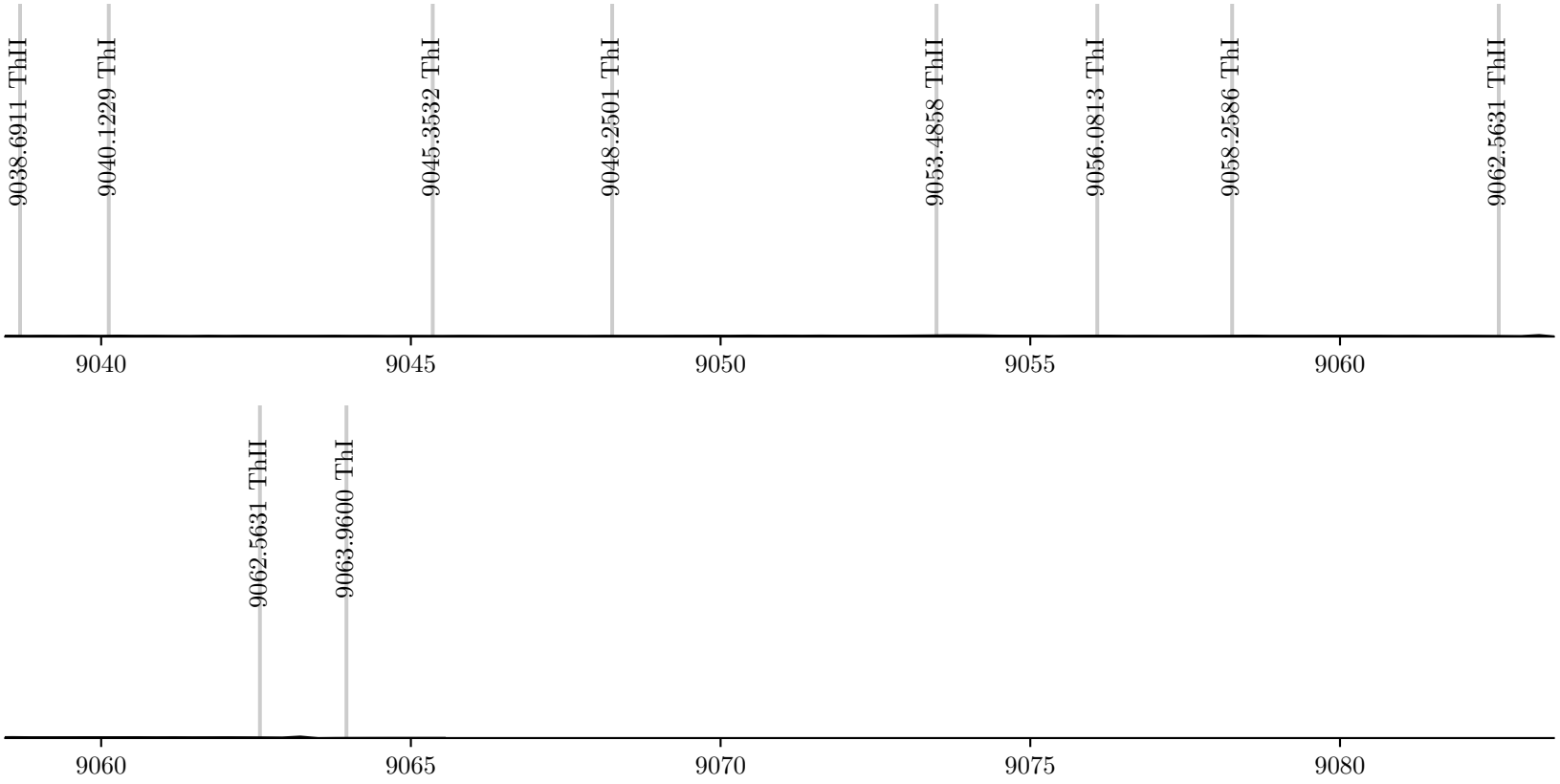

8739.7816 Th
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8749.1697 Th
8758.2434 ThI
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8761.6862 ArI
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8792.0572 ThI
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8799.0875 ArI
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8800


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8812.5111 ThI
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8817.7431 ThI
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8820.4210 ThI
8821.7586 ThI
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8821.7586 ThI
8829.6938 ThI
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8859.4123 ThII
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8860


8859.0181 ThI
8859.4123 ThII
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8862.2915 ThI
8866.7145 ThI
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8899.2971 ArII
8905.6581 ArII
8907.0331 ThII
8910.8566 ThI
8912.7739 ThI
8917.5099 Th
8919.7742 ThI
8920.2012 ArII
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8915
8920


8919.7742 ThI
8920.2012 ArII
8924.2459 ThI
8927.7293 ThI
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8941.6608 ThI
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8948.4211 ThI
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8955.8467 ThI
8957.9860 ThII
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8962.1468 ArI
8962.8942 ThI
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8959.2846 Th
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8964.0530 ThII
8967.0031 ThI
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8980


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9040.1229 ThI
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9040
9045
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9055
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9062.5631 ThII
9063.9600 ThI
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